

**ABET
Self-Study Report**

For the

**Chemical Engineering
Program**

At

The University of Illinois Chicago

Chicago, IL 60607



July 1, 2026

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**Program Self-Study Report
for
EAC of ABET
Accreditation or Reaccreditation**

BACKGROUND INFORMATION

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B. Program History

The College of Engineering was one of the original Colleges when University of Illinois at Chicago was founded in 1965 (originally named University of Illinois at Chicago Circle – renamed in 1982 after a merger with the University of Illinois at the Medical Center). Chemical Engineering started as a program within the Department of Energy Engineering. The other programs in the Department were Thermo-Mechanical Engineering and Fluids Engineering. In 1982 the College of Engineering was reorganized, and Chemical Engineering became one of its six constituent Departments. In 1986 the Department moved to 810 S Clinton Street (Chemical Engineering Building, CEB), with offices, laboratories and classrooms located conveniently in the same building. In 2001 the department began offering Chemical Engineering as a minor to students not enrolled in the Chemical Engineering undergraduate program. In 2003 the Department initiated the area of concentration in Biochemical Engineering. In 2022, the Department added 5 more areas of concentration, Process Automation, Energy and Environment, Polymers and Molecular Engineering, Nanotechnology, and Entrepreneurship. In 2019, the Department moved to the newly constructed Engineering Innovation Building (EIB) at 929 W Taylor Street. The Chemical Engineering Department is the main tenant with all offices, laboratories, and classrooms located in EIB. The Civil and Materials Engineering and Mechanical and Industrial Engineering Departments also have some faculty offices and laboratories in the building.

The last general review took place October 18-20, 2020. Since then, there has been some turnover in the faculty, but also an increase in the total number of faculty and affiliate faculty. Two faculty members moved to other academic institutions. Five faculty were promoted to full professors, and one faculty member was promoted to clinical professor. Five new faculty were hired: one full

professor, one associate professor, one clinical assistant professor, and two research assistant professors. One adjunct professor (from AbbVie) was hired as a co-instructor for the Biopharmaceutical Process Development and Manufacturing technical elective course (CHE 426). The Department also has four Professor Emeriti, and 19 adjunct/affiliate faculty, including from other departments in the university, other universities, and from industry.

Improvements have been made to the curriculum. Two courses were added to the curriculum. CHE 230 Molecular Systems in Chemical Engineering was added in 2022 and is designed to introduce several topics that are important to traditional and emerging fields in chemical engineering; molecular engineering, macromolecular systems, and nanomaterials. CHE 330 Polymer Science was added as a selective option with CHEM 346 Physical Chemistry II and gives our students the option to either learn about polymer science or focus on advanced topics in physical chemistry. In addition to the Biochemical Engineering concentration, five concentrations to the BS in Chemical Engineering degree were added in 2022: Polymers and Molecular Engineering, Energy and Environment, Nanotechnology, Process Automation, and Entrepreneurship. Two technical elective courses were also added; a course on the fundamentals of biopharmaceutical process development and manufacturing (CHE 426), and a course on engineering entrepreneurship (CHE 427). In addition, a General Education course was created in 2022, CHE 150 Climate Engineering for Global Warming. Note: this course was developed specifically for non-engineering majors to satisfy their Analyzing the Natural World requirement and is not part of our program requirements.

As part of the College of Engineering effort to improve the freshmen/transfer student Engineering Orientation course (ENGR100), 8-week major specific sections were added, with undergraduate mentors leading the lecture/discussion periods. In addition, the seminar series in the senior-level professional development course (CHE 499) was improved by adding more women engineering and academic-oriented speakers discussing their careers in a wide variety of industries and non-traditional areas (patent law, for instance). The Senior Design II (CHE 397) project experience is continually being improved by a team of industrial mentors who meet regularly with their student groups and suggest relevant and topical design projects each year. In addition to the College Design Expo, a Chemical Engineering Design Showcase was added where professional chemical engineers judge the senior design projects. Ethics and safety learning continues to be taught in the curriculum. In addition, the students have a more diverse set of research topics to choose from when taking the undergraduate research course (CHE 392) because of the new faculty that were hired. A CHE 392 course section has also been dedicated for the AIChE ChemE Car and ChemE Cube competitions to allow students to obtain technical elective credit while performing research for the annual competitions.

In addition, ChemEConnect, a student-led Department Action Team (DAT) within the Chemical Engineering department was created and acts as a departmental student advisory/community group to enhance the student experience. The team consists of undergraduate and graduate students, department staff, and faculty who meet regularly to discuss and address student feedback. For instance, ChemEConnect, in collaboration with the AIChE student chapter, has initiated student events like the Fall semester Welcome Back Picnic, the Spring semester End-of-Year banquet, and numerous professional development opportunities throughout the year.

C. Options

BS in Chemical Engineering — Biochemical Engineering Concentration

This concentration entails the completion of 8–10 semester hours distributed among CHE 422—Biochemical engineering (3 cr. hr.) as the technical elective and two elective courses in the non-major rubric category chosen from among BIOS 350—General Microbiology (3 cr. hr.), BIOS 351—Microbiology Laboratory (2 cr. hr.), CHEM 352—Introductory Biochemistry (3 cr. hr.) and CHEM 452—Biochemistry I (4 cr. hr.)

BS in Chemical Engineering — Energy and Environment Concentration

This concentration entails the completion of 9 semester hours distributed among the technical elective CHE 451—Renewable Energy Technologies (3 cr. hr.), one non-major technical elective CME 322 Environmental Engineering (3 cr.hr.) and a 2nd ChE technical elective from a selective list of courses (3 cr.hr.).

BS in Chemical Engineering — Entrepreneurship Concentration

This concentration entails the completion of 6 semester hours distributed among the technical elective CHE 427—Entrepreneurship in Engineering (3 cr. hr.), and an elective course selected from the following courses; ENTR 310, ENTR 200 ENTR 445 or a 400-level ChE technical elective approved by the director of undergraduate studies (3 cr.hr.).

BS in Chemical Engineering — Nanotechnology Concentration

This concentration entails the completion of 9 semester hours distributed among the technical elective CHE 455 Nanoscale Systems in Chemical Engineering (3 cr. hr.), an elective course selected from the following non-major courses; ECE 346, ME 418, PHYS 431, ECE 440, ECE 449, CHEM 458 or BIOE 485, and a second ChE technical from a selective list of courses (3 cr.hr.).

BS in Chemical Engineering — Polymers and Molecular Engineering Concentration

This concentration entails the completion of 9 semester hours distributed among the technical elective CHE 454 Molecular and Macromolecular Engineering (3 cr. hr.), an elective course selected from the following non-major courses; PHAR 422, PHAR 423, PHYS 450, CHEM 452, CHEM 458, PHYS 461, BIOE 485, and a second ChE technical from a selective list of courses (3 cr.hr.).

BS in Chemical Engineering — Nanotechnology Concentration

This concentration entails the completion of 9 semester hours distributed among the technical elective CHE 455 Nanoscale Systems in Chemical Engineering (3 cr. hr.), an elective course selected from the following non-major courses; ECE 346, ME 418, PHYS 431, ECE 440, ECE 449, CHEM 458 or BIOE 485, and a second ChE technical from a selective list of courses (3 cr.hr.).

BS in Chemical Engineering — Process Automation Concentration

This concentration entails the completion of 9 semester hours distributed among the technical elective CHE 433 Process Automation with Aspen Plus (3 cr. hr.), and two elective courses selected from the following non-major courses; ECON 120, IE 201, MATH 310, MATH 419 (6 cr.hr.).

Minor in Chemical Engineering (for non-ChE majors)

Students outside the Department of Chemical Engineering who wish to minor in Chemical Engineering must complete a specified sequence of classes totaling 16-18 semester hours (excluding prerequisite courses). This sequence includes three courses in Chemical Engineering:

CHE 210—Material and Energy Balances (4 cr. hr.),

CHE 301—Chemical Engineering Thermodynamics (3. cr. hr.),

CHE 321—Chemical Reaction Engineering (3 cr. hr.),

One fluid mechanics course chosen from among

CHE 311—Transport Phenomena I (3 cr. hr.)

ME 211—Fluid Mechanics I (4 cr. hr.)

and one heat or mass transfer course chosen from among

CHE 312—Transport Phenomena II (3 cr. hr.)

ME 321—Heat Transfer (4 cr. hr.)

CHE 313—Transport Phenomena III (3 cr. hr.)

Minors appear on a student's transcript but are not listed on the UIC diploma. Among the small number of Engineering minors, none have yet been minors in Chemical Engineering since 2004.

D. Program Delivery Modes

The Bachelor of Science in Chemical Engineering is offered to interested students in conventional day classes (lecture / laboratory / recitation). The program can be completed by full time students in four years by taking 15-18 credits per semester.

E. Program Locations

The UIC Chemical Engineering program is offered only on campus.

F. Public Disclosure

The Chemical Engineering Department's Program Educational Objectives (PEOs), Student Outcomes (SOs), annual student enrollment, and graduation data is available online to the public at the following URL.

<https://che.uic.edu/undergraduate-programs/che-major/>

G. Deficiencies, Weaknesses or Concerns from Previous Evaluation(s) and the Actions Taken to Address Them

For the 2020 accreditation cycle the ABET Final Statement (dated August 24, 2021) contained no Deficiencies, Weaknesses, or Concerns.

GENERAL CRITERIA

CRITERION 1. STUDENTS

The evaluation of Chemical Engineering students takes three forms:

1. Evaluation at the time of admission to the BSChemE degree
2. Evaluation of progress towards the BSChemE degree
3. Evaluation for completion of the BSChemE degree

These evaluation processes are discussed in individual sections below for both incoming freshman and transfer students.

A. Student Admissions

Students are admitted directly into the Chemical Engineering Program in the College of Engineering as: a new freshman; an internal transfer student from another UIC college; or a transfer student from an institution other than UIC. The general evaluation procedures for new freshman and internal transfer students are described below. Evaluation of transfer students from outside UIC is discussed later in this section.

Freshman students

The two primary factors used to determine the admissibility of a new freshman applying to the University of Illinois at Chicago are the high school percentile rank (HSPR) and the American College Test (ACT) composite score, if submitted. The University will also accept Scholarship Aptitude Test (SAT) scores, but SAT scores are converted to an ACT composite equivalent. UIC admission is now test-optional, and as such ACT/SAT scores are considered if submitted but are not required. Should an applicant submit more than one set of entrance exam scores, then the Office of Admissions will use the highest score when processing an application.

Before making the admission decision, a student's high school course work is checked to meet the specified subject pattern. All applicants must have successfully completed the following high school courses to be eligible for admission to UIC:

- A. Four years of English.
- B. Three years of Mathematics (must include algebra, geometry, advanced algebra/trigonometry);
- C. Three years of Science.
- D. Three years of Social Science.
- E. Two years of Language (other than English); and
- F. One year of an elective.

The College of Engineering provides the Office of Admissions the guidelines to be used for admission consideration of first year freshmen to the College. Applicants who demonstrate academic performance exceeding our admission guidelines are evaluated and accepted by the

Office of Admissions, and those who do not are referred to the College of Engineering for a holistic review. Applications which are referred to the College of Engineering are evaluated by a College of Engineering Admission Committee consisting of the Director of Admission and Records and the Associate Dean for Undergraduate Affairs. Admission is considered with evidence of strong high school grades, particularly in STEM courses, and high composite ACT scores and the mathematics subscale of the ACT, if provided. Table 1.1 shows the history of College of Engineering admissions standards for freshmen admissions for the past five years and indicates a gradual increase in incoming student quality as measured by the composite ACT score.

All entering freshmen are admitted with the condition that they must complete campus placement exams in math, in chemistry and in English composition. The results of these exams are critical in determining what courses students take during their first semester at UIC. If the results indicate that the student needs a preparatory course in math, or chemistry or English composition, then the student must register and pass the preparatory course.

Table 1.1 History of College of Engineering Admissions Standards for Freshmen Admissions for the Past Five Years

Academic Year	Composite ACT	Composite SAT	Percentile Rank in High School ¹	Number of New Students Enrolled
	AVG.	AVG.	AVG.	
2020	27.75	1255	83.79	556
2021	29.23	1245	76.87	842
2022	28.65	1281	77.16	880
2023	29.74	1283	77.69	1210
2024	28.03	1280	80.84	921
2025	28.97	1299	79.11	1060

¹ Admission is based on unweighted High School GPA and not Percentile Rank in High School

B. Evaluating Student Performance

Students are evaluated at the end of every semester during the College of Engineering Grade Review process. At that time, the Associate Dean for Undergraduate Affairs (ADUA) and/or the Director of the Academic Resource Center reviews the file of each student who has either a term Grade Point Average (GPA) or cumulative UIC GPA below 2.00 (A=4.00). In addition to monitoring the GPA of students, the College also uses the deficit point system to determine academic notice (probation) and drop decisions. The Deficit Point system has the advantage over the GPA system in that it clearly indicates the future academic performance that a student must achieve to return to clear status.

The following scale is on a per semester hour basis:

Letter grade	Grade points per credit hour	Deficit points
A	4	+2
B	3	+1
C	2 (2.0 minimum GPA at graduation)	0
D	1	-1
F	0	-2

If a student completes one 3-hour class and receives a grade of “D,” then this student would have a GPA of 1.00 and –3 deficit points. If another student takes four 3-hour classes and receives all D’s, then this student would also have a GPA of 1.00 but would have –12 deficit points. Although they both have the same GPA, the second student with –12 deficit points are having more serious academic difficulty. A student with negative deficit points must earn a positive number of deficit points in the future to bring the total back up to zero. Thus, a student with –12 deficit points must earn a combination of As and Bs for the number of hours necessary to achieve +12 deficit points to return to clear status.

Academic Notice (Probation) Rules

Rule 1. A less serious probation level is called Academic Notice Level 1 or 2.00 Pro. This notice is for any student whose Term GPA is below 2.00, but whose UIC Cumulative GPA is above 2.00. In the next semester, the student is expected to earn no grade less than C to continue.

Rule 2. Any student whose UIC Cumulative GPA falls below 2.00 is placed on Academic Notice Level 2 or 2.25 Pro. In the next semester, the student is expected to earn no grade less than a C and at least one B to continue. A student is not required to return to clear status in one semester. For example, if a student finishes the first semester with –10 deficit points, then finishes the second semester with 3 hours of B and 3 hours of C, this student will have reduced the total deficit points to –7. Although the student is still on probation, the student satisfied the probation conditions and is allowed to continue. At this rate of +3 deficit points per term, it will take the student 4 semesters to return to clear status, and the student is making progress towards that goal.

Rule 3. Graduation with a degree from the College of Engineering requires a minimum GPA of 2.00 in the major courses. Major courses are those required in the specific degree program as listed in the UIC Undergraduate Catalog. Rules 1 and 2 described above are also applied to the major courses.

Drop Rules

Rule 1. Any student who was on probation and did not satisfy the conditions of that probation may be dropped from the College of Engineering B.S. degree program.

Rule 2. A student who fails to make satisfactory progress toward a degree in the College of Engineering may be dropped. Examples of unsatisfactory progress are: -12 deficit points, excessive number of incomplete grades, failure to take courses required for the degree.

Rule 3. This rule applies to those students who had previously been dropped, were readmitted, and failed to meet the probation conditions necessary to continue in the College of Engineering. Students who have been dropped multiple times consult with an advisor to explore other career options. Only in rare cases will a student be readmitted after being dropped twice.

Prerequisite Monitoring and Enforcement

The importance and enforcement of prerequisite course requirements is advertised frequently during Advising Week and during registration periods. Many, but not all, of our courses have system-enforced prerequisite checking that prevents students from registering in a course for which they do not meet the prerequisites. We continue to expand system-enforced prerequisite checking each semester, but university limitations restrict the number of courses added per term. At the start of each semester, the Transfer Articulation and Degree (TADA) office compiles a list of College students whose enrollment violates system-enforced prerequisites. Usually, students are on this list due to failing or withdrawing from the prerequisite course. This list is sent to the Director of the Academic Resource Center, who then determines which students on the list have transfer credit that has not been posted yet, or who have permission to take the course without the listed prerequisite courses, which is granted under rare circumstances via a General Petition form approved by the Director of Undergraduate Studies (DUS). The remaining students are notified via email (cc to DUS, department head, course instructor, associate dean for undergraduate affairs) that they will be dropped from the course if they do not show proof of credit for the prerequisite course(s). The student generally has several days in which to respond to the email and/or make registration adjustments. In the third week, all identified students are removed from the courses for which they are registered improperly.

C. Transfer Students and Transfer Courses

Transfer applicants are encouraged to complete 60 semester (90 quarter) hours of transferable work at the time of enrollment and are advised to complete English Composition I and II; Calculus I, II, and III; Differential Equations; Physics I and II; Chemistry I. (Computer Science majors should have 12 semester hours (18 quarter hours) of lab science, which includes an 8-hour sequence).

Students must have earned at least a cumulative GPA of 2.50 (A=4.00), including all UIC and all transfer work.

Transfer Credit Evaluation Criteria

The state of Illinois has formal articulation agreements between state community colleges and senior public institutions. Credit from a community college is limited in the sense that the last 60 hours of the degree must be completed at a senior institution. In addition, the University maintains a general provision that either the first 90 or the last 30 hours of the degree must be earned at UIC.

The College of Engineering maintains an extensive evaluation program for courses available at 32 Chicago area colleges that a majority of UIC transfer students come from. The transfer articulation petition is available to prospective students on the COE website: <https://engineering.uic.edu/undergraduate/resources/>. The transfer information is housed on <https://transferology.com/index.htm> which conveniently allows students to review course equivalency information between institutions participating in the Illinois Articulation Initiative. Transfer credit is determined based on course equivalency.

Work successfully completed in other fully accredited institutions (either those approved by one of the regional accrediting associations or those approved by one of the agencies recognized by the National Commission of Accrediting) is generally accepted by the University on an hour-for-hour basis, as shown on the official transcripts received from those institutions. For consistency, credit awarded on the quarter hour basis is converted to the semester hour system by multiplying the number of quarter hours by 2/3.

Credit from institutions with provisional accreditation is accepted on a deferred basis until it is validated by satisfactory completion of additional work taken in residence at the University or in another fully accredited institution. Credit from unaccredited institutions is not accepted. However, knowledge in courses taken at such institutions may be awarded credit for the equivalent UIC course by successfully passing a UIC proficiency exam. No transfer credit is awarded for any course work completed at an institution external to UIC with a grade of D. Credit evaluation is done by the Office of the Associate Dean for Undergraduate Affairs in the College of Engineering. In many instances, descriptions, syllabi, etc. are sent to a faculty member in the department offering similar courses for their recommendation as to the equivalency to a UIC course and acceptance for transfer credit. The general principle used in accepting work from another college or university is that of “equivalency.” Whether a course taken elsewhere is equivalent to a course at UIC is determined by the college office or in the appropriate department at UIC. Catalog descriptions that show course content and prerequisites for a course are the primary means for determining equivalency. If the catalog description is not sufficient, syllabi, exams, computer programs, homework, textbooks, etc. may be requested from the student. Table 1.2 shows the Transfer students for the past five academic years. The data show a substantial increase in the number of transfer students entering the program during this period. This increase is driven largely by the increase in tuition costs, which encourages students to use lower cost community colleges to complete their pre-engineering course work.

Table 1.2 Transfer Students for Past Five Academic Years

Academic Year	Number of Transfer Students Enrolled
2020	427
2021	368
2022	318
2023	335
2024	348
2025	476

D. Advising and Career Guidance

The College of Engineering employs a mandatory advising system. Prior to the first semester of attendance, whether entering as freshmen or transferring from other schools, all students are advised by academic advisors from the Office of the Dean for Undergraduate Affairs. During their first semester, Electrical Engineering students are assigned to faculty advisors in the CHE department. Students are advised every semester until graduation. No student is allowed to register without meeting with their advisor, which is documented in the campuswide iAdvise system. Registration holds are utilized to ensure that students meet with their advisors every semester prior to registering for the next term.

Faculty advisors help students with curriculum matters, selection of electives, career path input, and other curriculum and career advice. The CHE DUS acts as an intermediary between faculty advisors and the Office of the Dean for Undergraduate Affairs. Students first are advised to discuss curriculum issues with the DUS for routine questions. The Office of the Dean for Undergraduate Affairs is staffed by professional academic advisors who support students with questions related to degree requirements, transfer credits, course substitution, special programs and other related issues.

Based upon discussions at a townhall meeting between COE students and the COE Dean, department heads and DUS, the COE implemented an advisor evaluation process starting in Fall 2006 to provide feedback to advisors on their performance in the advising process. In 2024, the responsibility for administering the advisor evaluations was moved to each individual department. In the Chemical Engineering department after their advising session, students fill out an online Advisor Evaluation form in the Chemical Engineering office. This allows us to record the students who complete the evaluation while the evaluations remain anonymous. Completing the Advisor Evaluation form is required for the academic hold to be removed so that a student can register for the next semester courses. Advisees rate their faculty advisors on a 5-point scale (1 lowest and 5 highest) as to their: 1) concern for the student, 2) knowledge of the curriculum, 3) availability during advising week, and 4) encouragement of the student, plus additional comments. Scores for all faculty members in each department are compiled and sent to all faculty within the department for feedback. The COE has also implemented an advisor award program for the top-rated advisor in each department as an incentive to improve advisor performance.

Figure 1.1 on the following page shows the online form that is used for the advisor evaluation. Table 1.3 shows average ratings for each survey question and an overall average from Fall 2024-Spring 2026. Table 1.4 shows the average of all questions for each faculty from 2024-2026. The advisor evaluation charts clearly show that the efforts of the CHE faculty are appreciated by CHE students.

Figure 1.1 Advisor Online Evaluation Form



**CHEMICAL ENGINEERING**
COLLEGE OF ENGINEERING

In a continued effort by the college to recognize and reward faculty, we ask you to fill out this short survey after your faculty advising meeting.

Please select between 1 and 5 for each of the statements regarding your faculty advisor: **1** for **STRONGLY DISAGREE** and **5** for **STRONGLY AGREE**.

What is your Major?

Chemical Engineering

Shows genuine concern for my academic and professional development.

1 - Strongly Disagree

2 - Disagree

3 - No Opinion

4 - Agree

5 - Strongly Agree

Has detailed knowledge of curriculum and provides good advice about courses.

1 - Strongly Disagree

2 - Disagree

3 - No Opinion

4 - Agree

5 - Strongly Agree

Was available at posted advising time during advising week.

1 - Strongly Disagree

2 - Disagree

3 - No Opinion

4 - Agree

5 - Strongly Agree

Encourages students to come at other times beyond the required meeting during advising week.

1 - Strongly Disagree

2 - Disagree

3 - No Opinion

4 - Agree

5 - Strongly Agree

Comments:

Table 1.3 Student evaluation of faculty advising: Average scores Fall 2024-Spring 2026.

Professor	Concern	Knowledge	Availability	Encouragement	Average
Faculty 1	4.00	4.43	4.00	3.57	4.00
Faculty 2	4.57	4.48	4.57	4.35	4.49
Faculty 3	4.51	4.64	4.77	4.54	4.62
Faculty 4	4.65	4.76	4.82	4.24	4.62
Faculty 5	3.94	3.06	5.00	3.94	3.98
Faculty 6	3.82	3.88	4.41	3.88	4.00
Faculty 7	5.00	5.00	5.00	5.00	5.00
Faculty 8	3.90	3.60	4.70	3.40	3.90
Faculty 9	3.76	4.00	4.19	3.52	3.87
Faculty 10	3.91	4.09	4.82	4.09	4.23
Faculty 11	4.72	4.72	4.69	4.55	4.67
Faculty 12	3.80	3.87	4.27	3.53	3.87
Faculty 13	4.41	4.52	4.76	4.52	4.55
Faculty 14	4.65	4.65	4.87	4.70	4.72
Faculty 15	4.59	4.71	4.74	4.41	4.61

Table 1.4 Student advisor evaluations Average over all questions (2024-2026)

Professor	F2024	Sp2025	F2025	Sp2026
Faculty 1	-	3.75	-	4.63
Faculty 2	4.07	4.90	4.15	4.92
Faculty 3	4.63	4.72	4.45	4.73
Faculty 4	4.61	4.88	3.92	5.00
Faculty 5	3.75	4.13	3.75	4.20
Faculty 6	3.81	3.56	4.92	3.96
Faculty 7	-	-	-	5.00
Faculty 8	-	4.13	3.75	3.86
Faculty 9	4.25	3.85	4.29	3.36
Faculty 10	-	-	4.15	4.29
Faculty 11	4.48	4.48	4.72	4.85
Faculty 12	4.58	4.25	1.75	3.96
Faculty 13	4.69	4.60	4.33	4.70
Faculty 14	5.00	4.94	4.63	4.54
Faculty 15	4.75	4.57	4.83	4.22
All Faculty	4.42	4.37	4.12	4.41

Career Guidance- UIC Engineering Career Center

Overview of services

- The Engineering Career Center (ECC) provides services and programs for students enrolled in the College of Engineering and employers who recruit UIC engineering students.
- The ECC consists of four full-time staff members and three part-time student assistants serving 6,000+ engineering students, meeting and working with students from freshmen to PhD candidates.
- The ECC assists students with the internship and full-time job search process, facilitating connections with employers nationally, regionally and locally.
- The University provides a job platform, Handshake, for both students and employers. Employers post openings; students upload their resumes and apply for positions.
- The ECC presents career outcomes information and specific program updates to College of Engineering advisory board members biannually via the Dean of Engineering.

Services and programs for students

- Students have access to a well-developed [website](#) with specific engineering career readiness information and event listings on Handshake.
- Students can schedule an appointment for an in-person one-on-one career advising meeting. They can also attend small group sessions for resume reviews.
- Students can reserve a private space in the Engineering Career Center to conduct virtual interviews, which have increased since the pandemic.
- The ECC curates our own social media, with approximately 1,500 followers in 2025. This keeps students informed about upcoming career events and provides tips for students' internship and job searches.
- The ECC delivers a biweekly electronic newsletter to all College of Engineering students. The newsletter contains relevant career topics, ways to interact with our Center, and upcoming events.
- The ECC works closely with engineering student organizations, presenting career readiness workshops to student members and arranging employer events, e.g. tech talks and tabling.
- Students attend two engineering- and computer science-specific career fairs per academic year, organized by the ECC. Students also receive marketing information on other career fairs conducted by campus-wide career offices.
 - More than 1,950 students attended the Sept. 2025 one-day career fair.
 - For all in-person fairs since 2019, the ECC has implemented Career Fair Tours as a low-stakes way for students to learn how to prepare for a career fair. Upwards of 80 students attend each fall, and the student response to this program has shown to increase the attendees' confidence when attending career fairs. The program was recognized in two publications of national organizations: the National Association of Colleges and Employers (NACE) and the Society for Human Resource Management (SHRM). The ECC administers a survey to students after each fair to

gather feedback for improvement. Several changes have been made based on the feedback, including adding additional methods to advertise the event and removing the need to sign up for a specific tour time in advance. We have also included at least one conversation with an employer as part of the tour, which students said was the most helpful part of the program.

- The ECC staff conducts presentations to multiple engineering classes, including professional development courses for graduating seniors and communication courses.
- The ECC conducted focus groups in 2021 to gather input from students for the continuous improvement of our services. Some of the feedback received from students included:
 - Students noted difficulty finding information on career services websites. UIC Career Services has purchased a contract with Uconnect to streamline career websites, making it easier to find content and resources for all majors on campus.
 - Students were not satisfied with the quality of career discussions happening in the classroom. The ECC increased the number of classroom presentations to cover five out of the six departments in the College of Engineering. The ECC has also increased the number of presentations to Engineering student groups.
 - Students expressed a desire to improve their interviewing skills. The Engineering Career Center offers one-on-one interview preparation appointments with students. Additionally, UIC Career Services purchased a contract with Stand Out to offer students 24/7 access to interview preparation, including virtual mock interviews with feedback.
 - Students discussed needing more assistance in developing their resumes. Aside from the one-on-one appointments and drop-in sessions offered, the ECC began an online resume submission system where students submit their resumes to us online and receive emailed feedback within 2-3 business days.
- Since 2014, the ECC has been responsible for implementing an early internship program where high-achieving students gain an experiential learning opportunity, the Guaranteed Paid Internship Program (GPIP). This unique program offers an on-campus research internship to those students who meet program requirements. Since 2020, the program has offered a summer internship to more than 890 engineering undergraduates.
 - Each summer, the ECC administers two surveys to both students and supervisors to evaluate intern performance and collect feedback about the program. The feedback is used for continuous improvement of program offerings, including which internships are offered in the future, and adjusting the timeline of the program to provide a longer internship length. In 2023, the ECC procured software to help administer the program, which decreased the time it takes to get the interns placed and start working.

Services and programs for employers

- Two career fairs per academic year. The ECC holds a September career fair and a February career fair. Attendance by employers has been strong over the past five years, with up to 180 employers attending a two-day career fair and 130+ attending a one-day fair.
- On-campus interview space immediately following each career fair and throughout the year is offered to employers with a cost to attend.
- Tech Talks and employer tabling – Employers contact the ECC to come to campus to inform students of opportunities with their company, to communicate their company

culture, and collect resumes. The ECC hosts and/or participates in 15-20 sessions occurring during the academic year.

Employer feedback and surveys

- Employers were surveyed after the Fall 2023 career fair regarding recruitment practices and graduate students' career readiness.
- Employer feedback is used to plan the upcoming year's events, to follow up with specific employers, as well as to modify existing services.
- The ECC has received internship evaluations from internship supervisors for graduate international students through Curricular Practical Training and for undergraduate students through GPIIP.
- Ongoing feedback from employers is obtained through conversations at events and while interviewing students in the career center. This information is shared with staff on an ongoing basis.

Professional development for career center staff

- The ECC holds memberships in several professional organizations for conferences, workshops, meetings, and resources for both personal development and to stay abreast of recruitment trends.
- Memberships in the following organizations currently exist among the staff:
 - National Association of Colleges and Employers.
 - Big10 Engineering Directors Group
 - America's Urban Campus
 - Society for Hispanic Professional Engineers; and
 - Chicago Career Professionals Network.
- In addition, two staff retreats are held each year for the purpose of planning for the upcoming year and discussing an educational topic.

First Destination Survey

- Graduating seniors with bachelor's degrees are emailed a survey upon graduation to determine what their first destination is after their degree, e.g., a job in industry or graduate school. The survey was created in collaboration with the UIC Office of Institutional Research (OIR) and campus Career Services.
- Follow-up surveying is implemented three months after graduation.
- Results of the survey are provided in aggregate form on the OIR website and updated each year:

<https://oir.uic.edu/surveys-2/uic-student-surveys/first-destination-survey/>

Collaboration with other departments within the College of Engineering

- The ECC works directly with the Communications department on website graphics, website content, the electronic newsletter, and TV monitors in various engineering buildings.
- The Advancement and Development office collaborates with the ECC on employers interested in hiring as well as giving.
- The academic advising office is often contacted for specific student issues/academic concerns.
- The ECC collaborates on events hosted through the Equity and Inclusion in Engineering Program (EIEP).

Collaboration with other departments on campus

- The ECC works closely with the central Career Services office, as well as career staff from the Business Career Center, Law School, College of Applied Health Sciences, College of Pharmacy, College of Liberal Arts and Sciences, and the Graduate college concerning the shared job platform Handshake, the hiring of staff, career fair volunteers, employer outreach, and general sharing of successful programs and services. This is done formally through the Career Leadership Council and informally.
- The ECC invites students from other UIC colleges to attend certain career fairs and employer recruitment events, where interdisciplinary collaboration meets the needs of the campus.

Transition from Blackboard to Canvas and Its Benefits for Students and Faculty

The University has initiated a strategic transition from Blackboard Learn to Canvas as part of an institution-wide effort to enhance the digital teaching and learning environment. Within the College of Engineering, this transition is being coordinated in collaboration with the Learning Technology Solutions (LTS) Learning Design team, which provides structured support to faculty during the migration of course materials, assessments, and instructional content. The Department of Electrical and Computer Engineering is actively participating in this initiative, with faculty working closely with LTS to ensure a smooth and effective transition of their courses.

This transition provides several important benefits that directly support student success and continuous improvement efforts. Canvas offers an intuitive and modern interface that improves ease of navigation for students, enabling clearer access to course materials, assignments, and grades. Enhanced communication tools, including integrated announcements, messaging, and calendar features, facilitate timely interaction between students and instructors. Additionally, Canvas supports mobile access more effectively, allowing students to engage with course content anytime and from any device, thereby increasing accessibility and flexibility.

From an instructional perspective, Canvas provides faculty with advanced tools for course design, content organization, and assessment. These include streamlined assignment creation, rubric-based grading, and integrated analytics that allow instructors to monitor student engagement and performance in real time. Such analytics support early identification of at-risk students and enable timely interventions, aligning with the program's commitment to student support and retention. The platform also facilitates more efficient management of course portfolios, which is particularly valuable for ABET accreditation processes, as it allows for easier documentation and retrieval of assessment artifacts, including assignments, student work samples, and grading rubrics.

Furthermore, the transition to Canvas promotes consistency across courses by encouraging the use of standardized templates and best practices in course design. This consistency enhances the overall student learning experience and ensures alignment with institutional and accreditation expectations. The support provided by the LTS Learning Design team, including training sessions and individualized consultations, helps faculty adopt effective pedagogical strategies and fully leverage the capabilities of the new system.

Overall, the migration to Canvas represents a significant improvement in the technological infrastructure supporting teaching and learning. It strengthens the department's ability to deliver high-quality education, supports data-informed continuous improvement, and enhances both

student engagement and faculty efficiency.

Transition from Blackboard to Canvas and Its Benefits for Students and Faculty

The University of Illinois System is currently undertaking a comprehensive Business Modernization and Transformation initiative to replace its legacy administrative system, Banner ERP, with a modern, [cloud-based Enterprise Resource Planning \(ERP\) platform](#). This system-wide effort is intended to significantly enhance the core administrative and academic processes that directly support students, including admissions, enrollment, registration, advising, academic records management, and financial services.

The new ERP platform will enable the University to streamline and standardize processes across campuses, reduce duplication and manual workarounds, and improve integration between systems. From a student perspective, this transformation is expected to result in more efficient and transparent services, improved access to accurate and up-to-date information, and a more intuitive and consistent user experience across administrative functions. In addition, the system will provide real-time data and analytics capabilities, allowing advisors, faculty, and administrators to better monitor student progress, identify potential challenges, and support timely interventions that promote student success and retention.

A key aspect of this initiative is its business-led and stakeholder-driven approach. Faculty, staff, and students are actively engaged in the design and implementation process to ensure that the new system reflects the actual needs and workflows of the University community. This inclusive approach helps ensure that student-related processes are not only efficient but also responsive and aligned with institutional priorities and accreditation expectations.

The ERP modernization effort is supported by executive leadership at the system level and is being implemented through a structured readiness and engagement program, which includes ongoing communication, training, and opportunities for stakeholder feedback. As the project progresses, it is expected to strengthen the University's ability to deliver high-quality student services, improve operational effectiveness, and support data-informed decision-making across all areas related to student support and success. More information about this initiative is available at the University of Illinois System ERP project website.

E. Work in Lieu of Courses

The College of Engineering does not allow credit towards any degree based on work or life experience. However, students may establish credit toward an undergraduate degree through the following examinations:

- ACT English/SAT Verbal.
- Advanced Placement (AP).
- International Baccalaureate (IB).
- College-Level Examination Program (CLEP); and
- Proficiency Examinations for Enrolled Students.

Students must submit official grade reports/examination results to the Office of Admissions before credit can be awarded. UIC will not award transfer course credit based on another institution's evaluation of test results. More information could be found on the university weblink: <http://www.uic.edu/ucatalog/AS.shtml#g>.

F. Graduation Requirements

All CHE students are continuously evaluated throughout their tenure in the CHE Department. This effort is aided by the Degree Audit Reporting System (DARS). This appraises students of what courses they have enrolled in, and which courses remain to be taken for the degree. During the student's senior year, the student is required to fill in a request to graduate checklist with the COE. The checklist includes the formal course requirements, as well as several additional requirements, including participation in Engineering EXPO in conjunction with CHE 396 (Senior Design I) and CHE 397 (Senior Design II).

Residency Requirements

In addition, a minimum residency requirement stipulating the number of consecutive credits that must be taken at UIC is in place and must be satisfied for graduation. The policy (which was most recently revised in Fall 2007) states:

“Either the first 90 semester hours or the last 30 semester hours of university work must be taken at UIC. In addition, at least one-half of the semester hours required in the student's major area of study must be completed at UIC. Concurrent attendance at the University of Illinois at Chicago and another collegiate institution or enrollment during the summer at another institution when approved by the student's college, does not interrupt the UIC enrollment residence requirement for graduation. Under exceptional circumstances, the enrollment residence requirement may be waived by the dean of the student's college upon petition of the student.”

Table 1.5 gives a summary of COE enrollments over the past six years, and Figure 1.2 gives detailed enrollments for chemical engineering. The Figure 1.2 shows that the enrollment decline of a few years ago has reversed. In part, this is due to extensive recruiting efforts that the chemical engineering department and college recruitment team have initiated, while ACT requirements were not lowered. Table 1.6 shows a sample of employment placements for chemical engineering graduates from 2020-2025.

Table 1.5 Enrollment Trends for Past Five Academic Years

	Year 2020	Year 2021	Year 2022	Year 2023	Year 2024	Year 2025
Full-time Students	3883	4023	4058	4558	4669	5424
Part-time Students	344	399	391	376	381	378
Student FTE ¹	3883	4023	4058	4558	4669	5424
Graduates	894	964	913	899	951	1254

¹ FTE = Full-Time Equivalent

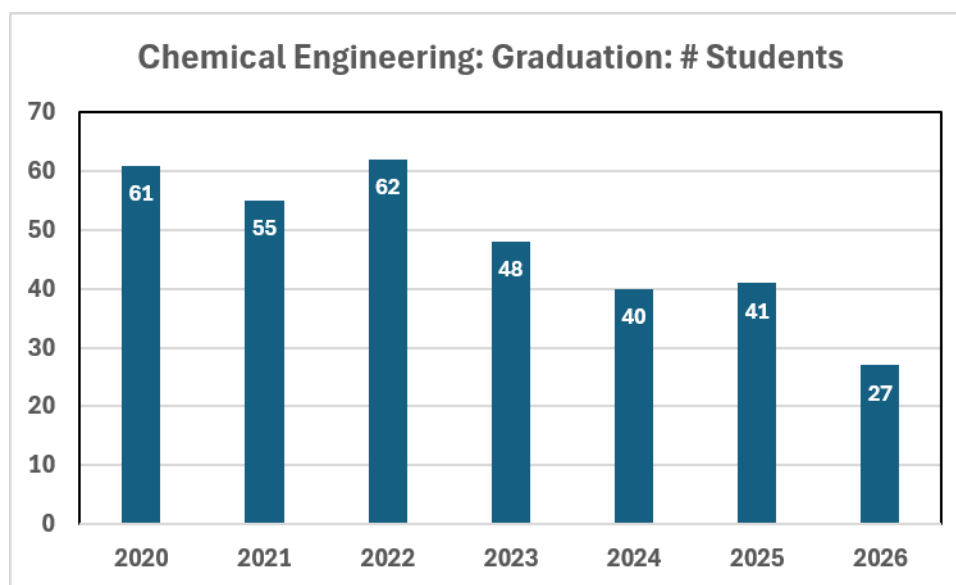
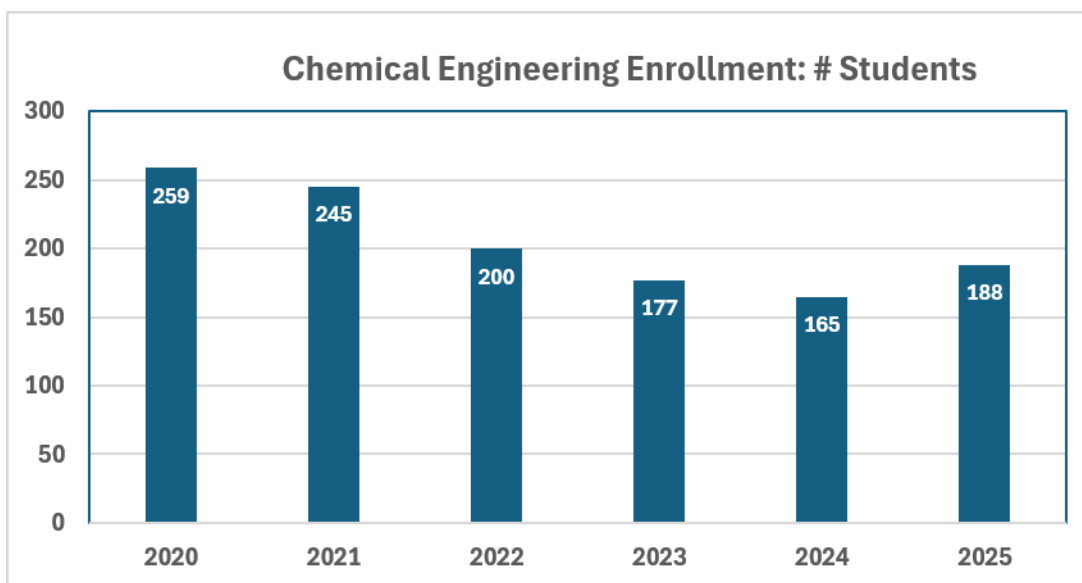


Figure 1.2 Undergraduate Enrollment and Graduation numbers for the Chemical Engineering program for the last 6 years

Table 1.4 25 Sample of Employment Placements

Numerical ID	Year Graduated	Initial or Current Employment / Other Placement
1	2020	Dow
2	2020	DuPont
3	2020	BASF
4	2020	3M
5	2020	LyondellBasell
6	2021	ExxonMobil
7	2021	Chevron Phillips Chemical
8	2021	Shell
9	2021	Huntsman Corporation
10	2021	Eastman Chemical Company
11	2022	Air Products
12	2022	ProLeiT
13	2022	ADM (Archer Daniels Midland)
14	2022	Procter & Gamble
15	2022	Kimberly-Clark
16	2023	Pfizer (Process/Manufacturing Ops)
17	2023	Merck & Co.
18	2023	Johnson Matthey
19	2023	Albemarle Corporation
20	2023	W. R. Grace & Co.
21	2024	Honeywell UOP
22	2024	Kraft-Heinz
23	2024	Marathon Petroleum
24	2024	Corteva Agriscience
25	2024	Lanza Jet
26	2025	Fluor Corporation (Process Engineering)
27	2025	Jacobs Engineering
28	2025	Bechtel (Chemical/Process Division)
29	2025	Worley
30	2025	Technip Energies

G. Transcripts of Recent Graduates

Student transcripts will be provided as requested by the EAC Team Chair.

CRITERION 2. PROGRAM EDUCATIONAL OBJECTIVES

A. Mission Statement

2.A.1. Scope and Mission Statement of the University of Illinois at Chicago (UIC)

The largest university in one of the world's most vibrant cities:

The University of Illinois at Chicago is an acclaimed research institution with 16 colleges dedicated to the discovery and distribution of knowledge.

Mission Statement of UIC

“UIC’s mission is to provide the broadest access to the highest levels of educational, research, and clinical excellence.”

The current UIC mission statement has been in place since a formal update was approved by the Board of Trustees resolution dated January 19, 2017. The recent leadership under Chancellor Miranda has introduced strategic priorities that support or operationalize the mission. These priorities reaffirm UIC’s commitment to student success; expand the research profile to position the university at the forefront of innovation and Chicago’s knowledge economy; engage with the region’s communities with a focus on educational and health equity; forge productive partnerships with businesses, nonprofits, and the government sector to create opportunities for faculty and student engagement; and elevate UIC as a destination to recruit and retain world-class faculty and staff.

The five Strategic Priorities are:

- Our top priority is **student success**, focusing on recruitment, retention, and graduation while fostering a sense of belonging and connection among all students—undergraduate, professional and graduate.
- **Research infrastructure** is the second priority, positioning UIC at the forefront of knowledge and discovery.
- Our third priority is **engagement with our community** to improve access to educational opportunities and clinical care in underserved communities, making our world more equitable, efficient and sustainable.
- Our fourth priority is **partnerships with businesses and nonprofits** to enhance faculty opportunities and strengthen student engagement and employment opportunities.
- **Recruiting and retaining the best faculty and staff** is our final priority. It matters that the people who make our university excel feel seen, valued and motivated.

Collectively, these priorities are advancing the university’s equity-focused mission to provide the broadest access to the highest levels of educational, research, and clinical excellence.

The mission statement is prominently displayed to all the UIC's constituencies at [UIC Mission](#) while the five strategic priorities are displayed at the following [UIC Strategic Priorities](#).

2.A.2. Mission Statement of the College of Engineering (COE)

The College of Engineering mission (Before Feb 2026)

“UIC Engineering’s mission is access to excellence and success. We educate students to innovate and to lead. We expand global knowledge of the engineering field through excellence in original research. In addition, as a public institution we prioritize our relationship with the city of Chicago and the state of Illinois. We foster strong local partnerships, provide crucial engineering expertise and facilities, and serve as a major economic driver for the region.”

The Priorities of the College of Engineering are:

- *Recruiting bright students with unique stories and talents;*
- *Attracting and retaining world-class teachers and researchers;*
- *Building top-tier facilities;*
- *Creating a strong alumni community; and*
- *Establishing strong partnerships with government, industry, and academia.*

The NEW College of Engineering mission published Feb 11th, 2026

The COE strategic planning process began in January 2025 and continued through the Fall 2025 semester, engaging members of the College community to gather input, capture ideas, and leverage collective strengths to advance key priorities. As an outcome of this effort, during the College of Engineering faculty meeting on February 11, 2026, the newly appointed Dean Lola Eniola-Adefeso announced the [new strategic plan](#), which includes an updated College of Engineering mission statement.

“Through our commitment to excellence, accessible education and impactful research, we prepare curious, proficient and diverse engineers and computer scientists to lead and advance upward mobility locally and globally.”

The College of Engineering mission statement is available to students and all other constituencies of the University of Illinois at Chicago College of Engineering at the following links:

[College of Engineering Strategic plan](#)

[College of Engineering Undergraduate Catalog](#)

[College of Engineering- About us](#)

The UIC College of Engineering [strategic planning process](#) was developed through a collaborative and inclusive effort involving faculty, staff, students, alumni, and external stakeholders. The process aimed to identify the College’s key strengths, differentiators, and future opportunities, while aligning its culture, mission, vision, and values. The resulting strategic plan, launched in February 2026, provides a clear roadmap to position the College as a

global leader in engineering education and research, with a strong emphasis on diversity, inclusion, and societal impact.

The plan defines a focused set of [strategic priorities](#) and actionable goals to guide decision-making, resource allocation, and long-term development over the next several years. It emphasizes stakeholder engagement, interdisciplinary collaboration, innovation in education and research, and strengthening partnerships with industry and the broader community to address real-world challenges.

Consistency of the College of Engineering Mission with the University Mission

The College of Engineering new mission directly supports and operationalizes the UIC mission by advancing broad access through accessible engineering education, strengthening institutional excellence through rigorous academic and research programs, and contributing to societal and economic mobility through impactful technological innovation locally and globally. The following table shows the mapping between the university mission and the college of engineering mission.

Table 2.1 UIC Mission to College of Engineering Mission Alignment Table

UIC Mission Component	College of Engineering Mission Element	Nature of Alignment
Broadest access	Accessible education	The College directly operationalizes UIC's access commitment by ensuring inclusive pathways into engineering and computing education.
Educational excellence	Commitment to excellence; prepare proficient engineers and computer scientists	The College advances institutional educational excellence through rigorous curricula, high academic standards, and strong student learning outcomes.
Research excellence	Impactful research	The College contributes to UIC's research mission by conducting high-impact engineering and computing research that addresses societal and technological challenges.
Clinical and societal excellence	Advance technological solutions that improve lives	Engineering innovations in health, infrastructure, computing, and sustainability support UIC's broader societal and public impact mission.
Institutional excellence at the highest level	Excellence in education and research	The College reinforces UIC's aspiration for top-tier performance through continuous improvement, innovation, and measurable impact.
Public mission and upward mobility	Advance upward mobility locally and globally	The College translates UIC's public urban mission into workforce development, economic mobility, and leadership preparation in Chicago and beyond.

On a yearly basis, every June, the office of Programs and Academic Assessment reviews the consistency between the University mission and the College mission as one of its main focuses for the HLC (Higher Learning Commission) Accreditation. Two representatives from the College of Engineering attend the university academic assessment annual meetings: the associate dean for undergraduate affairs and the College of Engineering director of educational assessment.

The College of Engineering mission statement is available online to students and all other College constituencies at the following URL:

<https://engineering.uic.edu/about/strategic-plan/#what-drives-us>

2.A.3. Mission statement of the Department Chemical Engineering

Chemical Engineering mission (Before April 2026)

“The mission of the Department of Chemical Engineering is to serve society through excellence in education, research, and public service. We provide for our students an education in chemical engineering, and we aspire to instill in them the attitudes, values, and vision that will prepare them for lifetimes of continued learning and leadership in their chosen careers. Through scholarship, the Department strives to generate new knowledge and technology for the benefit of the State of Illinois, the nation, and beyond.”

NEW Chemical Engineering mission

The Chemical Engineering department process to revise the mission statement began in March 2026 after the College of Engineering updated its mission statement. The process aimed to identify the department’s key strengths, differentiators, and future opportunities, while aligning its culture, mission, vision, and values. The mission statement was first discussed at the March faculty meeting, and several revisions were proposed. An online survey was then sent out to the faculty to obtain additional comments, which resulted in further discussions and revisions. At the April faculty meeting, the faculty approved the mission statement to the following statement:

“The mission of the Department of Chemical Engineering is to deliver an outstanding and accessible education to a diverse student body. We prepare skilled engineers to advance both fundamental and applied research, drive innovation, and lead the development of transformative technologies that enhance economic mobility and contribute to local and global prosperity”.

The Chemical Engineering new mission directly supports and operationalizes the College of Engineering mission by advancing broad access through accessible engineering education, strengthening institutional excellence through rigorous academic and research programs, and contributing to societal and economic mobility both locally and globally by advancing impactful technological solutions to improve lives.

Work in Progress: The department is now in the process of obtaining feedback on the new mission statement from our other stakeholders; the External Advisory Board, alumni, students, and external partners. Updates and feedback from our stakeholders will be presented at the ABET site visit.

B. Program Educational Objectives

In conjunction with the mission statement, the Chemical Engineering department is revising the Program Educational Objectives to align with the new College of Engineering and Chemical Engineering mission statements that were revised in March and April 2026. Given below are the Program Educational Objectives prior to May 2026 and the new Program Educational Objectives that have been proposed by the chemical engineering faculty. **Note:** the previous PEO's have been used in the Criterion 3 Student Outcomes and Criterion 4 Continuous Improvement.

Chemical Engineering Program Educational Objectives (as of June 2026)

PEO	Category	Description
1	Depth	In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.
2	Breadth	Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.
3	Professionalism and Service	In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment.
4	Life-Long Learning and Leadership	Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.

The Program Educational Objectives are prominently listed in the Undergraduate-ChemE Major link on the home page of the departmental website: www.che.uic.edu, or <https://che.uic.edu/undergraduate-programs/che-major/>. Additionally, the PEOs are included in all undergraduate course syllabi, and class time is devoted to explaining how the knowledge, skills and principles developed by the course in question contribute to the professional capacities represented by the PEOs.

NEW Chemical Engineering Program Educational Objectives (Work in Progress)

The Chemical Engineering department process to revise the PEO's began in May 2026 with feedback from the chemical engineering faculty to gather input and capture the department's overall mission and values, key strengths, and differentiators. As an outcome of this effort, the following Program Educational Objectives were proposed and approved by the faculty in June 2026. The new PEO's emphasize the department's educational objectives into four categories; professional impact, professional practice, service and leadership, and professional development. The consistency of the new PEO's with the College of Engineering Mission is given in Section 2.C.

Work in Progress: The department is in the process of obtaining feedback on the new Program Educational Objectives from our other stakeholders; the External Advisory Board, alumni, students, and external partners. Once this is completed, the department will update the PEO's and align our Student Learning Outcomes with the new PEO's. Continuing updates and feedback from our stakeholders will be presented at the ABET site visit.

PEO	Category	Description
1	Professional Impact	Apply fundamental principles, scientific and engineering knowledge, problem-solving, critical thinking, and creative design in professional engineering practice and non-engineering fields such as law, medicine, education and business.
2	Professional Practice	Contribute to industry, academia, government and entrepreneurial ventures and develop innovative solutions that address societal and technical challenges locally and globally, contributing to economic growth and upward mobility.
3	Service and Leadership	Practice engineering with clear communication, leadership, teamwork, and high standards of integrity, ethical responsibility, safety and protection of the environment, thereby creating a collaborative, diverse and inclusive culture in the workplace.
4	Professional Development	Reinforce skills and knowledge through graduate education, professional development courses, self-directed research, and on-the-job training and experience.

C. Consistency of the Program Educational Objectives with the Mission of the College of Engineering

Component of College of Engineering mission	Contributing PEO
Accessible Education	1
Commitment to excellence; prepare proficient engineers and computer scientists	1, 2, 3
Impactful Research	2,3,4
Advance technological solutions that improve lives	2,3
Excellence in education and research	1,4
Advance upward mobility locally and globally	2, 3

The healthcare component of UIC's mission (arising from the specific history of UIC and its medical school within the Chicago Metropolitan Area) is not directly served by the Chemical Engineering program and would not be expected among the objectives of typical chemical engineering programs. Recently however, there has been a strong collaboration between the Chemical Engineering Department and the College of Medicine on research studies such as nanomaterials, drug delivery, biocompatible materials, cancer detection instrumentation and pharmaceuticals (Professors Ying Liu and Vikas Berry). Students have contributed directly by performing undergraduate research in these areas. Also, graduates could be employed in related sectors, and hence the Chemical Engineering bachelor's degree program could contribute indirectly to medical technology and healthcare.

D. Program Constituencies

2.D.1. Program Constituencies.

Program Constituency	Vehicles of Consultation
Students	Course Evaluations (each semester) Academic advising (each semester) Senior Exit Survey (semester of graduation) ChemEConnect (ongoing)
Faculty	Faculty meetings, faculty retreats, department committees and task forces
Alumni	Alumni feedback surveys (ongoing) Alumni at large (department sponsored events, guest lectures, LinkedIn UIC Chemical Engineering group, email contact, telephone contact) Engineering Expo and ChE Senior Design Showcase judging (Spring semester)
Industrial and Professional	External Advisory Board meetings, surveys, correspondence AIChE Chicago Section (ongoing) Industrial mentors in Senior Design II (CHE 397) class Engineering Expo and ChE Senior Design Showcase judging (each spring semester)
University Administration	Educational Policy Committee meetings (bi-monthly)

Our students are critically important stakeholders in the curriculum and policies of the Chemical Engineering department, and we consult them regularly as summarized above. However, students are not in as strong a position as other constituents to determine the appropriate aspects and levels of professional accomplishment several years after graduation. Surely their feedback forms part of the conceptual milieu in which the department formulates policy. But we do not regard students formally as “constituents” in the review and updating of PEOs.

2.D.2. Meeting the Needs of Program Constituents and Other Stakeholders

Students. The Chemical Engineering program exists to educate and promote young engineers and launch them on successful careers in industry, post-graduate education and government service. In order that students understand the broader professional context toward which they are being educated, the PEOs are listed on the syllabus of each undergraduate course and class discussion is devoted to putting the performance indicators of the course in the professional and career context of PEOs. With this motivation, PEOs guide the professional development of our students and set targets for their knowledge, skills and principles.

Faculty. PEOs represent the ultimate goals of undergraduate teaching and learning, and thereby serve to organize our efforts in design, development, implementation and improvement of the curriculum toward professional success and effectiveness of our graduates.

Alumni. PEOs represent the cornerstone for professional success, effectiveness and leadership of our alumni.

Industry. For local and National industry that seeks to hire, develop and promote professional engineers, PEOs represent the spectrum of professional abilities and capacities of UIC Chemical Engineering graduates that make them desirable and enable them to contribute and lead. The industrial constituency is represented by the External Advisory Board (EAB) and also members of the AIChE Chicago Section.

The EAB has been assembled from among prominent industrial and corporate engineers from various sectors and localities, who complement each other's technical expertise. Since the last ABET review, 3 EAB members have stepped down; Michael Goluszka, LA-CO Industries; Dr. Diane Graziano, Argonne National Laboratory; and Michael Seacrist, SunEdison Semiconductor. Subsequently, 3 new members have joined the board; Dr. Saadet Acikgoz, Air Company, Dr. Shailendra Bordawekar, AbbVie, and Dr. Claus Daniel, Argonne National Laboratory.

Dr. Saadet Ulas Acikgoz

Director of Engineering
Air Company

Dr. Shailendra Bordawekar

Vice President, Synthetic Molecules CMC R&D
Development Sciences
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Dr. Tarun Chandra

Managing Principal
EmpiriQA LLC
Long Grove, IL 60047

Dr. Charles A Damianides

US Department of Energy
Executive Director, Center for Sustainable Fuels and Chemicals

Dr. Claus Daniel

Associate Laboratory Director, Advanced Energy Technologies
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Dr. S. S. Kumaran

Sr. Research Engineer
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Mr. Dennis O'Brien (Retired)

Group Manager, Refining and Petrochemicals Division
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Dr. Anil Oroskar

CTO, CFO
OroChem Technologies Inc.
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Dr. Gavin Towler

Senior Manager of Process Design Development
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25 E. Algonquin Road
Des Plaines IL 60017-5017

Dr. Anastasia Vaia

Global Vice President, Hydrogen and CCS
BP United Kingdom

The EAB is helping to steer the department's direction, structure and mission.

A number of practicing chemical engineers, members of the AIChE Chicago Section, partner with the department on (i) improving pedagogy (industrial mentors and guest lecturers in the Senior Design sequence, CHE 396, 397 – Senior Design), (ii) outreach activities (high schools, middle schools), and (iii) interactions with the AIChE Student Chapter. These industrial constituents include:

Senior Design Mentors:

Dennis O'Brien- retired, Jacob's Consultancy
Jon Prudhom- GTI Energy
Patrick Shannon- Retired from Middough
Dave Bahr- Worley Consulting
Mike Armstrong- Worley Consulting
Kiran Gupte- Retired from BP
Belma Demirel- BP
Annette Johnston- AAJ Project Consulting
Molly Ramsey -BP
Gavin Palmer- Valdes Engineering
Lance Baird- Lance Baird LLC
Ralph Sapko- Retired
Laura Weaver- Worley Consulting

Chemical Engineering Design Showcase Judges:

Steven Metgar- TM Process and Controls
Nina Jasmine Malapit (Alumnus)- Honeywell UOP

Peter Herena- CF Industries
Gabriela Bosak- Tetrapak
Ken Bicknell- Stepan
Anthony Taphanand (Alumnus)- Honeywell UOP
Iris Alvarado Santiago- Honeywell UOP
Izz Elmaazawi (Alumnus)- Bunge
Earl Washington- Factor_
Paul Diaz (Alumnus)- Constellation Energy
Maxum Cesar (Alumnus)- Lanza Jet
Jennifer Nacht (Alumnus)- Greer, Burns & Crain, Ltd.
Muhammad Abdelqader (Alumnus)- Greater Chicago Metropolitan Water Reclamation District

E. Process for Review of the Program Educational Objectives

Periodic review and updating of the PEOs involves all constituencies of the department. The PEOs are discussed periodically by faculty at faculty meetings, faculty retreats and with our industrial constituents at the meetings of the EAB, most recently in October 2025 and through online surveys. Most recently, the PEO's are being discussed due to the revision in our mission statement and PEO's to align with the new COE mission statement. The revisions are ongoing as the department reaches out to all our constituents to get feedback.

Alumni feedback is obtained where possible in person at return visits, such as alumni mixer events, judging at Engineering Expo, invited class lectures and student seminars, and other points of contact with the faculty. Online surveys are also used to obtain alumni feedback.

Our students are critically important stakeholders in the curriculum and policies of the Chemical Engineering department. The PEO's are given in course syllabi and in first class lectures. However, students are not in as strong a position as other constituents to determine the appropriate aspects and levels of professional accomplishment several years after graduation. Surely their feedback forms part of the conceptual milieu in which the department formulates policy. But we do not regard students formally as "constituents" in the review and updating of PEOs.

Finally, input from University administration is obtained from the Office of the Associate Dean for Undergraduate Affairs at regular, bi-monthly meetings of the COE Educational Policy Committee (EPC). This last avenue is a way for maintaining consistency between departmental PEOs and the UIC and COE mission statements.

For the ABET site visit we will present the following documentation on PEOs:

- Relevant agendas, notes and/or minutes from the EAB meetings, surveys, correspondence
- Correspondence, notes and/or minutes on faculty and EPC discussion of PEOs.
- Correspondence, surveys from the alumni.
- Surveys from students

Samples of this documentation are given below in Exhibits 2-1 to 2-4.

Exhibit 2-1.

External Advisory Board member email response on Mission Statement and PEO revision

 Outlook

Re: New UIC ChE Mission Statement and Program Educational Objectives

From tchandra@empirica.com <tchandra@empirica.com>

Date Wed 5/27/2026 2:56 PM

To Zdunek, Alan <zdunek@uic.edu>

 1 attachment (23 KB)

NEW Chemical Engineering Program Educational Objectivesv3 TC.docx;

CAUTION: External Sender

Hi Prof. Zdunek

Great to hear from you and thanks for reaching out. The Chem E Department's mission statement and PEOs are exceptionally well-aligned with both the CoE and the overall UIC mission statements. Please see attached with some minor suggestions and a comment for your consideration.

Best of luck with the forthcoming ABET accreditation review

Regards,

Tarun

On 2026-05-26 4:15 pm, Zdunek, Alan wrote:

Dear Tarun,

I hope this email finds you well. I have a favor to ask. The College of Engineering (COE) recently revised the College Mission Statement as part of the COE Strategic Plan Initiative. Because of this change, the ChE faculty revisited our Chemical Engineering mission statement and PEO's in order to align with the new College mission statement. I have attached a file with the new ChE mission statement and revised PEO's that was approved by the ChE faculty. I have included the previous mission statement/PEO's, and the UIC and College mission statements as a reference.

Please review at your earliest convenience and let me know if you approve of the revised statement and objectives and have any comments.

For your information, our 6-year engineering accreditation review (ABET) is scheduled for this October 2026 (ABET), with the ABET self-study report due July 1st. I will be adding the new mission statement and PEO's to our self-study report as a 'work in progress'.

Best regards,

Alan

Dr. Alan D Zdunek

Exhibit 2-2.

Faculty email response on Mission Statement

 Outlook

Re: ChE Mission Statement Survey- Please complete by Friday April 3rd.

From: Caracotsios, Michael <mcaracot@uic.edu>
Date: Mon 3/30/2026 8:50 PM
To: Zdunek, Alan <zdunek@uic.edu>

Alan,
How about this..

"The Department of Chemical Engineering is committed to excellence in affordable education, impactful research, and meaningful service to the community. We prepare a diverse body of proficient engineers who are ready to innovate, lead, and advance technology through a lifetime of learning and leadership, thereby strengthening upward mobility and contributing to both the local and global economy"

Best Regards
Michael

From: Zdunek, Alan <zdunek@uic.edu>
Sent: Monday, March 30, 2026 10:21 AM
To: Caracotsios, Michael <mcaracot@uic.edu>
Subject: Re: ChE Mission Statement Survey- Please complete by Friday April 3rd.

Hi Michael,

Thanks for the comments. Here is another version, let me know what you think. I wanted to keep the 'upward mobility and local/global economy' in the mission to align with the COE statement.

"The Department of Chemical Engineering is committed to excellence in accessible education, research, and public service. We prepare diverse and proficient chemical engineers to innovate, lead and advance engineering practice and technologies through a lifetime of continued learning and leadership that strengthens upward mobility and benefits the local and global economy.

Alan

Dr. Alan D Zdunek
Director of Undergraduate Studies
Clinical Associate Professor
University of Illinois at Chicago (UIC)
Department of Chemical Engineering (MC 110)
150 Engineering Innovation Building
929 W Taylor Street
Chicago, Illinois 60607

From: Caracotsios, Michael <mcaracot@uic.edu>
Sent: Saturday, March 28, 2026 8:09 AM
To: CHE-FACULTY@LISTSERV.UIC.EDU <CHE-FACULTY@LISTSERV.UIC.EDU>; Zdunek, Alan <zdunek@uic.edu>
Subject: Re: ChE Mission Statement Survey- Please complete by Friday April 3rd.

Alan,
I would replace "for lifetimes" with "for a lifetime" (I don't think anyone has more than one lifetime 😊)
Also personally I would strike the last sentence that begins with "Through scholarship..." I think it sounds kind of weird and as it is our mission statement is already long.

Best Regards
Michael

From: Chemical Faculty Listserv <CHE-FACULTY@LISTSERV.UIC.EDU> on behalf of Zdunek, Alan <zdunek@uic.edu>
Sent: Friday, March 27, 2026 4:04 PM
To: CHE-FACULTY@LISTSERV.UIC.EDU <CHE-FACULTY@LISTSERV.UIC.EDU>
Subject: ChE Mission Statement Survey- Please complete by Friday April 3rd.

Hello Colleagues,

As a follow up to our discussion at the March 13, 2026 faculty meeting, please provide feedback by next **Friday April 3rd** on the revisions that I made to the ChE Mission Statement to align with the new College of Engineering Mission Statement.

Please click on the link to access the ChE Mission Statement Survey: <https://forms.gle/VoiQuJiuUSjiEMXK9>

Thanks for your time and help on this.

Best,

Alan

Dr. Alan D Zdunek
Director of Undergraduate Studies
Clinical Associate Professor
University of Illinois at Chicago (UIC)
Department of Chemical Engineering (MC 110)
150 Engineering Innovation Building
929 W Taylor Street
Chicago, Illinois 60607

Exhibit 2-3.

Online survey responses from senior students on PEO's: Fall 2021 Aggregated scores

(1.) Greetings! Please fill out the short questionnaire below about our Program Educational Objectives (PEOs). We are interested in your input to help us review the PEO's and ensure that they remain consistent with our student's needs, the UIC mission and the ABET criteria (accreditation). In formulating your response, reflect on your own experiences in relation to the qualities of (1) depth, (2) breadth, (3) professionalism, (4) life-long learning that constitute the Program Educational Objectives of the Chemical Engineering program.

(1 = Weak, 2 = Moderate, 3 = Strong, 4 = Excellent)

3.31 Depth: In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.

3.24 Breadth: Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.

3.31 Professionalism and Service: In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment.

3.25 Life-Long Learning and Leadership: Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.

Exhibit 2-4.

Online survey responses from Chemical Engineering alumni on PEO's: Aggregated scores from 45 alumni

Alumni survey: ChE Program Educational Objectives

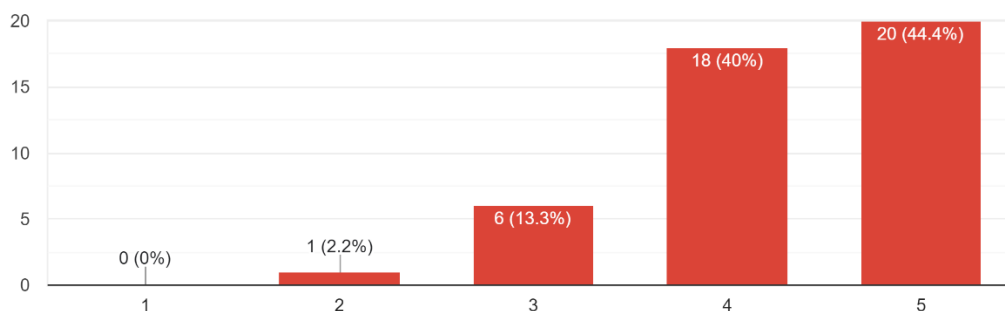
Greetings! Please fill out the short questionnaire below about our Program Educational Objectives (PEOs). We are interested in your input to help us review the PEO's and ensure that they remain consistent with our student's needs, the UIC mission and the ABET criteria (accreditation).

Please evaluate the Program Educational Objectives from 1 to 5, 1 being the lowest and 5 the highest. You may also leave a short comment at the end of the survey. In formulating your response, reflect on your own experiences in relation to the qualities of (1) depth, (2) breadth, (3) professionalism, (4) life-long learning that constitute the Program Educational Objectives of the Chemical Engineering program.

Thank you in advance for your time.

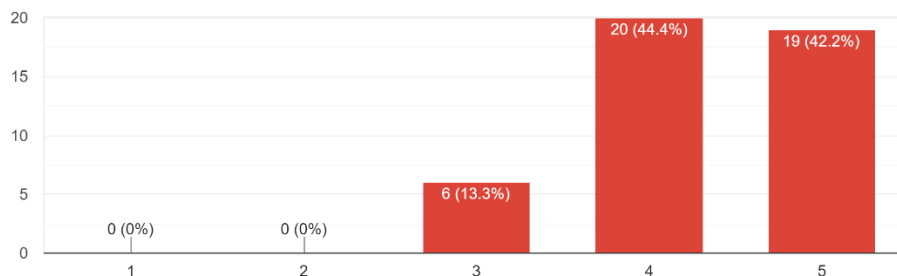
Depth: In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in app...alysis, creative design, and conceptual innovation.

45 responses



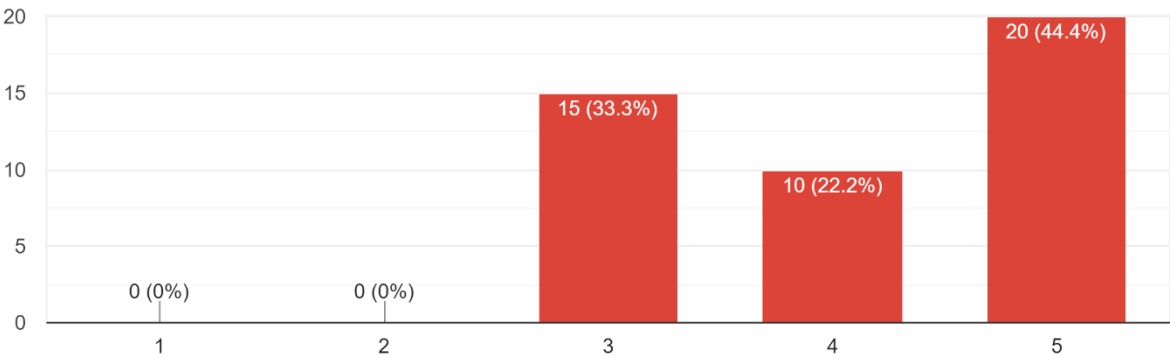
Breadth: Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public...te sectors, or in the pursuit of graduate education.

45 responses



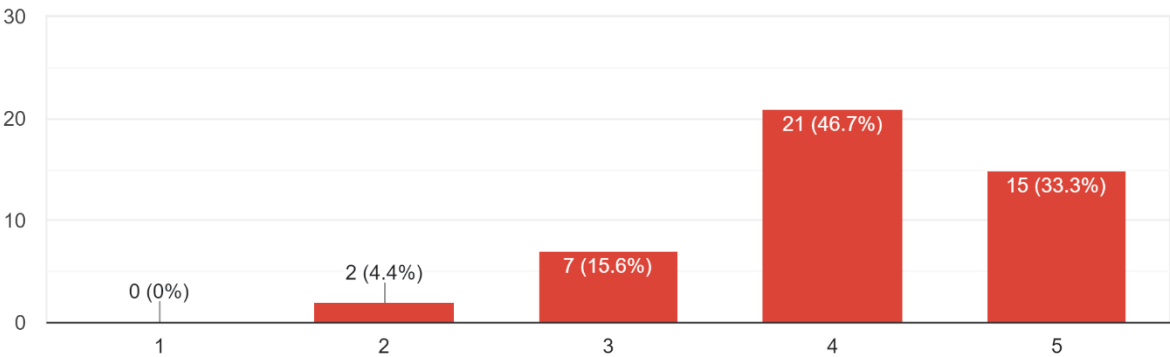
Professionalism and Service: In the service of industry, government, the engineering profession and society at large, graduates will function effective...sionalism, safety and protection of the environment.

45 responses



Life-Long Learning and Leadership: Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, grad... learning and demonstrate professional leadership.

45 responses



CRITERION 3. STUDENT OUTCOMES

A. Student Outcomes

Student Outcomes are prominently listed in the “ChemE Major” link on the home page of the departmental website: www.che.uic.edu and can also be accessed via the direct link; <https://che.uic.edu/undergraduate-programs/che-major/>. Additionally, relevant Student Outcomes are included in undergraduate course syllabi, and class time is devoted to explaining how the knowledge, skills and principles developed by the course in question contribute to the understanding and abilities represented by the Student Outcomes. The processes (Section 2.E.) for revisiting and updating PEOs also afford opportunities to reconsider Student Outcomes and amend them as necessary. The Student Outcomes coincide directly with the ABET Student Outcomes 1-7 as shown in Table 3-1.

Table 3-1. Student Outcome relationships to ABET Student Outcomes and the Program Educational Objectives.

SO	ABET	Description	PEO
1	1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	1
2	2	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	1
3	3	An ability to communicate effectively with a range of audiences.	3
4	4	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	2,3
5	5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	3
6	6	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	1
7	7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	4

B. Relationship of Student Outcomes to Program Educational Objectives

Table 3-1 also shows how the Student Learning Outcomes relate to the Program Education Objectives. Student Outcomes 1, 2 and 6 represent the core of fundamental engineering knowledge, skills and tools that prepare graduates to enact PEO 1 (depth) in their subsequent careers. Student Outcome 4 embodies the social, economic and political context in which graduates advance their careers and impact society (PEO 2). Student Outcomes 3, 4, and 5 establish the abilities of graduates to attain high professional and ethical standards in their careers and advance public safety and environmental protection, with clear communication and teamwork (PEO 3). Student Outcome 7 reflects the capacity of graduates to renew their knowledge and skills, enhance their innovation and lead (PEO 4).

C. Coverage of Student Outcomes in Courses

Coverage of Student Outcomes in any given course is classified using the following four-point scale:

- 4 = topic is main focus of course, grade largely dependent on demonstrated knowledge.
- 3 = course grade partially dependent on demonstrated knowledge on this topic.
- 2 = topic introduced in course, course grade not dependent on demonstrated knowledge.
- 1 = topic not covered specifically in course.

Tables 3-2 and 3-3 indicate the coverage of Student Learning Outcomes in each of the Chemical Engineering courses. Overall coverage of SLOs by ChE courses is summarized in Table 3-4: all SLOs are covered at levels 3 or 4 in at least two ChE courses. SLOs #1; An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, basic science and mathematics, receives this level of emphasis in 23 courses. SLO #2 and #3 receive this level of emphasis in 16, and 16 courses, respectively. Table 3-5 indicates the coverage of SLOs in each of the technical courses outside the Chemical Engineering department. Basic mathematics and science is covered in 14 non-engineering courses as shown in Table 3-5.

Table 3-2. Coverage of Student Outcomes 1-7 by Chemical Engineering Courses.

Course	(1) An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	(2) An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	(3) An ability to communicate effectively with a range of audiences.	(4) An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	(5) An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	(6) An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(7) An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
CHE 201	4	2	2	2	2	2	2
CHE 205	4	2	1	1	1	2	1
CHE 210	4	2	1	1	1	1	1
CHE 230	3	2	2	2	2	1	2
CHE 301	4	1	2	1	1	1	1
CHE 311	3	3	1	1	1	1	3
CHE 312	4	3	1	2	1	1	4
CHE 313	4	4	2	2	3	3	4
CHE 321	4	4	2	3	3	2	3
CHE 330	3	2	2	2	2	1	2
CHE 341	4	4	3	1	1	1	1
CHE 381	4	1	4	3	4	4	3
CHE 382	4	1	4	3	4	4	3
CHE 396	4	4	3	4	4	3	3
CHE 397	4	4	4	4	4	3	3
CHE 499	1	1	3	3	1	1	3

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;

3 = course grade partially dependent on demonstrated knowledge on this topic;

2 = topic introduced in course, course grade not dependent on demonstrated knowledge;

1 = topic not covered specifically in course.

Abbreviated, descriptive course titles

CHE 201 Intro Eng. Thermo

CHE 230 Molec Sys in ChE

CHE 312 Transport Phenom II

CHE 321 Reaction Engineering

CHE 382 Unit Ops Lab II

CHE 422 Biochem Engineering

CHE 451 Renewable Energy

CHE 205 Comp. Meth. In ChE

CHE 301 Chem Eng Thermo

CHE 313 Transport Phenom III

CHE 341 Process Control

CHE 396 Senior Design I

CHE 425 Nano Pharm App

CHE 453 Fund Electrochem

CHE 210 Mat. Energy Balance

CHE 311 Transport Phenom I

CHE 330 Polymer Science

CHE 381 Unit Ops Lab I

CHE 397 Senior Design II

CHE 433 Process Simulation: Aspen

CHE 499 Professional Develop. Seminar

Table 3-3. Coverage of Student Outcomes 1-7 by CHE 400-level Technical Elective Courses.

Course	(1) An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	(2) An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	(3) An ability to communicate effectively with a range of audiences.	(4) An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	(5) An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	(6) An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(7) An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
CHE 422	2	3	3	1	1	1	2
CHE 425	4	4	3	3	3	1	4
CHE 426	4	3	4	1	4	3	2
CHE 427	1	3	4	4	4	1	4
CHE 433	4	3	3	3	3	1	1
CHE 451	3	4	4	4	4	1	4
CHE 453	4	1	3	1	1	3	4
CHE 454	4	3	4	1	1	3	4
CHE 455	4	3	3	3	4	4	4
CHE 457	4	3	4	1	1	1	4

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;

3 = course grade partially dependent on demonstrated knowledge on this topic;

2 = topic introduced in course, course grade not dependent on demonstrated knowledge;

1 = topic not covered specifically in course.

CHE 422. Biochemical Engineering

CHE 425. Nanotechnology for Pharmaceutical Applications

CHE 426. Biopharmaceutical Process Development and Manufacturing

CHE 427. Entrepreneurship in Engineering

CHE 433. Process Simulation With Aspen Plus

CHE 451. Renewable Energy Technologies

CHE 454. Molecular and Macromolecular Engineering

CHE 455. Nanoscale Systems in Chemical Engineering

CHE 457. Colloidal and Interfacial Phenomena

Table 3-4. Summary of the Coverage of Student Outcomes in Chemical Engineering Courses.

Student Outcomes		Number of courses at the indicated level of coverage			
#	Description	4 Topic is main focus of course, grade largely dependent on demonstrated knowledge;	3 Course grade partially dependent on demonstrated knowledge on this topic;	2 Topic introduced in course, course grade not dependent on demonstrated knowledge;	1 Topic not covered specifically in course.
1	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	19	4	1	2
2	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	7	9	5	5
3	an ability to communicate effectively with a range of audiences.	8	8	6	4
4	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	4	7	5	10
5	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	8	4	3	11
6	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	3	6	3	14
7	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	9	7	5	5

Table 3-5. Coverage of Student Outcomes 1-7 by non-ChE technical courses.

Course	(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	(3) (an ability to communicate effectively with a range of audiences.	(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
MATH 180	4	1	1	1	1	1	2
MATH 181	4	1	1	1	1	1	1
MATH 210	4	1	1	1	1	2	1
MATH 220	4	1	1	1	1	2	1
PHYS 141	4	1	1	3	1	3	1
PHYS 142	4	1	3	1	1	3	1
CHEM 122/123	4	1	1	1	1	4	1
CHEM 124/125	4	1	1	1	1	4	1
CHEM 222	4	1	1	1	1	4	1
CHEM 232	4	1	1	1	1	1	1
CHEM 233	4	1	3	1	3	4	1
CHEM 234	4	1	1	1	1	1	1
CHEM 342	4	1	1	1	1	1	1
CHEM 346	4	1	1	1	1	1	1
CS 109	1	1	4	1	1	1	4
CME 260	4	4	1	1	1	3	1
CME 322	4	4	1	1	1	3	3
ECE 210	4	3	1	1	1	3	4

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;

3 = course grade partially dependent on demonstrated knowledge on this topic;

2 = topic introduced in course, course grade not dependent on demonstrated knowledge;

1 = topic not covered specifically in course.

Abbreviated, descriptive course titles:

MATH 180 Calculus I

MATH 220 Intro. Differential equations

CHEM 122/124 Gen Chem I

CHEM 232 Organic Chemistry I

CHEM 342 Physical Chemistry I

CME 260 Properties of Materials

MATH 181 Calculus II

PHYS 141 General Physics I

CHEM 124/125 Gen Chem II

CHEM 233 Organic Chem. Lab I

CHEM 346 Physical Chemistry II

ECE 210 Elect Circuit Analysis

MATH 210 Calculus III

PHYS 142 General Physics II

CHEM 222 Analytical Chemistry

CHEM 234 Organic Chemistry II

CS 109 Prog with MatLab

ENGR 100 Engineering Orientation

CRITERION 4. CONTINUOUS IMPROVEMENT

A. Student Outcomes

Oversight and conduction of assessment is ultimately the responsibility of the Undergraduate Committee (UGC) and its chair, the Director of Undergraduate Studies (DUGS). However, the overall assessment methodology has been developed in such a way that all faculty and some industrial representatives are involved.

4.A.1. Assessment Methods and Frequency of Administration.

The overall plan and schedule of Student Outcome assessment is summarized in Table 4-1. Direct assessment rests on a triad of methods, each of which probes the outcomes in different ways, each semester or annually, by different evaluators: Method #1- Grading rubrics in CHE courses, Method #2- Senior Written Examination, and Method #3- Engineering Expo judging rubrics. Table 4-2 shows how these three assessment methods cover the various Student Outcomes (SOs) and also map to the Program Educational Objectives (PEO). In summary, SOs 1-5 and 7 are covered by two or three direct assessment methods and SO 6 is covered by one assessment method (Method #1). Indirect assessment of the SOs is achieved via an Exit Survey that is filled out by graduating seniors (Method #4). Finally, in each course, end-of-semester student evaluations (Class Climate Evaluations) of teaching administered centrally by the Office of the Vice Provost for Faculty Affairs are used to help facilitate continuous improvement of courses by individual faculty (Method #5). These assessment and feedback methods and results are now described in detail.

Table 4-1. Student outcome assessment methods and frequency of administration.

Method	Description (Abbreviation)	Category	Schedule	Responsibility
1	Grading rubrics in CHE courses(Course Rubrics)	Direct / Formative	Selected courses each year	All faculty
2	Senior Written Examination	Direct / Summative	Fall semester before graduation	DUGS
3	Engineering Expo judging rubrics (Expo Rubrics)	Direct / Summative	Spring semester	Technical Judges
4	Senior Exit Survey	Indirect / Summative	Spring semester	DUGS
5	UIC Course Evaluations	Feedback	Each semester	All faculty

Table 4-2. Coverage of Student Outcomes by Direct Assessment Methods.

ABET SO	Description	PEO	<u>Method #1</u> Course Rubrics	<u>Method #2</u> Senior Written Exam	<u>Method #3 Expo</u> Rubrics
1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	1	X	X	
2	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	1	X	X	X
3	An ability to communicate effectively with a range of audiences	3	X		X
4	An ability to recognize ethical And professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	2,3	X	X	
5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	3	X	X	
6	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	1	X		
7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies	4	X	X	

Assessment Method #1 (Direct / Formative): Rubric-based assessment of a major assignment in required courses. In the courses selected to be assessed that semester, the professor formulates a detailed rubric to assess specific performance indicators (PIs) among the students on a major assessment assignment (e.g., exam or portion of exam, term paper, lab report, milestone presentation). These PIs are mapped to some or all of the relevant SOs for that course as specified in the syllabus (see Appendix A). Table 4-3 indicates the coverage and schedule of SO assessment by this mechanism. We cannot claim to cover every relevant SO for every course every semester, because the coverage in any given course hinges on the major assessment assignment being formulated by the professor. But the partially overlapping “collage” of SO “footprints” in all the courses does ultimately provide good coverage of the SOs overall. Furthermore, this approach regularly engages the entire faculty in the process of assessment and keeps assessment and continuous improvement uppermost in our minds. The SO assessment rubrics certainly do require more time and effort from professors than does the typical academic grading process for the assignments involved, but at a manageable level that encourages buy-in among the faculty. Significantly, we thereby also avoid the burden of additional assessment activities for students. The assessment information compiled by the professor may include (1) major assessment assignment, (2) assessment rubric, (3) calculations (e.g., Excel spreadsheet, when appropriate), and (4) summary of results. Exhibit 4-1 (Section 4.C) shows a sample of these documents for CHE 205 (Computational Methods in Chemical Engineering) in Spring 2022.

Table 4-3. Schedule of direct assessment of Student Outcomes by Method #1 (Course Rubrics).

SO	2026	2025	2024	2023	2022	F2020-2021
1	201 230 301 341 382	205 210 311 312 313 321 330 381 382	311 312 381 382	210 381 382	205 210 381 382 397	381 382
2	382 397	396 433 451	396 397		397	
3	382 422	381 382 455	381 382 425 453	381 382	330 381 382 397	381 382
4	382	381 382	381 382	381 382	381 382	381 382
5	382 397	381 382 396	381 382 396 397	381 382	381 382 397	381 382
6	382	381 382	381 382	381 382	381 382 397	381 382
7	382	381 382	381 382	381 382	381 382 397	381 382

Assessment Method #2 (Direct / Summative): Senior Written Exam. The Senior Written Examination is assigned in the Fall semester CHE 499 (Professional Development Seminar) of the academic year that the students are graduating. The Senior Written Examination questions are designed to assess; (1) numerical methods and computational/software proficiency, (2) engineering estimates and problem-solving skills, (3) career planning and potential for life-long learning, (4) professional ethics and broader impacts of technology knowledge, (5) process and laboratory safety, (6) interdisciplinary teamwork. Scoring is done with a detailed rubric that maps to the SOs. The assessment information compiled by the CHE 499 professor includes (1) assessment rubric, (2) calculations (e.g., Excel spreadsheet, and (3) summary of results. Exhibit 4-2 (Section 4.C) shows these documents for Fall 2021. It should be emphasized that the senior written exam portion constitutes a rigorous **direct** assessment of student knowledge, abilities and proficiencies, and is not based on self-assessment by students (i.e., indirect assessment). The schedule of direct assessment by Method #2 is shown in Table 4-4.

Assessment Method #3 (Direct / Summative): EXPO Judging rubrics from the annual Engineering EXPO and Chemical Engineering EXPO Design Showcase. Participation in the annual Engineering Expo has been a formal course requirement in CHE 397 (Senior Design II) since before the ABET accreditation cycle in 2008. At this event the design groups present posters on their projects to the UIC community, and their work and communication skills are evaluated by a panel of judges drawn from industry (including alumni) and faculty. In addition to the College Engineering Expo, an in-person Chemical Engineering Design Showcase was added in 2022 when it was called the **ChemE EXPO**. At that time, the College was only hosting a virtual EXPO due to the pandemic. In 2023, the College returned to an in-person EXPO format, however due to very positive feedback from the senior students and senior design mentors, the chemical engineering department resumed hosting the event in 2024, 2025, and 2026 under the name **Senior Design Showcase**. In the Senior Design Showcase, UIC chemical engineering alumni and non-UIC chemical engineers from industry judge the senior design projects. To augment the College-wide, standardized judging sheets (used to award prizes), the Chemical Engineering department developed a similar judging rubric with rating categories that map to our specific Student Outcomes. There is also a general comments section so that free format feedback can be cycled back to the students and CHE 397 instructor for continuous improvement of the course. The assessment rubric, sample judging form, and a rubric assessment for Spring 2021 are given in Exhibit 4-3 (Section 4.C). The schedule of direct assessment by Method #3 is shown in Table 4-4.

Assessment Method #4 (Indirect / Summative): Senior Exit Survey. A survey was developed by the Undergraduate Committee in consultation with the department faculty to obtain self-assessment of graduating seniors on all the SOs as well as feedback for improvement of the curriculum. From 2008 to 2021, an online survey system programmed in-house in PHP code was used, and a paper survey form was implemented in 2024. The Senior Exit Survey is given to the graduating seniors in the CHE 499 Professional Development course. The CHE 499 instructor analyzes the individual surveys and compiles a summary report that (i) calculates averages and (Satisfactory, Developing, Unsatisfactory) vectors for the Student Outcomes, and (ii) aggregates student responses for each of the feedback questions. A sample survey is provided in Exhibit 4-4 (Section 4.C). A schedule of the assessment by Method #4 is shown in Table 4-4.

Assessment Method #5 (Feedback): UIC Course Climate Evaluation. A university-wide Course Climate Evaluation survey, administered centrally by the Office of the Vice Provost for Faculty Affairs is sent to students at the end of each semester for each course that they are enrolled in. These student evaluations of teaching help facilitate continuous improvement of courses by individual faculty instructors. The UIC Course Climate Evaluation from CHE 382 Spring 2024 is provided in Exhibit 4-5 (Section 4.C). The schedule of assessment by this method is shown in Table 4-4.

Table 4-4. Schedule of assessment by Methods #2, # 3, #4, and #5.

Method	2026	2025	2024	2023	2022	2021
#2 (direct) Senior Written Exam		Fall	Fall	Fall	Fall	Fall
#3 (direct) Eng Expo Judging Rubrics	Spring	Spring	Spring		Spring	Spring
#4 (indirect) Senior Exit Survey	Spring	Spring				Fall 2020, 2021
#5 (feedback) Course Climate Evaluation	Spring Fall	Spring Fall	Spring Fall	Spring Fall	Spring Fall	Spring Fall

4.A.2. Expected levels of Attainment for the Student Outcomes.

Based upon our formulation of the course rubrics (Method #1), the Senior Written Examination (Assessment Method #2), and the Engineering Expo judging rubrics (Method #3), we align the rubric assessment levels of S (Satisfactory), D (Developing) and U (Unsatisfactory) roughly with letter grades A/B, C and D, respectively. The rubric assessment levels are given numerical values of: Satisfactory = 3, Developing = 2, and Unsatisfactory = 1. For each Student Outcome among our students, we are looking to achieve an average rating score above about 2.30, 65% or better attainment of Satisfactory (%S), and 85% or better attainment for Satisfactory or Developing ($\geq D$).

4.A.3. Evaluation of the Attainment of Student Outcomes.

Reduction of assessment data and their synthesis into higher-level information on attainment of SOs proceeds via four levels of analysis. The calculations are carried out by the faculty member(s) conducting the assessment and the Director of Undergraduate Studies, assisted by an assessment TA.

Level 1 analysis:

Reduction of data from assessment Rubrics.

For each of the three direct assessment methods (Table 4-1), individual students are assessed using Performance Indicators on rubric items such as exams questions, lab reports, oral presentations, individual and team projects (including computer coding, process simulation and process design), technical papers, safety quizzes, safety tutorials, and ethics assignments. Performance Indicators are given a “3” for Satisfactory, “2” for Developing, and “1” for Unsatisfactory. Each PI rubric score is then averaged over all the students, and a total PI average

for the rubric item is also calculated. The results for individual PI rubric items are then mapped onto the applicable SOs. One rubric item may (but need not) relate to more than one SO, and each SO may have contributions from several rubric items. This procedure produces direct assessment results on attainment of SOs for; (i) students in a particular course in a given semester (Assessment Method #1 – Course Rubrics), and (ii) a graduating senior class (Assessment Method #2 – Senior Written Examination; and Assessment Method #3 – Engineering Expo Judging rubrics).

Level 2 analysis:

Average and (S, D, U) attainment assessed with 4 methods by Student Outcome.

For each SO we combine the (S, D, U) results of all Level 1 analysis into a one-page summary report. This allows tracking of trends for each assessment method over time and also exposes similarities and differences between the methods. The summary reports for all SOs appear in Exhibit 4-6 (Section 4.C).

Level 3 analysis:

Average and satisfactory attainment for all Student Outcomes by method.

For each method we combine the average attainment rating and the percentage of students attaining between “Satisfactory” and “Developing” for all SOs year by year into a one-page summary report. This facilitates tracking SOs over time and also comparisons between SOs to determine which aspects of the curriculum are most in need of improvement. The summary reports for all methods appear in Exhibit 4-7 (Section 4.C.). The results of the indirect assessment are shown in Table 4-8.

Level 4 analysis:

Satisfactory attainment for all Student Outcomes.

For each SO we aggregate the Level 3 results from all three direct assessment methods into an overall percentage of students attaining; 1) Developing or higher, 2) Satisfactory. We also aggregate from all three direct assessment methods the rating scores for each SO. The Level 2 results from indirect assessment (Method #4) are carried over from the Level 3 analysis. The three summaries are shown in Tables 4-5, 4-6 and 4-7, and give us an overall, “bird’s eye” view of SO attainment and informs which outcomes need improvement. Bar charts of the numerical data in Tables 4-5 – 4.8 are also given in Exhibit 4-8 (Section 4.C).

Results: The direct assessment method summary (Level 4 analysis) shown in Tables 4-5, 4-6, and 4-7 indicate that the UIC Chemical Engineering undergraduates are sufficiently attaining the departmental Student Outcome criteria of 1) reaching a 65% or better attainment of a Satisfactory score, 2) achieving an 85% or better attainment of a Satisfactory or Developing score and realizing a rating score above 2.30/3.00. In fact, more than 83% are reaching 75% attainment of a Satisfactory score and over 90% of our students are reaching 70% attainment of a Satisfactory score. In addition, the average rating scores are well above the department criteria of 2.30/3.00, with average ratings above 2.60 for all student learning outcomes. The results from SO #1, SO #2, SO #3, SO#4, SO #5, SO #6 and SO #7 indicate that our students are sufficiently learning the core engineering, mathematics, science, experimentation, teamwork, communication and design skills. Satisfactory attainment of the safety/environmental and global, economic, societal and ethics performance indicators, as covered by SO #4, have been steady

above 80% Satisfactory attainment and reflect our efforts in this student learning outcome. Furthermore, SO #2 which encompasses the application of engineering design, also has high attainment and average rating values and reflects the effort by both our industrial mentors to instruct the students in chemical engineering industry design practice with an emphasis on following industry standards in their senior design projects, and by the design instructor in continuously improving the course from industry mentor and student feedback.

Table 4-8 presents the average scores from the Senior Exit Survey, which is a self-evaluation by the students to rate how they feel they have achieved learning the skills as given in the 7 student outcomes. All the ratings are between 5 = Excellent and 4 = Strong and correlate well with the three direct assessment method evaluations summarized in Tables 4-5, 4-6 and 4-7.

Table 4-5. Percentage of students attaining a rating of Satisfactory for all the SO's as determined by direct assessment (aggregate of Methods #1, #2, #3).

SO	2026	2025	2024	2023	2022	2021
1	70.0	72.9	74.7	76.8	77.9	80.8
2	76.6	74.2	65.0	87.8	87.5	76.7
3	77.4	83.7	72.9	80.5	78.5	80.9
4	79.5	87.0	88.1	80.6	84.2	89.6
5	76.6	88.1	83.1	88.3	86.6	88.2
6	76.9	84.0	77.3	78.0	77.1	83.1
7	84.6	89.3	79.4	85.2	82.6	78.5

Table 4-6. Percentage of students attaining a rating of Satisfactory or Developing for all the SO's as determined by direct assessment (aggregate of Methods #1, #2, and #3).

SO	2026	2025	2024	2023	2022	2021
1	90.0	93.9	92.5	92.9	95.4	97.1
2	100.0	99.0	99.2	93.9	97.4	100.0
3	100.0	99.6	99.1	97.5	98.2	97.4
4	91.0	97.7	96.9	92.5	95.6	98.5
5	100.0	97.7	99.6	96.8	98.3	97.6
6	100.0	98.8	94.9	94.5	96.6	95.6
7	100.0	100.0	99.3	98.6	97.7	99.0

Table 4-7. Average rating score for all the SO's as determined by direct assessment (aggregate of Methods #1, #2, and #3).

SO	2026	2025	2024	2023	2022	2021
1	2.65	2.64	2.61	2.63	2.67	2.77
2	2.77	2.71	2.67	2.74	2.85	2.77
3	2.78	2.83	2.75	2.75	2.79	2.80
4	2.71	2.84	2.90	2.73	2.81	2.88
5	2.79	2.86	2.84	2.85	2.87	2.86
6	2.77	2.85	2.73	2.70	2.74	2.73
7	2.94	2.90	2.79	2.85	2.80	2.84

Table 4-8. Average rating score for all the SO's as determined by indirect assessment Method #4 (Senior Exit Survey)

Note: a different rating system is used than Methods #1- #3:

(5 = Excellent, 4 = Strong, 3 = Moderate, 2 = Weak, 1 = Poor)

SO	2026	2025	2021	2020
1	4.30	4.19	4.08	4.53
2	4.16	4.02	4.05	4.43
3	4.53	4.32	4.08	4.60
4	4.18	4.19	4.04	4.48
5	4.47	4.26	4.08	4.55
6	4.41	4.32	4.08	4.46
7	4.56	4.20	4.08	4.60

B. Continuous Improvement

The loop of assessment, evaluation and feedback from constituencies has been closed in various ways to improve the program/curriculum, student learning experience, and individual courses. These cases are now described in detail. Ongoing and future improvements are also described in this section.

4.B.1. Ongoing Course Improvement using Feedback-based analysis.

We maintain an ongoing feedback loop for continuous improvement of existing courses by the following procedure (Assessment Method #5). At the end of every semester, students complete the Course Climate Evaluations administered centrally at the university level. Within two weeks or so the professor receives by email the compiled survey results. The professor reviews this student feedback from the evaluations and reflects on how he/she intends to improve that course for the next year (or semester). The Course Climate Evaluations represent a regular and extensive consultation with our student constituency with tangibly positive effects on their courses and learning experiences. Faculty respond to their course evaluations with recommendations and actions to improve their courses, which are captured through self-study statements and the Faculty Performance Profiles done every July. An example of a Course Climate Evaluation is shown in Exhibit 4-5. Faculty evaluation questions include; instructor teaching effectiveness, overall quality of the course, effective use of technology, course assignments/exams/quiz alignment with course content, and course workload. The overall teaching effectiveness evaluations for each CHE faculty for the last five years are shown in Figure 4.1. All chemical engineering instructors have ratings higher than 3.25, with >70% of the faculty rating higher than 4.00, indicating greater than satisfactory to excellent ratings by our students for our instruction methods. A sample self-study statement for CHE 311 (Transport Phenomena I) is shown in Exhibit 4-9 (section 4.C). Additional self-study statements written by faculty members and course recommendations and actions will be available for inspection during the upcoming ABET site visit.

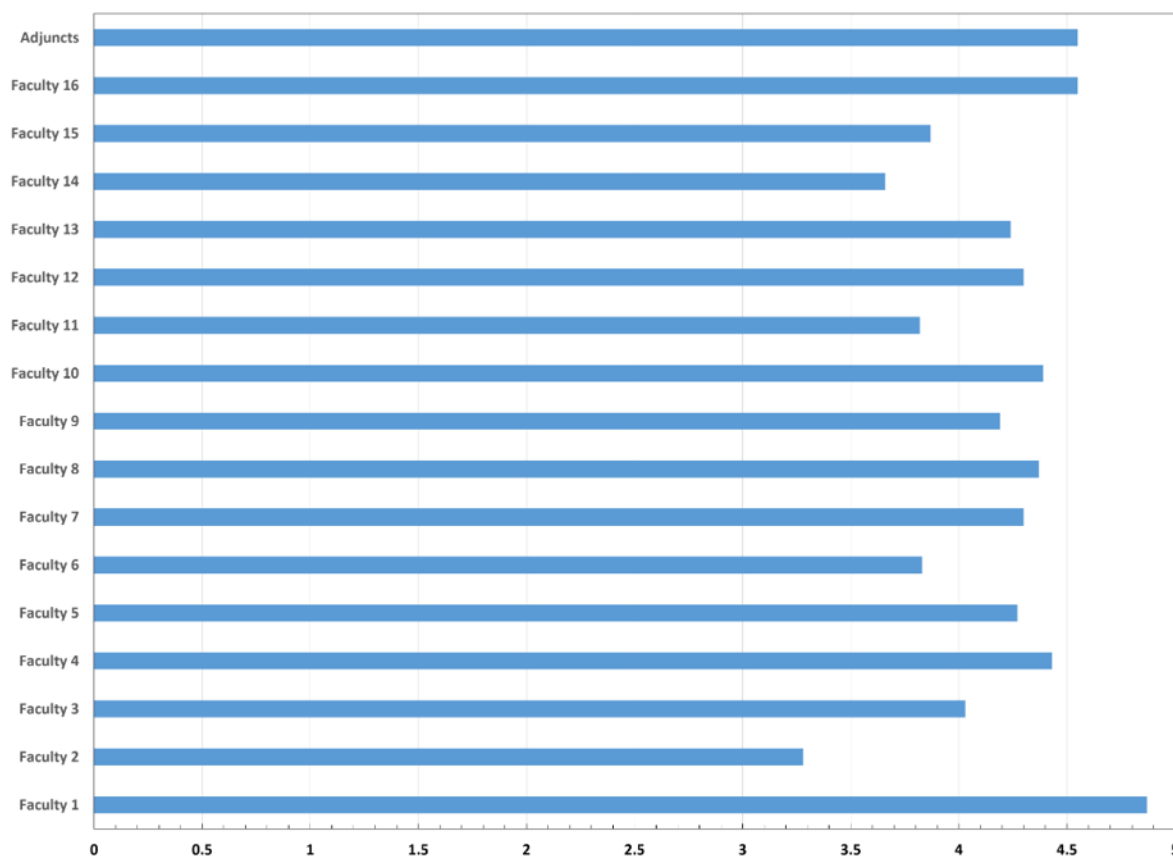


Figure 4.1. Overall teaching effectiveness scores for CHE faculty averaged over the last five years (2021-2026); from Course Climate Evaluations.

4.B.2. Improvement of CHE 381/382 Chemical Engineering Laboratory I and II from feedback-driven and rubric assessment analysis.

An improvement based on student feedback from the Course Climate Evaluations and rubric assessment analysis of SLO#1 has been made in the senior-level chemical engineering laboratory courses (CHE 381/382) that has closed the loop of assessment.

Improvement in SLO#1: Course improvement comments from the Course Climate Evaluations students suggested that the lab manuals needed updating in order to provide more information on how to properly analyze the experimental data. An analysis of the CHE 381/382 rubric assessment for SLO#1 in 2021 and 2022, shown in Table 4.1, confirmed lower than expected scores of students attaining a Satisfactory grade for performance indicator #7, which covers the experimental calculations and results (Final Lab report Results section). Consequently, starting in 2023, the instructor revised the lab manuals to provide better clarity on what analysis is required for the lab reports and also developed sample calculation sheets for several of the more complex experimental calculations in the humidification, gas absorption, heat transfer, distillation, process control, and membrane separation experiments. Subsequent rubric assessment analysis of this performance indicator from 2023 to 2026 shows that the percentage of students attaining a Satisfactory score increased, as well as the average rating score, with more than 80% of the students receiving a Satisfactory score. This is a case of assessment confirming

the need for improvement from student feedback and the assessment verifying the improvement achieved by making course information changes.

Table 4.1. Rubric assessment for SLO#1 Performance Indicator 7 from CHE 381/382.

	2021	2022	2023	2024	2025	2026
%S	47.5	60.4	48.1	69.2	89.2	84.6
%S+D	100	100	82.7	89.7	100	100
Average Rating	2.39	2.34	2.65	2.68	2.89	2.65

Ongoing improvement of lab report grading:

Additional course improvement comments from the Course Climate Evaluations students suggest that the teaching assistant (TA) grading of the lab report is uneven between different TA's and needs improvement. Although the instructor reviews the report grades from the TA's before posting, the instructor is in the process of creating a more detailed rubric for each section of the pre-lab and final lab reports in order to improve and provide more consistency in the grading.

4.B.3. Rubric assessment and feedback-based improvement of SLO #2: Engineering EXPO and Senior Design Showcase EXPO.

An improvement based on a rubric assessment analysis of SLO#2 from the Engineering EXPO/Senior Design Showcase judging scores and feedback from the judges and senior design team mentors has been made that has closed the loop of assessment. As noted in Section 4.A.1 Assessment Method #3, the senior design teams participate in the annual Engineering EXPO as part of their formal course requirement in CHE 397 (Senior Design II). In Spring 2022, the chemical engineering department initiated a Senior Design Showcase, a complimentary event to the Engineering EXPO but with chemical engineering senior design teams only. The Senior Design Showcase is now held annually from 2024-2026 before the Engineering EXPO. Although the Engineering EXPO judges represent a broader spectrum of judges with different expertise, the Senior Design Showcase judges consist of UIC chemical engineering alumni and other chemical engineers from industry with detailed knowledge of chemical process design. An assessment was made of SLO#2 using the Engineering EXPO scores from 2021-2022, and the Senior Design Showcase scores from 2024-2026. A decrease in the % Satisfactory attainment results was observed from 2022 to 2024, as shown in Table 4.2. Average ratings were satisfactory but had also decreased slightly. Discussions between the CHE 397 instructor and senior design mentors concluded that the Senior Design Showcase judges were judging the technical content more rigorously since they have more specific chemical engineering expertise. Feedback from the Senior Design Showcase judges indicated several areas for improvement; more detailed evaluation of design alternatives and constraints (economical, environmental, social, ethical), further analysis of design factors such as cost, manufacturability and sustainability, and better preparation to answer technical questions. The CHE 397 instructor worked with the senior design team mentors to provide direction and guidance during the class lectures and to improve these technical areas. Subsequent rubric assessment analysis from 2025 to 2026 shows that the percentage of students attaining a Satisfactory score has increased, as well as the average rating score, with 80% of the students receiving a Satisfactory score in 2026. A meeting held in Spring 2026 after the Senior Design Showcase with the senior design judges, senior design team mentors, CHE 397 instructor, and DUGS, concluded that the senior design

teams had made significant improvements in both technical and presentation quality of their projects due to better mentor guidance. Future improvements such as introducing a business framework, standardizing the process flow diagrams (PFDs) and keeping project topics open-ended were also recommended at this meeting and will be incorporated into the course next year.

Table 4.2. SLO #2 Attainment from Assessment Method #3: Engineering Expo Rubrics

Year	2026	2025	2024	2022	2021
% (3 = Satisfactory)	80.0	57.8	47.1	85.0	62.7
% > Developing	100.0	100.0	100.0	100.0	100.0
Average Rating	2.80	2.58	2.51	2.85	2.63

4.B.4. Feedback-based improvement of CHE 499- Professional Development Seminar course.

During the previous ABET review cycle (2015-2020), the CHE 499 Professional Development Seminar curriculum was revised to create a seminar series of industry-oriented chemical engineering speakers throughout the semester. Invited speakers, mainly chemical engineers working in industry, give talks on topics related to chemical engineering practice, professional development and job searching techniques. They come from a variety of industries such as oil/gas, pharmaceuticals, fine chemicals, food, patent law, alternative energy and semiconductors. The UIC Engineering Career Center also gives seminars on job-related topics. Student feedback from the Course Climate Surveys has been very positive about the structure of the seminar course. However, student feedback, both formally and informally, suggested inviting more women engineers as seminar speakers. The undergraduate student body in chemical engineering is approximately 40% women, so adding women speakers would be beneficial in providing more career experiences, advice and role models to our women students. Another suggestion made by the chemical engineering faculty during one of our faculty retreats was to add a speaker from academia. Although most of our undergraduate students pursue industry careers, there is a small but increasing population that are attending graduate school. An academic seminar speaker would provide another viewpoint on career choices and experiences in academia. Since 2022, 4 to 6 of the 12 seminar speakers have been women coming from a variety of industries, including non-traditional careers such as patent law. In addition, one academic-oriented speaker was added to the seminar series. The academic speakers have included PhD candidates, an engineer with a master's degree, and a chemical engineering faculty member. Student feedback has been very positive about these changes. These improvements, which started in Fall 2022, may also explain the upturn of SOs #4 and #7 scores in the Senior Exit Survey results (Method #4) shown in Table 4.3.

Table 4.3. SO#4 and SO#7 scores from Senior Exit Survey; Exhibit 4-7.

SO	2026	2025	2026
4	4.04	4.19	4.18
7	4.08	4.20	4.56

4.B.5. Rubric assessment and ongoing feedback-based improvement to CHE 230 course curriculum.

Two courses were added to the chemical engineering curriculum in 2022; CHE 230 Molecular Systems in Chemical Engineering, and CHE 330 Polymer Science. The courses are designed to introduce several topics that are important to traditional and emerging fields in chemical engineering; polymers, molecular engineering, macromolecular systems, and nanomaterials. A rubric assessment was performed on CHE 230 (SLO#1) in Spring 2026, and a rubric assessment of CHE 330 in Fall 2022 (SLO#3) and Fall 2025 (SLO#1). The results of the rubric assessments are shown in Table 4.4. Attainment of a Satisfactory score was achieved for a majority of the students, indicating that students are developing reasonable confidence in using mathematical relationships once the variables and governing equations are clearly defined. However, the Satisfactory attainment for CHE 230 is slightly below the 65% threshold criteria. The self-study report from the course instructor has provided several areas for improvement to CHE 230; including additional in-class examples that explicitly connect phase diagrams, microconstituents, and phase-rule calculations, placing more emphasis on helping students distinguish between phases and microconstituents, and more clearly identifying which values from a phase diagram are needed for a specific calculation. These improvements will be incorporated into the next offering of the course, with further assessment of closing the loop.

Student feedback on overlap with CME 260: Student feedback from the Course Climate Evaluation has indicated a perceived overlap between ChE 230 and CME 260 Properties of Materials. While some overlap is expected because of foundational topics such as phase behavior, thermodynamics, molecular structure, and material properties naturally support multiple chemical engineering courses, this feedback will be used to refine the course more broadly. The director of undergraduate studies, undergraduate committee member Vivek Sharma, department head Vikas Berry and Ezinne Achinivu-Ibagere, course instructor, have had several meetings and discussion about changing the topics in CHE 230. Course content was reviewed to identify areas where coordination with related courses may reduce unnecessary repetition while preserving useful conceptual reinforcement. Future offerings of ChE 230 will more clearly emphasize its distinct role as a molecular systems course by consistently connecting topics to molecular structure, bonding, microstructure, structure–property relationships, and engineering applications. An excerpt from the CHE 230 Course Climate Survey Evaluation showing student feedback and an email exchange between the faculty addressing this improvement is shown in Exhibit 4-10 (section 4.C).

Table 4.4. Course rubric assessment for CHE 230 and CHE 330.

	2022	2025	2026
	CHE 330 SLO #3	CHE 330 SLO#1	CHE 230 SLO#1
%S	79.7	65.0	60.0
%S+D	94.7	87.5	81.8
Average Rating	2.76	2.40	2.35

4.B.6. Program-Level Improvement: Addition of 5 Concentrations to the Curriculum.

The BS degree in Chemical Engineering has 6 optional concentrations: Biochemical Engineering, Energy & Environment, Entrepreneurship, Nanotechnology, Polymers & Molecular Engineering, and Process Automation. The last 5 concentrations were added in 2022 as a program-level improvement to the curriculum. Initial discussions about increasing the number of concentrations began in 2019 between the Department Head and the Director of Undergraduate Studies. They identified a need to offer more specialized studies in both traditional and emerging areas of chemical engineering, as well as in engineering-based entrepreneurship, ultimately deciding on the 5 concentration topics listed above because they are in areas where Department faculty have specific expertise both in research and in practice; molecular engineering, polymers, nanomaterials, energy, environment, process automation, and entrepreneurship. These specializations play a central role in biological, chemical/industrial, environmental and pharmaceutical processes and technologies and thus are important additions to the curriculum.

Timeline: The five proposed concentrations were discussed at the Chemical Engineering External Advisory Board meeting in June 2019 and were approved by the Board members. In addition, the results from a survey of chemical engineering alumni in Summer 2019 also indicated overwhelming support for the five proposed concentrations. Work then began with a faculty task force to develop the concentration curriculum; CHE required and elective courses, selective course options outside the CHE-rubric, and the proposal. After the proposal was approved by the Department, the College Education Policy Committee, and the Senate Educational Policy Committee (SCEP), the University Senate gave final approval in 2021. The concentrations were added to the curriculum, and students began signing up for the concentrations in 2022. The course requirement details for each concentration are outlined in section 5.A.2 and are also shown in detail on the Chemical Engineering Undergraduate website page: <https://che.uic.edu/undergraduate-programs/che-major/>.

Figure 4.2 shows the percentage and number of students pursuing a concentration from 2021-2026. There was an increase in students enrolled starting in 2022 when the 5 concentrations were added to the curriculum. There has also been an increase in the percentage of students pursuing a concentration, indicating a strong interest from our student population. Figure 4.2 also shows the breakdown of students taking the different concentrations. Biochemical Engineering, Process Automation and Energy & Environment have been the most popular, however, Polymer & Molecular Engineering has seen a recent increase in student enrollment.

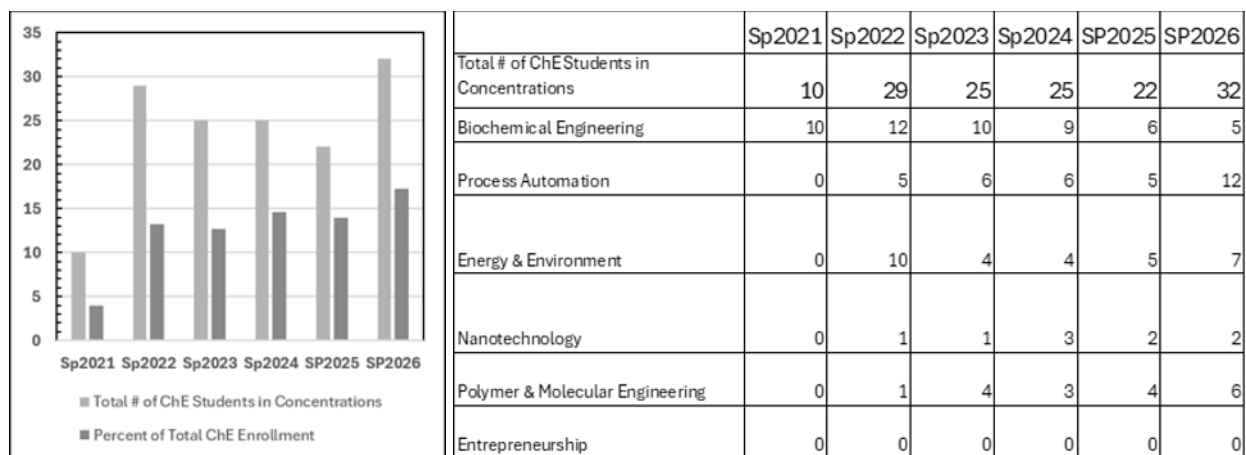


Figure 4.2 Total number and percentage of students in concentrations per academic year.

The 400-level courses taken in the various concentrations are: Biochemical Engineering- CHE 422 Biochemical Engineering; Energy and Environment- CHE 451 Renewable Energy Technologies and a second ChE technical elective; Entrepreneurship- CHE 427 Entrepreneurship in Engineering; Nanotechnology- CHE 455 Nanoscale Systems in Chemical Engineering and a second ChE technical elective; Polymers and Molecular Engineering- CHE 454 Molecular and Macromolecular Engineering and a second ChE technical elective (CHE 425, CHE 453, CHE 455); Process Automation- CHE 433 Process Simulation with Aspen Plus.

Although the original discussions on adding the concentrations were not actually precipitated by results of assessment, a course rubric assessment on SLO #2 and SLO #3 has been undertaken on the following 400-level courses that are part of these concentrations, as shown in Table 4.5. The analysis shows that the Satisfactory attainment threshold of 65% and average rating threshold of >2.30 is more than sufficiently being met in the 400-level chemical engineering concentration courses. Self-study reports from the instructors of these courses have provided suggestions for further improvements. These include; (CHE 451) adding course materials and class discussion on the benefits and socio-economic challenges between conventional and alternative energy solutions; (CHE 455) deepening the level of understanding of the underlying physics in nanoscale systems and encouraging more critical thinking by asking students to analyze limitations or interpret experimental scenarios.

Table 4.5 Course rubric assessment for 400-level courses taken in the concentrations.

	CHE 422	CHE 425	CHE 433	CHE 451	CHE 453	CHE 455
	SLO3	SLO3	SLO2	SLO2	SLO3	SLO3
%S	81.0	100.0	86.0	66.7	66.7	81.0
%S+D	100	100	91.2	100	100	100
Average Rating	2.82	3.00	2.49	2.63	2.67	2.81

4.B.7. Program-Level Continuous Improvements: Extracurricular Student Experiences-ChemE Connect, and AIChE student chapter initiatives.

Based on College of Engineering administration and student feedback, several improvements have been made at the program level to the chemical engineering extracurricular student experiences that will be described below.

Department Action Team/ChemE Connect

In 2023, the College of Engineering initiated several Department Action Teams within the engineering departments with the aim of improving the undergraduate community and student success. In the chemical engineering department, the Department Action Team consisted of a group of faculty, staff members, and students; Professors Betul Bilgin, Korosh Torabi, and Ezinne Achinivu-Ibagere, Laboratory and Computer Manager Jan Sagun, undergraduate student Emily Seriruk, and graduate student Nadia Nikolova. The team met regularly to discuss ideas on how to bring change from the bottom up, how to create sustainable change to the chemical engineering undergraduate education, and how to make meaningful changes in the department. A Town Hall meeting of faculty, students and staff was held in Fall 2024, and a department climate survey was sent to the chemical engineering undergraduate and graduate students and faculty/staff. A focus group of 9 students was also interviewed. Results concluded that a majority of students are comfortable with the climate in the department. Suggestions for improvement included; creating more chemical engineering clubs, increasing faculty/student interaction and providing more programs for first-year and transfer students. The team incorporated this feedback into their recommendations. The first initiative that the ChE DAT initiated was an End-of-the-Year Banquet held in April of 2024. The banquet provides an opportunity to share information about the department and the resources available, and increases student/faculty interaction, particularly for first-year and transfer students.

In 2024, the ChE Department Action Team changed their name to ChemE Connect. It is a student-led Department Action Team (DAT) within the Chemical Engineering department acting as a departmental student advisory/community group to enhance the student experience. The ChemE Connect team consists of the same three faculty members and staff member, but now have two undergraduate students, Joshua Olusoji and Anna Cai and one graduate student, Zac Adensaya. The ChemE Connect mission is to foster a stronger sense of community and belonging within the department, especially for our students but also for faculty and staff. They also aim to bring students together by providing them with both professional development events and a social circle to talk about things that aren't related to chemical engineering. For instance, in collaboration with the AIChE student chapter, ChemE Connect has hosted student events like the Fall semester Welcome Back Picnic in 2024 and 2025, the Spring semester End-of-Year banquet in 2025 and 2026, and numerous professional development opportunities throughout the year. These events also provide an opportunity to help students connect and build closer relationships. Additionally, the goal of ChemE Connect is to celebrate our community, its students, and their achievements. These initiatives from ChemE Connect have received very positive feedback from our students, faculty, staff, and external partners such as the senior design mentors and external advisory board. The ChemE Connect team remains active and will continue to support and improve the climate of the chemical engineering department. A description of the objectives of the ChE DAT is shown in Exhibit 4-12 (section 4.C).

AIChE Student Chapter Initiatives

The UIC student chapter of the American Institute of Chemical Engineers (AIChE) has historically been a strong student organization. Activities decreased due to the pandemic from 2020 to 2022, but recently the chapter has seen an increase in members and in chapter activities. Some of the recent student-led activities are listed below.

- Hosted North-Central Regional AIChE conference at UIC- April 2023.
- AIChE ChemE Car team: Started in 2018, won 1st place at North Central regional conferences (2021, 2022)
- AIChE ChemE Cube team: started in 2024
- Professional events: Q&A panels with industry professionals, plant visits, resume workshops
- Social events: Bowling night, Karaoke night, movie night, pot-luck dinner
- Volunteering: Chemical engineering department recruitment events
- Outreach:
- Fundraising: Raised over \$5000 to help fund ChemE Car/ChemE Cube teams and events
- Mentorship: AIChE facilitated mentorship between upperclassmen and lower classmen for Intro- ChE courses

4.B.8. Program-Level Ongoing Improvement of Safety and Ethics Content in the Curriculum.

Safety: The incorporation of safety-related material into the Chemical Engineering curriculum was initiated in the 2008-2014 ABET cycle and has continued up to the present, emphasizing the importance of safety training for students. The safety rubric is tracked and assessed using SO #4 (see Exhibit 4-6: Level 2 Analysis of Assessment Data). Specific safety tutorials from the AICHE website are now used regularly in several courses (CHE 201, 381, 382, 396 and 397) and include the following titles; “Introduction to Process Safety”, “Hazard Recognition”, “Dust Explosions”, and “Chemical Reactivity Hazards”. Safety assignments in the CHE 396 and CHE 397 Senior Design I and II courses are also used regularly. These safety-related assignments discuss the hazards associated with the specific processes that the student groups are designing and include a HAZOP analysis in the form of a report and presentation. Since 2015, two safety assignments have been incorporated into the CHE 381 and CHE 382 laboratory courses; an industrial R&D Chemical Hygiene Plan, and a Hazardous Chemical Safety Course reading assignment with a quiz to grade comprehension.

ChemE Car and ChemE Cube: As mentioned above, two teams of undergraduate students participate annually in the AIChE ChemE car and ChemE cube competitions. A major aspect of these competitions is a safety analysis of their project. The students complete an Engineering Design Package that includes a Job Safety Assessment Form, describing the major safety and chemical hazards involved with the design, a hazard analysis as well as identification of safety measures and safety controls. In addition, team members complete the AIChE Academy tutorial- ELA199 ChemE Car Safety Course as part of their safety training for the competition.

Undergraduate research. Finally, many of our undergraduates perform research with the faculty in their research laboratories through the CHE 392 Undergraduate Research course. These students are required to complete a UIC Environmental, Health and Safety Office (EHSO) Laboratory Training seminar in order to work in the labs. Faculty supervisors and graduate students also inform the students on the particular safety hazards associated with the research work in their laboratories.

Ethics: The incorporation of ethics-related material into the Chemical Engineering curriculum was initiated in the 2015-2020 ABET cycle and has continued up to the present, emphasizing the importance of ethics training for students. An assignment on ethics is regularly assigned in CHE 396. The AIChE Code of Ethics is covered in several courses (CHE 201, CHE 321 and CHE 396), where ethical situations related to engineering practice are given and discussion in class. Ethics as a rubric item is assessed in SO #4 and has shown steady scores of greater than 80% Satisfactory attainment in the senior written examination assessment (Exhibit 4-6 Level 2 SLO#4 analysis).

Examples of Student Output in Ethics Assignments and Safety Assignments are given in Exhibit 4-11.

4.B.9. Program-Level Future Improvements to the Curriculum: New Introductory Chemical Engineering course (CHE 101).

Based on discussions and feedback from our key constituencies and our faculty, a proposal is being drafted to modify our undergraduate program and add an introductory chemical engineering course to our curriculum; CHE 101 Introduction to Chemical Engineering and Sustainability. One of the goals of the course is to introduce first-year students to the chemical engineering discipline since students currently don't take a chemical engineering course until their third semester (Sophomore year). The other goal is to introduce sustainable engineering topics as applied in the chemical engineering discipline to prepare students for future technologies. The course will provide an overview of the principles and unit operations of chemical engineering as practiced in traditional industries as well as in sustainable technologies. Five key sustainable themes will be introduced: sustainable energy, sustainable health, sustainable food, sustainable air, and sustainable water. Students will explore challenges in these areas and the role of chemical engineering in developing sustainable products and processes. The topics covered will provide context to subsequent 200 and 200 level chemical engineering courses and the connection of chemical engineering to basic sciences and other engineering disciplines.

Since CHE 101 will be a 3-credit hour course, the program course requirements will need to be modified in order to keep the total credit hour requirement at 128 hours. Possible options that have been suggested are:

- Remove CHEM 234 as a requirement
- Create a Selective option between CHE 230 and CME 260
- Remove CHE 342 Physical Chemical I as a requirement

Faculty, External Advisory Board, and Alumni input is ongoing, and a task force consisting of several faculty members (including the DUGS) has been initiated to decide on the best option.

Continuous Improvement Documentation:

In addition to the course rubric assessment statements and self-study course statements written by faculty members, we will also present the following documentation in a web-format presentation system to support the continuous improvement cycles described above (4.B.1 to 4.B.9) for the ABET site visit:

- Minutes/presentations from departmental faculty meetings, faculty retreats
- Undergraduate committee reports.
- Minutes of College EPC meetings.
- Agenda/minutes from the Chemical Engineering Curriculum Day in 2022.
- Notes/minutes from the advisory board meetings.
- Results from the advisory board surveys.
- Results from the alumni surveys.
- ChemE Connect survey results, notes, presentation
- Notes/minutes from the chemical engineering education committee.
- Pertinent emails, discussion notes from faculty, external advisory board members, staff.

C. Additional Information

The following documents provide samples of the assessment, evaluation and continuous improvement procedures described in Sections 4.A and 4.B. A complete set of documents will be provided at the ABET site visit in a web format presentation system allowing easy access and cross-verification of information.

Exhibit 4-1.

Assessment Method #1 (Direct / Formative):

Rubric-based assessment of a major assignment in required courses.

Sample Description: Spring 2022 Assessment assignment, rubric and results for CHE 205.

ChE 205- Computational Methods in Chemical Engineering- Spring 2022

Student Learning Outcome 1

Rubric for assessment for Final exam, Problems, 1c, 2, 4, and 3a

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem 1c	Does not understand the connection between mathematical models and the system or process to be analyzed or designed	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem 2	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems
4. Translates academic theory into engineering applications Problem 4	Does not appear to grasp the connection between theory and the problem	Some gaps in understanding the application of theory to the problem and expects theory to predict reality	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality
6. Uses appropriate resources needed to solve problems Problem 3a	Uses appropriate resources to locate information needed to solve problems	Use limited resources to solve problems	Uses no resources to solve problems

Student Learning Outcome 1

Rubric for assessment for Final exam, Problems, 1c, 2, 4, and 3a

RESULTS BY PERFORMANCE INDICATOR

Rubric Item (PI)	1	2	4	6
Mapped SLO	1	1	1	1
Average	2.25	2.25	2.10	2.85
Std Dev	0.79	0.64	0.79	0.37

RESULTS BY STUDENT OUTCOME

SO	1
Average Score	2.36
Standard dev.	0.29

Analysis:

Average scores were fairly uniform across Performance Indicators (PI) 1, 2, and 4. The strongest score was for Performance Indicator 6; Using appropriate resources to solve problems. A lower score was observed for PI 4, Translating academic theory into engineering applications. There was a fairly even distribution between satisfactory and developing scores for PI's 1, 2 and 4. This data suggests that most students were strong in (i) using appropriate resources needed to solve problems, were developing in (ii) applying mathematical and scientific principles to formulate models, and (iii) solving engineering problems, but were weak in (iv) translating academic theory into engineering applications.

Exhibit 4-2.

Assessment Method #2 (Direct / Summative):

Senior Written Exam

Sample Description: Fall 2025 Senior Written Examination and rubrics.

(See next 5 pages)

BSCHE Senior Written Examination- Fall 2025

Scoring Rubrics Mapped to Student Learning Outcomes and Program Educational Objectives

Professionalism and Broader Knowledge

Question #	Performance Indicator (PI)	SLO	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1	1 Background research on career options and plans	7	No significant research. Unfamiliar with application deadlines, interview procedures and employer policies.	Has checked some information on specific universities or companies on the internet. Actively seeking job interviews or knows relevant deadlines for graduate schools.	Can quote employer policies (e.g., career tracks, promotions, sponsorship of advanced degrees). Has submitted applications, has job interviews scheduled, or already accepted for job or graduate school.
2a	2. Ethical and societal ramifications of technology	4	Unable to think of an example or cannot identify ethical ramifications of supplied technological topic.	Can independently come up with an example or identify an ethical ramification of supplied example.	Clearly explains ethical impact or social issues related to technological topic.
2c	3. Approach to ethical dilemmas	4	Unable to recognize ethical dilemma in hypothetical scenario. Purely ad hoc or “gut feel” approach toward ethical decisions. Does not know where or to whom to turn for guidance. Confounds fact-finding with ethical consideration.	Recognizes ethical dilemma in hypothetical scenario (or own experience). Logical considerations toward resolution. Can formulate steps to seek guidance.	Articulates resolution of dilemma as outcome of systematic thought process or ethical framework. Familiar with channels to air ethical concerns and/or cognizant officers.
2d	4. AIChE Code of Ethics	4	Unfamiliar with existence or content of code.	Knows where to find code and is familiar with some provisions.	Has read code more than once and can quote main provisions.
3	5. Multidisciplinary teamwork	5	Cannot name any collaborating engineering fields or explain context of collaboration.	Names one other type of engineer and articulates some aspects of collaboration.	Names two other types of engineer and clearly articulates example or specific contexts in which his/her work could intersect.

Technical Assessment

Question #	Performance Indicator (PI)	SLO	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
4	6. Description of numerical method and formulation of algorithm	1,2	Does not know how to proceed and/or unable to get started.	Describes most of numerical method but shows some gaps in understanding or is cannot formulate algorithm.	Describes numerical method correctly and successfully formulates algorithm.
4	7.Proficiency with Excel or MATLAB	1	Unfamiliar with Excel, MATLAB (and any other math package). Unable to get started or needs extensive consultation of on-line manual to get started. Incomplete or incorrect solution.	Able to do significant portion or most of problem with some quick consultation of on-line help.	Fluent in finishing problem completely with little or no consultation of online help.
5	8.Identification of system, surroundings and applicable flow streams	1,6	Does not mention system or surroundings or falters seriously in implementation.	Provides many details of system, surroundings and flow streams correctly	Complete and correct formulation of engineering system.
5	9.Effective problem solving skills	1	Unable to translate physical ideas into mathematical form. Ineffective solution.	Formulates mathematical problem substantially and makes substantial progress toward correct solution	Correct mathematical formulation and completion of correct solution, including units

Scoring Rubrics Mapped to Student Learning Outcomes and Program Educational Objectives

RESULTS BY PERFORMANCE INDICATOR:									
Rubric item (PI)	1	2	3	4	5	6	7	8	9
Mapped SLO	7	4	4	4	5	1,2	1	1,6	1
Average	2.88	2.91	2.86	2.84	2.92	2.91	2.83	2.86	2.84
Std Dev	0.33	0.28	0.34	0.55	0.40	0.28	0.37	0.34	0.35
TotalAverage	2.87								

RESULTS BY STUDENT OUTCOME ALL ASSESSMENTS						
	1	2	4	5	6	7
Average Score	2.86	2.91	2.87	2.92	2.86	2.88

Analysis:

Average PI scores showed most students were very strong in all Performance Indicators, for example, describing and performing solutions using numerical methods including using computer programming (SLO 1, 2). A large majority were strong in attaining new knowledge as

it pertains to career options and plans (SLO 7). A majority of students could identify ethical ramifications of technology and describing pathways towards resolution of ethical dilemmas (SLO 4). Most students had a satisfactory score in identifying the multi-disciplinary nature of engineering, and students consistently showed high scores in identifying specific examples of multi-disciplinary teamwork. (SLO 5).

Continuous improvements:

Average scores for SLO 1,2,6,7 showed a small but continuous improvement from Fall 2024 scores:

RESULTS BY STUDENT OUTCOME ALL ASSESSMENTS						
	1	2	4	5	6	7
Average Score	2.79	2.88	2.91	2.99	2.71	2.78
Standard dev.	0.10	N/A	0.04	N/A	N/A	N/A

BSCHE Graduate Written Examination – Fall 2025 Written examination conducted with graduating seniors in Chemical Engineering

Name:

Question 1

Career plans. (Scoring rubric was designed to assess potential for life-long learning.)

Where do you see yourself in 5-10 years, and what steps will you take to get there?

How have you researched your career plans and options (companies, graduate schools, PE certification, etc.)?

Does your prospective employer sponsor master's degree or MBA? Under which terms?

Question 2

Professional ethics and ethical ramifications of technology.

Can you name a recent news item or significant piece of legislation that impacts on technology or engineering, and comment on its ethical ramifications?

Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

Are you familiar with the AIChE Code of Ethics? Can you name specific provisions in it?

Which courses at UIC have contributed most to your views on professional ethics? How?

Question 3

Working in multidisciplinary teams

Can you name two other types of engineers that you might find yourself working with?

In what context or application would you be collaborating with them?

Question 4

Numerical methods and computational proficiency (Excel, MATLAB or Maple): choose one problem.

- (a) Generate a graph of the solution of the following ODE initial value problem.

$$\frac{dx}{dt} = \frac{1}{\tau} [x_{\text{in}}(t) - x(t)] - kx(t)^2, \quad x(0) = 2$$
$$x_{\text{in}}(t) = 1 + \varepsilon \sin(\omega t)$$

with the constants $\tau = 3.4$, $k = 1.2$, $\varepsilon = 0.1$ and $\omega = 5$. Can you think of a chemical engineering problem (or, at least, a course) in which such an ODE would arise?

- (b) Find a numerical value for the following definite integral.

$$\int_1^7 \frac{dx}{1 + \cos^3 x + \ln x}$$

Question 5

Engineering estimate.

The waterwheel shown in the following three pictures was observed to make one complete revolution every 15 seconds. Estimate the power output (in units of horsepower)?

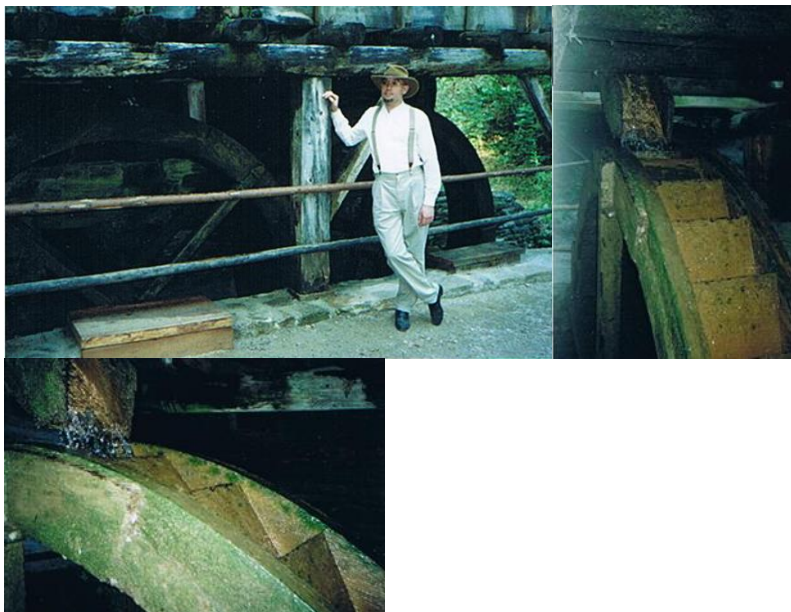


Exhibit 4-3.*Assessment Method #3 (Direct / Summative):**Judging rubrics from the annual Engineering Expo.***Sample Description: BSCHE Project Team: Judge Score Assessment- Spring 2021**

Scoring Rubrics Mapped to Student Learning Outcomes and Program Educational Objectives

Technical Quality

Performance Indicator (PI)	SLO	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Identification of project objective(s)	2	No design strategy; haphazard approach	Uses a design strategy with guidance	Develops a design strategy, decomposition of work into subtasks
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2	No consideration of economics, safety, and environment	Can use prior knowledge to design individual pieces of equipment competently when guided to do so	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	2	Cannot design processes or individual pieces of equipment without significant amounts of help	Can follow a previous example competently	Suggests new approaches and improves on what has been done before
4. Overall fulfillment of project objectives	2	No consideration of economics, safety, and environment	Design is done, but procedures and equations are not documented or referenced	Develops a solution that includes economic, safety, environmental and other realistic constraints
5. Project innovation	2	Unable to relate prior knowledge to the design problem	Can use prior knowledge to design individual pieces of equipment competently when guided to do so	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem
6. Project practicality in view of contemporary engineering standards/practice	2	No application of engineering and/or scientific principles	Includes only minor or cursory consideration of economic, safety, and environmental constraints	Applies engineering and/or scientific principles correctly to design practical processes

Presentation Quality

Performance Indicator (PI)	SLO	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
7. General professionalism and oral communication	3	Talk is poorly organized, e.g. no clear introduction or summary of talk is presented	Presents key elements of an oral presentation adequately, but "tell them" not clearly applied	Plans and delivers an oral presentation effectively; applies the principle of "(tell them)" -- well organized
8. Knowledge of the background material	3	Unable to relate prior knowledge to the design problem	Can use prior knowledge to design individual pieces of equipment competently when guided to do so	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem
9. Organization & quality of presentation material	3	Organizes materials in a logical sequence to enhance the reader's comprehension	Material organized well, but some sections and sub-sections are not identified clearly	Graphs, tables or diagrams are not easy to understand
10. Project description (see Program Book)	3	Text rambles, points made are only understood with repeated reading, and key points are not organized	Articulates ideas, but writing is somewhat disjointed, superfluous or difficult to follow	Articulates ideas clearly and concisely
11. Group dynamics/member participation	3	No interaction between team members. One member only talks	Interaction between team members, but one person leads talk	Discussion by all team members; all understand project
12. Quality, effectiveness, & use of visual aids	3	Multiple slides are unclear or incomprehensible. Presents well mechanically: · Makes eye contact · Can be easily heard · Speaks comfortably with minimal prompts (notecards) · No distracting nervous habits	Visual aides have minor errors or are not always clearly visible. Has some minor difficulties with the mechanical aspects of the presentation · Eye contact is sporadic · Occasionally difficult to hear or understand speaking	Uses visual aids effectively. Presents well mechanically: · Makes eye contact · Can be easily heard · Speaks comfortably with minimal prompts (notecards) · No distracting nervous habits

Rubric results: EXPO Spring 2026:

RESULTS BY PERFORMANCE INDICATOR: Technical and Presentation Quality											
Rubric item (PI)	1	2	3	4	6	7	8	9	11	12	
Mapped SLO	2	2	2	2	2	3	3	3	3	3	
Average	3.00	2.50	2.50	3.00	3.00	3.00	2.83	2.67	2.67	2.50	
Std Dev	0.00	0.55	0.55	0.00	0.00	0.00	0.41	0.52	0.52	0.55	
Average	2.77										

RESULTS BY STUDENT OUTCOME ALL ASSESSMENTS	2	3
Average Score	2.80	2.73
Standard dev.	0.27	0.19

EXPO 2020 PROJECT JUDGING FORM		Judge:	ID#:
Name: _____			
(print)		(signature)	
CUT HERE BEFORE PROVIDING TO DEPARTMENTS			
Check only one: ☐ Faculty/UIC Judge ☐ UIC Alumni Judge ☐ Industry Judge			
Project ID:	Category:	Department:	
Project Title:			
Technical Quality		(Poor)	(Excellent)
1. Identification of project objective(s)		1	2
2. Evaluation of design alternatives with realistic constraints (For example: economic, environmental, social, ethical, etc.)		3	4
3. Consideration of factors influencing the design (For example: cost, manufacturability, sustainability, maintainability, etc.)		5	6
4. Overall fulfillment of project objectives		7	8
5. Project innovation		9	10
6. Project practicality in view of contemporary engineering standards/practice		11	12
		13	14
		15	16
		17	18
		19	20
		21	22
		23	24
		25	26
		27	28
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		105	106
		107	108
		109	110
		111	

EXPO Judging Sheet used by the COE EXPO judges: 2025-2026

College of Engineering EXPO Evaluation Form

Project Name:

Evaluator Name:

Major:

1. Problem statement – clear and understandable

Does not meet expectations, Meets expectations, Exceeds expectations

2. Solution practicality (Is the solution practical in the context of engineering practice?)

Does not meet expectations, Meets expectations, Exceeds expectations

3. Presentation and communication skills

Does not meet expectations, Meets expectations, Exceeds expectations

4. Ability to answer technical questions appropriately

Does not meet expectations, Meets expectations, Exceeds expectations

5. Effectiveness of presentation material (poster, Powerpoint, handouts and other visual aids)

Does not meet expectations, Meets expectations, Exceeds expectations

6. Professional team-based presentation and interaction (more than 1 individual presented and answered questions, functioning as a unit, team members showed respect for one another)

Does not meet expectations, Meets expectations, Exceeds expectations

7. Comment on areas of excellence

Free response

8. Comment on areas of improvement.

Free response

Chemical Engineering EXPO Design Showcase Judging Sheet (2022, 2024-2026)

2026 CHEME EXPO JUDGING SHEET

Judge Name: Earl Washington

Team Number: 4

Team Members:

Technical Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following technical aspects:

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	5
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	4
4. Overall fulfillment of project objectives	5
5. Project practicality in view of contemporary engineering standards/practice	5
Total	23

Presentation Quality

Judges will be assigning a score of 1 (poor) through 5 (excellent) on the following presentation aspects:

1. General professionalism and oral communication	4
2. Knowledge of the background material	4
3. Organization & quality of presentation material	5
4. Group dynamics/member participation	5
5. Quality, effectiveness, & use of visual aids	5
Total	23

Comments: I like that this team made a byproduct into a profit center.
- Presentation board was well organized + visually appealing.
- Missing a Control Loop breakout

Exhibit 4-4.

*Assessment Method #4 (Indirect / Summative):
Senior Exit Survey.*

Sample Description: PDF printout of online survey (Fall 2020, Fall 2021)

(1.) Based upon your experiences as a Chemical Engineering student at UIC, please rate your skills in the following categories, using the following criteria:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

- 1.50 Ability to apply knowledge of mathematics, science and engineering.
- 1.63 Ability to design and conduct experiments, as well as to analyze and interpret data.
- 1.67 Ability to design a system, component, or process to meet desired needs.
- 1.39 Ability to function on multi-disciplinary teams.
- 1.50 Ability to identify, formulate, and solve engineering problems.
- 1.26 Understanding of professional and ethical responsibility.
- 1.45 Ability to communicate effectively.
- 1.46 Broad education necessary to understand the impact of engineering solutions in a global and societal context.
- 1.40 Recognition of the need for, and an ability to engage in life-long learning.
- 1.85 Knowledge of contemporary issues.
- 1.57 Ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.
- 1.73 Overall, do you feel that your undergraduate degree will prepare you for your career?

(2.) Which courses or topics were the most interesting or useful?

(3.) Which courses or topics were the least interesting or useful?.

(4.) What courses/topics should we consider deleting or adding to the curriculum?

(5.) Do you have any additional comments or suggestions?.

(6.) How many years/months did it take for you to complete your B.S. degree in chemical engineering?

(7.) Did you work part-time while you were a student? If yes, then approximately how many hours per week?

(8.) On average, how much many hours did you spend on each ChE course outside of class?

(9.) Did you participate in any summer research/internship program at UIC or at other schools? Did you complete an internship at a company? What company? Was this a useful experience? What specifically did you learn that was most useful?

(10.) Have you found a job? If so, in which company and industry? Please indicate your job title or specialization.

(11.) Are you planning to attend a graduate or a professional school? If so please list the name of the school and the specific program (M.S./Ph.D/M.D.) that you will enroll in, or those programs that you have looked into.

(12.) Do you have a strategy in mind for continued professional development and/or continue learning? For example, do you intend to take the FE/PE exam at some point in your career? If so, when?

SENIOR EXIT SURVEY paper form 2024-2025: Student Survey Example from 2025

Anoo
Venkataramanan

SENIOR EXIT SURVEY

As a senior student, we would like you to complete this self-assessment survey and to provide feedback for improvement of our curriculum for the ongoing ABET accreditation of our chemical engineering program,. Thank you!

(1.) Based upon your experiences as a Chemical Engineering student at UIC, please rate your skills in the following categories, using the following criteria:
(5 = Excellent, 4 = Strong, 3 = Moderate, 2 = Weak, 1 = Poor)

- 5 Ability to apply knowledge of mathematics, science and engineering.
- 5 Ability to design and conduct experiments, as well as to analyze and interpret data.
- 4 Ability to design a system, component, or process to meet desired needs.
- 5 Ability to function on multi-disciplinary teams.
- 5 Ability to identify, formulate, and solve engineering problems.
- 5 Understanding of professional and ethical responsibility.
- 5 Ability to communicate effectively.
- 5 Broad education necessary to understand the impact of engineering solutions in a global and societal context.
- 5 Recognition of the need for, and an ability to engage in life-long learning.
- 5 Knowledge of contemporary issues.
- 4 Ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.
- 5 Overall, do you feel that your undergraduate degree prepared you for your career?

(2.) Which courses or topics were the most interesting or useful?

CHE 201, CHE 205, CHE 210, and CHE 313 were the most interesting and useful classes in my opinion. Each class taught us the fundamentals of chemical engineering. They weren't too difficult to understand and material is crucial for understanding for our future.

Flip over for Questions #3-#5

(3.) Which courses or topics were the least interesting or useful?

Outside of the CHE classes, I think that Organic Chemistry II is the least useful. I don't think the material after Organic Chemistry I is beneficial to our learning.

(4.) What courses/topics should we consider deleting or adding to the curriculum?

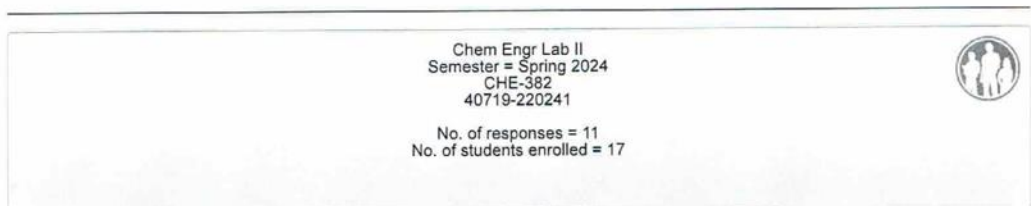
As I previously stated, I don't think that Organic Chemistry II is crucial for our curriculum. I think that introducing another ECE course would benefit us more.

(5.) Do you have any additional comments or suggestions?

Not currently. Thank you to all my professors and UIC for providing me with a great education!

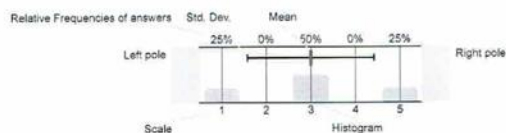
Exhibit 4-5.

Assessment Method #5 (Feedback):
UIC Course/Instructor Evaluation Survey.
Sample Description: CHE 382 Spring 2024



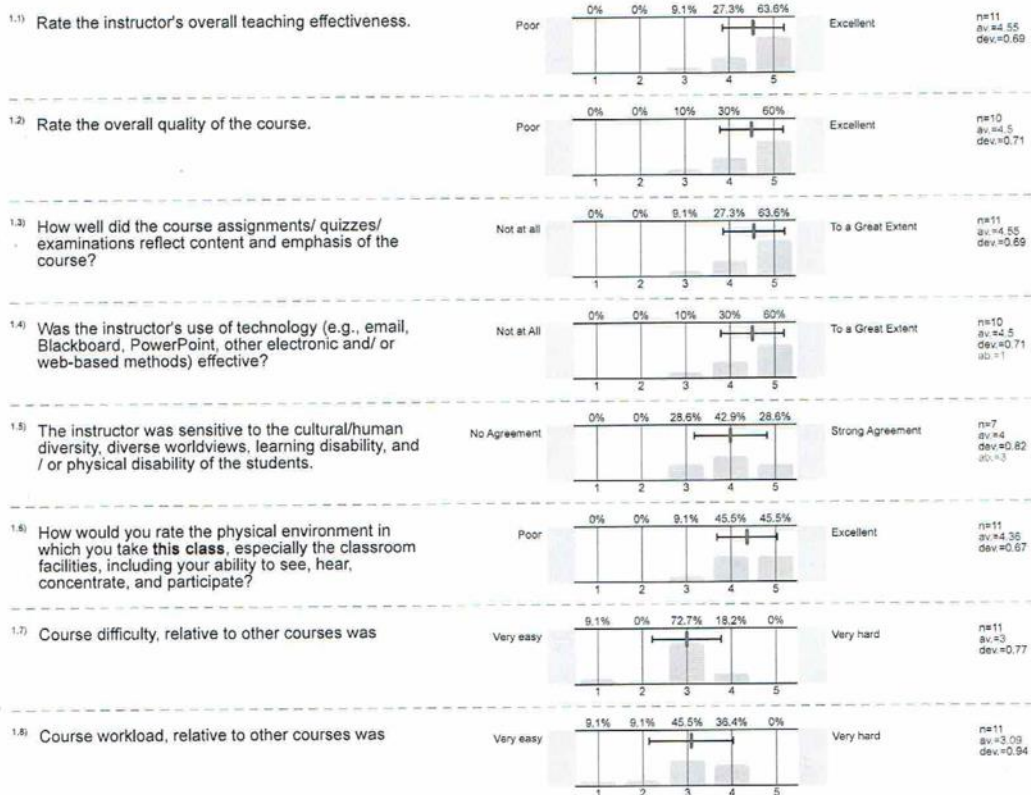
Legend

Question text



n=No. of responses
av=Mean
dev=Std. Dev.
ab=Abstinence

1. Instructor - Faculty Evaluation Questions



2. OPEN ENDED QUESTIONS

2.1) Please comment on specific characteristics of the course that were most beneficial to you:

- Dr Zdunek is a caring professor who engage effectively regarding any issue related or not related to the course.
- Dr. Zdunek is always there to answer questions and help as needed
- Professor Zdunek is always helpful and willing to assist with questions regarding experimental procedures and lab reports. Upon making a mistake in the lab that could have done irreversible damage to a level sensor in the distillation column he noticed my mistake and respectfully made me aware that I was acting without taking note of warning signs on the equipment and was encouraging in the after math of my mistake. Good professor.

Zach Hinton

- lab was fun and Zdunek is helpful as always

2.2) Please comment on specific aspects of the course that need improvement:

- Grading was much too harsh for undergraduate students. In addition, there were major discrepancies in grading. Some reports would get points off for things that were not marked down in previous reports.
- Have more office hours in case students have equations on parts of the lab calculations and analysis
- Not sure how to fix this issue but we were stuck with a lab mate the entire semester that was a true leech regarding workload concerning conducting the labs and writing the reports alike. We were stuck with this individual because we were kind enough to let him join our group the first day of lab when none of us knew him. We worked as a group of 3 the entire semester. I am sure it has happened before but it really sucked.

Zach Hinton

- show less favoritism in grading, be more consistent when grading, allow options for individual grading

Exhibit 4-6.*Level 2 analysis of assessment data.***Attainment of Student Outcome #1****An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.**

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2026	2025	2024	2023	2022	F2020 2021
% (3 = Satisfactory)	70.0	60.7	66.8	74.0	70.7	74.3
% (2 = Developing)	20.0	26.5	20.7	18.9	21.7	20.9
% (1 = Unsatisfactory)	10.0	12.8	12.5	7.1	7.5	4.9
% > Developing	90.0	87.2	87.5	92.9	92.5	95.1
Average Rating	2.65	2.42	2.43	2.53	2.51	2.68

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)		85.0	82.5	79.6	85.0	87.3
% (2 = Developing)		15.0	12.5	13.3	13.2	11.8
% (1 = Unsatisfactory)		0.0	2.5	7.1	1.8	0.9
% > Developing		100.0	97.5	92.9	98.2	99.1
Average Rating		2.86	2.79	2.72	2.83	2.85

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

*Note different rating system than above Methods #1- #3:**(5 = Excellent, 4 = Strong, 3 = Moderate, 2 = Weak, 1 = Poor)*

Year	2026	2025	2024	2023	2022	2021 2020
Average Rating	4.30	4.19				4.08 4.53

Attainment of Student Outcome #2

An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)	73.1	72.8	61.4		90.2	
% (2 = Developing)	26.9	24.1	38.6		3.9	
% (1 = Unsatisfactory)	0.0	3.1	0.0		5.9	
% > Developing	100.0	96.9	100.0		94.1	
Average Rating	2.73	2.68	2.61		2.82	

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)		92.0	85.0	87.8	87.3	90.6
% (2 = Developing)		8.0	12.5	6.1	10.9	9.4
% (1 = Unsatisfactory)		0.0	2.5	6.1	1.8	0.0
% > Developing		100.0	97.5	93.9	98.2	100.0
Average Rating		2.86	2.88	2.74	2.88	2.91

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)	80.0	57.8	47.1		85.0	62.7
% (2 = Developing)	20.0	42.2	52.9		15.0	37.3
% (1 = Unsatisfactory)	0.0	0.0	0.0		0.0	0.0
% > Developing	100.0	100.0	100.0		100.0	100.0
Average Rating	2.80	2.58	2.51		2.85	2.63

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(5 = Excellent, 4 = Strong, 3 = Moderate, 2 = Weak, 1 = Poor)

Year	2026	2025	2024	2023	2022	2020 2021
Average Rating	4.16	4.02				4.05 4.43

Attainment of Student Outcome #3**An ability to communicate effectively with a range of audiences.****Assessment Method #1 (Direct / Formative): Course Rubrics**

Year	2026	2025	2024	2023	2022	F2020 2021
% (3 = Satisfactory)	81.5	85.2	75.8	80.5	75.3	75.5
% (2 = Developing)	18.5	13.9	22.3	17.0	21.1	19.3
% (1 = Unsatisfactory)	0.0	0.9	1.9	2.5	3.6	5.2
% > Developing	100	99.1	98.1	97.5	96.4	94.8
Average Rating	2.82	2.84	2.79	2.75	2.76	2.74

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)	73.3	82.2	70.0		81.7	86.3
% (2 = Developing)	26.7	17.8	30.0		18.3	13.7
% (1 = Unsatisfactory)	0.0	0.0	0.0		0.0	0.0
% > Developing	100.0	100.0	100.0		100.0	100.0
Average Rating	2.73	2.82	2.75		2.82	2.86

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)	73.3	82.2	70.0		81.7	86.3
% (2 = Developing)	26.7	17.8	30.0		18.3	13.7
% (1 = Unsatisfactory)	0.0	0.0	0.0		0.0	0.0
% > Developing	100.0	100.0	100.0		100.0	100.0
Average Rating	2.73	2.82	2.75		2.82	2.86

Assessment Method #4 (Indirect / Summative): Senior Exit Survey*Note different rating system than above Methods #1- #3:**(5 = Excellent, 4 = Strong, 3 = Moderate, 2 = Weak, 1 = Poor)*

Year	2026	2025	2024	2023	2022	2020 2021
Average Rating	4.53	4.32				4.08 4.60

Attainment of Student Outcome #4

An ability to recognize ethical, safety, and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2026	2025	2024	2023	2022	F2020 2021
% (3 = Satisfactory)	79.5	84.7	83.7	78.9	82.2	92.4
% (2 = Developing)	11.5	14.3	10.9	14.9	11.5	4.5
% (1 = Unsatisfactory)	9.0	1.1	5.4	6.3	6.4	3.1
% > Developing	91.0	98.0	94.6	93.8	93.6	96.9
Average Rating	2.71	2.81	2.80	2.73	2.79	2.88

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)		89.3	92.5	82.3	86.1	86.8
% (2 = Developing)		8.0	6.7	8.8	11.5	13.2
% (1 = Unsatisfactory)		5.3	0.8	8.8	2.4	0.0
% > Developing		97.3	99.2	91.2	97.6	100.0
Average Rating		2.87	2.99	2.73	2.83	2.88

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(5= Excellent, 4 = Strong, 3 = Moderate, 2 = Weak, 1 = Poor)

Year	2026	2025	2024	2023	2022	2020 2021
Average Rating	4.18	4.19				4.08 4.55

Attainment of Student Outcome #5

An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2026	2025	2024	2023	2022	F2020 2021
% (3 = Satisfactory)	76.6	80.2	68.7	80.7	76.7	76.4
% (2 = Developing)	23.4	19.2	30.4	17.0	19.9	18.7
% (1 = Unsatisfactory)	0.0	0.6	0.9	2.3	3.4	4.9
% > Developing	100.0	99.4	99.1	97.7	96.6	95.1
Average Rating	2.79	2.80	2.69	2.78	2.76	2.72

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)		96.0	97.5	95.9	96.4	100.0
% (2 = Developing)		0.0	2.5	0.0	3.6	0.0
% (1 = Unsatisfactory)		4.0	0.0	4.1	0.0	0.0
% > Developing		96.0	100.0	95.9	100.0	100.0
Average Rating		2.92	2.99	2.92	2.97	3.00

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(5 = Excellent, 4 = Strong, 3 = Moderate, 2 = Weak, 1 = Poor)

Year	2026	2025	2024	2023	2022	2020 2021
Average Rating	4.47	4.26				4.08 4.55

Attainment of Student Outcome #6

An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2026	2025	2024	2023	2022	F2020 2021
% (3 = Satisfactory)	76.9	83.9	74.5	82.4	70.6	77.5
% (2 = Developing)	23.1	13.8	22.8	14.7	24.3	15.6
% (1 = Unsatisfactory)	0.0	2.4	2.7	2.8	5.0	6.9
% > Developing	100.0	97.6	97.3	97.2	95.0	93.1
Average Rating	2.77	2.83	2.75	2.77	2.66	2.61

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)		84.0	80.0	73.5	83.6	88.7
% (2 = Developing)		16.0	12.5	18.4	14.5	9.4
% (1 = Unsatisfactory)		0.0	7.5	8.2	1.8	1.9
% > Developing		100.0	92.5	91.8	98.2	98.1
Average Rating		2.86	2.71	2.63	2.82	2.85

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(5 = Excellent, 4 = Strong, 3 = Moderate, 2 = Weak, 1 = Poor)

Year	2026	2025	2024	2023	2022	2020 2021
Average Rating	4.41	4.32				4.08 4.60

Attainment of Student Outcome #7

An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2026	2025	2024	2023	2022	F2020 2021
% (3 = Satisfactory)	84.6	90.5	81.2	86.6	79.7	75.9
% (2 = Developing)	15.4	9.5	17.4	12.5	17.5	22.0
% (1 = Unsatisfactory)	0.0	0.0	1.4	0.9	2.8	2.1
% > Developing	100.0	100.0	98.6	99.1	97.2	97.9
Average Rating	2.94	2.91	2.79	2.85	2.76	2.80

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)		88.0	77.5	83.7	85.5	81.1
% (2 = Developing)		12.0	22.5	14.3	12.7	18.9
% (1 = Unsatisfactory)		0.0	0.0	2.0	1.8	0.0
% > Developing		100.0	100.0	98.0	98.2	100.0
Average Rating		2.88	2.78	2.84	2.84	2.87

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2026	2025	2024	2023	2022	2021
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(5 = Excellent, 4 = Strong, 3 = Moderate, 2 = Weak, 1 = Poor)

Year	2026	2025	2024	2023	2022	2020 2021
Average Rating	4.56	4.20				4.08 4.60

Exhibit 4-7.*Level 3 analysis of assessment data.***Direct Assessment Method #1: Direct / Formative
Course Rubrics****3 = S = Satisfactory****2 = D = Developing****1 = U = Unsatisfactory**

Percentage of students attaining S

SO	2026	2025	2024	2023	2022	2021
1	70.0	60.7	66.8	74.0	70.7	74.3
2	73.1	72.8	61.4		90.2	
3	81.5	85.2	75.8	80.5	75.3	75.5
4	79.5	84.7	83.7	78.9	82.2	92.4
5	76.6	80.2	68.7	80.7	76.7	76.4
6	76.9	83.9	74.5	82.4	70.6	77.5
7	84.6	90.5	81.2	86.6	79.7	75.9

Percentage of students attaining S and D

SO	2026	2025	2024	2023	2022	2021
1	90.0	87.2	87.5	92.9	92.5	95.1
2	100.0	96.9	100.0		94.1	
3	100.0	99.1	98.1	97.5	96.4	94.8
4	91.0	98.0	94.6	93.8	93.6	96.9
5	100.0	99.4	99.1	97.7	96.6	95.1
6	100.0	97.6	97.3	97.2	95.0	93.1
7	100.0	100.0	98.6	99.1	97.2	97.9

Average rating among students

SO	2026	2025	2024	2023	2022	2021
1	2.65	2.42	2.43	2.53	2.51	2.68
2	2.73	2.68	2.61		2.82	
3	2.82	2.84	2.75	2.75	2.76	2.74
4	2.71	2.81	2.80	2.73	2.79	2.88
5	2.79	2.80	2.69	2.78	2.76	2.72
6	2.77	2.83	2.75	2.77	2.66	2.61
7	2.94	2.91	2.79	2.85	2.76	2.80

**Direct Assessment Method #2: Direct / Summative
Senior Written Examination**

3 = S = Satisfactory

2 = D = Developing

1 = U = Unsatisfactory

Percentage of students attaining S

SO	2026	2025	2024	2023	2022	2021
1		85.0	82.5	79.6	85.0	87.3
2		92.0	85.0	87.8	87.3	90.6
3						
4		89.3	92.5	82.3	86.1	86.8
5		96.0	97.5	95.9	96.4	100.0
6		84.0	80.0	73.5	83.6	88.7
7		88.0	77.5	83.7	85.5	81.1

Percentage of students attaining S and D

SO	2026	2025	2024	2023	2022	2021
1		100.0	97.5	92.9	98.2	99.1
2		100.0	97.5	93.9	98.2	100.0
3						
4		97.3	99.2	91.2	97.6	100.0
5		96.0	100.0	95.9	100.0	100.0
6		100.0	92.5	91.8	98.2	98.1
7		100.0	100.0	98.0	98.2	100.0

Average rating among students

SO	2026	2025	2024	2023	2022	2021
1		2.86	2.79	2.72	2.83	2.85
2		2.86	2.88	2.74	2.88	2.91
3						
4		2.87	2.99	2.73	2.83	2.88
5		2.92	2.99	2.92	2.97	3.00
6		2.86	2.71	2.63	2.82	2.85
7		2.88	2.78	2.84	2.84	2.87

**Assessment Method #3 – Direct / Summative
Engineering Expo Rubrics**

3 = S = Satisfactory

2 = D = Developing

1 = U = Unsatisfactory

Percentage of students attaining S

SO	2026	2025	2024	2023	2022	2021
1						
2	80.0	57.8	47.1		85.0	62.7
3	73.3	82.2	70.0		81.7	86.3
4						
5						
6						
7						

* Expo judging performed remotely due to pandemic

Percentage of students attaining S and D

SO	2026	2025	2024	2023	2022	2021
1						
2	100.0	100.0	100.0		100.0	100.0
3	100.0	100.0	100.0		100.0	100.0
4						
5						
6						
7						

Average rating among students

SO	2026	2025	2024	2023	2022	2021
1						
2	2.80	2.58	2.51		2.85	2.63
3	2.73	2.82	2.75		2.82	2.86
4						
5						
6						
7						

Assessment Method #4 – Indirect / Summative
Senior Exit Survey

Average rating among students

Note: Different rating system than above Methods #1- #3:

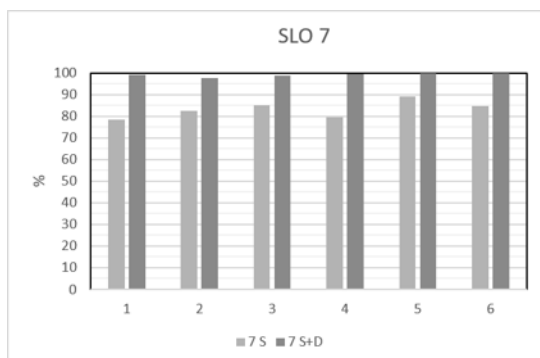
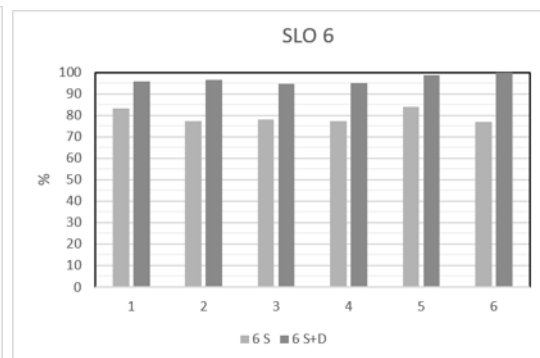
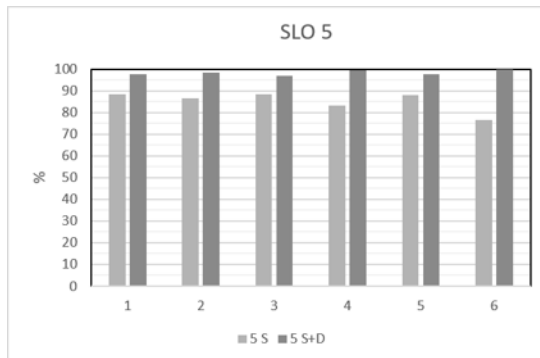
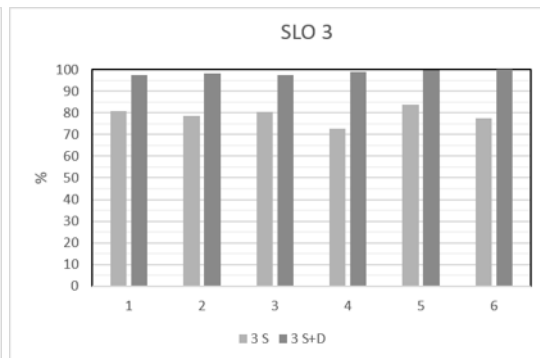
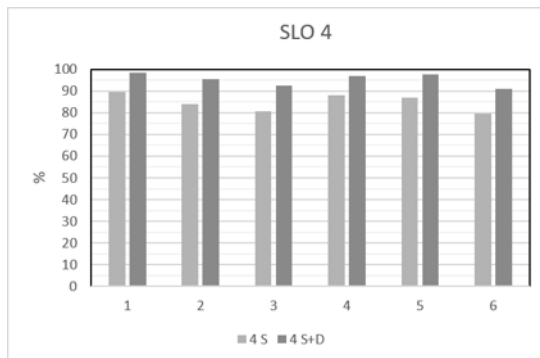
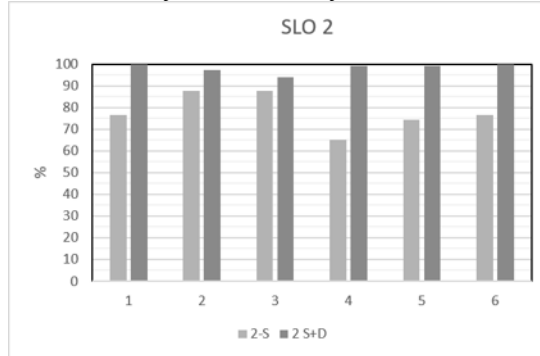
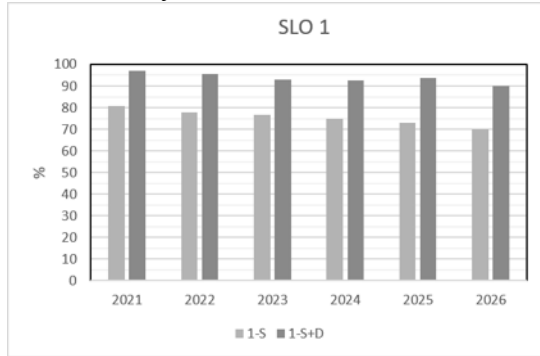
(5 = Excellent, 4 = Strong, 3 = Moderate, 2 = Weak, 5 = Poor)

SO	2026	2025	2024	2023	2022	2020 2021
1	4.30	4.19				4.08 4.53
2	4.16	4.02				4.05 4.43
3	4.53	4.32				4.08 4.60
4	4.18	4.19				4.04 4.48
5	4.47	4.26				4.08 4.55
6	4.41	4.32				4.08 4.46
7	4.56	4.20				4.08 4.60

Exhibit 4-8.

Level 4 analysis of assessment data:

Bar charts of Table 4.5 and 4.6 results: % attainment of *S* and *S+D* for each SLO.



Level 4 analysis of assessment data:

Bar charts of:

Table 4.7 Average ratings for Methods #1, #2, #3

Table 4.8 Average rating for Senior Exit Survey

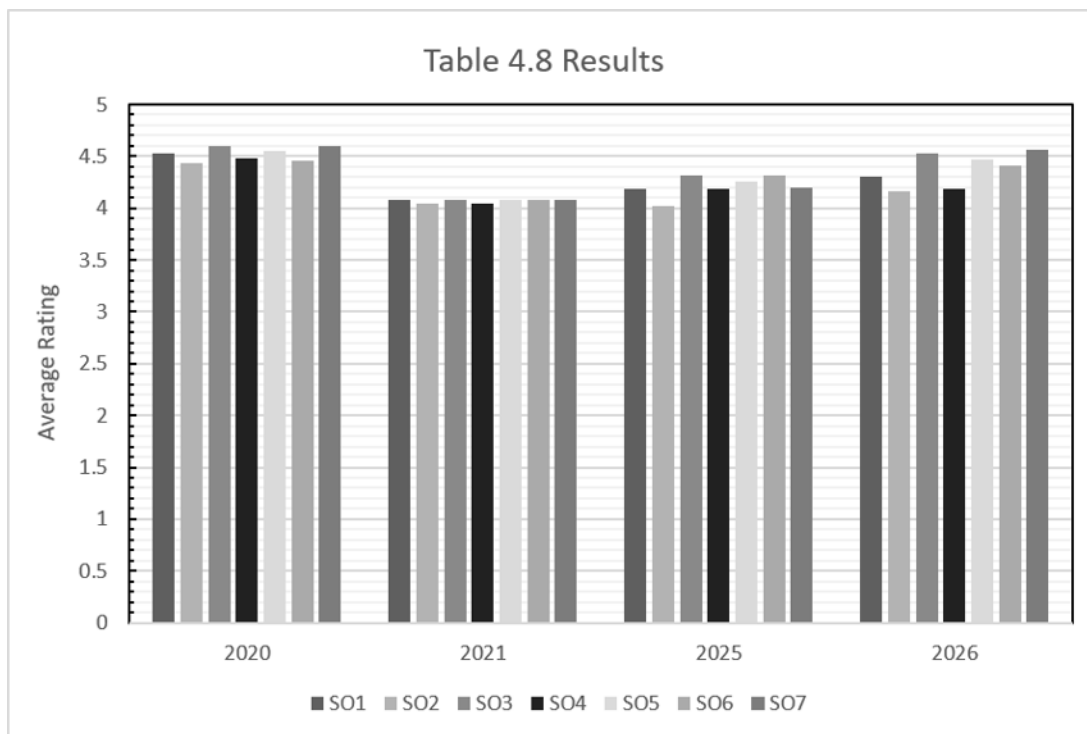
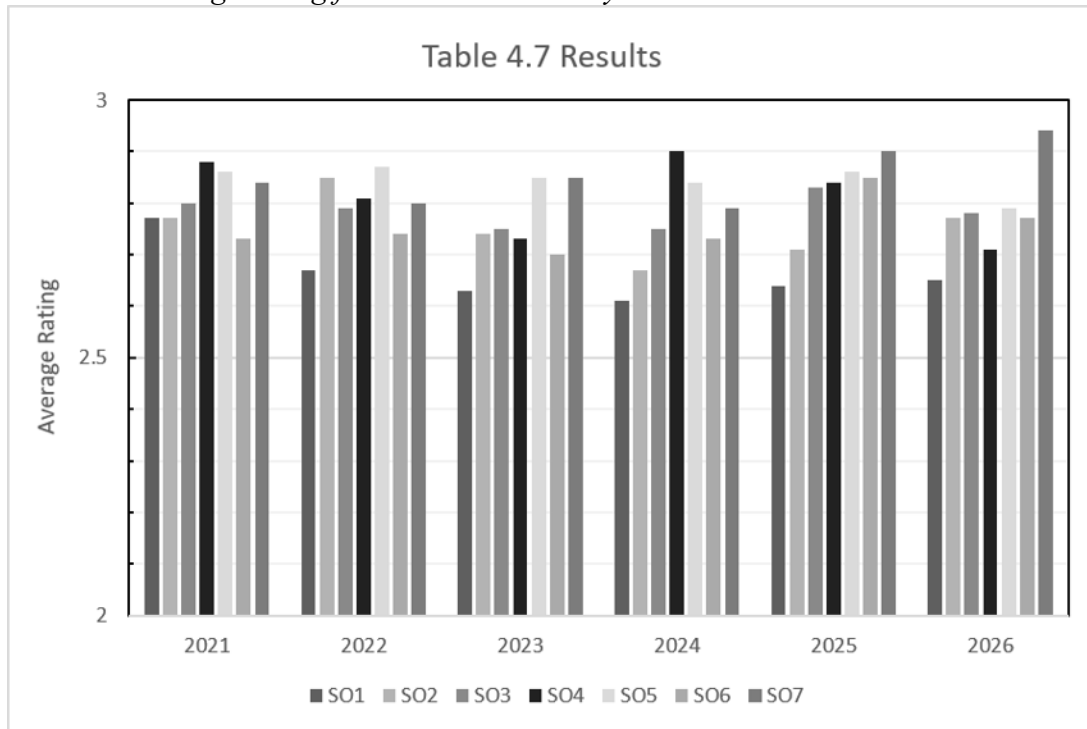


Exhibit 4-9.

Continuous improvement – closing the feedback loop on assessment:

Sample Dates: Fall 2025

Sample Description: Faculty self-study for CHE 311 Transport Phenomena I.

1. Self-Study Evaluation (Findings and Interpretation)

The assessment of Student Outcome 1 (SO1) in CHE 311 was conducted using the comprehensive final exam as a direct measure of student ability to apply principles of momentum transport and fluid mechanics to solve complex engineering problems.

The average exam score of 70.97 indicates that, overall, students demonstrated an acceptable level of competency in core topics such as fluid statics, conservation of momentum, dimensional analysis, and internal/external flow. However, the standard deviation of 17.19 suggests a notable spread in performance, indicating inconsistent levels of mastery across the cohort.

The performance indicator average of 2.56/3.0 exceeds the program's attainment benchmark (2.0), confirming that SO1 is achieved for this course. The distribution of student performance further supports this conclusion, with (21/36) performing at or above the developing level and demonstrating satisfactory proficiency.

A subset of students struggles with fundamental concepts and/or the integration of multiple principles required for solving complex fluid mechanics problems. In particular, variability in performance may be attributed to:

- Difficulty in translating physical systems into mathematical models
- Weaknesses in prerequisite knowledge (e.g., differential equations)
- Challenges in multi-step problem-solving under exam conditions

Additionally, the relatively high variability in exam scores indicates that while high-performing students excel, others are not consistently reaching the expected level of competency.

2. Continuous Improvement Actions

Based on the assessment findings, the following targeted actions will be implemented to improve student learning and reduce performance variability:

A. Strengthening Conceptual Foundations

- Introduce short, focused concept reviews (e.g., mini-lectures or pre-class modules) on key prerequisite topics such as force balances, units/dimensions, and differential analysis.
- Incorporate concept-check questions during lectures to reinforce understanding of fundamental fluid mechanics principles (e.g., pressure variation, shear stress, velocity profiles).

Exhibit 4-10.

Continuous improvement – Excerpt of Student Feedback on CHE 230 course from Climate Survey Evaluation

Ezinne
Achinivu
Molecular Systems in
Chem Eng Semester =
Spring 2024
C

**2. OPEN ENDED QUESTIONS**

Content is really similar to CME 260 so already knew a lot of stuff

Email exchange discussing CHE 230/CME 260 overlap with faculty.

Re: Overlapping Course Content (CHE 230, CHE 330, CME 260)

Chemical Faculty Listserv on behalf of Zdunek, Alan
CHE-FACULTY@LISTSERV.UIC.EDU

A couple more comments on this issue:

@Betul- Thanks for sharing the students' concerns on this.

It appears we need to revisit this again as we haven't 'closed the loop' on the overlap. I will set up a meeting as Vivek suggested with the instructors who are teaching/have taught 230/330/454.

Background information:

Ezinne brought up the issue in Fall 2024 and Vikas, Ezinne, Vivek and I had discussions on reducing the overlap between 230 and 330, and 230/260 back in January 2025.

Our discussions centered around:

Currently covering many topics in 230, with some being too advanced for a 200-level course. Need to integrate 230, 330 and 454 more effectively.

There is no textbook on molecular systems in chemical engineering, any suggestions? Also, guidance is needed on creating a set of homework assignments.

Thanks,

Alan

Dr. Alan D Zdunek
Director of Undergraduate Studies
Clinical Associate Professor
University of Illinois at Chicago (UIC)
Department of Chemical Engineering (MC 110)
150 Engineering Innovation Building
929 W Taylor Street
Chicago, Illinois 60607

Exhibit 4-11.*Continuous improvement – Safety and Ethics Examples***Ethics Questions from the Senior Written Examination- Fall 2025****Professional ethics and ethical ramifications of technology.**

Can you name a recent news item or significant piece of legislation that impacts on technology or engineering, and comment on its ethical ramifications?

Lack of AI regulation, This worries me as the lack of AI controls or limits can lead to AI attempting to replace key safety measurements in order to save on money. And The defunding of the USCSB can lead to more fatal accidents.

Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

Not in chem e or UIC but in a previous internship I had to decide on if it was appropriate to reclassify a shopping zone in Lincoln Square. The issue was if Rezoning from B1 to B2 was worth the gain since B2 allows for alcohol consumption in restaurants but allows for things that are considered immoral “! I am not joking when I say this is in Chicago Zonung code.” Like tattoo parlors. Since a school and a elderly center are next to the zone, We decided on finally adding it after almost 100 years of litigation against a old prohibition era code.

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

I relied on my Mentors experience and working with community through meets and organizing with those in favor and against by finding a middle ground to rezone the area. This taught me to solve ethical dilemmas you need both Experience and systematic methodology to solve complicated ethical dilemmas.

Are you familiar with the AIChE Code of Ethics? Can you name specific provisions in it?

Yes. The one that stuck the most with me is “Accept responsibility for their actions, seek and heed critical review of their work and offer objective criticism of the work of others. “

Which courses at UIC have contributed most to your views on professional ethics? How?

321 Reaction Engineering, 381 Unit Ops Lab, 499 Senior Seminar and 396 Senior Design. All have brought up safety as key focus that I would like to have more time to explore in. These classes taught me how the ethical concern of being engineer in safety for the environment is important as a lot of climate change is done from Chemical Engineering but is being Fixed by Green Chemical Engineering.

SAFETY QUIZ:CHE382-Chemical Engineering Laboratory II

Please answer the following questions using the information from the 6 selected chapters of the JT Baker Hazardous Chemical Safety document; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards and Compressed Gases given in Safety Assignment I. Note: Answers from another source may not be considered correct. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words. (5 points per question)

1. What is the definition of a toxic chemical?

It's any chemical that by its chemical action may cause damage to structure or disturbance to function when injected, inhaled, absorbed, applied to, injected into or developed within the body in relatively small amounts.

2. Name the 4 types of toxicity damage from a toxic chemical:

Local toxicity, systemic toxicity, acute toxicity, chronic toxicity

3. Name two parameters that can affect toxicity:

One parameter is the mode of entry, such through the skin, eye, respiratory tract, or digestive system. Another parameter is the physical condition of victim, such as them being ill or their body being damaged from a previous assault.

4. What is the PEL (Permissible Exposure Limit)?

The PEL is the maximum exposure recommended by NIOSH. These are usually derived from the TWA values given by ACGIH and are enforced by OSHA.

5. What is the DOT (Department of Transportation) definition of a compressed gas?

Compressed gases are defined as those nonflammable materials which have a psia greater than 40 at 70 °F or a psia greater than 104 at 130 °F, or those flammable materials which have a greater psia than 40 at 100 °F

6. a) How does one determine how much material remains in a non-liquified gas cylinder?

How much material left in a non-liquefied gas cylinder is directly proportional to the pressure inside the cylinder, so reading the cylinder pressure gauge informs of the amount of material left inside the cylinder.

b) On page 13-10, which pressure gauge on the two-stage regulator will give you the cylinder pressure reading?

The high-pressure gauge (left gauge closest to cylinder) shows the cylinder pressure reading

7. How does one determine how much material remains in a liquified gas cylinder?

How much liquefied gas remaining in a cylinder can only be determined by weighing it. The net weight of the contents is noted on a card attached to the cylinder

8. Why do insidious hazards represent a problem to workers?

Because the workers may never be aware that a problem exists until chronic, systemic poisoning has occurring.

9. What are the four general categories of insidious hazards?

The categories of insidious hazards are Unanticipated Chemical Reactions, Unanticipated Toxicity, Improper Faculty Design, and Improper Handling of Equipment and Chemicals

10. Name 2 control measures that can be implemented to reduce insidious hazards in the laboratory.

Two control measures are to provide adequate ventilation in the form of hoods and forced air, as well as having efficient cleanup agents available for spills.

11. What is the definition of a reactive chemical?

Reactive chemicals are substances that enter into violent reactions during which the spontaneous liberation of heat and/or gases is too rapid to be safely dissipated by the surroundings. This makes the reaction lose control and either the vessel bursts, an explosion occurs, and several more possible incidents.

12. Name the two groups and their sub-headings used to categorize reactive chemicals:

The two groups are:

- Unstable Components: Explosive, and Others.

- Unstable Mixtures: Explosives , Water-Reactive Compounds, Air-Sensitive Compounds, Other Mixtures of Oxidizing Agents with Reducing Agents

13. List two precautionary measures for handling and/or storing reactive chemicals.

Using the chemicals with added inhibitors and Storing a minimum quantity of the material and do not store near an area populated by workers.

14. How are corrosive chemicals generally defined?

Corrosive chemicals are substances that, by direct chemical action, are corrosive to metal or can injure body tissue.

15. How do corrosive chemicals act on body tissues?

They act on body tissue through direct contact with the skin or eyes, inhalation, or ingestion.

16. True or False: Concentrated alkaline solutions have a more corrosive effect on tissue than most strong inorganic acids.

True

17. Name the six categories of common corrosive liquids.

Inorganic Acids, Organic Acids, Other Inorganics, Caustic Solutions, Organic Solvents, Other Organics

18. What category of corrosive solids presents the greatest hazard because of their frequent use in industry?

Caustic alkalis present the greatest hazard

19. Name the three components that make up the FIRE TRIANGLE:

Fuel, oxidizer, and energy are the three components of the fire triangle

20. What category of flammables is the most common source of fires in industry?

Flammable Solvents are the most common source of fires in the industry

Exhibit 4-12.

Continuous improvement – Extracurricular Activities: Chemical Engineering News Item describing Objective of the ChE Department Action Team. Additional student and faculty/staff feedback will be provided during the review.

Department Action Team aims to build and improve undergraduate community and success Meghan King | Posted on October 27, 2023:

<https://che.uic.edu/news-stories/department-action-team-aims-to-build-and-improve-undergraduate-community-and-success/>

Chemical engineering department faculty hope to bring change from the bottom up through the Department Action Team, a group of faculty, staff members, and students who aim to create sustainable change to undergraduate education and make meaningful change in the department.

“If you want to start bottom-up change, looking from an outsider perspective only doesn’t work, you have to get others involved who have an insider perspective,” said Clinical Associate Professor Betul Bilgin.

Bilgin is working with Clinical Assistant Professor Korosh Torabi, Assistant Professor Ezinne Achinivu, Laboratory and Computers Manager Jan Sagun, undergraduate student Emily Seriruk, and graduate student Nadia Nikolova.

The group also hopes to increase student belonging and build community. They aim to bring students together by providing them with both professional development events and a social circle to talk about things that aren’t related to chemical engineering.

Nikolova, who also completed her undergraduate degree at UIC, shared that sometimes students tend to stick within their own groups, whether organizations, social, or major-related. She hopes that the Department Action Team’s efforts will expand their events to include commuter students, more first-year students, and transfer students.

“Having a space for students is important, whether it’s just hanging out or working on class materials, there’s a lack of space for these students to develop into a cohesive cohort,” Seriruk said. “Students tend to stay in their clicks during classes and then as soon as class is over, everyone heads home or goes their separate ways. So, we’re hoping that with these events that will be at the same time every week or every other week, we will give them a space to get together and get to know each other outside of lectures.”

Bilgin said that the faculty and staff appreciate how much these students contribute, enriching their conversations with many different perspectives.

Seriruk agreed, saying that there’s a mutuality that exists within the group.

“We’re not in the background; faculty and staff are constantly asking ‘what do you think as undergrads?’ and they truly value students’ opinions,” she said.

CRITERION 5. CURRICULUM

A. Program Curriculum

5.A.1. Study Plan.

Details of our Bachelor of Science in Chemical Engineering curriculum can be found in Table 5-1 at the end of this section. The program is a semester-based program that requires 8 semesters (4 years) to complete. A typical degree plan is shown in Figure 5.3 and a flowchart of pre-requisites for required courses within the ChE curriculum is shown in Figure 5.4.

5.A.2. Alignment of the Program Curriculum with Program Educational Objectives.

The Chemical Engineering PEOs (Section 2.B) describe career success of our graduates based upon depth, breadth, professionalism and skills in life-long learning — acquired in a rigorous, innovative, and inclusive learning environment. As noted in Section 2B, the Program Educational Objectives are in the process of being revised, however the 2020-2026 program curriculum has followed the previous PEO's, so the alignment with these PEO's are given below. Once the new PEO's have been finalized, a new alignment will be made with the program curriculum by the Undergraduate Committee and approved by the faculty.

Depth. Table 5-1 demonstrates that students are required to take 45 credit hours in advanced chemical engineering courses (including the ChE technical elective) plus 9 credit hours of supporting engineering courses. Students also have a selective option choice between CHE 330 and CHEM 346. These core courses cover a wide range of topics beginning with an Introduction to Thermodynamics (CHE 201, 3 hours) typically taken by the Fall semester of the sophomore year. This is followed by Computational Methods in Chemical Engineering (CHE 205, 3 hours), Molecular Engineering in Chemical Engineering (CHE 230, 3 hours), and the Material and Energy Balances course (CHE 210, 4 hours). CHE 205 is a distinctly chemical engineering specific mathematics and computation course, carefully formulated to preview key concepts from the core courses and develop the relevant problem-solving skills. CHE 230 was added in 2022 and is also a distinctly chemical engineering specific materials course, providing fundamental concepts in molecular engineering and materials chemistry. CHE 210 introduces the critical concepts of control-volume balances as well as basic problem-solving skills for problems that may contain many variables and unknowns. The introduction of these tools in the sophomore year allows the students to build a base of engineering design and computational expertise that they will further develop as they advance through the junior level courses and ultimately apply during their senior year in the Unit Operations Laboratory (CHE 381, 382) and Senior Design sequence (CHE 396, 397). In the junior year, students take the first two courses in Transport Phenomena (CHE 311 and CHE 312, 6 hours), which cover fluid mechanics (momentum transfer), heat transfer, and mass transfer. In addition, students take an advanced course in Chemical Engineering Thermodynamics (CHE 301, 3 hours) in which students learn to apply equilibrium criteria to multiphase systems. The course in Chemical Reaction Engineering (CHE 321, 3 hours) which deals with the design of reactors and a third course in Transport phenomena (CHE 313, 3 hours) that covers separation processes complete the junior-level core in Chemical Engineering.

Senior-level courses complete the major in chemical engineering subject areas and complement the engineering science courses which dominate the junior year. These courses are a two-course sequence of Unit Operations Laboratory (CHE 381 and 382, 4 hours) in which transport, separation, and process control experiments are performed, data are analyzed and laboratory reports are written. The chemical process control course (CHE 341, 3 hours) introduces control theory and uses examples from all the previous courses. The polymer science course (CHE 330, 3 hours) was added as a selective option with CHEM 346 Physical Chemistry II and gives our students the option of focusing on polymers as engineering material (CHE 330), or quantum mechanics (CHEM 346). Finally, the capstone two course Senior Design sequence (CHE 396 and 397, 7 hours) focuses on design of chemical processes and plants. In these courses, students demonstrate their ability to independently apply chemical engineering principles, cost estimation, and economic evaluation methodologies, flow sheet synthesis and process simulation, optimization techniques, and report writing and oral presentation skills to address contemporary chemical engineering design problems. The Senior Design II course (CHE 397, 4 hours) involves a team-oriented design project which is supervised by one or two industrial mentors. Senior Design and Unit Operations Laboratory require our students to demonstrate and synthesize a depth of knowledge in many important chemical engineering areas, including troubleshooting equipment, designing experiments, plant design, economics, process safety, and ethics, as well as building teamwork, communication and presentation skills. In addition, students are required to take a technical elective course (3 hours) which may be any 400-level chemical engineering graduate course or an independent research course (CHE 392, 3 hours) in which the student can pursue in depth a particular topic of personal interest under the direct supervision of a departmental faculty member. Figure 5.1 shows the number of students that have performed undergraduate research as a CHE 392 course. The 400-level course topics range from Process Simulation with Aspen Plus, a key skill used in Senior Design II, Nanotechnology for Pharmaceutical Applications, Biopharmaceutical Process Development and Manufacturing, and a host of other topics. Students may also pursue an optional concentration in one of six areas; Biochemical Engineering, which requires CHE 422 (3 hours); Polymers and Molecular Engineering which requires CHE 454 Molecular and Macromolecular Engineering and a second ChE technical elective; Energy and Environment, which requires CHE 451 Renewable Energy Technologies and a second ChE technical elective; Nanotechnology, which requires CHE 455 Nanoscale Systems in Chemical Engineering and a second ChE technical elective; Process Automation, which requires CHE 433 Process Simulation with Aspen Plus; and Entrepreneurship, which requires CHE 427 Entrepreneurship in Engineering. The last 5 concentrations were added to our curriculum in 2022. All six concentrations also require supporting courses from outside of chemical engineering. Figure 5.2 shows the percentage and number of students pursuing a concentration from 2021-2026. The rigorous sequence of courses and optional concentrations described above attests to the depth of knowledge with which Chemical Engineering students emerge upon graduation and upon which they build their professional careers.

Breadth. The curriculum also requires students to demonstrate a breadth of knowledge. This includes a thorough grounding in chemistry and a working knowledge of advanced topics in chemistry such as organic, physical, and analytical chemistry. The courses that address the Program Educational Objective of breadth begin in the freshman year with an introduction to General Chemistry (CHEM 122/123 and 124/125, 10 hours). In the sophomore year students take

a sequence of three courses in organic chemistry (CHEM 232, 233, and 234, 8 hours) which consist of two lecture courses and one laboratory course. In the junior year the chemistry courses are completed with a combination lecture/laboratory course in Analytical Chemistry (CHEM 222, 4 hours) and a course in physical chemistry (CHEM 342, 3 hours). As mentioned above, CHEM 346 Physical Chemistry II is a selective option with CHE 330 and gives our students the option of studying quantum mechanics. Students must also demonstrate a high level of mathematical skills in a four-course sequence (MATH 180, 181, 210, and 220: 14 hours) taken during the freshman and sophomore years that involves differential and integral calculus, vector and multivariable calculus, differential equations and other advanced topics in mathematics. These mathematics skills are used throughout the curriculum. Also, during the freshman and sophomore years a two-course sequence in physics (PHYS 141 and 142: 8 hours with a lab element) requires our students to demonstrate a basic grounding in mechanics and electromagnetism.

In addition to the basic science courses described above, our students must also demonstrate their breadth of technical knowledge by taking engineering courses outside of Chemical Engineering. These are a course in Properties of Materials (CME 260: 3 hours), a course in Computer Programming (CS 109: 3 hours with a lab element) and a course in Electrical Circuit Analysis (ECE 210: 3 hours with a lab element). These courses add breadth by giving our students insight into other related engineering disciplines and preparing them for the multidisciplinary career environment in which they will work.

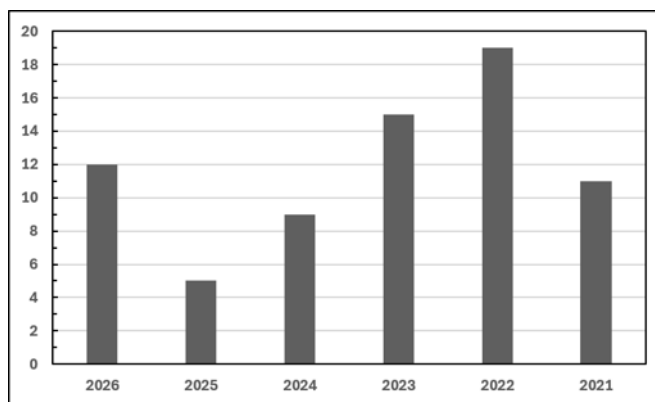


Figure 5.1. Number of undergraduate students in CHE 392 undergraduate research courses

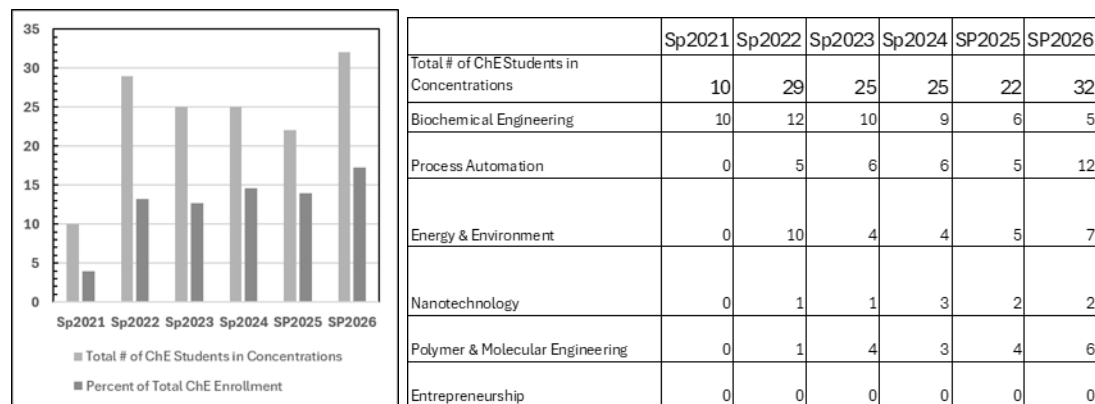


Figure 5.2 Total number and percentage of students in concentrations per academic year.

Furthermore, about 10% of our students also complete a minor degree in another discipline. The most popular minors are mathematics and computer science, with business analytics, biological science, biomedical engineering, industrial engineering, entrepreneurship and philosophy also being considered. These minor degrees also add breadth by giving our students insight into other related disciplines and preparing them for the multidisciplinary career environment in which they will work.

Professionalism. In the freshman year our students take a general Engineering Orientation course (ENGR 100: 0 hours) course that introduces the development of a career in engineering. The students learn about the engineering curriculum at UIC, including the various student organizations available to them. They also are introduced to the Engineering Career Center and learn about engineering careers of all varieties, and good professional skills and habits. An important part of ENGR 100 are the major-specific breakout sessions where the students meet faculty and upperclassmen in their major and participate in a technical project related to their discipline. This activity was added to bring them into the engineering community during their first semester, increase the retention of the students and help them begin to identify as engineers. The design course sequence described above (CHE 396 and 397: 7 hours) is also integral to our students' development of professionalism. The design courses bring in speakers from industry who discuss current issues and describe the requirements of a professional career in chemical engineering. Topics concerning safety, professional ethics, design strategies, business strategies, are discussed. In addition, the students are required to take a professional development seminar (CHE 499, 0 hours) which includes presentations by practicing chemical engineers and career development professionals. The students are exposed to the diverse career paths and industries where chemical engineers are employed, such as food processing, pharmaceuticals, oil & gas, specialty chemicals, and business/law. In addition, a variety of issues such as engineering ethics are covered and culminate in a written senior examination. Finally, all our students are required to participate in the College of Engineering's Engineering Expo where the final design projects are presented and judged by engineers from industry. This requires our students to connect their design projects and the basic engineering science and design principles applied to the projects to the complex modern world and the many issues that must be addressed in any real-world application of engineering. In addition to the College of Engineering Expo, the department has initiated the Chemical Engineering Design Showcase that is conducted before the College Expo. The Design Showcase present their final design projects to chemical engineering specific judges from industry. The addition of the ChE Design Showcase has allowed the students to obtain comments and suggestions from judges that are more aligned with the chemical and allied industries. The Chemical Engineering Design project titles for Spring 2026 are listed below:

Chemical Engineering design project titles- Spring 2026

CHE.01	CRACK'D: Polypropylene Waste to Ethylene and Propylene
CHE.02	Clean Fuel, Cleaner Skies - Ethanol to Jet Fuel
CHE.03	Fischin' for Cleaner Skies
CHE.04	Fluoro Dynamics – Protect and Pioneer Our Future
CHE.05	From CO2 Emissions to Olefins
CHE.06	StarBound - From Biomass to Beyond

Life-Long Learning. We address this Program Educational Objective with a curriculum that forces students to develop the knowledge, skills and mental orientation that will equip them to be life-long learners. For example, in the Senior Design sequence, students learn to write a literature review, which requires that they begin to explore the vast research literature available. They must synthesize knowledge from literature and apply it to their specific design project. During the semester, the students are asked to give presentations detailing the progress on their designs and thereby develop their oral communication skills. Finally, they receive feedback not only from the course instructor and industrial mentor, but also from professional engineers who participate as judges at the Engineering Expo and Chemical Engineering Design Showcase. Also, students discuss many current issues in chemical engineering including development of new energy sources and products. These topics often become the major theme for many of the design projects. Finally, our students are required to participate in one-on-one advising sessions with a faculty member every semester. During the advising sessions, students are asked to plan their career or academic objectives and to consider how to best prepare to achieve their goals. This one-on-one mentoring is a valuable part of providing a learning environment that is suited to the individual student and to promote the skills and knowledge needed to pursue life-long learning. As mentioned above, the Professional Development Seminar (CHE 499) brings in industrial, professional, and academic speakers to open our students' vistas on unexpected career paths, real-life experience in subfields of chemical engineering, and interaction with other types of engineers. Finally, career options and advanced degrees relevant to pharmaceuticals, medicine, renewable energy, environmental engineering, nanotechnology, process automation, molecular engineering, and entrepreneurship are enhanced by the six concentrations.

5.A.3. Support of the Attainment of Student Outcomes by the Curriculum and Prerequisites.

Section 3.C has detailed how attainment of Student Outcomes is facilitated by both Chemical Engineering courses (Tables 3-3, 3-4, and 3-5) and technical courses outside our department (Tables 3-6). Within the ChE program each Student Outcome receives significant coverage by multiple courses, and this is bolstered by further treatment in the non-ChE portion of the curriculum.

5.A.4. Prerequisite Structure of Required Courses.

Figure 5-3 shows the prerequisite structure of required courses within the ChE curriculum.

5.A.5. Meeting the Requirements of Depth of Study for General and Program Criteria.

It is shown in Table 5.1 that our programs course requirements meet or exceed the minimum credit hours and distribution. How the Chemical Engineering program stands in relation to ABET requirements is detailed as follows.

- (a) **ABET Requirement.** *“A minimum of 30 semester credit hours (or equivalent) of a combination of college-level mathematics and basic sciences with experimental experience appropriate to the program.*

Our program features 47 credit hours (50 with selective option CHEM 346) of college level mathematics and basic sciences. This is distributed among mathematics (4 courses, 14 hours), physics (2 courses, 8 hours), chemistry (7 courses, 25 hours or 8 courses, 28 hours). Laboratory experience is integrated into the General Chemistry (CHEM 122/123 and 124/125, 10 hours) and General Physics (PHYS 141, 142, 8 hours) and accounts for half of Analytical Chemistry (CHEM 222, 4 hours). In addition, CHEM 233 (2 hours) is a dedicated Organic Chemistry Lab course.

(b) ABET Requirement. “A minimum of 45 semester credit hours (or equivalent) of engineering topics appropriate to the program, consisting of engineering and computer sciences and engineering design, and utilizing modern engineering tools.

Our program has 57 hours of engineering topics (or 54 hours with selective option CHEM 346 instead of CHE 330), distributed as follows.

Sophomore Engineering courses outside major (9 hours): ECE 210, CME 260, CS 109.

Sophomore Chemical Engineering courses (13 hours): CHE 201, 205, 210 230.

Junior Chemical Engineering Courses (15 hours): CHE 301, 311, 312, 313, 321.

Senior Chemical Engineering Courses (20 hours): CHE 330, 341, 381, 382, 396, 397, 4XX.

(c) ABET Requirement. “A broad education component that complements the technical content of the curriculum and is consistent with the program educational objectives.

Our program features 6 hours of English composition, 15 hours of General Education courses divided amongst five broad categories, and 3 additional hours of elective credit outside the chemical engineering rubric that can also be used for general education. Many students use the latter courses to study a foreign language or take business courses. We also have many students who opt for a minor in Mathematics since only two additional courses (one in the form of the free elective) are necessary to complete that requirement.

(d) ABET Requirement: “A culminating major engineering design experience that 1) incorporates appropriate engineering standards and multiple constraints, and 2) is based on the knowledge and skills acquired in earlier course work.

5.A.6. Major Design Experience.

The major design experience in our program is composed of CHE 396 and 397, which are the two capstone design courses. The first course (CHE 396) in the sequence focuses on the design and economic analysis of chemical processes. This includes cost estimation and economic evaluation methodologies and well as design of individual processing units. Students become more familiar with process flow simulation, although they have already had experience with the Aspen Plus in several previous courses (at a minimum in CHE 205, CHE 301 and CHE 313), and in the technical elective, CHE 433 Process Simulation with Aspen Plus. The design of unit operations relies on the Chemical Engineering core and basic engineering science courses from the junior year. These include chemical engineering thermodynamics where the students are taught the basics of equation of state models and activity coefficient models upon which to base their design simulations. In addition, students utilize the transport phenomena sequence (CHE 311, 312, and 313) which

supports the design of heat exchangers, piping systems, and separation processes by supplying the estimates of the transport properties and the insight provided by the engineering analysis of these unit operations. The reactor design course (CHE 321) furnishes an understanding of basic reactor modeling which is relevant for many process units. CHE 396 begins the conceptual transition from theory to practice, from “ideal” to “real”. Operational consequences of connecting process units together are explored, as are pumping and piping, fouling, instrumentation, controls, process safety, occupational health, sustainability, optimization regarding energy and raw materials, and engineering ethics. Since 2008, we have moved to define the spring semester design projects late in CHE 396, to enable better time flow in CHE 397.

In the second design course (CHE 397) students are assigned to work on a major design project in small teams of three or four students each. Each team is assigned an industrial mentor, a practicing design engineer who provides feedback and guidance in mandatory weekly meetings carried out on-site. Within a well-structured framework of report milestones, the teams give multiple presentations describing their coalescing designs throughout the semester in front of their classmates and the team mentors to trace their progress and build detail into the final design. The overall design of a process flow sheet and the control and optimization of the process is supported by the previous design course (CHE 396), the mass and energy balance course (CHE 210) and the process control course (CHE 341). The milestones culminate in the final written and oral reports (co-judged by the industrial mentors) as well as presentation of the completed project at the annual Engineering Expo, where it is judged by practicing industrial engineers and the Chemical Engineering Design Showcase, where it is judged by practicing chemical engineers. Projects are reviewed by judges and the instructor to be such that the designs meet modern and realistic design constraints and standards. It is essential that projects are reviewed by practicing engineers. Throughout the semester, guest speakers are invited to give presentations on several aspects of the chemical industry and career development. Students also have an opportunity to meet informally with visitors to discuss their own interests. One advantage of extensive industrial contact and observation of individual students is the opportunity for networking. Senior Design often leads to job offers for our graduating seniors.

5.A.7. Summary of Program Curriculum Changes

Four courses were added to the curriculum. CHE 230 Molecular Systems in Chemical Engineering was added in 2022 and is designed to introduce several topics that are important to traditional and emerging fields in chemical engineering; molecular engineering, macromolecular systems, and nanomaterials. CHE 330 Polymer Science was added as a selective option with CHEM 346 Physical Chemistry II in Fall 2020/Spring 2021 and gives our students the option to either learn about polymer science and engineering or focus on advanced topics in physical chemistry. Two technical elective courses were also added; a course on the fundamentals of biopharmaceutical process development and manufacturing (CHE 426), and a course on engineering entrepreneurship (CHE 427). In addition, a CHE 392 course section has also been dedicated for the AIChE ChemE Car and ChemE Cube competitions to allow students to obtain technical elective credit while performing research for the annual competitions.

Note: The addition of CHE 230 and CHE 330 does not change the 128-credit hour requirement for the BS Chemical Engineering major. CHE 230: In 2022, the Chemistry department changed the credit hours for CHEM 232 and CHEM 234 from 4 hours to 3 hours and our department changed the credit hours for CHE 396 Senior Design I from 4 hours to 3 hours. This created a deficit of 3 credit hours that was then filled with the addition of CHE 230 (+ 3 credit hours) into the curriculum. In addition, since CHE 330 Polymer Science is a selective option with CHEM 346, it does not increase the total credit hours in our program.

In addition to the Biochemical Engineering concentration, five concentrations to the BS in Chemical Engineering degree were added in 2022: Polymers and Molecular Engineering, Energy and Environment, Nanotechnology, Process Automation, and Entrepreneurship. A detailed description of this curriculum change is given in 4.B.6.

5.A.8. Cooperative Education

Although it is not required, many of our students do participate in co-op and internship programs. Our location in a major metropolitan center with extensive chemical, materials, energy, food, healthcare, consulting, electronics industries, to name a few, offers our students many opportunities for a meaningful industrial experience during their undergraduate education. However, course credit is not offered for these activities.

5.A.9. Material to be Available for Review at the Site Visit.

The Program Evaluator will have access to an extensive sampling of course materials (syllabi, handouts, hard copies or hyperlinks of textbooks, chemical engineering laboratory I and II lab manuals, homework assignments, quizzes, exams) and corresponding student work. We will also provide extensive documentation of assessment methods/data, evaluation procedures, feedback and continuous improvement, along with detailed information on how course content supports attainment of Student Outcomes. This material will be organized in a web format that facilitates efficient examination and cross-checking of the material.

In conclusion, with 47 hours of basic mathematics and science, 57 hours of engineering topics and 21 hours of general education with an additional 3 hours of free electives, UIC's Chemical Engineering program satisfies ABET requirements.

B. Course Syllabi

Appendix A provides the syllabi of all courses used to satisfy the mathematics, science, and chemical engineering discipline-specific requirements required by Criterion 5.

Table 5-1 Curriculum Chemical Engineering

Course (Department, Number, Title) List all courses in the program by term starting with the first term of the first year and ending with the last term of the final year.	Indicate whether course is Required, Elective or a Selected Elective by an R, an E or an SE. ¹	Subject Area (Credit Hours)			Last Two Terms the Course was Offered: Year and, Semester, or Quarter	Maximum Section Enrollment for the Last Two Terms the Course was Offered ²
		Math & Basic Sciences	Engineering Topics; Check if Contains Significant Design (✓)	Other		
Freshman Year, First Semester						
ENGL 160 English Composition I	R			3	Fall 2025 Spring 2026	24 24
MATH 180 Calculus I	R	4			Fall 2025 Spring 2026	120:25 120:25
CHEM 122/123 Matter & Energy and Foundations of General Inquiry I	R	5			Fall 2025 Spring 2026	324:25:24 324:25:24
ENGR 100 Engineering Orientation: ChE Sections	R		0		Fall 2024 Fall 2025	24 24
General Education Core I: Understanding the Past	SE			3	Fall 2025 Spring 2026	
Freshman Year, Second Semester						
CS 109 Programming for Engineers with MatLab	R		3		Fall 2025 Spring 2026	268:30 294:30
MATH 181 Calculus II	R	4			Fall 2015 Spring 2026	115:23 115:23
CHEM 124/125 Chemical Dynamics & Foundations of General Inquiry II	R	5			Fall 2025 Spring 2026	286:23 286:23

PHYS 141 General Physics I	R	4			<i>Fall 2025</i> <i>Spring 2026</i>	<i>192:24</i> <i>192:24</i>
Sophomore Year, Third Semester						
MATH 210 Calculus III	R	3			<i>Fall 2025</i> <i>Spring 2026</i>	<i>115:23</i> <i>115:23</i>
CHEM 232 Structure and Function	R	3			<i>Fall 2025</i> <i>Spring 2026</i>	<i>287:24</i> <i>287:24</i>
PHYS 142 General Physics II	R	4			<i>Fall 2025</i> <i>Spring 2026</i>	<i>192:24</i> <i>192:24</i>
ENGL 161 English Composition II	R			3	<i>Fall 2025</i> <i>Spring 2026</i>	<i>24</i> <i>24</i>
CHE 201 Intro to Engineering Thermodynamics	R		3		<i>Fall 2025</i> <i>Spring 2026</i>	<i>41</i> <i>15</i>
Sophomore Year, Fourth Semester						
MATH 220 Intro to Differential Equations	R	3			<i>Fall 2025</i> <i>Spring 2026</i>	<i>132:22</i> <i>132:22</i>
CHE 210 Material & Energy Balances	R		4√		<i>Fall 2025</i> <i>Spring 2026</i>	<i>34</i> <i>33</i>
CHE 205 Computational Methods in Chemical Engineering	R		3		<i>Fall 2025</i> <i>Spring 2026</i>	<i>27</i> <i>28</i>
CHE 230 Molecular Engineering in Chemical Engineering	R		3		<i>Spring 2025</i> <i>Spring 2026</i>	<i>47</i> <i>57</i>
CHEM 222 Analytical Chemistry	R	4			<i>Fall 2025</i> <i>Spring 2026</i>	<i>96:24</i> <i>96:24</i>
Junior Year, Fifth Semester						
CHE 311 Transport Phenomena I	R		3√		<i>Fall 2025</i> <i>Spring 2026</i>	<i>35</i> <i>16</i>
CHE 301 Chemical Engineering Thermodynamics	R		3		<i>Fall 2025</i> <i>Spring 2026</i>	<i>31</i> <i>20</i>

CHEM 342 Physical Chemistry I	R	3			Fall 2024 Fall 2025	62:62 69:69
CHEM 233 Synthesis Techniques Lab	R	2			Fall 2025 Spring 2026	22 22
CHEM 234 Chemical Synthesis	R	3			Fall 2025 Spring 2026	146:25 146:25
General Education Core II				3	Fall 2025 Spring 2026	
Junior Year, Sixth Semester						
CHE 312 Transport Phenomena II	R		3√		Fall 2025 Spring 2026	12 35
CHE 313 Transport Phenomena III (Separation Processes)	R		3√		Fall 2025 Spring 2026	15 34
CHE 321 Chemical Reaction Engineering	R		3√		Fall 2025 Spring 2026	12 52
ECE 210 Electrical Circuit Analysis	R		3		Fall 2025 Spring 2026	84:28 98:29
CME 260 Property of Materials	R		3		Fall 2025 Spring 2026	140 146
General Education Core III	SE			3	Fall 2025 Spring 2026	
Senior Year, Seventh Semester						
CHE 381 Chemical Engineering Laboratory I	R		2		Fall 2024 Fall 2025	16:16:7 16:10
CHE 396 Senior Design I	R		3√		Fall 2024 Fall 2025	39 27
Selective Option: CHE 330 Polymer Science (ENG) or CHEM 346 (Math/Science)	SE	(3)	(3)		Fall 2024 Fall 2025	34:21 40:26

CHE Technical Elective, e.g. CHE 422 Biochemical Engineering, CHE 433 Process Simulation with Aspen Plus, CHE 494-Selected Topics in Chemical Engineering		R		3√		Fall 2025 Spring 2026	18 8
CHEM 499 Professional Development Seminar		R		0		Fall 2024 Fall 2025	42 27
General Education Core IV		SE			3	Fall 2025 Spring 2026	
Senior Year, Eighth Semester							
CHE 382 Chemical Engineering Laboratory II		R		2		Spring 2025 Spring 2026	15:24 14:13
CHE 397 Senior Design II		R		4√		Spring 2025 Spring 2026	40 27
CHE 341 Chemical Process Control		R		3		Spring 2025 Spring 2026	39 25
Elective Outside the Major Rubric		SE			3	Fall 2025 Spring 2026	
General Education Core V		SE			3	Fall 2025 Spring 2026	
<i>Add rows as needed to show all courses in the curriculum.</i>							
TOTALS (in terms of semester credit hours)			47 Hours (50 Hours)	57 Hours (54 Hours)	24 Hours		
Note: Total credit hours in Math/Science or Engineering depend on the CHE330/(CHEM346) option							
OVERALL TOTAL CREDIT HOURS FOR COMPLETION OF PROGRAM		128					
Total must satisfy minimum credit hours	Minimum Semester Credit Hours		30 Hours	45 Hours			

1. **Required** courses are required of all students in the program, **Elective** courses (often referred to as open or free electives) are optional for students, and **Selected Elective** courses are those for which students must take one or more courses from a specified group.
2. For courses that include multiple elements (lecture, laboratory, recitation, etc.), indicate the maximum enrollment in each element. For Selected Elective courses, indicate the maximum enrollment for each option.

Instructional materials and student work verifying compliance with ABET criteria for the categories indicated above will be required during the campus visit.

SAMPLE
SEMESTER COURSE SCHEDULE
UIC Chemical Engineering

FRESHMAN			Total Hours = 15				Total Hours = 16
ENGL 160	3	English Composition I		CS 109 (= MATH 180)	3	Programming for Eng w/ MatLab	
MATH 180	4	Calculus I		MATH 181 (MATH 180)	4	Calculus II	
CHEM 122	4	Matter and Energy and Foundations		CHEM 124 (CHEM 122)	4	Chemical Dynamics and	
CHEM 123	1	of General Inquiry I		CHEM 125 (CHEM 123)	1	Foundations of Chem Inquiry II	
ENGR 100	0	Engineering Orientation		PHYS 141 (MATH 180)	4	Gen. Physics - Mech.	
HUM/SOC	3	General Education Course					
SOPHOMORE			Total Hours = 16				Total Hours = 17
MATH 210 (MATH 181)	3	Calculus III		MATH 220 (MATH 210)	3	Intro to Differential Equations	
CHEM 232 (CHEM 124/125)	3	Structure and Function		CHE 205 (=201, =MATH 210)	3	Comput. Methods in ChE	
PHYS 142 (MATH 181, PHYS 141)	4	Gen. Physics II-Elec. & Mag		CHE 210 (=201, =CHE 205)	4	Material & Energy Balances	
CHE 201 (MATH 181, PHYS 141)	3	Intro to Thermodynamics		CHE 230 (CHEM 232)	3	Molecular Engineering in CHE	
ENGL 161	3	English Composition II		CHEM 222 (CHEM 124/125)	4	Analytical Chemistry	
JUNIOR			Total Hours = 17				Total Hours = 18
CHE 311 (MATH 220, CHE 205)**	3	Transport Phenomena I		CHE 312 (CHE 311)	3	Transport Phenomena II	
CHE 301 (CHE 201 & 205)	3	ChE Thermodynamics		CHE 313 (CHE 301)	3	Transport Phenomena III	
CHEM 342 (PHYS 142, MATH 210)*	3	Physical Chemistry I		CHE 321 (= CHE 301, CHE 210, MATH 22)	3	Chem. Reaction Eng.	
CHEM 234 (CHEM 232)	3	Chemical Synthesis		CHE 260	3	Property of Materials	
CHEM 233 (= CHEM 232)	2	Synthesis Techniques Lab		HUM/SOC	3	General Education Course	
HUM/SOC	3	General Education Course		ECE 210 (MATH 180)	3	Elec. Circuit Analysis	
SENIOR			Total Hours = 14				Total Hours = 15
CHE 396 (CHE 312, 313, 321)	3	Senior Design I		CHE 397 (CHE 396)	4	Senior Design II	
CHE 381 (CHE 312)	2	Chemical Eng. Lab I		CHE 341 (CHE 312, 313, 321 & MATH 220)	3	Chemical Process Control	
CHE 330	3	Polymer Science		CHE 382 (CHE 381, 313)	2	Chemical Eng. Lab II	
HUM/SOC	3	General Education Course		ELECTIVE	3	Outside Major	
CHE ELECTIVE	3	400 Level ChE course or Equiv.		HUM/SOC	3	General Education Course	
CHE 499 (Fall semester only)	0	Prof. Development Seminar					
Total Hours in Major = 128							

Legend:

Milestone Courses: Must be taken in the specified semester to be able to graduate in 4 yrs.

() Prerequisites

(=) Concurrent registration

(*) Credit or concurrent registration in CHEM 222

(**) Credit or concurrent registration in CHE 210

ENGINEERING COURSES

ENGR. 100	0	Engineering Orientation
CS 109	3	Programming for Eng w/ MatLab
ECE 210	3	Elec. Circuit Analysis
CME 260	3	Properties of Materials
ENGLISH		
Engl. 160	3	English Composition I
Engl. 161	3	English Composition II
MATHEMATICS		
Math 180	4	Calculus I
Math 181	4	Calculus II
Math 210	3	Calculus III
Math 220	3	Intro. to Differential Equations
PHYSICS		
Phys 141	4	General Physics I
Phys 142	4	General Physics II
ELECTIVES		
Understanding the Past	3	
Exploring World Cultures	3	
Understanding Individuals & Society	3	
Understanding US Society	3	
Understanding Creative Arts	3	
Elective Outside Major	3	

CHEMISTRY COURSES

CHEM 122-Lecture, CHEM123-Lab	5	Matter & Energy & Foundations of Chem Inquiry I
CHEM 124-Lecture, CHEM125-Lab	5	Chem Dynamics Foundations of Chem Inquiry II
CHEM 222	4	Analytical Chemistry
CHEM 232	3	Structure and Function
CHEM 233	2	Synthesis Techniques Lab
CHEM 234	3	Chemical Synthesis
CHEM 342*	3	Physical Chemistry I
CHEM 346 (selective with CHE 330)	3	Physical Chemistry II
CHEMICAL ENGINEERING SEQUENCE		
CHE 201	4	Intro to Thermodynamics
CHE 205	3	ComputMethods in ChE
CHE 210	4	Material & Energy Balances
CHE 230	3	Molecular Systems in ChE
CHE 301	3	Thermodynamics
CHE 311**	3	Transport Phenomena I
CHE 312	3	Transport Phenomena II
CHE 313	3	Transport Phenomena III
CHE 321	3	Chemical Reaction Eng.
CHE 330 (selective with CHEM 346)	3	Polymer Science
CHE 341	3	Chemical Process Control
CHE 381	2	Chemical Eng. Lab I
CHE 382	2	Chemical Eng. Lab II
CHE 396	3	Senior Design I
CHE 397	4	Senior Design II
CHE 499 (Fall Semester Only)	0	Prof. Development Seminar
CHE Elective	3	

Figure 5.3. Typical degree plan for the Chemical Engineering Program

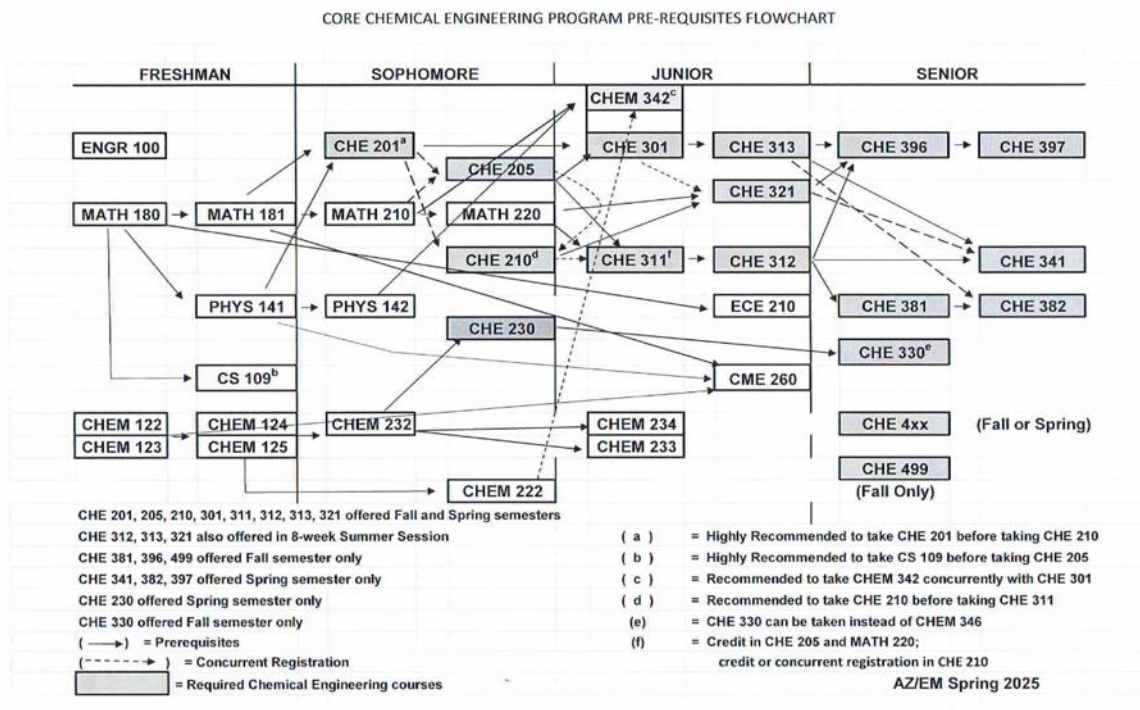


Figure 5.4. Pre-requisite structure of required courses within the ChE curriculum.

CRITERION 6. FACULTY

A. Faculty Qualifications

Most professors in the department have a PhD in Chemical Engineering with the rest having PhD's in closely aligned disciplines such as polymer, environmental, and mechanical engineering, and physics. Thirteen professors have industrial or national lab experience, and four professors have established their own companies. Several have had post-doctoral experience outside the US and are also very familiar with the educational system in Europe and Asia. In addition, during the past five years the Department has appointed one additional clinical professor for a total of 4 clinical track professors. Teaching is their main responsibility (from 50% to 100% each semester – two to three classes), and two research professors. Three of these professors have spent most of their careers in industry.

The normal course load of a research active faculty member is four classes per year. The normal course load for clinical teaching faculty is six courses per year. However, under college rules faculty who actively participate in the training of PhD students get a reduced course load of three classes and in some cases for faculty members with heavy research activities, the load is two classes. Overall, we are moving towards 2.5 courses per year for all research active faculty (2 classes one year, and 3 classes the next). Faculty are allowed to “buy out” a class, but in the past five years it has not happened in the Department.

B. Faculty Workload

Expertise among the faculty is sufficiently diverse so that any course in the undergraduate curriculum could be taught by at least two of the current faculty members. This teaching versatility has been fostered by the diverse educational backgrounds of faculty members both in traditional chemical engineering and related fields, in addition to both long and short teaching experiences. Research in support of teaching is also encouraged and pursued by most of the faculty through the undergraduate research course (CHE 392) and through the University Honors College. A summary of the distribution of faculty workload summary and Activity Analysis are given in Tables 6.1 and 6.2. We would like to point out that while Table 6.2 shows the actual time being spent by the faculty in different activities as per university policies, Table 6.1 also takes into consideration the actual impact of the activity in judging its level, so there may not necessarily be a direct correspondence between the two Tables.

C. Faculty Size

The Department currently has 189 undergraduate students. In academic year 2025-2026, there were eleven tenured/tenure-track (TT) faculty (one is 25% in the Chemical Engineering department, with the Bioengineering department being the main appointment). There are also four non-tenure track (NTT) clinical professor faculty, and two research professors and one visiting research professor. Tables 6.1 lists these faculty members and their teaching and other responsibilities.

The current faculty workload, as described earlier in Section 6.B, permits more than adequate advising and educational interactions with students. It also provides enough depth of expertise to permit faculty sabbatical leaves for professional development without adverse effects on teaching. Given below is a short summary of the faculty involvement with students, university service, professional development, and interactions with industry and professional practitioners.

6.C.1. Assistant Professors.

Dr. Ezinne Achinivu-Ibagere is an active member of the American Institute of Chemical Engineers (AIChE) and the American Chemical Society (ACS) and regularly attends conferences in bioprocessing and green chemistry. She has served on proposal review panels for the National Science Foundation (NSF) and the Department of Energy (DOE) and is an active reviewer for several professional journals, including Green Chemistry, Industrial & Engineering Chemistry Research, and ACS Sustainable Chemistry & Engineering. Dr. Achinivu-Ibagere's research focuses on the sustainable conversion of lignocellulosic biomass into renewable fuels, bioplastics, and value-added chemicals using ionic liquid-based pretreatment and bioprocess engineering, with recent publications featuring graduate and undergraduate student co-authors as lead contributors. She is a co-founder of ChemE Connect, a departmental initiative dedicated to strengthening sense of belonging among students, faculty, and staff within the UIC Chemical Engineering community, and she actively mentors students in her laboratory in experimental design, biomass characterization, and sustainable process development. Dr. Achinivu-Ibagere currently serves on the CHE Graduate Committee and the CHE Advisory Committee and is a member of the newly founded UIC Catalyst Center (UIC CAT), which advances interdisciplinary research across campus. She also serves on the Advisory Board of the U.S. DOE Bioenergy Technologies Office (BETO) Agile BioFoundry, where she contributes to shaping national priorities in biomanufacturing and engages directly with industry partners and professional practitioners to connect fundamental research to scalable applications in the bioeconomy.

6.C.2. Associate Professors.

Dr. Sangil Kim is an active member of the Electrochemical Society. He has served as a panelist for the National Science Foundation (NSF) and Department of Energy (DOE) for review of proposals. He is an Editorial Board member of Membranes and a Review Editor for Frontiers in Membrane Science and Technology journals and actively reviews for several journals such as the Advanced Materials, ACS Energy Letters, and Industrial & Engineering Chemistry Research. Dr. Kim's research has focused on developing new membrane materials, understanding functioning mechanisms of the nanostructured device, and applying for biomedical engineering, energy-efficient water purification and gas separation, and energy storage. He has led as a PI or co-PI multidisciplinary research projects sponsored by DTRA, DOE, ARPA-E, and NSF. He currently serves as a member of the CHE Safety Committee.

Dr. Shafigh Mehraeen is a Principal Investigator and an associate professor at the UIC, where he leads research funded by the NSF, DoD, and NIH. His work utilizes computational modeling to advance water remediation technologies and develop cell-free drug delivery systems for retinal repair. An active contributor to the scientific community, Dr. Mehraeen serves as a panelist for the NSF CBET division and is a frequent reviewer for high-impact journals such as Physical Review

Letters, ACS ES&T, and Langmuir. He holds active memberships in ACS, AIChE, and ASME, and currently serves on the UIC Chemical Engineering Graduate Admissions Committee. Committed to workforce development, he also leads initiatives providing research experiences for high school educators.

Dr. Lewis E. Wedgewood is an active member of the American Institute of Chemical Engineers (AIChE) and the Society of Rheology (SOR) and has participated in the Midwest AIChE conferences. He is currently a member of the department's Advisory Committee which reviews department academic policy and financial expenditures. He is also on the department Undergraduate Committee which modifies the curriculum and develops new courses. He has been a panelist for the National Science Foundation for review of proposals and is an active reviewer for journals such as the Physics of Fluids.

6.C.3. Full Professors.

Dr. Vikas Berry is the Satish C. & Asha Saxena Professor and Head of Chemical Engineering at the University of Illinois Chicago, where he has been on the faculty since 2014. He received his Ph.D. in Chemical Engineering from Virginia Tech in 2006, following his M.S. from the University of Kansas in 2003 and B.Tech. from the Indian Institute of Technology Delhi in 1999. Dr. Berry is internationally recognized for pioneering contributions to graphene-based quantum materials and bio/nano technologies, including the development of four distinct methods for graphene synthesis and production. His work on graphene bio interfaces has served as a key foundation for biological applications of graphene. He is a recipient of the NSF CAREER Award, the Sigma Xi Outstanding Junior Scientist Award, and the Big 12 Fellowship, and previously held the William H. Honstead Professorship at Kansas State University. His research has been published in leading journals including Nature Photonics, Nature Communications, Nano Letters, Advanced Materials, and Small, and has been featured in major outlets such as The Economist, The Wall Street Journal, and Science News. Dr. Berry serves on the editorial boards of Scientific Reports and the Journal of Nanoscience Letters, and his research is supported by the NSF, DoD, and industry.

Dr. Brian Chaplin is an active member of the American Chemical Society and organizes research symposia annually. Dr. Chaplin has served on proposal review panels for the National Science Foundation and Environmental Protection Agency. He is a frequent reviewer of several professional journals and Associate Editor for ACS ES&T Engineering. Dr. Chaplin has several externally funding research grants focused on water treatment and is recognized as a leader in the field of electrochemical water treatment. He has utilized this expertise to form meaningful collaborations with other academic researchers, consultants, municipalities, and industrial end-users. For example, he is a research thrust leader for the Great Lakes ReNEW NSF Innovation Engine, which is a center focused on research and commercialization of water treatment technologies within the Great Lakes region.

Dr. Ying Liu holds the Dr. Satish C. & Asha Saxena Professorship in the Department of Chemical Engineering. She is also affiliated with the Departments of Pharmaceutical Sciences and Biomedical Engineering at UIC. She currently serves on the advisory board of NSF's ChemMatCARS and on the steering committee of AAAS Pharmaceutical Sciences subcommittee.

Previously, she served as the Interim Associate Dean of Research in College of Engineering and Director of Graduate Studies in the Department of Chemical Engineering. Ying Liu has garnered several prestigious awards throughout her career, including National Science Foundation CAREER Award (2014), UIC Researcher and Scholar of the Year Rising Star Award (2016), and UIC Graduate Mentor Award (2017). Ying Liu's research focuses on soft-matter physics and the knowledge-based design of delivery platforms for therapeutic drugs and cells. The Liu research group is currently advancing AI-guided design and optimization of nano formulations for drug and gene delivery.

Dr. Anh Ngo is a professor at University of Illinois at Chicago, Chemical Engineering department. He works on a wide range of topics including quantum materials with strong electronic correlations, renewable energy materials, catalysis and nanoscale devices. Anh Ngo's research is also focused on computational models for materials, then combining those materials with experimental characterization to help design new materials and devices. Specifically, his expertise lies in predictive modeling of materials at multiple length and time scales ranging from quantum to meso-scale. State-of-the-art computational approaches he uses include *ab initio* electronic structure methods, quantum chemical calculations, density functional theory, quantum Monte Carlo methods, molecular dynamics, coarse-graining methods, battery cell level modeling, advanced sampling techniques, and development of materials models using Artificial Intelligence/machine learning methods. Dr. Anh Ngo has received multidisciplinary research funding from the DOE, DOD, and NSF. He has served on proposal review panels for the DOE Office of Critical Minerals and Energy Innovation, Manufacturing and Energy Supply Chains, and the Office of Energy Efficiency and Renewable Energy (EERE), including the Bipartisan Infrastructure Law (BIL) Battery Materials Processing and Battery Manufacturing programs. He is a member of the U.S. Battery500 Consortium and the Low-cost Earth-abundant Na-ion Storage (LENS) Consortium. He has also served as a member of the CHE Graduate Committee.

Dr. Vivek Sharma is an active member of the Society of Rheology (SOR), the American Institute of Chemical Engineers (AIChE), the American Physical Society (APS), the Adhesion Society, the Society for Engineering Sciences (SES), and the American Chemical Society (ACS). He organizes and chairs sessions at several national meetings: regularly at SOR Meetings, ACS Colloids, AIChE (2013-2026), and APS (2014-2026), and occasionally at ACS, Adhesion Society, and the U. S. National Congress on Theoretical and Applied Mechanics. Dr. Vivek Sharma regularly attends meetings of these professional societies, presents papers and posters at their conferences, and has organized short courses for SoR and APS DPOLY. He was elected as a member-at-large for APS DPOLY (2023-2026), AIChE Area 1J: Fluid Mechanics programming committee, and serves on the ACS Colloids conference organizing committee, the SoR centennial committee and the AIChE awards committee, and in the past has served as social media chair for SoR, APS DPOLY, and APS DSOF. In addition, Dr. Sharma has presented at Gordon Research Conferences (GRCs) on Adhesion, Polymer Physics, and Colloidal, Macromolecular and Polyelectrolyte Solutions (CMPS), served as a discussion leader in Polymer Physics & CMPS, plus participated in several GRC meetings. He was recently elected vice-chair for CMPS GRC; he will organize it in 2030. Dr. Sharma also co-organized the Telluride Workshop in Molecular Engineering in Soft Matter in 2023 and 2025 and served as an invited instructor for the Food and Soft Matter Workshop held in Cargese, Corsica, FR, in 2024 and 2026. Dr. Sharma has served on several panels for the National Science Foundation (NSF) and

actively reviews papers for over a dozen scientific journals. He was the Guest editor for a special issue on Food as Soft Matter. Dr. Sharma has given over hundred invited talks at conferences, in industry and in universities, won the 3M Nontenured Faculty Award (2019), TA Distinguished Young Rheologist Award (2017) UIC's COE Teaching Award in 2017, 2023 and 2026, and UIC COEs Mentoring Award in 2022. Dr. Sharma's students have won over a dozen prizes and recognitions at the GRC meetings, APS March Meeting (including Padden Award), AIChE and ACS Colloids meetings. Dr. Sharma has held visiting professor appointments at the University of Chicago, Stanford University, and ESPCI Paris.

Dr. Meenesh R. Singh is a Professor of Chemical Engineering at the University of Illinois Chicago (UIC), where he directs the Materials and Systems Engineering Lab (MaSEL) and the UIC Catalyst (CAT) Center. His research develops advanced computational and experimental tools to address critical challenges, including carbon sequestration, nitrogen cycle management, clean water access, and advanced therapeutics. Dr. Singh earned his B.E. from Sardar Patel University (2005), MTech. from IIT Bombay (2008), and Ph.D. from Purdue University (2013), and was a Postdoctoral Fellow at the Joint Center for Artificial Photosynthesis (UC Berkeley/LBNL). His work has resulted in 100+ publications, 8 patents (2 licensed), 3 book chapters, 5 open-source tools, 130 conference presentations, and 35 invited talks, supported by ~\$9.2M in external funding and collaborations with multinational companies. He has mentored Ph.D. graduates now at Massachusetts Institute of Technology, Lawrence Livermore National Laboratory, and AbbVie, and currently advises Ph.D. students, postdocs, and undergraduates. He serves on editorial boards of multiple journals, including Chemical Engineering Research & Design, Sustainable Chemistry, Challenges, Sustainability, and Current Trends in Chemical Engineering and Process Technology, and is a reviewer for 15+ journals. Dr. Singh has chaired AIChE sessions in separation and reaction engineering and advises the UIC AIChE Chem-E-Car team (nationally placed, 2019). His honors include the AIChE George Klinzing Award and multiple best paper awards. His work has been featured in major outlets including CNN, IEEE Spectrum, and Quanta Magazine. He is also CTO of eN-RAMPS.

Dr. Christos Takoudis (25%, Home Department: Bioengineering) is an active participant in meetings of the American Institute of Chemical Engineers, the American Chemical Society, the American Vacuum Society, and the Materials Research Society. He maintains active research collaborations with industrial partners (e.g., Kurt J. Lesker) in addition to federally funded projects. Portions of his recent research have been highlighted on social media and in professional outlets by the American Institute of Physics and the American Vacuum Society. He is also engaged in consulting activities, serves as a panelist for funding agencies, and regularly reviews manuscripts for professional journals.

6.C.4. Clinical and Research Professors.

Dr. Said Al-Hallaj is the Founder and CEO of NETenergy, Chief Battery Scientist at Beam Global company (Nasdaq: BEEM, BEEMW), Co-founder and former CEO of AllCell Technologies LLC (acquired by Beam Global), and a Research Professor of Chemical Engineering at the University of Illinois at Chicago (UIC). Dr. Al-Hallaj earned his bachelor's and master's degrees in chemical engineering from the Jordan University of Science and Technology (JUST) and a Ph.D. in chemical engineering from the Illinois Institute of Technology (IIT). Said co-authored a book

entitled “Hybrid Hydrogen Systems” and has published several book chapters and numerous numbers of peer reviewed and conference journal papers with >16,000 literature citations. Said is a serial entrepreneur, and his R&D effort has led to the successful commercialization of several clean-tech technologies.

Dr. Betul Bilgin is a Clinical Professor in the Department of Chemical Engineering at the University of Illinois Chicago (UIC), where she has been a faculty member since 2014. She teaches Senior Design I and II, Introduction to Thermodynamics, Computational Methods, and Transport Phenomena III, and has led the development of these courses using active, project-based, and industry-integrated approaches. She has strengthened the Senior Design program by incorporating structured design processes and emphasizing safety, ethics, economic analysis, and environmental and societal considerations, while expanding industry mentorship and engagement. Her research focuses on engineering education, including engineering judgment, identity development, data science integration, and AI-supported learning. Dr. Bilgin is actively involved in student mentoring, advising the AIChE Student Chapter, and leading outreach, diversity, and curriculum initiatives. She also serves nationally as Director of the ASEE Chemical Engineering Division and 2nd Vice Chair of the AIChE Education Division. Her contributions to engineering education have been recognized with multiple awards, including the ASEE Ray Fahien Award (2024), UIC College of Engineering Harold Simon Award (2022), UIC Teaching Recognition Program Award (2021), and AIChE “35 Under 35” and ASEE “20 Under 40.”

Dr. Michael Caracotsios is a Clinical Associate Professor in the Department of Chemical Engineering, following a distinguished 30-year career with UOP/Honeywell prior to his retirement in 2015. He teaches and develops course materials and computational tools for Chemical Engineering Thermodynamics (CHE 301), Process Simulation and Design with ASPEN Plus (CHE 433), and Chemical Process Control (CHE 341). He is also the President and Founder of AthenaVisual, Inc., which promotes the use of AthenaVisual Plus, a state-of-the-art engineering software platform for mathematical modeling, parameter estimation, model discrimination and validation, and sequential optimal experimental design. Dr. Caracotsios has supervised numerous undergraduate research projects that have led to the development of advanced computational tools in Thermodynamics, Fluid Mechanics, Heat Transfer, Chemical Equilibrium, Process Control, and Reactor Design. These tools are fully integrated into his teaching, enabling students to analyze and solve complex, real-world problems drawn from his extensive industrial experience. He received his Ph.D. in Chemical Engineering from the University of Wisconsin–Madison. Dr. Caracotsios is the author of two patents, twelve peer-reviewed publications, and numerous presentations at national and international conferences. He is also co-author of the book *Computer-Aided Design of Chemically Reactive Systems*, published by John Wiley & Sons (2006). A strong advocate for accessible education, Dr. Caracotsios actively develops and distributes open instructional materials. He has authored comprehensive lecture notes and computational resources for courses in Thermodynamics, Process Simulation, and Process Control. He recently completed the first edition of a thermodynamics textbook, supported by a \$2,500 award from the University Library, with final publication planned for Fall 2026. Dr. Caracotsios regularly invites industry experts to deliver seminars on topics including process design, control, safety, and leadership. He is also engaged in outreach efforts, visiting local high schools to promote chemical engineering and highlight its impact on society. In addition, he organizes student visits to industrial facilities, providing direct exposure to professional practice. These experiences have included visits to

Novaspect (an Emerson Impact Partner in Process Automation), the Dresden and LaSalle nuclear power plants in Illinois, and EATS 2025 at McCormick Place in Chicago—a major food and beverage technology exposition that provides access to leading equipment and technology suppliers, networking opportunities, and interactive industry demonstrations. He is actively involved in undergraduate student activities within the department. Dr. Caracotsios is also an active member of the American Institute of Chemical Engineers (AIChE).

Dr. Naveen Dandu is a Research Assistant Professor at the University of Illinois Chicago. His research lies at the intersection of computational chemistry, materials science, and artificial intelligence, with a focus on developing physics-informed, data-driven approaches for energy materials, nanostructures, and electrochemical systems. His work integrates density functional theory (DFT), ab initio molecular dynamics, and machine learning to accelerate the discovery and design of next-generation materials for applications such as batteries, catalysis, and nanoscale devices. Dr. Dandu is an active member of the American Chemical Society and contributes extensively to the scientific community as a reviewer for leading journals including Journal of the American Chemical Society, Angewandte Chemie, Journal of Chemical Information and Modeling, Energy & Environmental Science: Batteries, and iScience. He engages in interdisciplinary collaborations across academia, national laboratories, and industry. His research has been closely aligned with U.S. Department of Energy initiatives, including collaborations with Argonne National Laboratory (since 2019) and the Joint Center for Energy Storage Research (from 2019 to 2021), where he has contributed to data-driven electrolyte discovery and advanced materials modeling. His group develops scalable computational frameworks that bridge quantum, atomistic, and continuum scales, enabling predictive modeling alongside experimental validation. In addition to his research, Dr. Dandu is actively involved in teaching and mentoring undergraduate and graduate students. He is committed to interdisciplinary education and to advancing the role of artificial intelligence in scientific discovery.

Dr. Roshan Nemade is a Research Assistant Professor in Chemical Engineering at the University of Illinois Chicago, where he is actively engaged in teaching, mentoring, and research. His university service includes serving as a judge for 3MT thesis and poster presentations at the graduate symposium, reviewing for Chemical Engineering Research and Design, and contributing to laboratory training, safety practices, and student research activities. He regularly mentors undergraduate and graduate students through semester and summer research programs, supporting their development in experimental techniques, data analysis, and scientific communication. Dr. Nemade maintains continuous professional development through participation in technical conferences, NSF I-Corps commercialization training, and advanced instrumentation certification at the UIC Nanotechnology Core Facility. His work involves strong collaboration with industry and professional practitioners, including partnerships with Lawrence Berkeley National Laboratory, Spraying Systems Co., and industrial collaborators in advanced materials manufacturing. Through these efforts, he integrates academic research with real-world engineering applications and technology commercialization.

Dr. Korosh Torabi is actively engaged in student mentorship and development as a co-adviser for the AIChE student chapter, where he co-leads the ChemE Car and ChemE Cube teams, and he founded the ChemE-Connect committee to foster a strong sense of community and belonging within the chemical engineering department at the University of Illinois Chicago. His university

service includes serving on the ChE department undergraduate curriculum committee and judging the ChE graduate research symposium at UIC, both of which support academic quality and student scholarship. At the professional level, Dr. Torabi has served on the Education Committee of the Biophysical Society and has organized and chaired multiple sessions at the AIChE Annual Meeting, advancing professional development and engagement within the discipline. He also maintains active connections with industry and professional practitioners through his role on the advisory board of the Engineering Technology program at Henry Ford College, contributing to curriculum alignment with workforce needs and strengthening collaboration between academia and industry.

Dr. Alan D. Zdunek is a Clinical Associate Professor and the Director of Undergraduate Studies (DUGS) in the Chemical Engineering department. He served as the Director of the Engineering Success Program (ESP) in the College of Engineering from 2018-2026 and has been an active member of the COE Education Policy Committee since 2015. He was also the faculty advisor for the UIC AIChE student chapter from 2015-202. As DUGS, Dr. Zdunek is involved in all aspects of the undergraduate program, from curriculum development, student extracurricular activities, professional development, chemical engineering alumni relations, external advisory board relations, and high school/community college recruitment. He was a co-PI on two research projects; an NSF grant developing advanced materials for solid-oxide fuel cells, and a UIC Proof-of-Concept grant studying the feasibility of a water desalination technology. Recently he has been a co-PI on an NSF IUSE proposal to promote engineering identity development through mentoring networks. He has also supervised several undergraduate research projects. Community service activities include giving “What is Chemical Engineering” presentations at local high schools and community colleges and recruitment events on campus. Utilizing his 17 years of industrial experience, Dr. Zdunek consults in the areas of corrosion engineering and electrochemical technology. He is an active member of the American Institute of Chemical Engineers (AIChE), American Chemical Society (ACS) and has served as the Chairman of the ECS Chicago section.

6.C.5. Current/Past National Science Foundation CAREER/NYI Awardees Among the Faculty.

Dr. Brian Chaplin 2015-2020

Dr. Ying Liu, 2014-2019

Dr. Vikas Berry, 2011-2017

6.C.6. Faculty Teaching, Student Advising, and Other Notable Awards.

- Dr. Vivek Sharma
 - Padden Award from the American Physical Society, 2025
 - John H. Dillon Medal from the American Physical Society, 2023
 - COE Faculty Teaching Award (2017, 2022, 2023, 2025)
- Dr. Betul Bilgin
 - UIC Teaching Scholar (Spring 2017)
 - American Institute of Chemical Engineers (AIChE) 35 under 35 award in education category (August 2017)
 - American Society for Engineering Education (ASEE) 20 under 40 award (June 2018)

- UIC Teaching Sustainability Initiative Spring (2020 and 2024)
 - UIC Teaching Recognition Program (TRP) Award (Spring 2021)
 - UIC COE Harold Simon Award (Spring 2022)
 - UIC COE Advising Award (Spring 2023)
 - UIC COE Teaching Award (Spring 2022-24)
 - ASEE Ray Fahien Award (June 2024)
- Dr. Alan Zdunek
 - COE Faculty Teaching Award (2016, 2017, 2018, 2025)
 - COE Faculty Advising Award (2024)
- Dr. Shafigh Mehraeen
 - 2018, 2019, 2023, College of Engineering Teaching Award, UIC
- Dr. Vikas Berry
 - Listed in the Top 2% cited researchers since 1960 (Stanford Rankings), 2021
 - Rudolph A. Marcus Award, 2017
- Dr. Meenesh Singh
 - Faculty Entrepreneur Leadership Program (FELP) Cohort Member, 2025
 - UIC's Outstanding Global Scholar, 2024
 - UIC Proof-of-Concept Phase-II Award, 2023
 - UIC CTRI Award, 2023
 - UIC COE Advising Award, 2022
 - UIC Proof-of-Concept Phase-I Award, 2021
 - UIC CTRI Award, 2021
 - UIC COE Research Award, 2021
- Dr. Said Al-Hallaj
 - Stanford Top 2% researchers in Chemical Engineering
 - Top 10 papers at the MIT A+B 2024 Energy Symposium
 - Illinois Tech Alumni Professional Achievement Award, 2022
- Dr. Ying Liu
 - Sino-American Pharmaceutical Professionals Association Leadership Award, 2023
 - Chicago Biomedical Consortium, Catalyst Award, 2020
 - UIC Graduate Mentor Award, UIC, IL, 2017
 - UIC Researcher and Scholar of the Year, Rising Star Award, UIC, IL, 2016
- Dr. Ezinne Achinivu-Ibagere
 - AAAS Science & Technology Policy Fellowship
- Dr. Roshan Nemade
 - Postdoc Small Scholarship Award at University of Illinois Chicago, 2024.
 - Award for Graduate Research (AGR) at University of Illinois Chicago, 2022.

Other awards given to the faculty are listed in their Faculty Vitae in Appendix B.

6.C.7. Changes in Faculty.

Dr. Anh Ngo was hired in the Fall of 2020 as an Associate Professor, Dr. Ezinne Achinivu-Ibagere was hired in Fall 2022 as an Assistant Professor, Dr. Korosh Torabi as hired as a Clinical Assistant Professor in 2023, and Dr. Roshan Nemade was hired as a Research Assistant Professor in 2024. Five faculty were promoted to Professor; Dr. Brian Chaplin (2022), Dr. Ying Liu (2022), Dr. Anh Ngo (2025), Dr. Vivek Sharma (2025, and Dr. Meenesh Singh (2025). Dr. Betul Bilgin was promoted to Clinical Professor in 2024, and Dr. Alan Zdunek was promoted to Clinical Associate Professor in 2020. Two faculty were also promoted to Associate Professor; Dr. Sangil Kim (2022) and Dr. Shafigh Mehraeen (2024). In addition, Dr. Ying Liu became the Dr. Satish C. & Asha Saxena Professor in 2022. Dr. Gang Cheng left the department in 2023.

D. Professional Development.

Details of the professional development activities of each faculty in the Department are provided in the short bio summaries given above in sections 6.C.1-6 and in faculty CV's in Appendix B. Highlights of faculty professional development activities are summarized below:

Summary of Faculty Professional Development Activities

- Dr. Vivek Sharma
 - Chairs technical sessions at Society of Rheology, ACS Colloids, AIChE, American Physics Society, US National Congress on Theoretical and Applied Mechanics.
 - Gordon Research Conference (GRC) presenter and attendee: Adhesion, Polymer Physics; Colloidal, Macromolecular and Polyelectrolyte Solutions (CMPS)
 - Vice-Chair for GRC CMPS conference
 - Co-organized Telluride Workshop in Molecular Engineering in Soft Matter
 - Invited instructor for Food and Soft Matter Workshop in Cargese, Corsica, France.
- Dr. Betul Bilgin
 - Serves as Director of the ASEE Chemical Engineering Division
 - 2nd Vice Chair of the AIChE Education Division
 - Regularly attends American Society of Engineering Education meetings.
- Dr. Alan Zdunek
 - Attended One UIC Alignment for Enrollment Success Conference, September 2025
 - Attended IES Abroad Global Institute, Barcelona, Spain, 2024
 - Attended IES Abroad Engineering Consortium, Dublin, Ireland, 2022.
- Dr. Shafigh Mehraeen
 - Mentor Training Workshop, UIC (2025)
 - Panelist for NSF CBET Division
 - Reviewer for Physical Review Letters, ACS ES&T, Langmuir
- Dr. Vikas Berry
 - Serves on editorial boards of Scientific Reports, Journal of Nanoscience Letters

- Leading research collaboration programs with Argonne National Laboratory, University of Chicago, other international academic partners
 - Regularly attends AIChE national meetings
- Dr. Meenesh Singh
 - Strong collaboration with 7+ multinational companies
 - Chief Technology Officer at eN-RAMPS LLC. (UIC Start-up)
 - Actively involved in prototyping and scale-up of 4 technologies
- Dr. Said Al-Hallaj
 - Leadership: Board Member, Armour College of Engineering at Illinois Tech
 - Industry Advisory Board for Mechanical Engineering at UIC
 - Commercialization Fellow for UIC Office of Technology Management
- Dr. Ying Liu
 - Advisory board member of NSF's ChemMatCARS
 - Steering committee member of AAAS Pharmaceutical Sciences subcommittee
 - Served as Interim Associate Dean of Research in College of Engineering
- Dr. Ezinne Achinivu-Ibagere
 - Regularly attends AIChE and ACS conferences in bioprocessing and green chemistry
 - Advisory Board member of the U.S. DOE Bioenergy Technologies Office (BETO) Agile BioFoundry
 - NSF and DOE proposal review panelist
- Dr. Sangil Kim
 - Editorial Board member of Membranes and a Review Editor for Frontiers in Membrane Science and Technology journals
 - Active reviewer for several journals: Advanced Materials, ACS Energy Letters, and Industrial & Engineering Chemistry Research
 - NSF and DOE proposal review panelist
- Dr. Brian Chaplin
 - Research thrust leader for the Great Lakes ReNEW NSF Innovation Engine.
 - Organizes American Chemical Society annual research symposia
 - Associate Editor for ACS ES&T Engineering
- Dr. Michael Caracotsios
 - Actively develops and distributes open instructional materials
 - Completed 1st edition of a thermodynamic textbook
 - Developed advanced computational tools in thermodynamics, fluid mechanics, heat transfer, chemical equilibrium, process control and reactor design
- Dr. Naveen Dandu

- Active reviewer for leading journals including Journal of the American Chemical Society, Angewandte Chemie, Journal of Chemical Information and Modeling, Energy & Environmental Science: Batteries, and iScience
 - Engages in interdisciplinary collaborations across academia, national laboratories, and industry; U.S. Department of Energy initiatives, including collaborations with Argonne National Laboratory (since 2019) and the Joint Center for Energy Storage Research
- Dr. Roshan Nemade
 - NSF I-Corps commercialization training
 - Advanced instrumentation certification at the UIC Nanotechnology Core Facility
 - Strong collaboration with industry and professional practitioners, including partnerships with Lawrence Berkeley National Laboratory, Spraying Systems Co.
- Dr. Korosh Torabi
 - Served on the Education Committee of the Biophysical Society
 - Organized and chaired multiple sessions at the AIChE Annual Meeting, advancing professional development and engagement.
 - Advisory board member of the Engineering Technology program at Henry Ford College, contributing to curriculum alignment with workforce needs and strengthening collaboration between academia and industry.
- Dr. Anh Ngo
 - Proposal review panelist for: DOE Office of Critical Minerals and Energy Innovation, Manufacturing and Energy Supply Chains; Office of Energy Efficiency and Renewable Energy (EERE); Bipartisan Infrastructure Law (BIL) Battery Materials Processing and Battery Manufacturing programs.
 - Member of the U.S. Battery500 Consortium and the Low-cost Earth-abundant Na-ion Storage (LENS) Consortium.

The department places much emphasis on the development of all faculty members, especially junior and mid-career faculty, as well as the clinical (non-tenure teaching track) faculty. A formal mentor is appointed for each Assistant Professor appointed in the Department. The mentor works closely with the faculty members in developing better and more effective teaching strategies, as well as developing exciting new research ideas and provides continuous feedback on the faculty members' progress towards tenure. A special program has been set up in the department to support CAREER proposals from assistant professors (biweekly meetings with mentors, mock panel review). For mid-career faculty, although a formal mentor is not appointed, the Department Head holds regular meetings with them to enhance both their teaching and research effectiveness. This includes arranging meetings with representatives of both industrial and governmental funding agencies and alerting them when funding is available for these activities through university-wide programs as well as from external agencies. The department also provides seed funding to both early career and mid-career faculty to initiate research projects. In addition, the department provides funding to faculty to visit funding agencies and present talks at professional meetings. As part of its contribution to Faculty Development, the University has provided a new personal

computer to all faculty on a 3/4-year cycle as well as to new faculty members when they first join. The University additionally shares the cost of startup costs of new faculty with the College and the Department through various programs. The department encourages faculty members to be active in professional societies and supplements the cost of attending their meetings when they do not have travel funds available in grants. In addition, the department provides funds to faculty members to visit granting agencies and covers other related costs of grant proposal development. For faculty members with 25% appointments in the department, the primary responsibility of faculty development is that of their home departments. Finally, to ensure that all faculty have adequate computing resources to ensure their professional development, the Department provides software needed to fulfill their teaching and professional responsibilities.

E. Authority and Responsibility of Faculty

Course Creation, Modification and Evaluation. New courses being developed can be initiated by an individual faculty member, the undergraduate committee in the Department, which consists of 3- 4 faculty members, from discussions at department curriculum meetings, recommendations from the external advisory board, and from student input. Faculty members initiating a new course first consult with the head of the department or the director of undergraduate studies to provide reasons for developing the course. The head of the department may initiate a task force consisting of the faculty member and other faculty members with similar expertise to further develop a new course outline. The task force prepares a detailed outline of the course with hours to be spent on each subject/topic, suggests an appropriate textbook, and submits a detailed proposal to the Undergraduate Committee. Once approved by the Undergraduate Committee, it is discussed by the entire faculty during a faculty meeting. After approval by the Department, the Director of Undergraduate Studies creates a UIC Course Request Form (CRS) and presents the new course proposal to the College Education Policy Committee, which meets bi-monthly and has one member from each department. The Associate Dean of Undergraduate Affairs is an ex-officio member of this Committee. After discussions, revisions, and approval by the College Educational Policy Committee, it is forwarded to the Office of the Provost and Vice-Chancellor for Academic Affairs. That office seeks advice from the Senate Educational Policy Committee (SCEP). After approval from SCEP it is sent to the University Senate for final approval for the following academic year. As a sidenote, new, 400-level courses most often begin as CHE 494 Selected Topics in Chemical Engineering. This is a temporary course designation that allows the faculty to develop and teach a course for several semesters before the department decides on making it a permanent course in the curriculum.

New and existing courses are evaluated by the instructor using end-of-semester student evaluations (Class Climate Evaluations) that are administered centrally by the Office of the Vice Provost for Faculty Affairs. The course instructor reviews this student feedback from the evaluations and reflects on how he/she intends to improve that course for the next year (or semester). The Course Climate Evaluations represent a regular and extensive consultation with our student constituency with tangibly positive effects on their courses and learning experiences. Faculty respond to their course evaluations with recommendations and actions to improve their courses. If needed, the course instructor may meet with the department head or director of undergraduate studies to discuss the student feedback and actions to improve the course. For new courses, the task force

that initially developed the course may also be consulted to provide valuable input and recommendations to the evaluation process.

Program Educational Objectives and Student Learning Outcomes.

The Student Learning Objectives and Program Educational Objectives represent the ultimate goals of undergraduate teaching and learning, and thereby serve to organize our efforts in design, development, implementation and improvement of the curriculum toward professional success and effectiveness of our graduates. Hence, all faculty play an important role in both the definition and attainment of the SLOs and PEOs. The Program Educational Objectives are periodically discussed and reviewed at faculty meetings and faculty retreats. If there is a consensus that the PEO's need to be revised, further discussion is initiated by the director of undergraduate studies and new statements, and wording are studied and then proposed to the faculty. After further discussion, the faculty approve the first set of revisions and the new PEO's are sent to our additional stakeholders (external advisory board, alumni, industrial partners) for their input and recommendations before final approval by the faculty.

The Director of Undergraduate Studies periodically initiates a course instructor review of the Student Learning Outcomes applicable to each chemical engineering course to determine if the SLO's are up to date with the topics being taught, or if they need to be revised. The course instructors teaching each chemical engineering course review the SLO's and make recommendations to change or keep them the same. The Performance Indicators (PI) that are connected to the course SLO's, are used by the faculty to develop course assignments (quizzes, exams, projects, presentations) in order to test and attain the Student Learning Outcomes, and ultimately the Program Educational Objectives for each specific course. As shown in section 4.A.1, the process for assessment and continuous improvement of the SLO's is the responsibility of the Director of Undergraduate Studies, however, the assessment methodology has been developed so that all faculty are involved in the process.

Table 6-1. Faculty Qualifications

Chemical Engineering

Faculty Name	Highest Degree Earned- Field and Year	Rank ¹	Type of Academic Appointment ² T, TT, NTT	FT or PT ³	Years of Experience			Professional Registration/ Certification
					Govt./Ind. Practice	Teaching	This Institution	
E. Achinivu-Ibagere	PhD (ChE & BioMolecE) 2014	AST	TT	FT	5	4	4	
S. Al-Hallaj	PhD (ChE), 1999	P	NTT	PT	15	9	9	
V. Berry	PhD (ChE) 2006	P	T	FT	1	19	12	
B. Bilgin	PhD (ChE) 2014	P	NTT	FT	0	12	12	
M. Caracotsios	PhD(ChE) 1986	ASC	NTT	FT	30	41	11	
B. Chaplin	PhD (ChE), 2007	P	T	FT	4	16	13	
N. Dandu	PhD (Chem) 2016	AST	NTT	FT	3	3	3	
S. Kim	PhD (ChE) 2007	ASC	TT	FT	15	4	4	
Y. Liu	PhD (MAE), 2007	P	T	FT	0	18	18	
S. Mehraeen	PhD (MAE), 2011	ASC	TT	FT	2	14	10	
R. Nemade	PhD (ChE) 2023	AST	NTT	FT	3	1	1	
Anh Ngo	PhD (Phys) 2010	P	T	FT	12	2	2	
V. Sharma	PhD (Polym. Sci. & Eng.), 2008	P	T	FT	0.5	13	13	
M.R. Singh	PhD (ChE), 2013	P	T	FT	3	10	10	
C. Takoudis	PhD (ChE), 1982	P	T	FT	1	45	30	
K. Torabi	PhD (ChE) 2011	AST	NTT	FT	0	12	3	
L. E. Wedgewood	PhD (ChE), 1988	ASC	T	FT	0	36	28	
A. Zdunek	PhD (ChE), 1990	ASC	NTT	FT	17	18	16	

1. Code: P = Professor ASC = Associate Professor AST = Assistant Professor I = Instructor A = Adjunct O = Other
2. Code: TT = Tenure Track T = Tenured NTT = Non Tenure Track
3. At the institution
4. The level of activity, high, medium or low, should reflect an average over the year prior to the visit plus the two previous years.

Table 6-2. Faculty Workload Summary

Chemical Engineering

Faculty Member (name)	PT or FT ¹	Classes Taught (Course No./Credit Hrs.) Term and Year ²
E. Achinivu-Ibagere	FT	CHE 230 (3) Spring 2022, 2023, 2024,2025 CHE 330 (3) Fall 2022, 2024; Spring 2023 CHE 392(1-3) Spring 2022-2026, Fall 2022-2025
S. Al-Hallaj	PT	CHE 427 (3) Fall 2020-2025, CHE 451 (3) Fall 2020-2025 CHE 392 (1-3) Spring 2026
V. Berry	FT	CHE 321 (3) Spring 2021, 2022, 2023 CHE 392 (1-3) Fall 2020-2023, Spring 2021, 2023-2026 CHE 527 (4) Spring 2026
B. Bilgin	FT	CHE 396 (3) Fall 2020- 2025 CHE 397 (4) Spring 2021-2026 CHE 201 (3) Fall 2020- 2025; Spring 2021-2026 , CHE 205 (3) Spring 2021- 2026; CHE 313 (3) Summer 2022-2026 CHE 392 (1-3) Spring 2023, 2024
M. Caracotsios	FT	CHE 301 (3) Fall 2020-2025; Spring 2021-2026 CHE 311 (3) Fall 2021, 2022 CHE 341 (3) Fall 2020-2025 CHE 433 (3) Spring 2021-2026

		CHE 392 (3) Fall 2020, Spring 2022, 2026
B. Chaplin	FT	CHE 205 (3) Fall 2023; CHE 312 (3) Fall 2020; CHE 453 (3) Fall 2021, 2024 CHE 510 (4) Spring 2021
N. Dandu	FT	CHE 454 (3) Spring 2023
S. Kim	FT	CHE 313 (3) Fall 2021, 2023, 2025; Spring 2021-2026 CHE 392 (1-3) Fall 2020, 2022-2024, Spring 2021,2022, 2026 CHE 494 (3) Fall 2025, Spring 2023 CHEM 501 (4) Fall 2020-2023
Y. Liu	FT	CHE311 (3) Spring 2021 CHE 392 (1-3) Fall 2021, 2024, Spring 2024 CHE 422 (3) Spring 2023, 2026 CHE425 (3) Spring 2021, 2022, 2023 CHE 457 (3) Fall 2020, 2021 CHE 527 (4) Fall 2023, Spring 2025
S. Mehraeen	FT	CHE 205 (3) Fall 2020-2022, 2024, 2025 CHE 438 (3) Spring 2022, 2024 CHE 531 (4) Spring 2021-2026 CHE 545 (4) Fall 2021-2025
R. Nemade	FT	CHE 230 (3) Spring 2026 CHE 330 (3) Fall 2025 CHE 392 (1-3) Fall 2025
A. Ngo	FT	CHE 455 (3) Fall 2021, 2022; Spring 2023-2026
V. Sharma	FT	CHE 311 (3) Fall 2020, Spring 2022 CHE 392 (1-3) Spring 2022, 2026 CHE 454 (3) Spring 2021, 2024, 2025 CHE 457 (3) Fall 2023, Spring 2026 CHE 494 (3) Fall 2024 CHE 512 (4) Spring 2022, 2024
M.R. Singh	FT	CHE 321 (3) Fall 2020-2022, 2024, 2025 CHE 150 (3) Fall 2023-2025, Spring 2024,2025 CHE 392 (1-3) Fall 2021, 2023, 2024 Spring 2021-2024, CHE 494/426 (3) Spring 2021-2026

		CHE 527 (4) Spring 2021, 2022, 2026
C. Takoudis	FT	CHE 312 (3) Fall 2021-2025; CHE 313 (3) Fall 2020 CHE 392 (1-3) Spring 2022, 2023, 2026, CHE 514 (4) Sp 2021
K. Torabi	FT	CHE 311 (3) Spring 2023, 2025, 2026 CHE 312 (3) Fall 2023-2025, Spring 2023-2026, Summer 2024-2026 CHE 321 (3) Fall 2023, Spring 2024-2026, Summer 2026 CHE 501 (4) Fall 2025; CHE 502 (4) Fall 2024
L. E. Wedgewood	FT	CHE 210 (4) Fall 2020-2025, Spring 2021-2026 CHE 312 (3) Spring 2021, 2022, 2024 CHE 511 (4) Spring 2023, 2025; CHE 512 (4) Spring 2026 CHE 520 (4) Fall 2020-2025
A. Zdunek	FT	CHE 381 (2) Fall 2020-2026; CHE 382 (2) Spring 2020- 2026, CHE 499 (0) Fall 2020-2026

1. FT = Full Time Faculty or PT = Part Time Faculty, at the institution
2. For the academic year for which the Self-Study Report is being prepared.
3. Program activity distribution should be in percent of effort in the program and should total 100%.
4. Indicate sabbatical leave, etc., under "Other."
5. Out of the total time employed at the institution.

CRITERION 7. FACILITIES

A. Offices, Classrooms and Laboratories

The chemical engineering department is housed in the Engineering Innovation Building (EIB). The department moved to this facility in summer 2019. The 57,500-square-foot-building is at the corner of Morgan and Taylor Streets on UIC's east campus and is shared by the Chemical Engineering, Civil and Materials, and Mechanical Engineering Departments; the Chemical Engineering department is the largest tenant. The EIB is a state-of-the-art facility designed by Dewberry architects and built by Berglund Construction in 2018-2019, and was LEED Gold certified in 2021. The 7-year-old building features high-tech classrooms and laboratories, including vibration-dampened nanotechnology labs, biomedical research rooms, molecular engineering research labs, and a highly attuned darkroom for complex fluid imaging. Here follows a summary of each of our program's facilities in terms of their ability to support the attainment of the student outcomes and to provide an atmosphere conducive to learning.

7.A.1. Offices.

The Engineering Innovation Building contains 16 offices for chemical engineering faculty located on the first and second floors, two offices for teaching assistants (EIB 264 and 276), and one central office for the administrative staff (EIB 226). Faculty offices have a 43" LED 4KTV monitor mounted on the wall for student discussions. There is secure WIFI connectivity throughout the building.

7.A.2. Classrooms.

There are two classrooms in EIB, Rooms 124 and 271. Both rooms are equipped with moveable tablet armchairs, high-definition video projectors and an Integrated Multimedia lectern with an installed computer, laptop connectivity, a document camera, a Blu-Ray player and internet connectivity. Room 124 has a state-of-the-art, lecture board containing a MultiTouch Video Wall System of eight, 55" LCD video wall displays and a moveable whiteboard. Room 271 has a 75" LED 4KTV monitor for lectures as well as a whiteboard. Most chemical engineering courses are taught in these two classrooms, but other university classrooms with similar features are also utilized to provide the best class schedule for the students.

7.A.3. Laboratory Facilities.

Unit Operations Laboratory. The Unit Operations Laboratory for CHE 381 and 382 is located on the first floor of EIB in Room 128. A unique feature is that can be viewed from classroom EIB 124 by a large viewing window. The unit operations laboratory contains 12 functional chemical engineering process demonstrators supporting hands-on instruction of chemical engineering principles such as heat, mass and momentum transfer, fluid properties, process control, and separations processes such as distillation, evaporation, absorption, and extraction. The laboratory space is specifically designed as an instructional chemical engineering laboratory

with dedicated workspaces for the process equipment and easy access to utilities such as 120V and 220V electricity, industrial hot and chilled water, purified water, compressed air, and vacuum lines. Eight moveable lab stations and two permanent lab bench areas are located throughout the lab and for the analytical equipment used in the process experiments. A new 48kW electric steam generator was installed in December 2019 and is plumbed directly to the appropriate process equipment. There are two fume hoods and four dropdown snorkel lines for chemical vapor control. The laboratory is furnished with two (2) computers connected to the distillation column and process control demonstrators, respectively. High-speed Wi-Fi and ethernet connections to the UIC network are both available in the lab. Most students use personal laptop computers, tablets, or other devices.

ChemE Car/ChemE Cube Laboratory. The EIB 231 laboratory on the second floor is used by the two student teams that participate in the AIChE ChemE Car and ChemE Cube annual competitions. The laboratory has 6 moveable lab stations, one fume hood, and compressed air lines, as well as access to 120V and 220V electricity. The ChemE Car and ChemE Cube teams use this laboratory to develop and test their prototype models.

Engineering Makerspace. <https://makerspace.uic.edu/>

The University of Illinois Chicago (UIC) Engineering Makerspace is a premier, dedicated fabrication and prototyping facility that supports the academic and professional development of engineering students, faculty, and student organizations within the College of Engineering (CoE). Located in the Roosevelt Road Building on the UIC campus, the Makerspace comprises over 25,000 square feet of program space equipped with more than 100 machines and hundreds of hand tools, facilitating a wide range of design, fabrication, and testing activities fundamental to modern engineering education and innovation.

The Makerspace serves as an interdisciplinary hub where students engage in hands-on, experiential learning that complements classroom instruction. It directly supports coursework in additive and subtractive manufacturing, CNC machining, electronics, and digital fabrication, including formal academic offerings such as ENGR 111 (introductory additive manufacturing), ENGR 112 (subtractive machining and safety), and project-based courses like ENGR 214 where students design and build functional prototypes.

A broad array of fabrication technologies is available within Makerspace. The Digital Fabrication Lab houses multiple 3D printers (e.g., Voron Trident, Stratasys Objet30 Prime, Formlabs Form3+), CNC mills, laser cutters, and waterjet systems. Supporting technologies include textiles and vinyl cutters, electronics assembly and testing stations, 2D/3D scanning and photography equipment, and silicone molding and resin casting tools. Traditional Machine Shop capabilities include vertical mills, lathes, carpentry tools, and manual and CNC equipment. The facility also includes collaborative workshop areas and common project spaces that encourage teamwork and knowledge sharing.

Access to the Makerspace is structured to prioritize current College of Engineering students and staff, aligning with educational objectives and ensuring that resources are available to support accredited program outcomes. Projects associated with coursework or recognized student

organizations are the primary focus of use, with procedures defined for project submission and scheduling. The facility enforces comprehensive safety and training policies, including required safety training for tool use and staff supervision for many machines, to ensure user competency and mitigate risk.

Makerspace's role in the educational mission of the CoE is significant. It enables students to apply theoretical knowledge to real-world problems through iterative design and fabrication, strengthens their technical communication and problem-solving skills, and fosters an innovation-oriented culture. This environment directly contributes to the attainment of student outcomes by providing resources for prototyping, testing, and multidisciplinary collaboration beyond traditional laboratories.

In summary, the UIC Engineering Makerspace is a vital facility within the College of Engineering that enhances educational quality and supports the department's commitment to student success, through well-maintained, current tools; safety-oriented operations; and broad accessibility for curricular and co-curricular activities.

UIC Innovation Center. <https://innovationcenter.uic.edu/>

The University of Illinois Chicago (UIC) Innovation Center is a cross-disciplinary hub that connects students, faculty, and industry partners to advance experiential learning, interdisciplinary collaboration, and applied innovation. The Innovation Center's mission is to drive innovation forward through hands-on engagement and partnerships that ignite student success, deepen academic-industry ties, and generate real-world solutions to complex challenges. Serving as a central bridge between academia and industry, the Innovation Center engages participants from more than 28 disciplines across the university community. It has collaborated with over 38 unique corporate partners and trained more than 1,200 students in collaborative solution development.

At the heart of the Innovation Center's activities are labs, classes, and workshops that offer structured, experiential opportunities for students and faculty to work on real-world problems. These activities are designed to foster a human-centered approach to innovation, emphasizing empathy, creative problem solving, iterative testing, and teamwork. Labs operate as ongoing collaborative environments where interdisciplinary teams engage in sustained innovation work. Examples include **Medical Device Labs** and **User Experience Labs**, where students and industry partners co-develop solutions that address healthcare, usability, and systems challenges. These labs allow participants to apply classroom knowledge to tangible projects, encouraging technical proficiency, design thinking, and professional readiness.

Classes and programmatic offerings at the Innovation Center reinforce critical and creative thinking through project-based learning. Courses such as **Interdisciplinary Solution Development (IXD)** and **Entrepreneurial Solution Development (EXD)** bring together students from diverse majors to solve industry-sponsored problems or to identify and pursue unmet needs independently. These opportunities help students deepen analytical skills, build professional competencies, and navigate complex, multi-stakeholder environments.

The Innovation Center also hosts workshops and short-term engagements that enhance collaboration, build practical skills, and foster networks among students, faculty, and external partners. Through these immersive experiences, participants learn structured approaches to innovation while contributing to meaningful outcomes for corporate and community stakeholders. For students, the Innovation Center is an accessible platform to gain real-world experience, explore career pathways, and engage with industry mentors. It provides internships, networking opportunities, and forums for ideation and prototype development, enhancing students' readiness for the modern workforce.

In summary, the UIC Innovation Center exemplifies a commitment to integrated learning, practical problem solving, and strategic partnerships. By fostering an environment where academic inquiry meets industry needs, the center supports the university's educational objectives and contributes to a skilled, innovative, and industry-ready student population, strongly aligning with the expectations of our student outcomes regarding professional skills, experiential learning, and collaborative engagement.

7.A.4. Additional Facilities.

The Student Lounge (Room EIB 232) is located on the 2nd floor on the northeast corner of the EIB building. The lounge has floor-to-ceiling windows on two sides with excellent ambient lighting and features a kitchenette with sink, microwave and refrigerator, several work/lunch tables, a sofa, a 75" LED 4KTV, and an electric piano. A library/reading room (Room 276) with a 43" LED 4KTV monitor and moveable whiteboard, a theory lab (Room 230), and a meeting/teleconferencing room (Room 224) with a 75" LED Multi-touch Display are also used by our students. There is an all-gender bathroom on the second floor (Room 238), and a lactation room located on the second floor (Room 262) that also serves as a meditation/prayer room. A large foyer in the front of the building has five, 75" LED 4KTV monitors and is used periodically for student and alumni events. An outside area behind the building features a grass lawn and three picnic tables with chairs. Computers and laboratories are serviced by a small machine and electronics shop on site (Room 170). The College of Engineering machine shop and a glassblower are in the Science and Engineering Laboratory building (SEL) located across the street from EIB.

B. Computing Resources

The Chemical Engineering department provides a special license (150) and installation of Aspen V14 Process Simulation software to the students that can be installed on the student laptops. The department also has 22 rental laptops running Windows 11 Enterprise for the students in case they are in need of a PC-Windows computer. Aspen is specific to Windows operating system. The university also provides complimentary installation of Microsoft Office 365, MATLAB, and Adobe products to the students, faculty, and staff. The Chemical Engineering department also has a B/W laser printer for student used in EIB 276, a main office printer in the department office (EIB 224) for b/w and color printing jobs, and a 36-inch poster printer for student poster presentations for conferences and the senior design EXPO.

In the Unit Operations Laboratory, there are 2 computers that are dedicated to running the distillation column and process control demonstrators. High-speed WIFI and ethernet connections to the UIC network are both available in the lab. Most students use personal laptop computers, tablets, or other devices.

Aside from computers dedicated to student use in the Engineering Innovation Building (EIB), computer labs are distributed throughout the campus – although the vast majority are concentrated on the east campus within easy walking distance of EIB. General information and a schedule for specific computer labs can be found in the following website:

<http://accc.uic.edu/service/computer-labs>

All these computers are installed with Excel and MATLAB and Microsoft Office 365, which are the main software packages that Chemical Engineering students would need.

For advanced computing needs and high-performance computing (HPC) on campus, the students have access to two clusters, Extreme and SABER. Extreme is a powerful machine with 3,500 cores, 24.5 Terabytes of memory and 1.25 Petabytes of local raw storage, of which 275 Terabytes is raw fast scratch storage. The cluster is composed of traditional, high memory and GPU computing nodes: 200 HPC worker nodes, 160 x Generation I: Intel Xeon E5-2670 @ 2.60 GHz, cache size of 20MB with 16 cores in each node, 128GB RAM and 1TB local storage; 40 x Generation II: Intel Xeon E5-2670 @ 2.50 GHz, cache size of 20MB with 20 cores per node, 128 GB RAM and 1TB local storage; 3 x Generation I High Memory Worker Nodes: Intel Xeon E5-4650L @ 2.60GHz, cache size of 20MB with 32 cores per node with 1TB RAM; 50 x Generation II GPU Worker Nodes; 3 x Login Nodes; and 2 x Admin Nodes. The traditional HPC and high memory worker nodes currently run CentOS 6.9. The GPU worker nodes currently run CentOS 7.5.

The Shared Analytics & Big-data Enterprise Resource (SABER) was introduced into the Advanced Cyberinfrastructure for Education and Research (ACER) HPC resource offerings in May 2017. In addition to traditional HPC workloads, this NSF-funded cluster provides access to various Big-Data/Analytics software packages. The SABER cluster consists of traditional high-performance computing nodes; 76 x HPC Nodes with Intel Xeon E5-2650 @ 2.20GHz; 24 x Cores/Node with 30 MB cache, 128GB RAM, 1TB Local Storage. Software on each node is CentOS 7.3.

C. Guidance

Undergraduates performing research in faculty laboratories are required to attend a UIC Environmental Health and Safety Office (EHSO) laboratory safety training course before working in the laboratory. Information on the safety courses available can be found at: <http://ehso.uic.edu/training/introduction-to-laboratory-safety-training/>

Undergraduates enrolled in the unit operations laboratory (CHE 381 and 382) are provided training and guidance by the instructor and teaching assistants on proper usage and safety precautions before they begin their experimental work on the chemical process demonstrators. This is initially done in the first lab period of the semester, and thereafter during ‘pre-lab’ laboratory periods. Additional guidance and oversight is provided while the lab teams are performing the experiments and afterwards to ensure proper equipment shut-off procedures.

Makerspace staff provide guidance to students before they can use the machining tools and equipment. Three courses for more detailed instruction are also offered; ENGR 111 Additive Manufacturing, ENGR 112 Subtractive Machining and ENGR 214 Advanced Manufacturing Methods.

7.C.1. Support Policies.

All computing support policies for computer account usage, network access, are other support policies can be found at: <https://accc.uic.edu/about/policies/>

All ChE students, faculty and staff should read and agree with all these policies before using computing resources.

7.C.2. Accessibility.

ACCC also work closely with the Disability Resource Center to ensure accessibility resources are available to the UIC community. More information can be found following this link:

<http://accc.uic.edu/service/learning-environments/accessibility>

D. Maintenance and Upgrading of Facilities

7.D.1. Maintenance.

Routine maintenance and an ongoing improvement program for the Chemical Engineering Unit Operations Laboratories (ChE 381 and 382) are conducted on a continuous basis throughout the year. Suggestions for improvement are taken from student assessment (e.g., Senior Exit Survey and CHE 381/382 course evaluations), and faculty input. Maintenance and improvements are funded by the student laboratory fee, and if needed, by department or college funds. It includes the following:

7.D.2. Selected Ongoing Improvements- Unit Operations Laboratory.

Year	Experiment	Status
2020	General	Individual safety glass storage using blue bins (pandemic initiative)
2020	General	Tablet stands and tablets allowing remote student access during lab period (pandemic initiative)
2021	Distillation	Repaired faulty reboiler level sensor wiring
2022	Distillation	Upgraded ethanol measurement with densitometer (Snap 41 portable ethanol meter)
2022	Fluidized Bed	Purchased sand with more uniform mesh size (40-100 mesh)
2022	General	Relocated fluidized bed, viscosity and membrane separation experiments
2023	Fluid Flow in Pipes	Flow Valve Replacement (UIC facilities)
2023	Viscosity	Replumbed compressed air stainless steel tubing before regulator
2023	Gas Absorption	Installed steel-reinforced hose in compressed air line for better safety
2023	Humidification	Upgraded temperature measurements: Installed Omega DP9800 Thermocouple Temperature monitor
2023	Liquid Extraction	Added high-traction drainage mats around apparatus
2024	Fluid Flow in Pipes	Replaced pressure tube fittings for better differential pressure accuracy
2024	General	Purchased 4 Fisherbrand Isotemp stir plates for titration experiments
2024	Humidification	Installed new air regulator and water flow valve
2025	General	Purchased portable air compressor (220 psi/30 gallon) and connected to fluidized bed experiment after house compressed air supply went down
2025	Membrane Separation	Purchased new oxygen sensors (Max O2 and Teledyne Analytical brands)
2026	Gas Absorption	Repaired column- re-caulked and re-installed
2026	Liquid Extraction	Upgraded pump for faster mineral oil recycle back to feed tank

EIB Facility Improvements:

The building compressed air equipment was replaced with a new unit in 2025.

The building deionized water system was repaired for more reliable and consistent operation and quality in 2025.

7.D.3. Routine and Periodic Maintenance.

Weekly: Students rinse, clean and dry all labware (beakers, flasks, graduated cylinders, buckets, pipettes, test tubes, racks, trays). Students clean up common lab bench areas, especially around scales/balances, and put away any equipment used during lab period. The student lab teams shutdown their experiments, clean up the apparatus and bench area, and leave the experiment ready for the next lab team. Water, air, steam, pumps and individual electric power boxes are turned off. Instructor/TA performs an inspection at the end of each lab period to make sure all experiments are shutdown correctly; chemicals are properly stored and equipment is put away.

Semester: Faculty/staff perform a run-through of the lab experiments at the beginning of the semester before the first lab period to make sure the experiments are working properly. An inventory is performed at the end of the semester to determine if lab supplies/chemicals/gases need to be ordered for the next semester. Equipment is periodically cleaned, recalibrated, refilled or replaced by faculty/staff during the semester. For example, capillary tubes in the viscometer experiment are cleaned out with a solvent to remove Carbopol polymer residue left inside. Manometers are checked to make sure they are filled correctly. The boiler is turned on in the morning before the lab commences and a blowdown procedure is followed at the end of the day. The UIC facilities team also periodically checks the steam generator to make sure it is working correctly. Safety equipment is maintained in good working order; safety spill kits and chemical spill pads are inspected and replaced when needed, the laboratory safety manual is updated annually, MSDS binder is updated when needed, first-aid kits and eye showers are inspected weekly, an ample supply of nitrile lab gloves, heat-resistant gloves, chemical-resistant gloves and sharps gloves are maintained in the lab, and safety glasses are checked for breakage and new glasses ordered if necessary.

The professor teaching the Unit Operations Laboratory and the staff responsible for lab and computer maintenance regularly assess laboratories and equipment, keep the departmental administration apprised of any issues, and make updates as necessary.

7.D.4. Upgrading Facilities.

Here is a summary of some of the significant upgrades related to student instruction.

Classrooms: The two classrooms in the Engineering Innovation Building are equipped with Integrated Multimedia lecterns with an installed computer, laptop connectivity, a document camera, a Blu-Ray player and internet connectivity to facilitate the use of software demonstrations and computer visuals during instruction. Room 124 has a state-of-the-art, lecture board containing a MultiTouch Video Wall System of eight, 55" LCD video wall displays. This new technology allows instructors to illustrate complex and abstract concepts much easier than can be accomplished using the whiteboard.

We anticipate that instructors will be able to achieve the desired learning outcomes on a wider group of students, as some students learn most effectively through visualization.

Offices: Faculty offices include a 43-inch or 55-inch 4KTV LED display mounted to the wall that can be connected to the faculty's laptop or desktop computer. This allows better communication

with students during in-office meetings when discussing coursework, undergraduate research projects and while advising.

Laboratory: Instructional laboratory upgrades have been described in the previous sections.

E. Library Services

A public research facility, the University Library at the University of Illinois at Chicago serves an urban campus of more than 33,000 students and 11,000 faculty and staff members, as well as the Chicago metropolitan area and the State of Illinois.

The Richard J. Daley Library is located at 801 South Morgan on UIC's East Campus and is the main university library. The Richard J. Daley Library serves nine colleges, including Engineering. Patrons have access to more than 2.2 million volumes, 634 databases, 504,341 e-books, and over 60,000 current journal titles. 381 journals are included in the category of Chemical Engineering, such as AIChE journal, Chemical Engineering Science, Colloid and Polymer Science, Colloids and Surfaces, Combustion and Flame, Journal of Catalysis.

The Library of the Health Sciences (LHS) located at 1750 West Polk is one of the largest health sciences libraries in the nation and continues to support education, research, and clinical practice. With more than 725,000 volumes and 5,200 journals.

The availability of the books and journals can be checked on the library website. The available books can be borrowed from the library, and the journal articles can be either copied in the library or downloaded if the e-copies are available. If an item is not available from UIC library, it may be accessed by requesting interlibrary loan. The items can be renewed in the library or online after signing up for an account.

The faculty members can submit course reserves and recommend purchases of books or subscription of journals by contacting the reserves staff or filling in the online form.

Besides UIC libraries, the Chemical Engineering department has a small library located in our building. The reference books can be used in the library. The library can also be reserved as a study room.

Overall Comment on Facilities

Aside from the above specifics on facilities dedicated to the Chemical engineering program, it is worth mentioning that since transitioning to our new building, we have been able to keep the strong community feeling between the faculty, students and staff that was present in our previous location. Faculty are never more than a few feet away from classrooms, the labs and the student lounge, and our student community benefits from a significant esprit de corps that translates into a more complete educational experience for our students.

CRITERION 8. INSTITUTIONAL SUPPORT

A. Leadership

The undergraduate program in Chemical Engineering is administered by a four-member Undergraduate Committee, the department head, and the undergraduate program/student advisor. The Undergraduate Committee is chaired by a faculty member designated as the Director of Undergraduate Studies (DUGS). The DUGS is responsible for monitoring the effectiveness of the curriculum to meet the declared objectives and outcomes of each course, as well as the overall curriculum. The DUGS responsibility also entails oversight of all assessment procedures and coordination. The department head and DUGS are responsible for keeping abreast of chemical engineering educational trends, including recommending and implementing new areas and topics into the undergraduate curriculum. The undergraduate program/student advisor is responsible for program operations, such as the semester course schedule, providing course and program information to the students, communicating with the College of Engineering Academic Resource Center, and resolving student issues.

The department has implemented a process to review our overall curriculum, update current undergraduate courses, and develop new courses and concentrations to enhance our students' learning experience. The Undergraduate Committee conducts formal and informal surveys and receives input from our stakeholders; students, alumni, members of the External Advisory Board, Engineering Expo judges, and senior design mentors. Curriculum improvement recommendations are discussed in the undergraduate committee and department head, and then at departmental faculty meetings. Specific curriculum meetings may be held between the department head, director of undergraduate studies, and the director of graduate studies. In some instances, the undergraduate program coordinator and a faculty member with an interest in the specific curriculum change are also included. Any ideas developed in these meetings are discussed in the undergraduate or graduate committees before going to the faculty for input and approval.

If needed, curriculum and course changes are brought to the College Education Policy Committee for approval and then to the university Senate committee on education policy (SCEP) for approval and implementation. The DUGS and Undergraduate Committee also consider the approval for requests for modifications in the curriculum of individual students under special circumstances. In addition, the committee approves new courses developed by individual faculty members before they are submitted to the entire faculty for their approval.

The department initiated a curriculum review meeting in 2018 that is held every 4 years; the last one was in 2022. We also discuss our curriculum in depth at our annual summer faculty retreats, during the External Advisory Board meetings, meetings with Alumni Groups, as well as with representatives of organizations that hire significant number of our graduates. If issues concerning either courses or the overall curriculum are raised in these meetings, these issues are handled by the Undergraduate Committee or faculty task forces. The curriculum meetings, faculty retreats, and undergraduate committee meetings and faculty task forces have helped develop several improvements to our curriculum.

For instance, one such improvement that was implemented in 2022 was the addition of 5 new concentrations to our curriculum. These include (a) Energy and Environment, (b) Polymer and Molecular Engineering, (c) Nanotechnology, (d) Process Automation, and (e) Entrepreneurship. Six new courses have also been added that align with the new concentrations; (a) CHE 230

Molecular Systems in Chemical Engineering, (b) CHE 330 Polymer Science, (c) CHE 426 Biopharmaceutical Process Development and Manufacturing, (d) CHE 454 Molecular and Macromolecular Engineering (e) CHE 455 Nanoscale Systems in Chemical Engineering, and (f) CHE 457 Colloid and Interfacial Phenomena. Another curriculum improvement is to offer CHE 312, CHE 313 and CHE 321 in the summer session so that transfer students can stay on track to graduate the next Spring semester. The Fall and Spring semester course schedules have now been revised to minimize ChE/non-ChE course time conflicts, another step in streamlining the path towards graduation for our students.

B. Program Budget and Financial Support

The institutional budget for the Department of Chemical Engineering consists of two parts, personnel and the operating costs. The personnel costs are those for salaries of faculty, union staff members, academic professionals and teaching assistants. The budgeted amount for these salaries increases each year by an average percentage mandated by the University. These increases have so far been to retain qualified personnel in the department. The budget of the College is currently in part tied to the enrollment in Engineering and the tuition revenue generated by the enrolled students. While the Department budget is not tied directly to enrolments, it is heavily influenced by it. The department has had stable student enrolments in the last few years. A few years ago, we increased the number of TAs in the department from 9 to 16 to reflect an increase in our enrollments. The budget for operating costs has not increased for the last few years and due to poor statewide economic conditions and has in fact decreased slightly in the last fiscal year. The department has three accounts where it generates additional revenue: (a) The Indirect Cost account (the fraction of the overhead funds available from research grants that is returned to the department), (b) Gift account (gift funds given by benefactors of the department) and (c) Educational Program account (tuition revenue from other educational programs, such as China 3+2, India-MS programs). These additional funds have therefore also been used to ensure that the program objectives are met.

Some examples of how these funds have been used in meeting program objectives are given below:

- (i) Updating Departmental Computing Services: The funds have been used to purchase site licenses for software needed by our students and faculty for their classroom activities (ASPEN, CHEMCAD. Etc.).
- (ii) New Unit Operations Equipment: Digital densitometer, laser tachometer, digital balances, portable air compressor, oxygen sensors, digital thermocouple readers, digital hygrometer, and titration equipment.
- (iii) Professional Development Supplement: The faculty receive professional development funds from the university (~\$900 for non-tenure-track and \$1200 for tenure-track). The above-mentioned funds have been used to supplement the funds from university for teaching workshops, and conference visits.
- (iv) Startup Packages: The department hired three assistant professors and one associate professor in 2016. The departmental share of the startup funds is covered by these funds.

Financial Framework

The current sources of financial support for the Department are the State of Illinois with additional revenue mentioned above. The state funding covers faculty and staff salary, as well as some additional non-permanent positions such as Teaching Assistants and part-time student workers.

This portion of the budget is fixed each year by the University -- changes made are generally consistent with the support provided to the University by the State of Illinois. In addition, the Department receives a portion of the Indirect Cost Recovery (ICR) from research grants housed in the department, or those in which the department faculty participate. Currently the University returns 47.5% of the overhead cost to the College of Engineering, and this is then divided equally between the College and the Department. The Department actively works with the Development Office of the College and University to raise gifts from alumni and local and national businesses with an interest in our Department and University. These gifts when they are discretionary are also used to fund special projects in the department including sometimes emergency repairs to equipment in the UG Laboratories that has not been budgeted. Our development activities have resulted in one major gift and several smaller gifts. The major gift is being used for an endowed professorship position in the department.

While there has been no increase in state budget, the alternate revenue streams have seen an increase driven by a steady increase in our student enrollments and our international programs. This together with careful budgetary management in general by the Dean's office has allowed us to provide a high-quality education for the undergraduate students.

In 2019, moved into our new \$45M building (Engineering Innovation Building) that houses the entire department. The building has a state-of-the-art classroom, unlike any other with a large touchscreen. Further, the room is annexed to our unit-operations laboratory via a large window through which students in the lecture room can see the various equipment; a model that was inspired by the medical observatory classrooms. The building also has student learning space spread throughout the building, including a computational room, library, lounge, and atrium seating space. College alumni donor funding for the scholarships and computer software is another example of support for our program. Strong support enables effectiveness in teaching and learning. The College conducts surveys of teaching and advising every semester, and the Chemical Engineering department usually ranks the highest among these surveys. Another measure of the quality of instruction we have provided for our students is the Annual College of Engineering EXPO. In the past years, our students have won a higher share of awards. The department has also initiated a Chemical Engineering Design Showcase with chemical engineering specific judges that complements the College of Engineering EXPO.

While there has been no increase in state budget, alternate revenue streams have seen an increase driven by a steady increase in our international programs. This has allowed us to persist in providing a quality education for the undergraduate students.

C. Staffing

The support staff in the department consists of a Business/Office Manager, an Undergraduate/Graduate Student and Program Coordinator, an Assistant to the Head, Lab/Computer Manager and an Assistant Director of Communications. In addition, the department generally hires one or two undergraduate students for operational support. By way of professional development, the support staff attend university workshops and have won a number of service and recognition awards.

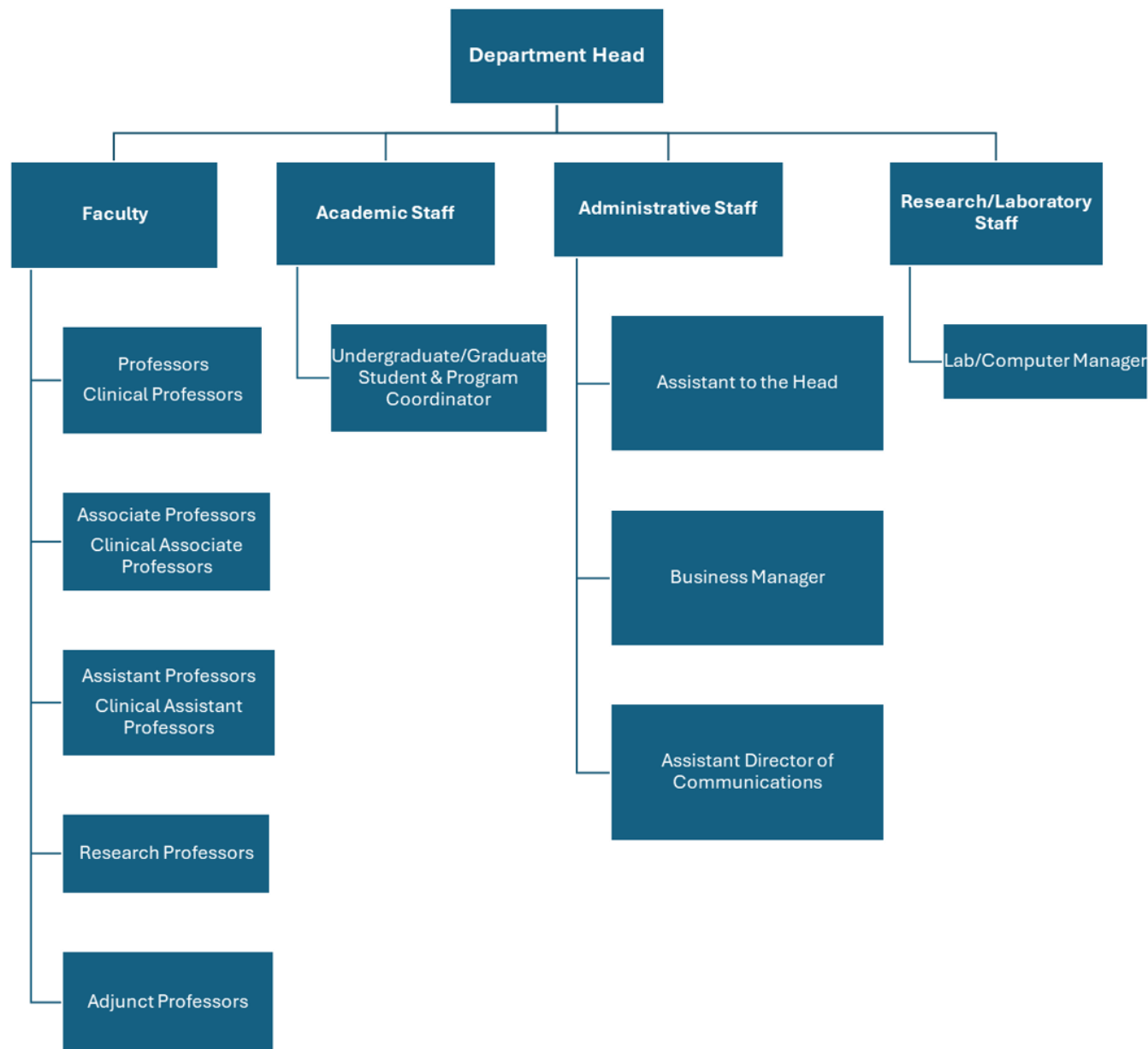


Figure 8.1. Chemical Engineering Faculty and Staff Organizational Chart.

D. Faculty Hiring and Retention

The Department has been successful with support from the College and the University to recruit several excellent additions to our faculty. The following table shows the faculty hiring conducted over the last six years in our department:

Year	Name	Position
2020	Anh Ngo	Associate Professor
2022	Ezinne Achinivu	Assistant Professor
2022	Naveen Dandu	Research Assistant Professor
2023	Korosh Torabi	Clinical Assistant Professor
2024	Roshan Nemade	Research Assistant Professor

These new faculty members have broadened the teaching and research programs of the department. Since 2020, we have had one move of Professor Gang Cheng. We currently have 5 NSF CAREER awardees and 1 AAAS Science & Technology Policy Fellow.

E. Support of Faculty Professional Development

8.E.1. Promotion and Tenure Seminars.

The university conducts several professional development seminars, hosted by the Office of Faculty Affairs, and coordinated by the vice-provost for faculty affairs, Nancy Freitag. Some examples of such seminars over the last year include:

1. New Faculty Orientation
2. Expanding our Global Connections: International Research, Collaboration, Study Abroad and Campus Experiences
3. Innovation at UIC
4. Navigating the Research Landscape
5. Mentoring in Research Education and Career Development
6. Faculty Search Committee Training
7. Diversity Education
8. Mid-Career Writing
9. Research Training

8.E.2. Mentoring Program.

There are university-wide and department-wide mentoring programs. Mentoring is important to retain faculty, improve morale and enhance performance generally. The university wide mentoring program supplements the efforts of individual departments, providing assistance and guidance to their probationary faculty members. The Office of Faculty Affairs coordinates the process by which a junior faculty may select a mentor from a list of senior faculty who have volunteered their services. In general, junior faculty can ask for and receive guidance on how to get things done within the university and how to chart a productive research path.

The Chemical Engineering department appoints a faculty mentor for all new Assistant Professors as part of the support program. The mentor/mentee are expected to meet at least once in two weeks in a formal manner; which most of the mentoring occurs in an informal manner; the mentor does not have any extraordinary influence on personnel matters concerning the faculty member. The mentor is expected to help the faculty member in writing proposals, papers and any other help he may wish to seek. In addition, the department has a CAREER committee that thoroughly guides assistant professors submitting NSF-CAREER proposals from May to July. For mid-career faculty a mentor is generally not appointed, but senior faculty members have open discussions with faculty to suggest research topics and upcoming research funding opportunities, to enhance their professional development.

8.E.3. Workshops on increasing teaching effectiveness.

Several workshops are offered each year for professional development in teaching through the Center for the Advancement of Teaching Excellence (CATE). The CATE composed of faculty members who have been recognized for their demonstrated teaching excellence, provides a mechanism by which faculty, staff, and administrators can work collaboratively toward the improvement of instruction and the advancement of learning in UIC's culturally pluralistic community. Through the development of programs, policies, resources and rewards, CATE strives to encourage every member of the academic community to develop interests, abilities, and achievements as both teachers and learners. CATE offers various programs and resources to support excellent teaching and learning throughout the UIC community including:

- Teaching Recognition Program (TRP) and Award for Excellence in Teaching (AET) are programs that recognize and reward documented excellence of UIC faculty in instruction.
- Departmental Teaching Excellence Award (DTEA) recognizes the commitment to teaching and documented instructional excellence of departments and other academic units.
- CATE provides publications, reports, links, and other instructional resources for faculty, staff, and administrators. The Council seminars are coordinated by the Vice-Provost for Faculty Affairs. **Registration is free for these seminars and they are widely publicized. In most cases faculty members decide on their own which seminars would enhance their teaching abilities, while often a colleague will suggest one of these to a junior colleague.**

8.E.4. Sabbatical leaves for professional development.

Faculty may apply and expect to receive sabbatical leaves for the purpose of study or research, as recommended by the head of the department and with the concurrence of the Dean of the College of Engineering.

F. Respectful Environment

The Department of Chemical Engineering at the University of Illinois Chicago (UIC) is committed to fostering a respectful, inclusive, and equitable environment for all students, faculty, staff, and administrators. A culture of professionalism and mutual respect is essential for student

learning, well-being, and achievement of the program's student outcomes. UIC maintains comprehensive university-wide policies that support a safe and respectful community, including the UIC Student Code of Conduct, the Non-Discrimination Policy, Title IX compliance procedures, and clear mechanisms for reporting and addressing grievances or misconduct. These policies are actively reinforced at the college and departmental levels to ensure accountability, fairness, and transparency. Within the Department of Chemical Engineering, several mechanisms, activities, and services specifically foster a respectful and supportive environment:

- **Academic Advising and Mentorship:**
Faculty mentors provide individualized academic, research, and career guidance in a supportive and student-centered manner. Advising practices emphasize respect, fairness, confidentiality, and constructive communication.
- **Inclusive Student Engagement:**
Department-sponsored activities—including AIChE chapter events, Chem-E-Car, undergraduate research programs, alumni engagement events, and peer-mentoring initiatives—promote teamwork, leadership, and professional conduct in diverse student groups.
- **Faculty and Staff Professional Development:**
Faculty and staff participate in UIC's Office for Access and Equity workshops, College of Engineering training, and departmental meetings focused on respectful communication, inclusive teaching, conflict resolution, and culturally responsive practices.
- **Support Services and Resource Accessibility:**
The department collaborates closely with campus units such as the Counseling Center, the Disability Resource Center, and the Office of Diversity, Equity, and Engagement to ensure that students have accessible pathways to academic, mental-health, and wellness support.
- **Inclusive Classroom and Laboratory Practices:**
Faculty are encouraged to integrate inclusive pedagogy, equitable participation strategies, and transparent expectations while maintaining a safe, respectful, and professional atmosphere in both classroom and laboratory settings.
- **Community Building and Culture of Respect:**
Departmental town halls, advising forums, graduate-student check-ins, and regular communication from departmental leadership reinforce community norms, encourage open dialogue, and provide channels for feedback, ensuring all voices are heard.

Through these established mechanisms—policies, activities, and support services—the Department of Chemical Engineering actively sustains an environment built on mutual respect and professionalism. This environment enhances student engagement, supports academic and research excellence, and ensures the continuous attainment of the program's student outcomes.

PROGRAM CRITERIA

The Chemical Engineering Program Criteria require that the curriculum must:

- a.) Provide a thorough grounding in the basic sciences including chemistry, physics, and/or biology, with some content at an advanced level, appropriate to the objectives of the program.
- b.) Include the engineering application of these basic sciences to the design, analysis, and control of chemical, physical, and/or biological processes, including economic, environmental, global, and societal issues as well as the hazards associated with chemical processes.

Coverage of these areas in the Chemical engineering curriculum was discussed above under Criterion 5. The course work that addresses the program criteria are briefly recapped in this section. Examples of student work will be available during the on-site visit that will demonstrate a working knowledge of these topics. Chemical engineering students begin their chemistry courses during their freshman year with general chemistry (two lecture/laboratory courses: Chem 122/123 and 124/125: 10 credit hours). Organic chemistry (two lecture courses: Chem. 232 and 234: 6 credit hours, and one laboratory course: Chem 233: 2 credit hours) is typically taken in the sophomore year, while physical chemistry (two lecture courses: Chem 342 and 346(selective): 6 credit hours) and analytical chemistry (one lecture course: Chem 222: 4 credit hours) fall in the junior year.

Chemical engineering courses begin with an introduction to thermodynamics (ChE 201: 3 credit hours) in the fall semester of the sophomore year. This is followed by a computational methods course (ChE 205: 3 credit hours), a material and energy balance course (ChE 210: 4 credit hours), and a materials course (CHE 230: 3 credit hours). CHE 205 develops mathematical background specifically addressing chemical engineering applications and general problem-solving skills using Excel, VBA, MATLAB and Aspen Plus process simulation. CHE 210 introduces the critical concept of balances as well as basic problem-solving skills to be applied and further developed throughout the curriculum. CHE 230 introduces several topics that are important to traditional and emerging fields in chemical engineering; molecular engineering, macromolecular systems, and nanomaterials. In the junior year, students take chemical engineering thermodynamics (ChE 301: 3 credit hours) and three transport phenomena courses (ChE 311, 312 and 313: 9 credit hours) which cover fluid mechanics, heat and mass transfer and separation processes. CHE 330 (selective) covers an introduction to polymer science. In the spring semester of junior year, the students take chemical reaction engineering (ChE 321: 3 credit hours) which includes design work. Senior level courses complete the major chemical engineering subject areas and complement the design experience. These courses are the two chemical engineering laboratory courses (ChE 381: 2 credit hours and ChE 382: 2 credit hours); chemical process control (ChE 341: 3 credit hours), and the two senior design courses. Preparation for engineering practice culminates in the design course sequence (ChE 396 and 397: 7 credit hours) taken in the senior year. In these courses the students demonstrate their ability to independently apply chemical engineering principles, cost estimation and economic evaluation methodologies, flow sheet synthesis and process simulation, optimization techniques, and report writing and oral presentation skills, to contemporary chemical engineering design problems. Industrial mentors provide guidance to the design teams and instruct the students in engineering

practice and industrial standards. Content in the above courses is further augmented by a host of technical electives such as Process Simulation with Aspen Plus, Biochemical Engineering, Renewable Energy Technologies, Nanotechnology for Pharmaceutical Applications, Biopharmaceutical Process Development and Manufacturing, Molecular and Macromolecular Engineering, Fundamentals of Electrochemistry. Syllabi for all required and elective courses are given in Appendix A.

Coverage of ethics was discussed in Section 4.B.8. Assignments on ethics are covered in several courses in the curriculum; CHE 396, 397 as well as CHE 201, 341 and 433. Safety and hazards associated with chemical processes was discussed in Section 4.B.8. Mandatory student completion of AIChE safety tutorials and relevant class discussion are integrated into the Chemical Engineering curriculum in CHE 381, 382, 396 and 397. Student knowledge and proficiency in laboratory and chemical process safety are assessed directly the curriculum in CHE 381, 382, 396 and 397. Further safety-related information is given in the next section.

ACCREDITATION POLICIES AND PROCEDURES MANUAL

Institutional Level

Introduction and Governance Framework

The University of Illinois Chicago (UIC) operates within a rigorous, multi-tiered institutional governance framework that ensures all laboratory environments, research facilities, and instructional spaces are managed safely, ethically, and in full compliance with state and federal regulations. As a Carnegie R1 research-intensive institution with over \$509 million in sponsored research awards in FY23, UIC maintains an extensive technical infrastructure comprising millions of gross square feet of specialized spaces, including active engineering and scientific laboratories. Ultimate oversight of corporate safety, fiscal accountability, and legal compliance resides with the University of Illinois Board of Trustees and its dedicated Audit, Budget, Finance, and Facilities Committee.

Day-to-day enforcement, regulatory oversight, and infrastructure maintenance are operationalized through specialized campus units under the Office of the Vice Chancellor for Research (OVCR) and the Office of the Vice Chancellor for Administrative Services. Specifically, the Office of Research Integrity and the Office of Sponsored Programs (OSP) maintain regulatory compliance, verify project safety parameters, and manage protocols involving biohazards, recombinant DNA, and technical equipment. Academic requirements, laboratory-specific curricula, and course prerequisites are governed directly by the university faculty through the UIC Senate Committee on Educational Policy (SCEP) and college-level educational committees, ensuring that safety parameters are seamlessly woven into student learning outcomes.

Additionally, programs within the College of Engineering adhere strictly to the specialized criteria outlined by ABET. ABET criteria mandate that program learning outcomes are systematically evaluated and used as continuous input to maintain premier safety, educational quality, and hazard management protocols.

Safety Policies and Procedures

UIC's comprehensive safety matrix is established via system-wide and campus-specific policies designed to eliminate institutional hazards and ensure operational integrity. Key guiding mechanisms include:

- **System Policy on Integrity in Research and Publication:** Provides the absolute foundational standard for professional and regulatory compliance across all laboratory activities.
- **Biohazard and Recombinant DNA Compliance Protocols:** Managed directly under the OVCR's compliance wing, these policies strictly govern the acquisition, use, storage, and disposal of biological agents, hazardous chemicals, and embryonic materials.
- **Contact/Credit Hour Policy Integration:** UIC's standardized Contact/Credit Hour Policy explicitly details instructional expectations for laboratory courses, ensuring appropriate faculty-to-student ratios, adequate supervised instruction hours, and mandated safety briefings prior to practical work.
- **Syllabus Mapping Requirements:** In accordance with policies from Faculty Affairs and the Center for the Advancement of Teaching Excellence (CATE), every laboratory course is structurally required to issue an official syllabus outlining student safety obligations, emergency procedures, and operational boundaries.

Efforts to Keep Laboratories Clean and Free of Hazards

UIC maintains a continuous, systematic process for upgrading, monitoring, and sanitizing its academic and scientific infrastructure. The university tracks laboratory and facility conditions via the Office of Academic Program Review and Assessment (APRA). Through its 8-year program review cycles and continuous cross-campus feedback loops, APRA synthesizes facility health summaries and routes them to campus stakeholders to prompt required maintenance actions.

Recent proactive interventions based on institutional assessments include:

- **Classroom and Lab Infrastructure Alignment:** Systematic modifications to facilities are conducted using targeted master plans, ensuring campus building renewals and classroom upgrades address critical physical space limitations.
- **Modified Cleaning and Sanitation Procedures:** Implementing rigorous, specialized space allocations in academic and active-learning technical spaces helps prevent cross-contamination or mechanical hazards.
- **Targeted Strategic Upgrades:** Multi-million-dollar capital investments in state-of-the-art facilities ensure built-in technical safeguards, proper fume hood distributions, and localized hazard isolation fields. Between 2017 and 2023, UIC added 20,274 square feet in teaching labs and 18,346 square feet in research labs. This includes the \$43 million Engineering Innovation Building (housing Chicago's only high-bay structural research laboratory) and a new \$118 million multi-use Computer Science Center (Computer Design, Research and Learning Center).

Student Safety Training

Safety training at UIC is a mandatory prerequisite that scales with a student's degree level and degree-adjacent technical exposures:

- **Undergraduate Foundation Training:** Students participating in undergraduate research (such as those within the Guaranteed Paid Internship Program across 42 engineering and scientific labs) receive structured mentorship and localized lab-safety onboarding before activating their positions.
- **Responsible Conduct of Research (RCR) Curations:** The Office of Research Integrity curates continuous, mandatory RCR and lab-safety frameworks, offering specialized instructional modules and workshops tailored to technical research infrastructures.
- **Advanced Graduate and Specialized Training:** Specialized safety compliance training modules are systematically verified through digital workflows like the UIC Research portal before any physical experimentation or disclosure authorization is cleared. Completion of mandatory developmental experiences, such as specialized academic integrity or laboratory compliance webinars, is regularly utilized to reinforce safety expectations.

Personal Protective Equipment (PPE) Used by Students

The deployment and use of Personal Protective Equipment (PPE) are rigorously managed through course-level requirements and external accreditation expectations. PPE requirements are formalized via two primary methods:

- **Syllabus-Based Mandates:** In compliance with instructional policies, laboratory course coordinators must explicitly list mandatory PPE configurations directly within the course deliverables.
- **Laboratory-Specific Access Constraints:** In alignment with specialized accreditation metrics (such as those governing ABET-accredited engineering disciplines), students are restricted from accessing chemical, structural, or biological bays unless full, compliant PPE is actively utilized according to course learning parameters.

Enforcement of Safety Policies and Procedures

Safety enforcement at UIC is maintained through absolute institutional accountability, combining early intervention alerts with severe administrative consequences for non-compliance:

- **Faculty and Supervisory Accountability:** Deans and Department Executives are operationally authorized under university statutes to enforce safety mandates. If standard safety procedures or equipment protocols are ignored, the Office of the Vice Chancellor for Research has explicit authority to enact compliance restrictions, manage risk boundaries, or audit project authorization parameters.
- **Student Disciplinary Frameworks:** Student failure to strictly comply with laboratory protocols, instructor safety parameters, or universal facility guidelines is handled directly under institutional student misconduct frameworks. Violations are treated with appropriate severity, utilizing an established administrative pipeline that includes formal reviews, initial fact assignments, and mandatory developmental safety interventions.

Routine Safety Inspections

UIC maintains a multi-layered verification loop to ensure all laboratory spaces consistently satisfy local, state, and federal baseline regulations:

- **APRA and SCEP Program Evaluation:** The Office of Academic Program Review and Assessment (APRA) coordinates regular, formal space and operational health checks as part of its systematic evaluation sequence. Academic and research facilities are comprehensively examined using rigorous rubrics that map space condition, structural adequacy, and personnel behavior during the 8-year review cycles.
- **Specialized Accreditation Peer Reviews:** For technical engineering programs, specialized ABET Peer Reviews occur sequentially, accepting self-studies and external peer review reports to ensure all research and instructional laboratories actively support student learning while maintaining documented, continuous improvement loops.
- **External Regulatory Audits:** As a state agency and federal contractor, UIC's infrastructure is subject to independent safety and compliance inspections conducted by state agencies and certified independent third parties, ensuring that internal controls protect assets and maintain regulatory standards.

Chemical Engineering Department Level Safety Practice

The chemical engineering department has a Safety Committee that consists of 2-3 faculty members and the laboratory/computer manager, Jan Sagun. The committee keeps tracks of all safety records, information and training for the research and undergraduate laboratories.

Chemical Engineering Undergraduate Laboratory Courses

CHE 381/CHE 382. Undergraduates enrolled in the unit operations laboratory (CHE 381 and 382) are provided safety information in the CHE 381 and CHE 382 course syllabus, and training and guidance by the instructor and teaching assistants on proper usage and safety precautions before they begin their experimental work. This is initially done in the first lab period of the semester, and thereafter during ‘pre-lab’ laboratory periods. All students enrolled in CHE 381 and CHE 382 read and sign an initial Safety Checklist form during the first week of the lab course. Additional guidance and oversight during the semester on experiment-specific chemical and safety hazards are provided while the lab teams are performing each experiment and afterwards to ensure proper equipment shut-off procedures.

Undergraduate Laboratory Safety Equipment, Practices and Documentation

There are two fume hoods and four dropdown snorkel lines for chemical vapor control in the laboratory. 3 chemical spill kits are located accessible locations. Personal protective equipment used in the lab include; nitrile lab gloves, chemical resistant gloves, sharps-resistant gloves, heat-resistant gloves, safety glasses, safety goggles, laboratory coats, and a 4-wheel gas cylinder cart. Safety documentation in the laboratory includes a binder of MSDS chemical sheets, and the UIC Laboratory Safety Plan. Fume hoods, eyewash stations and the safety shower are inspected yearly by the UIC Environmental Health and Safety Organization (EHSO). Eyewash stations are inspected weekly during the semester by the instructor or teaching assistant. Safety signage inside the lab includes labels indicating the location of acid, caustic, and flammable chemical storage cabinets, signs indicating that safety glasses must be worn at all times, that no food or drinks are allowed in the lab, and hot temperature hazards.

Routine laboratory hygiene practices

Weekly: Students rinse, clean and dry all labware (beakers, flasks, graduated cylinders, buckets, pipettes, test tubes, racks, trays). Students clean up common lab bench areas, especially around scales/balances, and put away any equipment used during lab period. The student lab teams shutdown their experiments, clean up the apparatus and bench area, and leave the experiment ready for the next lab team. Water, air, steam, pumps and individual electric power boxes are turned off. Instructor/TA performs an inspection at the end of each lab period to make sure all experiments are shutdown correctly; chemicals are properly stored and equipment is put away.

Semester: Faculty/staff perform a run-through of the lab experiments at the beginning of the semester before the first lab period to make sure the experiments are working properly. An inventory is performed at the end of the semester to determine if lab supplies/chemicals/gases need to be ordered for the next semester. Equipment is periodically cleaned, recalibrated, refilled or replaced by faculty/staff during the semester. For example, capillary tubes in the viscometer experiment are cleaned out with a solvent to remove Carbopol polymer residue left inside. Manometers are checked to make sure they are filled correctly. The boiler is turned on in the morning before the lab commences and a blowdown procedure is followed at the end of the day. The UIC facilities team also periodically checks the steam generator to make sure it is working correctly. Safety equipment is maintained in good working order; safety spill kits and chemical spill pads are inspected and replaced when needed, the laboratory safety plan is updated annually, MSDS binder is updated when needed, first-aid kits and eye showers are inspected weekly, an ample supply of nitrile lab gloves, heat-resistant gloves, chemical-resistant gloves and sharps gloves are maintained in the lab, and safety glasses are checked for breakage and new glasses ordered if necessary.

ChemE Car/ChemE Cube Laboratory. The EIB 231 laboratory on the second floor is used by the two student teams that participate in the AIChE ChemE Car and ChemE Cube annual competitions. The laboratory has 6 moveable lab stations, one fume hood, and compressed air lines, as well as access to 120V and 220V electricity. The ChemE Car and ChemE Cube teams use this laboratory to develop and test their prototype models. The students complete an Engineering Design Package that includes a Job Safety Assessment Form, describing the major safety and chemical hazards involved with the design, a hazard analysis as well as identification of safety measures and safety controls. In addition, team members complete the AIChE Academy tutorial- ELA199 ChemE Car Safety Course as part of their safety training for the competition.

Undergraduate research projects. Undergraduates performing research in faculty laboratories are required to attend a UIC Environmental Health and Safety Office (EHSO) laboratory safety training course before working in the laboratory. Information on the safety courses available can be found at: <https://ehso.uic.edu/training/introduction-to-laboratory-safety-training/>. The faculty supervising the undergraduate students provide training and guidance by the instructor on safety hazards and precautions in their research labs before the students begin their experimental work.

APPENDICES

Appendix A – Course Syllabi

A.1. Chemical Engineering Courses.

CHE 201

1. Course number and name

CHE 201. Introduction to Thermodynamics

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; 50-minute optional discussion section per week; Engineering

3. Instructor's or course coordinator's name

Betul Bilgin

4. Textbook, title, author, publisher, year

Fundamentals of Engineering Thermodynamics, Moran/Shapiro/Boettner/Bailey, 9th Edition (Zybooks version is used)

5. Specific course information

a. Brief description of the content of the course (catalog description)

Work and energy; conversion of energy; theory of gases and other states of matter; applications to energy conversion devices. Second law of thermodynamics, entropy, and equilibrium, with applications.

b. Prerequisites or co-requisites

Math 181, Calculus II and Phys 141, General Physics I (Mechanics).

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

Students will learn to:

- Apply energy balances for closed systems and control volumes.
- Evaluate thermodynamic efficiencies of compressors, turbines, power plants, refrigeration and heat pump cycles.
- Identify phases at a given state and find thermodynamic properties such as internal energy, specific volumes from steam tables.
- Draw and interpret pressure-enthalpy (P-H), pressure-specific volume, temperature specific volume, temperature-entropy charts.
- Relate thermodynamic properties via partial derivate, Maxwell's relations.
- Interpret phase diagrams of binary systems.
- Apply the entropy balance and identify if the process is reversible or irreversible.

Project

One, semester long project related to thermodynamics concept. Students will be working with groups and are expected to design a product or process with the focus of sustainability. Project includes research phase, modeling phase, prototype phase and testing phase. Design competition will be held among all the groups. All the groups are required to create Adobe Spark webpage for their design.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(3) an ability to communicate effectively with a range of audiences.		X		
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.		X		
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X		

7. **Brief list of topics to be covered**

- I. Introduction
- II. Fundamentals (Chapter 1)
- III. The First Law (Chapter 2)
- IV. Properties of Pure Fluids (Chapter 3)
- V. Control Volume Analysis (Chapter 7)
- VI. The Second Law of Thermodynamics (Chapter 6)
- VII. Entropy (Chapter 4)
- VIII. Thermodynamic Relations (Chapter 11 Section 1 through 5)
- IX. Power Cycles (Chapter 9; select sections)

AIChE Academy eLearning tutorials:

Students are expected to complete two AIChE Academy eLearning tutorials online and submit their certificates of completion to the instructor. They can choose two of the Level I Curriculum as listed below:

- An Introduction to Managing Process Safety Hazards
- Introduction to Process Safety
- Hazard Recognition
- Identifying & Minimizing Process Safety Hazards
- Lab Safety

CHE 205

1. Course number and name

CHE 205. Computational Methods in Chemical Engineering

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Shafiqh Mehraeen, Betul Bilgin

4. Textbook, title, author, publisher, year

Numerical Methods for Chemical Engineers Using Excel, VBA, and MATLAB, by V.J. Law. ISBN: 978-1-4665-7534-9, CRC Press, 2013.

a. Other supplemental materials

VBA programming; tutorials on the internet.

Consult with the following websites:

[http://excelvbatutor.com/vba tutorial.html](http://excelvbatutor.com/vba%20tutorial.html)

[http://www.homeandlearn.org/basic mathematical operators.html](http://www.homeandlearn.org/basic-mathematical-operators.html)

[http://www.automateexcel.com/vba-code-examples.](http://www.automateexcel.com/vba-code-examples)

5. Specific course information

a. Brief description of the content of the course (catalog description)

Computational methods and software relevant to unit operations. Excel spreadsheets (curve fitting, heat conduction), Matlab, Aspen Plus (process simulation), algorithms and object-oriented concepts in chemical engineering.

b. Prerequisites or co-requisites

Credit or concurrent registration in CHE 201 and MATH 210.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Formulate a chemical engineering problem as a mathematical model and select appropriate solution method.
- Write and modify VBA scripts for specific computational purposes and run them.
- Use Excel functionality to tabulate thermophysical property correlations and equations of state.
- Analyze 1d (linear) and 2d (bilinear) interpolation from data tables with VBA macros.
- Calculate vector coordinates for changes of basis vectors with Excel.
- Solve ordinary differential equations (ODE) with Excel.
- Solve optimization design problems with Excel Goal Seek and Solver functionalities and VBA macros.
- Solve partial differential equations (PDE).
- Carry out error analysis of data and regression analysis.

- Find root of a function using bisection, secant, fixed-point iteration, and Newton's method.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.		X		

7. Brief list of topics to be covered

- Ch. 1: Roots of a single nonlinear equation
- Ch. 2: Visual Basic for applications programming
- Ch. 3: Linear algebra and systems of linear equations
- Ch. 4: Numerical differentiation and integration
- Ch. 5: Ordinary differential equations (initial value problems)
- Ch. 6: Ordinary differential equations (boundary value problems)
- Ch. 8: Partial differential equations
- Ch. 7: Regression analysis and parameter estimation
- Ch. 9: Linear programming, nonlinear programming, nonlinear equations, and nonlinear regression using Solver

CHE 210

1. Course number and name

CHE 210. Material and Energy Balances

2. Credits and contact hours

Credit hours: 4; Contact hours: 150 min/wk. lecture, 50 min/wk. discussion; Engineering

3. Instructor's or course coordinator's name

Lewis Wedgewood

4. Textbook, title, author, publisher, year

Introduction to Chemical Processes: Principles, Analysis, Synthesis by Regina M Murphy, McGraw-Hill, 1st edition, New York, 2007

5. Specific course information

a. Brief description of the content of the course (catalog description)

Material and energy balances applied to chemical systems. Introduction to chemical and physical properties. Introduction to the use of computers for chemical process calculations.

b. Prerequisites or co-requisites

Credit or concurrent registration in CHE 201 and CHE 205.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Synthesize a block flow diagram from a process description
- Identify and modify reaction pathways to achieve specific production while minimizing waste and occurrence of hazardous compounds
- Perform a degree of freedom analysis to determine degree of process specification
- Perform component, element, and total mass (or molar) material balances
- Quantify and make appropriate use of system performance specifications for reactors, splitters, separators
- Correctly identify modes of operation for process units (steady-state, transient, batch, continuous) and simplify the general material balance equation appropriately.
- Perform unit conversions as necessary.
- Calculate reaction equilibrium constants and determine fractional conversion and selectivity in equilibrium reactions.
- Use model equations and graphical data to determine composition in vapor-liquid phase equilibria
- Formulate process analysis problems as a system of linear equations
- Use tabulated data or model equations to evaluate changes in state properties
- Perform energy balances to determine basic heating/cooling requirements of a reactive process

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X	X	
(3) an ability to communicate effectively with a range of audiences.	X			
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	X			
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	X			
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	X			

7. Brief list of topics to be covered

- Raw materials, reaction path synthesis
- Generation-consumption analysis
- Material balance equations
- Process flow sheeting
- Degree of freedom analysis
- Reactor materials balance
- Stream composition and system performance (conversion, selectivity, yield)
- Recycle, purge
- Chemical reaction equilibrium and chemical kinetics
- Separation technologies
- Phase equilibria
- Energy balances, Energy flows
- Energy data and model equations
- Energy balances on reactive processes

CHE 230

1. Course number and name

CHE 230. Molecular Systems in Chemical Engineering

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Roshan Nemade, Ezinne Achinivu

4. Textbook, title, author, publisher, year

Materials Science and Engineering: An Introduction, Willam D. Callister Jr and David G. Rethwisch, Wiley

5. Specific course information

a. Brief description of the content of the course (catalog description)

Introduction to fundamental concepts in molecular engineering and materials chemistry; properties of molecular systems and applications of macromolecules, bio-macromolecules and nanomaterials in energy, medicine, environment and technology.

b. Prerequisites or co-requisites

CHEM 232.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

The main objective of the course is to familiarize the students with the fundamental concepts of Materials Science and Engineering which will be used as background knowledge for the understanding of specialized problems in the field of Chemical Engineering. At the end, students will be able to: Understand the fundamental and essential concepts used in polymeric materials for Chemical Engineering applications.

1. Understand the fundamental and essential concepts used in materials chemistry and molecular systems
2. Classify the molecular systems, nanomaterials, and materials systems.
3. Know the basic methods to synthesize and characterize materials systems
4. Understand the basic properties that characterize the behavior of molecular systems
5. Select appropriate type of material for specific application
6. Offer different approaches to modify structure/microstructure to get desired properties

One semester long project to reflect the three components of molecular engineering related to Polymer Design, Processing and Application. Students will be working with groups and are expected to prepare a case study. Each group will be assigned one molecular system application to perform a critical review to explore products material properties and performance requirements from a molecular system lens.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(3) an ability to communicate effectively with a range of audiences.		X		
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.		X		
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X		

7. Brief list of topics to be covered

- Introduction to Molecular Systems
- Molecular Structure & Defects
- Diffusion & Material Properties from molecular structure
- Molecular Interactions: Influence on Macroscopic Properties
- Engineering Functional Materials
- Introduction to Macromolecules
- Nanoscale Systems
- Sustainability, Circular Materials & commercializing products with molecular engineering

CHE 301

1. Course number and name

CHE 301. Chemical Engineering Thermodynamics

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Michael Caracotsios

4. Textbook, title, author, publisher, year

Fundamentals of Chemical Engineering Thermodynamics, Themis Matsoukas, Prentice Hall, 2013.

a. other supplemental materials

- Chemical Engineering Thermodynamics Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020
- J. M. Smith, H. C. Van Ness and M. M. Abbott, Introduction to Chemical Engineering Thermodynamics, 7th Edition, McGraw-Hill, 2005.
- S. I. Sandler, Chemical and Engineering Thermodynamics, 3rd Edition, John Wiley & Sons, Inc., 1999.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Review of first and second laws of thermodynamics; Equations of state; Multi-component systems and phase equilibria; Chemical reaction thermodynamics and reaction equilibria.

b. Prerequisites or co-requisites

CHE 201 and CHE 205.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

Understand the basic concepts of Internal Energy, Gibbs Energy, Entropy and Enthalpy.

Apply control volume analysis to a variety of situations related the phase and chemical equilibrium and solve problems utilizing computer software.

Understand the theory of thermodynamics of ideal and non-ideal solutions

Understand the applicability and use of various equations of state in describing the thermodynamic state of pure fluids and mixtures thereof.

Determine equilibrium conditions between phases using equations of state and liquid activity coefficients.

Determine equilibrium conditions between reactants and products using equilibrium constants and Gibbs energy minimization concepts.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
ability to communicate effectively with a range of audiences.		X		

7. **Brief list of topics to be covered**

REVIEW AND BACKGROUND: Introduction to Thermodynamics in Chemical Engineering, Elements of Differential and Integral Calculus, Numerical Methods for Algebraic and Differential Equations, Mass and Energy Balances in the Context of Thermodynamics, Tutorials on Programming with VBA and Athena Visual Studio

THERMODYNAMIC PROPERTIES OF SINGLE-COMPONENT FLUIDS: PVT Behavior of Single-component Fluids, Gibbs Phase Rule for Single-component Fluids, The Ideal Gas Behavior, Cubic and Virial Equations of State, Fugacity and Fugacity Coefficients from Equations of State, Thermodynamic Properties of Single-Phase and Two-Phase Systems

THERMODYNAMIC PROPERTIES OF MULTI-COMPONENT MIXTURES: Partial Molar Properties Excess and Residual Thermal Properties, the Gibbs Phase Rule for Multi-component Mixtures, PVT Behavior of Multi-component Mixtures, Cubic Equations of State and Mixing Rules, Fugacity and Fugacity Coefficients from Equations of State, Liquid Activity Coefficients for Non-ideal Solutions, Thermodynamic Properties of Single-Phase and Two-Phase Systems

MULTICOMPONENT PHASE EQUILIBRIUM: Criteria for Phase Equilibrium, Vapor-Liquid Equilibrium: The Fundamental Problem, Flash Calculations: PT, PH, PV, TH, TV and HV Calculations, VLE Data Analysis: Activity Coefficients from Experimental Data, Liquid-Liquid Equilibrium: The Fundamental Problem

CHEMICAL REACTION EQUILIBRIUM: Criteria for Equilibrium in Chemical Reactions, Gibbs Free Energy and Equilibrium Constants, Evaluation of Equilibrium Constants from Thermodynamic Data, Calculation of Equilibrium Conversion for Single and Multi-reaction Systems

CHE 311

1. Course number and name

CHE 311. Transport Phenomena I

2. Credits, contact hours and categorization

Credit hours: 3; Contact Hours: 150 min/week lecture; Engineering

3. Instructor's or course coordinator's name

Korosh Torabi

4. Text book, title, author, publisher, year

R.B.Bird, W.E. Stewart & E. N. Lightfoot, Transport Phenomena, 2nd Ed., Wiley (2006)

5. Specific course information

a. Brief description of the content of the course (catalog description)

Momentum transport phenomena in chemical engineering. Fluid statics. Fluid mechanics; laminar and turbulent flow; boundary layers; flow over immersed bodies.

b. Prerequisites or co-requisites

Credit or concurrent registration in CHE 205, CHE 210, and MATH 220.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Performance indicators of student learning.

- Use mathematical language to perform vector and tensor operations relevant to transport phenomena.
- Comprehend the fundamental physical principles of momentum transport and apply the principle of conservation of momentum to solve basic fluid dynamics problems.
- Set up the differential momentum balance and specify its boundary conditions for various flow problems.
- Solve ordinary differential equations and apply boundary conditions to determine steady-state one-dimensional flow velocity and stress profiles.
- Solve partial differential equations using the methods of separation of variables and combinations of variables to determine the velocity profiles of flow fields with two independent variables.
- Comprehend the physical significance and mathematical formulation of empirical expressions for turbulent momentum transport.
- Understand the definition of friction factors to solve various flow problems.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

Week	Topics	Book Chapter
1	Conservation laws, flux, and balance equations Math Review	Chapter 0, Appendix A and Math Handout
2	Continuum fluid mechanics and Stress tensor Math review	Appendix A and Math Handout
3	Newton's law of viscosity and the combined momentum flux Math review	Chapter 1
4-5	Introduction to shell momentum balance Flow of a falling film	Chapter 2
6	Flow through a circular tube and modified pressure Creeping flow around a sphere	Chapter 2
7	The equation of continuity and the equations of motion Drifting time derivative and Reynolds transport theorem	Chapter 3
7-8	Solving flow problems using the equations of change	Chapter 3
9-10	Unsteady flow and the method of combination of variables Unsteady flow and method of separation of variables	Chapter 4
11	(March 23-27) Spring break. No classes.	
12	Introduction to turbulent flow and time-smoothed equations of change	Chapter 5
13-14	Friction factors for flow in tubes	Chapter 6
14	Friction factors for flow around submerged objects Pressure drop in packed beds	Chapter 6
15	Macroscopic momentum and mechanical energy balance	Chapter 7
16	Review and problem-solving	

CHE 312

1. Course number and name

CHE 312. Transport phenomena II

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Korosh Torabi, Christos Takoudis

4. Textbook, title, author, publisher, year

Required (KT) –Transport Phenomena, Revised 2nd Edition by R. B. Bird, W. E. Stewart and E. N. Lightfoot. John Wiley & Sons, Inc (2006).

Required (CT) -- Theodore L. Bergman, Adrienne S. Lavine, *Fundamentals of HEAT and MASS Transfer*, Wiley, Eighth Edition, 2019

5. Specific course information

a. Brief description of the content of the course (catalog description)

Heat and mass transport phenomena. Heat conduction, convection and radiation. Heat exchanger design. Diffusion. Mass transfer coefficients.

b. Prerequisites or co-requisites

CHE 311

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

The broad objectives of this course are:

(a) To develop clear understanding of the basic mechanisms of heat and mass transfer in physical processes encountered in daily living experiences and in chemical engineering practice; and

(b) To develop the skills to describe heat and mass processes in terms of mathematical models, make appropriate assumptions to simplify the models, and perform calculations to obtain qualitative and/or quantitative information which can lead to better design or better understanding of the processes.

- a. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(3) an ability to communicate effectively with a range of audiences.	X			
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	X			
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Introduction to Conduction
- One-Dimensional Steady-State Conduction
- Two-Dimensional Steady-State Conduction
- Transient Conduction
- Introduction to Convection
- Heat Transfer in External Flows
- Heat Transfer in Internal Flows
- Free Convection
- Heat Exchangers
- Radiation Heat Transfer
- Radiation Exchange Between Surfaces
- Diffusion Mass Transfer
- Diffusion – Homogeneous Chemical Reaction Systems
- Diffusion – Heterogeneous Chemical Reaction Systems
- Transient Diffusion and Applications
- Diffusion and Chemical Reaction Inside a Porous Catalyst
- Mass Transfer Design Calculations
- Multimode Heat and Mass Transfer

CHE 313

1. Course number and name

CHE 313. Transport phenomena III

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Sangil Kim

4. Textbook, title, author, publisher, year

Wankat, P. C., Separation Processes Engineering: Includes Mass Transfer Analysis, 5th edition, Prentice Hall, 2022. -OR-

Christie John Geankoplis, Allen Hersel, Daniel H. Lepek, Transport processes and Separation Process Principles, 5th edition, Prentice Hall, 2018

a. Other supplemental materials

Seader, J.D et al., Separation Processes Principles, 3rd edition, John Wiley & Sons, New Jersey, 2011.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Mass transfer and phase equilibria. Multistage separations; applications in distillation; extraction; absorption and drying; membrane separation.

b. Prerequisites or co-requisites

CHE 301

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Be familiar with separation principles, design, separation mechanisms, processes, and applications
- Understand the underlying thermodynamic and mass transfer principles
- Understand the operation of simple and multistage batch distillation systems, and McCabe-Thiele method
- Discuss the differences between batch and continuous operation
- Explore use of various types of separation techniques
- Explore use of alternative separation techniques to the existing ones

a. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.		X		
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			X	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Introduction to separation processes
- Flash Distillation
- Binary distillation
- Multicomponent distillation
- Batch distillation
- Stage and packed column design
- Absorption and stripping
- Liquid-liquid extraction
- Membrane separations

CHE 321

1. Course number and name

CHE 321. Chemical Reaction Engineering

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Korosh Torabi, Meenesh Singh

4. Textbook, title, author, publisher, year

"Elements of Chemical Reaction Engineering",
Fifth, Fourth or Third Edition, by Fogler, H. Scott

5. Specific course information

a. Brief description of the content of the course (catalog description)

Kinetics of homogeneous single reactions. Ideal reactors: batch, stirred tank and plug flow systems. Conversion and yield in multiple reactions. Design and optimization of reactors: non-isothermal, catalytic, and non-ideal reactors.

b. Prerequisites or co-requisites

CHE 210, CHE 301, MATH 220.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

1. Define the rate of chemical reaction. Apply the mole balance equations to a batch reactor, continuous stirred tank reactor, plug flow reactor, and packed bed reactor.
2. Define conversion and space time. Write the mole balances in terms of conversion. Determine reactor sizes (volume, catalyst weight).
3. Understand rate laws and the Arrhenius equation. Describe homogeneous, heterogeneous, elementary, nonelementary and reversible reactions.
4. Apply reaction stoichiometry to molar and volumetric flow rates of a species and conversion.
5. Describe the CRE algorithm to solve chemical reaction engineering problems. Apply this algorithm to design isothermal reactors.
6. Determine reaction order and rates from experimental data of batch or flow reactors.
7. Understand integral and differential methods to identify rate parameters.
8. Describe yield and selectivity. Apply CRE algorithm to design the reactor with multiple reactions for maximal selectivity.
9. Modify CRE algorithm to design non-isothermal reactors (adiabatic and non-adiabatic) and optimize the reactor staging.
10. Analyze multiple steady states and optimal operation of reactors.
11. Define a catalyst, derive a catalytic reaction mechanism and rate-limiting step
12. Obtain rate expressions from quasi-equilibrium hypothesis.

13. Understand catalyst deactivation mechanism Analyze heterogeneous catalytic reactors
14. Define residence time distribution (RTD), RTD functions
15. Evaluate non-ideal reactor models.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.		X		
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.		X		
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

Reaction fundamentals

Reactor Types, General Mole Balances, Conversion, Reactor Sizing, Stoichiometric Tables

Isothermal Reactor Design

SOFTWARE - MATLAB for Reactor Design

Pressure Drop in Reactors

Unsteady State Operation, Rate Analysis

Multiple Reactions

General Energy Balance & CSTR, PFR Energy Balance, Batch Energy Balance

Equilibrium Conversion

Non-Isothermal Reactor Example

Multiple Steady States in a CSTR/Multiple Steady States/Multiple Reactions

Summary of Non-Isothermal Reactors

Catalytic Reactions, Langmuir-Hinshelwood Rate Laws

Langmuir-Hinshelwood Rate Laws, Heterogeneous Reactors, Mass Transfer

External and Internal Mass Transfer Limitations

Effectiveness Factor and Falsified Kinetics

Residence Time Distribution

Zero-Parameter Flow Models, Non-Ideal Flow and combination of ideal reactors

CHE 330

1. Course number and name

CHE 330. Polymer Science

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Roshan Nemade, Ezinne Achinivu

4. Textbook, title, author, publisher, year

Polymer Chemistry, Second Edition / Edition 2 by Paul C. Hiemenz, Timothy P. Lodge, Tim Lodge

5. Specific course information

a. Brief description of the content of the course (catalog description)

Overview of polymer science and engineering. Introduction to polymers as an engineering material; basic concepts of properties; synthesis and structure of polymeric materials, polymer production, property prediction and performance in products.

b. Prerequisites or co-requisites

CHE 230 or CME 260 and CHEM 232.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

The main objective of the course is to familiarize the students with the fundamental concepts of Polymer Science from the lens of Material Science and Engineering which will be used as background knowledge for the understanding of specialized problems in the field of Chemical Engineering. Students will be able to:

1. Understand the fundamental and essential concepts used in polymeric materials for Chemical Engineering applications.
2. Know the detailed methods to synthesize and characterize the polymeric materials.
3. Understand the properties that characterize the behavior of polymeric materials.
4. Select appropriate types of polymeric material for specific application.
5. Offer different approaches to modify structure/microstructure of polymeric materials to get desired properties

One semester long project to reflect the three components of molecular engineering related to Polymer Design, Processing and Application. Students will be working with groups and are expected to prepare a presentation. Each group will be assigned one polymer to perform a critical review on the synthesis mechanisms followed by selecting a consumer product to explore products material and performance requirements.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(3) an ability to communicate effectively with a range of audiences.		X		
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.		X		
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X		

7. Brief list of topics to be covered

- Introduction to Polymer Science
- Polymer Classification
- Methods and Intro to Mechanisms of Polymerization
- Characterization of polymers
- Thermodynamics of polymers
- Overview of Advanced characterization tools
- Polymer processing
- Polymer application

CHE 341

1. Course number and name

CHE 341. Chemical Process Control

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Michael Caracotsios

4. Textbook, title, author, publisher, year

Process Dynamics and Control, D.E. Seborg, T.F. Edgar, D.A. Mellichamp, and F.J. Doyle, Wiley, 2011.

a. other supplemental materials

- Chemical Process Control Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020
- T.E. Marlin, Process Control: Designing Processes and Control Systems for Dynamic Performance, McGraw-Hill, 2000; B.A. Ogunnaike and W.H. Ray, Process Dynamics, Modeling and Control, Oxford, 1994; J.B. Riggs and M. N. Karim, Chemical and Bio-Process Control, Ferret Publishing, Austin, Texas, 2016; G. Stephanopoulos, Chemical Process Control: An Introduction to Theory and Practice, Prentice Hall, 1984.
- SOFTWARE: Athena Visual Studio, EXCEL based Thermodynamic Calculator CHE301 Calculator.xlsm; EXCEL based Process Control tool CHE341 Loop Simulator.xlsm

5. Specific course information

a. Brief description of the content of the course (catalog description)

Analysis and design of chemical processes and control systems. Feedback and feedforward controllers. Stability, tuning, and simulation of P-I-D controllers. Introduction to the control of chemical plants and digital process control.

b. Prerequisites or co-requisites

MATH 220 and CHE 312 and CHE 313 and CHE 321.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Understand the basic concepts of process dynamic modeling for a variety of chemical engineering systems.
- Know how to develop empirical models from process data using the step response methodology
- Understand the theory of feedback and feed forward control and determine optimal controller settings for stable and robust performance.
- Know when to select and how to implement cascade and feed forward controllers

- Apply available computer software (Matlab, Excel or Athena Visual Studio), or use Laplace transformation theory to solve process dynamic models and investigate their open and closed loop response to various external disturbances.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.			X	

7. **Brief list of topics to be covered**

INTRODUCTION TO PROCESS CONTROL IN CHEMICAL ENGINEERING: Elements of Differential and Integral Calculus, Numerical Methods for Algebraic and Differential Equations, Mass and Energy Balances in the Context of Process Control, Tutorials on Programming with VBA and Athena Visual Studio

CONTROL OF INDIVIDUAL UNITS: Reactors, Heat Exchangers, Distillation Columns, Compressors, Other Unit Operations

CONTROL CONCEPTS AND DYNAMIC MODELING: Development of Dynamic Process Models, Solution of Dynamic Models and Open Loop Response, Laplace Transforms and Transfer Functions, Linearization of Process Dynamic Models, Process Identification from Plant Data

FEEDBACK CONTROL FUNDAMENTALS: P, PI and PID Controllers, Block Diagrams and Routh-Hurwitz Stability Criterion, Instrumentation Hardware and Representation

FEEDBACK CONTROLLER DESIGN AND TUNING: Criteria for Controller Tuning, PID Controller Tuning Methods, Guidelines and Troubleshooting, Common Control Loops

ADVANCED CONTROL SYSTEMS AND MIMO SYSTEMS: Feedforward and Cascade Control, Interaction and Loop Pairing and Decoupling, Relative Gain Array Control of Common Industrial Process Units, Model Predictive Control of Linear Processes, Advanced Process Control and Real-Time Optimization

CHE 381

1. Course number and name

CHE 381: Chemical Engineering Laboratory I

2. Credits, contact hours and categorization

Credit hours: 2; Contact hours: 4 lab hours per week; Engineering

3. Instructor's or course coordinator's name

Alan Zdunek

4. Textbook, title, author, publisher, year

Unit Operations of Chemical Engineering, W.L. McCabe, J.C. Smith, P. Harriott, 7th Edition, McGraw-Hill, 2005.

Other supplemental materials: Provided online in the UIC Blackboard course folder to help aid in successfully preparing and carrying out the experiments. Material includes Lab report templates, lab presentation templates, laboratory safety information, technical writing guidelines, error analysis methodology, specific articles and files related to the various experiments including vendor information on the experimental apparatus and components, theory and empirical correlations, engineering charts, and material and physical properties.

5. Specific course information

- a. Brief description of the content of the course (catalog description) Heat and momentum transfer operations associated with chemical processes. These include heat exchangers, fluid properties and fluid flow. Technical report writing, computer calculations.
- b. Prerequisites or co-requisites CHE 312.
- c. Indicate whether a required, elective, or selected elective.
Required.

6. Specific goals for the course

Provide students with a hands-on opportunity to work with physical equipment in a laboratory environment.

- a. Specific outcomes of instruction.
 - 1 Solve complex, open-ended engineering problems creatively, in a team setting.
 - 2 Integrate knowledge and skills from prior chemical engineering courses.
 - 3 Design and conduct realistic experiments, analyze data, draw concrete conclusions and make relevant recommendations.
 - 4 Communicate effort and findings effectively through the written and oral word.
 - 5 Master and apply safety analysis and safety practices in a laboratory environment.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	X			
(3) an ability to communicate effectively with a range of audiences.				X
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.				X
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

Specific lab experiments performed:

- 1) Viscosity: Non-Newtonian fluids
- 2) Batch Distillation
- 3) Fluid Flow in Pipe Systems
- 4) Evaporation
- 5) Humidification
- 6) Membrane Separation

List of Topics:

Experimental verification of theoretical principles, familiarization with engineering equipment and exposure to an engineering sense, teamwork skills; participation, leadership and team member assessment, time value; experimental planning, implementation, and completion, experimental skills; design, process safety analysis, data collection, error analysis, written and oral technical communication, integrated problem-solving.

Other topics:

Week #1: General lab safety information, technical writing and technical presentation tips.

Week # 2: Industrial Chemical Hygiene Plan review

Week #10: AIChE Academy eLearning tutorial: ELA962 Chemical Reactivity Hazards

CHE 382

1. Course number and name

CHE 382: Chemical Engineering Laboratory II

2. Credits, contact hours and categorization

Credit hours: 2; Contact hours: 4 lab hours per week; Engineering

3. Instructor's or course coordinator's name

Alan Zdunek

4. Textbook, title, author, publisher, year

Unit Operations of Chemical Engineering, W.L. McCabe, J.C. Smith, P. Harriott, 7th Edition, McGraw-Hill, 2005.

Other supplemental materials: Provided online in the UIC Blackboard course folder to help aid in successfully preparing and carrying out the experiments. Material includes; Lab report templates, lab presentation templates, laboratory safety information, technical writing guidelines, error analysis methodology, specific articles and files related to the various experiments including vendor information on the experimental apparatus and components, theory and empirical correlations, engineering charts, and material and physical properties.

5. Specific course information

- a. Brief description of the content of the course (catalog description) Heat, momentum and mass transfer operations associated with chemical processes; these include distillation columns, reactors, humidifiers, and evaporators. Technical report writing, computer calculations.
- b. Prerequisites or co-requisites CHE 381 and credit or concurrent registration in CHE 313.
- c. Indicate whether a required, elective, or selected elective.
Required.

6. Specific goals for the course

Provide students with a hands-on opportunity to work with physical equipment in a laboratory environment.

- a. Specific outcomes of instruction.
 - 1 Solve complex, open-ended engineering problems creatively, in a team setting.
 - 2 Integrate knowledge and skills from prior chemical engineering courses.
 - 3 Design and conduct realistic experiments, analyze data, draw concrete conclusions and make relevant recommendations.
 - 4 Communicate effort and findings effectively through the written and oral word.
 - 5 Master and apply safety analysis and safety practices in a laboratory environment.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	X			
(3) an ability to communicate effectively with a range of audiences.				X
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.				X
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

Specific lab experiments performed are:

- 1) Fluidized Bed
- 2) Continuous Distillation
- 3) Heat Transfer to a Stirred Tank
- 4) Process Control: Level Control
- 5) Gas Absorption in a Packed Bed
- 6) Liquid-Liquid Extraction

List of Topics:

Experimental verification of theoretical principles, familiarization with engineering equipment and exposure to an engineering sense, teamwork skills; participation, leadership and team member assessment, time value; experimental planning, implementation, and completion, experimental skills; design, process safety analysis, data collection, error analysis, written and oral technical communication, integrated problem-solving.

Other topics:

Week #1: General lab safety information, technical writing and technical presentation tips.

Week # 2: JT Baker Hazardous Chemical Safety Course; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards, Compressed Gases

Week #10: AIChE Academy eLearning tutorial: ELA992 Dust Explosion Control

CHE 396

1. Course number and name

CHE 396: Chemical Engineering Design I

2. Credits and contact hours

Credit hours: 4. Contact hours: Two, 75 min lecture periods; Engineering.

3. Instructor's or course coordinator's name

Betul Bilgin

4. Textbook, title, author, publisher, year

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012.

a. Other supplemental materials

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009

Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker

5. Specific course information

a. Brief description of the content of the course (catalog description)

Introduction to modern, process design and development, engineering economics, and report writing. Design and cost of equipment relating to materials handling to heat transfer, mass transfer, and reactors.

b. Prerequisites or co-requisites

CHE 312, CHE 313, CHE 321

c. Indicate whether a required, elective, or selected elective.

Required

6. Specific goals for the course

a. Specific outcomes of instruction.

Provide a real-world perspective on process design

Introduce design and selection of equipment

Give insight into practical constraints that govern design decisions

Help you translate knowledge into practice & start solving real problem

Improve presentation, analytical thinking and team working skills

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.			X	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			X	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

- Chemical Design Introduction
- Flowsheet development
- Utilities and Energy Recovery
- Process simulation
- Process control
- Materials of construction
- Capital cost
- Operating cost
- Economic Analysis
- Process Hazard, Professional ethics
- Professional Ethics Presentations
- General Site Consideration

CHE 397

1. Course number and name

CHE 397: Chemical Engineering Design II

2. Credits and contact hours

Credit hours: 4. Contact hours: Two, 75 min lecture periods; Engineering.

3. Instructor's or course coordinator's name

Betul Bilgin

4. Textbook, title, author, publisher, year

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012.

a. Other supplemental materials

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009

Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker

5. Specific course information

a. Brief description of the content of the course (catalog description)

Application of principles and design methodology of chemical engineering to the design of large-scale chemical processes and plants. A major design project is assigned for solution and presentation at the College Engineering Expo by students working in small groups with industry mentors. Extensive computer use required.

b. Prerequisites or co-requisites

CHE 396

c. Indicate whether a required, elective, or selected elective.

Required

6. Specific goals for the course

a. Specific outcomes of instruction.

This course involves the application of all prior CHE concepts to the chemical process design. Design needs to be analyzed from safety, environmental impact, feasibility, risk and economics perspectives. Course aims to equip students with the skill to evaluate open-ended design problems conceptually, analytically and practically.

Students will be able to:

- Develop design plan and project charter for each design phase
- Work effectively with team members
- Present effectively technical material based on audience
- Identify key components in market analysis and construct design basis
- Analyze competing processes
- Write technical content in a way that is appropriate to the audience
- Apply economic analysis
- Identify potential safety concerns in the process and formulate strategies to mitigate those

- Simulate full process with Aspen Plus or SuperPro
- Size process equipment

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.				X
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			X	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

- Process description, Design basis, BFD, PFD Flowsheet development
- Equipment list and sizing
- Utilities demand summary for entire plant
- Process simulation
- Process control strategy, process parameters
- Materials of construction
- Plant Cost Analysis (NPV, sensitivity analysis, payback period, capital cost, Aspen economic analysis)
- Environmental and safety concerns, process hazards, mitigation.

CHE 422

1. Course number and name

CHE 422. Biochemical Engineering

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Ying Liu

4. Textbook, title, author, publisher, year

Bioprocess Engineering: Basic Concepts by Michael L Shuler, Fikret Kargi, Matthew DeLisa, Prentice Hall, 3rd Edition, 2017.

a. Other supplemental materials

Molecular Biology of the Cell, 6th Edition, Bruce Alberts, Alexander D. Johnson, Julian Lewis, David Morgan, Martin Raff, Keith Roberts, Peter Walter, Garland Science, 2014.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Enzyme-catalyzed and microbially mediated processes. Free and immobilized enzymes. Batch and continuous cell cultures. Transport phenomena in microbial systems and fermentation processes. Design of biological reactors.

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction.

To be able to describe and understand recombinant DNA, fermentation and separation methods such as PCR, filtration and centrifugation, cell rupture, and purification techniques such as adsorption, solvent extraction, membrane filtration, chromatography (affinity, ion, hydrophobic, reverse phase), precipitation, and crystallization. To apply fundamental principles for the separation of biopharmaceuticals via chemical engineering unit operations.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.		x		
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			x	
(3)	an ability to communicate effectively with a range of audiences.			x	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	x			
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	x			
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	x			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		x		

7. Brief list of topics to be covered

The course is aimed at senior undergraduate and graduate students and is designed to move engineers and other non-biologists past the initial language and knowledge barriers of the Biochemical Engineering field.

- Fundamental recombinant DNA techniques
- Genetically engineered biomacromolecules,
- General molecular biology techniques; Polymerase Chain Reaction (PCR) basics, site-directed and random mutagenesis, display techniques, DNA sequencing, DNA and protein gel electrophoresis, cutting and splicing DNA, protein expression, DNA and RNA hybridization techniques, and Western blots.
- Protein engineering and protein design approaches; knowledge-based design, computational protein design, and directed evolution.
- Examples of engineered proteins with novel structural and functional properties
- Fermentation (basic principles of fermentation processes, scaling-up),
- Product recovery and separation (cell disruption, sedimentation, filtration/centrifugation, solvent extraction, adsorption, precipitation), product purification (adsorption, ultrafiltration, crystallization) and
- Final separation and advances in bioseparation strategies.
- Process design; creating engineered strains and cell culture/fermentation. Purification of the end product for energy, biotechnology or biomedical applications.

CHE 425

1. Course number and name

CHE 425 Nanotechnology for Pharmaceutical Applications

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Ying Liu

4. Textbook, title, author, publisher, year

Drug Delivery: Engineering Principles for Drug Therapy Topics in Chemical Engineering, W. Mark Saltzman, Oxford, 2001.

a. Other supplemental materials

Selected reading materials from high-impact journals and a few textbooks, including but not limited to, Science, Nature, JACS, PRL, Biomacromolecules, ACS Nano, Nano Letters, Journal of Colloids and Interfacial Science.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Emerging role of nanostructures in drug development and delivery. Principles of nanostructure formation, characterization, surface functionalization, in vitro and in vivo transport, and visualization.

b. Prerequisites or co-requisites

CHE 301 and CHE 311 and CHE 312; or consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will be able to:

- Understand the fundamental and essential concepts used in nanotechnology, synthesis, fabrication and characterization.
- Analyze thermodynamic and kinetic control of the nanoparticle formation and stabilization
- Analyze drug loading, encapsulation efficiency, and transport
- Discuss nanomaterial selection, design and fabrication.
- Discuss nanomaterial impact on the environment
- Apply written and oral communication skills to a final term paper and presentation

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3)	an ability to communicate effectively with a range of audiences.			X	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Factors that impact the development ability of drug candidates.
- Evolution of controlled drug delivery systems.
- Advanced Nanotechnologies and Micro technologies for Drug Delivery
- Self-assembled delivery vehicles: basic theoretical considerations and modeling concepts.
- Nanoparticle characterization
- Liposomes as intravenous solubilizers.
- Polymeric nanoparticle drug delivery systems.
- Nano suspensions with enhanced drug dissolution rates of poorly water-soluble drugs.
- Pharmaceutical Fundamentals
- Physiological, Biochemical, and Chemical barriers to Oral Drug Delivery.
- Cell culture models for drug transport studies.

CHE 426

1. Course number and name

CHE 426. Biopharmaceutical Process Development and Manufacturing

2. Credits, contact hours and categorization

Credit hours: 3 or 4. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Nandkishor Nere and Meenesh Singh

4. Textbook, title, author, publisher, year

- a) D. J. am Ende and M. T. am Ende (Editors), "Chemical Engineering in the Pharmaceutical Industry: Active Pharmaceutical Ingredients", 2nd Edition, Hoboken, NJ, Wiley, 2019.
- b) D. J. am Ende and M. T. am Ende (Editors), "Chemical Engineering in the Pharmaceutical Industry: Drug Product Design, Development, and Modeling", 2nd Edition, Hoboken, NJ, Wiley, 2019.
- c) F. Jameel, S. Hershenson, M. A. Khan, and S. Martin-Moe (Editors), "AAPS Advances in the Pharmaceutical Sciences Series: Quality by Design for Biopharmaceutical Drug Product Development (Vol 18)", Springer, 2015.
- d) F. Jameel, and S. Hershenson (Editors), "Formulation and Process Development Strategies for Manufacturing Biopharmaceuticals", Hoboken, NJ, Wiley, 2010.

Other supplemental materials

- a) N. K. Pandit and R. P. Soltis, "Introduction to the Pharmaceutical Sciences: An Integrated Approach", 2nd Edition, Lippincott Williams & Wilkins, 2012.
- b) N. W. Warne and H-C. Mahler (Editors), "AAPS Advances in the Pharmaceutical Sciences Series: Challenges in Protein Product (Vol 38)", Springer, 2015.

5. Specific course information

- a. Brief description of the content of the course (catalog description)
Introduction to Biopharmaceuticals Process Development and Manufacturing. Drug Substance (Active Pharmaceutical Ingredients) Development. Drug Product Development. Enabling Technologies in Biopharmaceutical Industry. Biopharmaceutical Manufacturing.
- b. Prerequisites or co-requisites
Consent of the instructor.
- c. Indicate whether a required, elective, or selected elective.
Elective.

6. Specific goals for the course

- a. Specific outcomes of instruction.
To be able to understand manufacturing processes for both the small molecules and biologics. Learn how to apply fundamentals of chemical engineering unit operations and

processes to design and scale-up processes to make both the drug substances and drug products. Use integrated engineering principles to solve practical problems of process development in biopharmaceutical industry. Work in collaboration with peers to solve problems assigned through the team project and communicate through interactive project presentations.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(3)	an ability to communicate effectively with a range of audiences.				X
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	X			
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			X	
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X		

7. Brief list of topics to be covered

The course is aimed at senior undergraduate and graduate students spanning multiple disciplines and is designed to introduce the practical world of process development and manufacturing of pharmaceutically active molecules and drug products.

- Introduction to drug discovery, drug delivery and development.
- Bio/Analytical Chemistry Tools and Technologies
- Active pharmaceutical ingredients (API)-Solid Form Screening and Selection
- Essentials of API Process Development
 - Fundamentals of chemical reaction engineering (CRE)
 - Applications of CRE in pharmaceutical industry
 - Crystallization process development
 - Extraction, Distillation, Filtration, Washing and Drying
- Biopharmaceutical Process Scale-up

- Essentials of Biologics Process Development
 - Upstream (bioreactor) process development
 - Downstream (purification) process development
- Essentials of Small Molecule Drug Product Formulation and Process Development
 - Formulation Screening and Selection
 - Blending, Roller compaction, Wet Granulation, Film coating
 - Spray Drying, Hot Melt Extrusion, Tableting and Encapsulation
- Essentials of Biologics Drug Product Development
 - Formulation, overall process and lyophilization
 - Freeze-Thawing, Pooling, Reconstitution and Filling
- Advanced Topics
 - Applications of Predictive Modeling
 - Continuous Processing
 - Emerging Technologies-Electrochemistry, Photochemistry, Bio-catalysis, Allied separations, Green chemistry
- cGMP and key Considerations for the Safety and Cleaning in Manufacturing
- Team project presentations and peer feedback

CHE 427

Entrepreneurship and Engineering

1. Course number and name

CHE 427 Entrepreneurship and Engineering

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: One, 165 min/wk. lecture; Engineering.

3. Instructor's or course coordinator's name

Said Al-Hallaj

4. Textbook, title, author, publisher, year

Selected publications, reports and handouts.

5. Specific course information

a. Brief description of the content of the course (catalog description)

The course provides an introductory view into the various aspects involved in technology commercialization (i.e. taking an idea from the lab to the marketplace).

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Understand fundamentals and basic concepts of intellectual property, confidentiality of Non-Disclosure Agreements (NDA), patent applications, copyrights, Know-how
- Learn about UIC resources for entrepreneurship; Office of Technology Management, Invention Center, Business Plan Competition, Illinois Venture
- Learn how to develop a business development strategy, business plan, marketing strategies, finance and accounting
- Understand the fundamentals and basic concepts of starting a business, legal structure (LLC, Inc., operating agreements, governance)
- Learn concepts and practical strategies for fundraising, grants, seed/angel investment, venture capital, strategic partnerships
- Present a Case Study with final project term paper and presentation

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	X			
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(3)	an ability to communicate effectively with a range of audiences.				X
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Anderson Entrepreneurship Quiz
- Protecting your intellectual property (IP) (Guest Lecturer: Program Manager, UIC-OTM)
- UIC Resources Overview (Guest: Program Manager, UIC-OTM)
- Business Development Strategy
- Why Marketing Study is Important (Guest Speaker: Evergreen Climate Innovations)
- Financials Basics
- Starting Your Own Business (is it a good idea?)
- How to build an effective team
- Corporate Entrepreneurship
- Pitching your idea
- Fundraising Strategy (are you ready to pitch your idea?)

CHE 433

1. Course number and name

CHE 433. Process Simulation with Aspen Plus

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Michael Caracotsios

4. Textbook, title, author, publisher, year

Process Simulation with Aspen Plus Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020.

a. other supplemental materials

- Ralph Schefflan, Teach yourself the Basics of Aspen Plus, 2nd Edition, Wiley. (2016).
- Gavin Towler and Ray Sinnott, Chemical Engineering Design, 2nd Ed, Elsevier Ltd. (2013).
- Lorenz T. Biegler, Ignacio E. Grossmann and Arthur W. Westerberg, Systematic Methods of Chemical Process Design, Prentice Hall International Series, December (1999).
- T. F. Edgar and D. M. Himmelblau, Optimization of Chemical Processes, McGraw-Hill, Inc.(1988).
- A. W. Westerberg, H. P. Hutchison, R. L. Motard and P. Winter, Process Flowsheeting, 1st Edition, Cambridge University Press (1979).
- SOFTWARE: Athena Visual Studio from www.AthenaVisual.com; EXCEL, ASPEN Plus and optionally MATLAB and ASPEN Hysys.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Main principles of Process Simulation (Material and Energy Balances) relevant to chemical and biological processes. Process Simulation or Flowsheeting is defined as the use of commercially available products (like ASPEN Plus, ASPEN Hysys, UniSim, VMG, ChemCad, gProms, Pro II) to perform steady-state Heat and Weight Balances, Sizing and Cost calculations for chemical, biological, petrochemical and refining processes.

b. Prerequisites or co-requisites

CHE 312 and CHE 313 and CHE 321; or consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Select and justify choices for physical property models required for accurate heat and weight balance calculations.

- Design and model unit operations such as chemical reactors, heat exchangers, heaters and coolers, pumps, compressors and turbines, mixers, distillation and packed columns and user specific operations.
- Combine units operations in a cohesive Flowsheet that describes existing or conceptual processes.
- Familiarize themselves with computational aspects such as design specs, calculator blocks, sensitivity analyses, flow sheet convergence
- Perform optimization studies.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(3) an ability to communicate effectively with a range of audiences.			X	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	

7. Brief list of topics to be covered

INTRODUCTION TO PROCESS FLOWSHEETING IN CHEMICAL ENGINEERING:

Elements of Differential and Integral Calculus, Numerical Methods for Algebraic and Differential Equations, Mass and Energy Balances in the Context of Process Simulation, Tutorial on Programming with ASPEN Plus and Athena Visual Studio

THERMODYNAMICS AND PROCESS UNIT OPERATIONS: Vapor-Liquid Equilibrium and Property Methods, VLE Data Regression, Mixers and Splitters, Two and Three Phase Flash Blocks, Heaters, Coolers and Condensers, Distillation, Pumps, Compressors and Turbines, Example Flowsheets and Design Specifications

PROCESS FLOWSHEETING FUNDAMENTALS: Recycle Stream Selection and Tear Equations, Convergence Methods for Tear Equations, Calculator and Transfer Function Blocks, Simultaneous Recycle Convergence and Design Specifications, Sensitivity Analysis and Case Studies, Sequential Modular vs. Equation Based Solution Methods

CHEMICAL KINETICS AND REACTOR UNIT OPERATIONS CSTR: Plug Flow and Fixed Bed Reactors, Conversion and Equilibrium Reactors, Example Flowsheets with Design Specifications, Excel Interface for Custom Reactor and Physical Models

COMPLEX UNIT OPERATIONS AND ADVANCED SIMULATION TOPICS: Packed columns for Absorption and Stripping Operations, Rate based vs. Equilibrium Models, Homogeneous and Heterogeneous Azeotropic Distillation, Process Optimization and Pinch Analysis for Energy Optimization, Heat Exchangers and Heat Exchanger Network Optimization, Solids Handling and Electrolyte Systems

CHE 451

1. Course number and name

CHE 451 Renewable Energy Technologies

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: One, 165 min/week lecture; Engineering

3. Instructor's or course coordinator's name

Said Al-Hallaj

4. Text book, title, author, publisher, year

"Hybrid Hydrogen Systems for Stationary and Transportation Applications", Said Al-Hallaj and Kristofer Kiszynski, 1st Edition, 2011, Springer Publishing, ISBN 978-1-84628-466-3.

5. Specific course information

a. Brief description of the content of the course (catalog description)

The course will cover fundamentals of renewable energy technologies such as solar, wind and biomass. In addition, course will provide an introduction to various energy storage technologies such as batteries and fuel cells. Finally, the hydrogen economy promises and challenges such as hydrogen storage and transportation will be discussed and analyzed.

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will be able to:

- Understand the fundamentals and essential concepts used in renewable energy technologies such as solar, wind, biomass.
- Understand and analyze energy storage technologies such as batteries, fuel cells and supercapacitors.
- Address key ideas, as well as technical and economic issues related to renewable energy in a final project term paper.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3)	an ability to communicate effectively with a range of audiences.				X
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Introduction (Renewable Energy and Sustainability)
- Solar Energy Calculations and Solar Photovoltaics (PV)
- Building Integrated Photovoltaics
- Wind Energy Basics and Applications
- Bioenergy
- Hydrogen Economy and Hydrogen Storage
- Fuel Cells
- Energy Storage for Smart Grid Applications
- Geothermal Energy
- Hydrogen Economy: Production and Infrastructure
- Optimization of Renewable Hydrogen Energy Systems
- Control of Hybrid Fuel Cell/Battery Systems
- Solar Desalination and Hybrid Fuel Cell/Desalination

CHE 453

1. Course number and name

CHE 453 Fundamentals of Electrochemistry

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Brian P. Chaplin

4. Textbook, title, author, publisher, year

(Required) Bard, A. J.; Faulkner, L. R., *Electrochemical Methods: Fundamentals and Applications*. Second ed.; John Wiley and Sons: 2001; p 833.

a. Other supplemental materials

Bockris, J. O. M.; Reddy, A. K.; Gambrá-Aldeco, M., *Modern Electrochemistry*. 2nd ed.; Plenum Press: New York, 2000.

Orazem, M. E.; Tribollet, B., *Electrochemical Impedance Spectroscopy*. John Wiley and Sons: 2008; p 525.

A. Lasia, Electrochemical Impedance Spectroscopy and Its Applications, In: *Modern Aspects of Electrochemistry*, B. E. Conway, J. Bockris, and R.E. White, Eds., Kluwer Academic/Plenum Publishers, New York, 1999, Vol. 32, p. 143-248.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Introduction to the fundamentals of electrochemistry and its application in a variety of technologies (i.e., batteries, fuel cells, electrolysis cells). Includes methods for the analysis of cells using electrochemical techniques.

b. Prerequisites or co-requisites

Graduate or advanced undergraduate standing.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction.

1	Use basic Excel cell functionality to tabulate and graph electrochemical data derived from cyclic voltammetry and electrochemical impedance spectroscopy experiments, and interpret with mathematical models
2	Solving differential equations by the Laplace transform technique, with application to the diffusion equation
3	Solve partial differential equations numerically and apply to analyze linear sweep and cyclic voltammetry data
4	Using complex variable notation to derive equivalent circuit model fits of experimental electrochemical impedance spectroscopy data and compare to computer generated fits using Gamry software

5	Review a published paper in the areas of electrochemistry or electrochemical technologies, and prepare a presentation that summarizes and critiques the paper.
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b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	X			
(3) an ability to communicate effectively with a range of audiences.			X	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	X			
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	X			
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			X	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Week #1. Introduction and overview of electrochemical cells and electrode processes
- Week #2-3. Thermodynamics of electrochemical cells
- Week #4-6. Kinetics of electrochemical reactions
- Week #7. Mass transfer in electrochemical systems
- Week #8. Adsorption and double-layer structure
- Week #9-11. Electrochemical methods including potential step and sweep methods and controlled current techniques
- Week #12-13. Numerical simulations applied to potential step and sweep methods
- Week #14. Electrochemical impedance spectroscopy of electrochemical cells
- Week #15. Student presentations

CHE 454

1. Course number and name

CHE 454 Molecular and Macromolecular Engineering

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Vivek Sharma

4. Text book, title, author, publisher, year

(1) Polymer Chemistry, Second Edition, Paul C. Hiemenz, Timothy P. Lodge, CRC Press, 2007.

(2) Polymer science, A Materials Science Handbook, Volume I & II, Jenkins, A.D. North, Holland Publishing Co., Amsterdam and London; American Elsevier Publishing Company, New York, 1972.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Advanced course in polymer science and engineering. Polymerization, polydispersity, molecular configuration, solution properties, thermodynamics, glass and rubbery states, crystallization, viscoelasticity, elastic properties, multiphase systems.

Prerequisites or co-requisites

CHE 330.

b. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Understand the fundamental and advanced concepts used in macromolecular systems for chemical engineering applications
- Learn the detailed methods used to synthesize and characterize macromolecular systems
- Understand the structural, viscoelastic, mechanical and thermal properties of macromolecules and their characterization methods.
- Reinforce the ability to communicate effectively through individual papers and presentations in class.

a. Explicitly indicate which of the student outcomes are addressed by the course.

	1	2	3	4
(7) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				x
(7) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			x	
(7) an ability to communicate effectively with a range of audiences.				x
(7) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	X			
(7) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	x			
(7) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			x	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				x

7. **Brief list of topics to be covered**

- Introduction to Plastics and Polymer materials
- Molecular Structures, Microstructures, and Polymerization
- Plastics Manufacturing
- Introduction to Composites
- Composites: Materials and Manufacturing
- Thermoplastics
- Composite Processing
- Introduction to 3D printing
- Printing in plastic, metal, glass, wood, concrete and other materials
- Bioprinting
- Environmental Aspects of Polymer-based Materials and Material processing

CHE 455

1. Course number and name

CHE 455 Nanoscale Systems in Chemical Engineering

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Anh Ngo

4. Textbook, title, author, publisher, year

"Introduction to Nanoscience", S.M. LINDSAY. ISBN-10: 0199544212

a. Other supplemental materials

- a) Introduction to Nanomaterials and Devices by Omar Manasreh.
- b) Introduction to Nanoscience and Nanotechnology, Chris Binns
- c) Nanomaterials and Nano chemistry, by Bréchnignac, C., Houdy, P., Lahmani, M

5. Specific course information

a. Brief description of the content of the course (catalog description)

Basic principles associated with nanoscience and nanotechnology; fabrication and synthesis, size dependent properties, characterization; applications of materials at nanometer length scales; emphasis on recent technological breakthroughs.

b. Prerequisites or co-requisites

CHE 230; and Graduate or advanced undergraduate standing.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Explain the history of nanotechnology.
- Understand the nanoscale paradigm in terms of properties at the nanoscale dimension.
- Demonstrate a working knowledge of nanoscience/nanotechnology principles and applications
- Apply key concepts in materials science, chemistry, physics, biology, and engineering to the field of nanotechnology.
- Problem solving in homework/exams and by giving a presentation to the class.
- The student will be able to gain the ability to communicate effectively.

b. Explicitly indicate which of the student outcomes are addressed by the course.

	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				x
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			x	
(3) an ability to communicate effectively with a range of audiences.			x	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			x	
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				x
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.				x
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				x

7. Brief list of topics to be covered

Basic concepts and modern methods in major developments of nanotechnology and nanomaterials will be introduced. Tools and principles relevant to nanoscale materials will also be introduced. Current and future nanotechnology applications in engineering, materials, physics, chemistry, electronics, and energy will be discussed.

- Physics and Chemistry of Nanomaterials
- Application of Quantum Mechanics to Nanomaterials
- Electronic Structure of Nanomaterials
- Characterization Tools
- Nano-manufacturing
- Applications
 - Nanomaterials in engineering, physics, chemistry
 - Nanomaterial Examples – Energy
 - Nanomaterial Examples – Medicine

- Nanomaterial Examples – Environmental applications
 - Nanomaterial Examples – Electronics
- Implications, Ethics, & Safety of Nanomaterials

CHE 457

1. Course number and name

CHE 457 Colloidal and Interfacial Phenomena

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Vivek Sharma

4. Textbook, title, author, publisher, year

Lecture notes, assignments and extra reading material form bulk of the class content.

Sample resource materials:

1. Capillarity and Wetting Phenomena: Drops, Bubbles, Pearls, Waves, Pierre-Gilles de Gennes, Francoise Brochard-Wyatt and David Quere, Springer, 2004.
2. The Physics of Foam, Denis Weaire, Stefan Hutzler, Oxford University Press, 1999.
3. Soap Bubbles: Their Colors and Forces which Mold Them, C. V. Boys, Dover 1959
4. The Science of Soap Films and Soap Bubbles, C. Isenberg, Dover 1992
5. Physical Chemistry of Surfaces, 6th edition, Arthur W. Adamson & Alice P. Gast, John Wiley & Sons, 1997.
6. Soap Films: Studies of their Thinning & a Bibliography, Karol J. Mysels, Kozo Shinoja and Stanley Frankel, 1957.
7. The Colloidal Domain, 2nd ed., D. Fennel Evans and Hakan Wennerstrom, Wiley-VCH 1999.
8. Interfacial Transport Processes and Rheology, Darsh Wasan, Edwards and Howard Brenner, Butterworth-Heinemann, 1991
9. Intermolecular and Surface Forces, Jacob Israelachvili, Academic Press, 3rd edition, 2011.
10. Colloidal Dispersions by W. B. Russel, D. A. Saville and W. R. Schowalter, Cambridge University Press, 1988.
11. Froth: The Science of Beer by Mark Denny, Johns Hopkins University Press, 2009.
12. Fizzics: The Science of Bubbles, Droplets and Foams by F. Ronald Young, Johns Hopkins University Press, 2011.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Interfacial phenomena in practice: soap bubbles, emulsions, foams, detergents; surface tension driven flows, Marangoni effect; interfacial rheology; colloids, emulsions, 3D-printing; interfacial thermodynamics, biological interfaces.

Prerequisites or co-requisites

CHE 301, CHE 312.

b. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction.

1. Demonstrate knowledge of capillarity and wetting (surface tension, contact angle, Laplace pressure, disjoining pressure, interfacial thermodynamics, super hydrophobicity, surfactants, spreading kinetics).
2. Apply the conservation principles and flux laws to model transport processes central to interfacial science and engineering.
3. Apply the physical and mathematical concepts underlying colloidal, emulsion & foam stability, thermodynamics and dynamics to solve new problems “by analogy”.
4. Understand the significant role played by fizzes and interfacial phenomena in daily life and in typical engineering practice.
5. Develop a fundamental understanding of experimental methods use for measuring surface tension, interfacial rheology & for studying surface-tension driven flows, pattern formation and instabilities.
6. Prepare a well-researched written research proposal and an oral presentation.

a. Explicitly indicate which of the student outcomes are addressed by the course.

	1	2	3	4
(7) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				x
(7) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(7) an ability to communicate effectively with a range of audiences.				X
(7) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	X			
(7) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	x			
(7) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				x

7. Brief list of topics to be covered

1. Introduction: Soap bubbles, soda pop, emulsions, foams, fog, dew, beer froth, shaving foam, whipped cream & detergents, “Ben Franklin stilled the waves” & historical perspective
2. Capillarity & deformable interfaces
3. Physicochemical hydrodynamics
4. Interfacial rheology: fundamental concepts & techniques
5. Interfacial Shear & Dilatational Viscosity/Modulus
6. Thin film hydrodynamics: Viscoelastic interfaces
7. Biological Interfaces
8. Thermodynamics of interfaces
- 9 Liquid crystalline phases
10. Imaging & spectroscopy of interfaces & interfacial phenomena
11. Industrial applications

CHE 499

1. Course number and name

CHE 499. Professional Development Seminar

2. Credits, contact hours and categorization

Credit hours: 0; Contact hours: 10-12, 50-minute seminars/lectures per semester; Engineering

3. Instructor's or course coordinator's name

Alan Zdunek

4. Textbook, title, author, publisher, year

None

5. Specific course information

a. Brief description of the content of the course (catalog description)

Weekly seminars provide career information about chemical engineering and professional job search skills. Students provide evaluations of their engineering knowledge and educational experience in the Chemical Engineering Department.

b. Prerequisites or co-requisites

Open only to chemical engineering seniors, must be taken in the Fall semester of their last year of study.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Interact with practicing chemical engineers
- Learn about traditional and non-tradition chemical engineering career paths, both in academia and industry
- Learn job searching skills; networking, resume writing, interviewing skills, salary negotiation
- Learn what industries hire chemical engineers
- Learn about their role as UIC Chemical Engineering alumni
- Relate their UIC educational experience in an exit survey

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(3) an ability to communicate effectively with a range of audiences.			X	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

Invited speakers provide information on industries hiring chemical engineers, job-related experience, traditional and non-traditional chemical engineering career paths, and career advice. Career development information from Engineering Career Center. Student evaluation of their educational experience in the Chemical Engineering Department including senior exit survey and Senior written examination.

Selected Seminar Speakers from past years:

Job-Searching Skills: Marytza Aranda, UIC Engineering Career Center
Career and Life Skills: Robert Orent, Stepan Company
Safety/Environment: Lissa Aquino, CSL-Behring
Biopharmaceuticals: Chris Ciszek, CSL-Behring
Process Engineering/Research & Development: Jeff Tyska, Honeywell/UOP
Food Process Engineering: Lillian Wong, Carissa Tsouchlos, Tetrapak
Consumer Products: Priyanka Sharma, Kazmira LLC
Entrepreneurship: Said Al-Hallaj, All-Cell Technology
Sustainable Process Engineering: Aimen Lateef, LanzaTech
Oil/Gas Field Services/Engineering: Roxana Nicezyporuk, Honeywell/UOP
Process Control-Food: Thomas Scheller, ProLeiT
Patent Attorney: Jennifer Nacht, Greer, Burns, Crain Law
Academia: Michael Bentel, Clemson University, Ebin Joseph, UIUC
Nuclear Industry: Julie Ann Kim, Constellation Energy
Non-ChE: Lorena Pereira, Underwriter, FM Global
Petrochemicals: Mamtu Chandnana, INEOS Chemicals

A.2. Other Engineering Courses.

CS 109

1. Course number and name

Programming for Engineers with Matlab

2. Credits, contact hours, and categorization.

3 credit hours with a total of 5 contact hours per week(3 hours per week and 2 hours labs).

3. Instructor's or course coordinator's name

Sara Riazi

4. Textbook, title, author, and year

Online zybooks for textbook link.

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Program design and problem-solving using MATLAB; numeric computation; data types and operators; control structures; functions; file I/O; arrays and structures; engineering applications: matrices and equation solution; programming assignments.

b. Prerequisites or co-requisites

Credit or concurrent registration in MATH 180

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Develop an understanding of program design using data types, operators, control structures, functions, file I/O, arrays and structures.	(7)
CLO2: Demonstrate the understanding of the aforementioned topics through computer projects with engineering applications.	(3), (7)

7. Brief list of topics to be covered

- Preliminaries
 - Syntax, program execution, program structure
 - Variable declarations and simple data types
 - Assignment and simple I/O statements
- Control structures
 - Sequencing
 - Alternation

- Loop constructs
- File I/O
 - Format instruction
 - File operations
- Arrays and Structures
 - Declaration, initialization, manipulation
 - Examples of algorithms
- Functions
 - Syntax and return values
 - Parameter passing
- Applications
 - Roots of an equation
 - Integration
 - Differentiation
- Programming in the MATLAB environment

CME 260

1. Course number and name

Properties of Materials – CME 260

2. Credits and contact hours

3 hours with a total of 3 contact hours per week(lecture).

3. Instructor's or course coordinator's name

Matthew Daly

4. Instructional Materials

James F. Shackelford, Introduction to Materials Science for Engineers, 8th Edition ISBN: 9780133826654

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

Brief description of the content of the course (catalog description)

CME 260 is an introductory course in materials engineering designed to familiarize students with the relationships between processing, structure, and properties of materials used in the manufacturing of devices. The course provides a foundational understanding of how material characteristics influence performance and functionality in engineering applications.

Prerequisites or co-requisites

CHEM 122 and CHEM 123 and PHYS 141 and MATH 181.

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Acquire a fundamental understanding of how structure underpins the emergent physical properties of materials	(1)
CLO2: Understand the practical aspects of materials processing and its influence on microstructure and properties	(2)
CLO3: Ability to perform materials selection based on evaluation of engineering design constraints and considerations (e.g., strength, processing, manufacturability)	(2)
CLO4: Ability to formulate and solve complex materials engineering problems based on the application of materials science concepts and applied mathematics	(1)
CLO4: Ability to review experimental data from materials testing to distinguish and evaluate the performance of engineering materials	(6)

7. Brief list of topics to be covered

- Classifications of materials
- Atomic bonding and atomic packing
- Densities, motifs, and crystal structures
- Crystal positions, directions, and planes
- X-ray diffraction
- Imperfections in lattices
- Kinetics and diffusion (Fick's laws)
- Mechanical behavior of materials
- Failure of materials
- Thermal behavior of materials
- Electronic properties of materials
- Phase rule and phase diagrams
- Steel phase diagram and lever rule
- Equilibrium and non-equilibrium microstructures
- Materials selection and processing considerations

CME 322

1. Course number and name:

CME 322: Environmental Engineering

2. Credits and contact hours:

3 credit hours; 3 contact hours per week, Engineering

3. Instructor's or course coordinator's name: (In past 5 years)

Amid Khodadoust

4. Instructional Materials:

- Introduction to Environmental Engineering, by Mackenzie Davis and David Cornwell, 6th Ed., McGraw-Hill, 2022.

5. Specific course information

a. brief description of the content of the course (catalog description)

The principles of environmental engineering systems for the removal of environmental pollutants using engineered systems and for the monitoring of environmental pollutants in natural systems are discussed in this course.

b. prerequisites or co-requisites

prerequisite: CHEM 122/123 ; prerequisite or corequisite: CME 211

c. indicate whether a required, elective, or selected elective (as per Table 5-1) course in the program

6. Educational objectives for the course (e.g. The student will be able to explain the significance of current research about a particular topic.)

- identify different pollutants in the environment (ABET Student Outcomes 1,7)
- write mass balance equations for pollutants (ABET Student Outcome 1)
- identify and measure environmental quality parameters (ABET Student Outcomes 1,6)
- design water treatment units (ABET Student Outcomes 1,2)
- design wastewater treatment units (ABET Student Outcomes 1,2)
- identify solid waste recycling/disposal (ABET Student Outcomes 1,7)

Explicitly indicate which of the student outcomes listed in Criterion 3 or any other outcomes are addressed by the course.

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

7. Brief list of topics to be covered

Week	Lecture No.	Topic	Chapter Sections
Week 1	Lecture 1	introduction- environmental systems; pollution; regulations	1.1, 1.5
	Lecture 2	pollutants (dissolved chemicals, solid particles) fluids, fluid flow rate, fluid container systems	2.1-2.3
Week 2	Lecture 3	mass balances: pollutant mass, pollutant concentration	2.1-2.3
	Lecture 4	mass balances: mass flow rate, mass transformation rate	2.1-2.3
Week 3	Lecture 5	mass balances: engineered and natural systems	2.1-2.3
	Lecture 6	mass balances: reactor systems	2.1-2.3
Week 4	Lecture 7	water chemistry: chemical reactions, pH, strong acids/bases	5.2-5.3
	Lecture 8	water chemistry: ions, salts, and precipitation reactions	5.3
Week 5	Lecture 9	water chemistry: weak acid/base reactions and carbonate system	5.3-5.4
	Lecture 10	water chemistry: alkalinity water treatment: overview, water treatment plant	5.4 6.1
Week 6	Lecture 11	water treatment: total hardness, ion-exchange softening	6.3
	Lecture 12	water treatment: lime-soda softening	6.3
Week 7	Lecture 13	water treatment: turbidity and colloidal solids - coagulation	6.2
	Lecture 14	MIDTERM EXAMINATION #1	
Week 8	Lecture 15	water treatment: coagulation and flocculation; mixing	6.4
	Lecture 16	water treatment: sedimentation	6.5
Week 9	Lecture 17	water treatment: sedimentation ; filtration	6.5-6.6
	Lecture 18	water treatment: filtration water treatment: disinfection	6.6 6.7
Week 10	Lecture 19	water pollution: dissolved oxygen – organics in water	7.3
	Lecture 20	water pollution: biochemical oxygen demand (BOD)	7.3
Week 11	Lecture 21	water pollution: BOD test, oxygen demand and DO in rivers	7.3
	Lecture 22	MIDTERM EXAMINATION #2	
Week 12	Lecture 23	wastewater treatment: overview, wastewater treatment plant wastewater treatment: pretreatment	8.2-8.4 8.5
	Lecture 24	wastewater treatment: primary treatment wastewater treatment: secondary treatment (microbiology)	8.6 8.7
Week 13	Lecture 25	wastewater treatment: secondary treatment (activated sludge) wastewater treatment: aeration tank (activated sludge)	8.7 8.7

	Lecture 26	wastewater treatment: secondary clarifier (activated sludge)	8.7
Week 14	Lecture 27	wastewater treatment: oxidation ponds, trickling filters	8.7
	Lecture 28	sludge treatment: thickening, de-watering, stabilization	8.11
Week 15	Lecture 29	air pollution	9.5
	Lecture 30	municipal solid waste (MSW): composition, recycling municipal solid waste: disposal of MSW in landfills	11.1,11.6 11.4
Week 16	Finals Week	<i>FINAL EXAM</i>	

ECE 210

1. Course number and name

ECE 210, Electrical Circuit Analysis

2. Credits, contact hours, and categorization.

3 credit hours, with a total of 5 contact hours per week (2 hours lecture and 3 hours lab)

3. Instructor's or course coordinator's name

Ning Jin

4. Textbook, title, author, and year

Instructional Materials

Circuit Analysis and Design, 3rd edition

Published by Michigan Publishing, © 2025

Fawwaz T. Ulaby, Michel M. Maharbiz, and Cynthia M. Furs

ISBN-13: 9781607859239

5. Specific course information

a. Brief description of the content of the course (catalog description)

Linear circuit analysis: networks, network theorems, dependent sources, operational amplifiers, energy storage elements, transient analysis, sinusoidal analysis, frequency response, filters.

b. Prerequisites or co-requisites

Credit or concurrent registration in MATH 180

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Analyze dc circuits containing resistors, independent and dependent voltage and current sources using Kirchhoff's and Ohm's Laws.	(1), (2), (7)
CLO2: Analyze complex resistive circuits using voltage division, current division, series and parallel resistor combinations and nodal analysis.	(1), (2), (7)
CLO3: Analyze arbitrarily complex circuits using nodal and mesh analysis.	(1), (2), (7)
CLO4: Analyze arbitrarily complex circuits containing operational amplifiers.	(1), (2), (7)
CLO5: Determine the transient response of first-order RC and RL circuits.	(1), (2), (7)
CLO6: Determine the response of arbitrarily complex RLC circuits using Laplace Transform. Filters, BW and resonant frequency	(1), (2), (7)
CLO7: Determine the steady-state dc-response by resistive circuits analysis.	(1), (2), (7)
CLO8: Demonstrate and verify analysis of circuits by performing experiments and simulations that validate and illustrate concepts	(1), (2), (6), (7)

7. Brief list of topics to be covered

- Units, notation; charge, current, voltage, power; elements
- Ohm's Law; Kirchhoff's Laws; equivalent circuits
- Linear Circuits
- Nodal analysis and Nodal analysis examples
- Mesh analysis and Mesh analysis examples
- Thevenin and Norton Equivalent Circuits and examples
- Maximum power transfer theorem
- Op Amp characteristics; negative feedback; ideal model
- Inverting amp; summing amp; difference amp; buffer
- Periodic/non-periodic waveforms; capacitors; inductors
- Response of RC circuit and RL circuits
- Power factor; maximum power transfer
- Unit impulse function, Laplace Transform technique
- Properties of the Laplace Transform
- Circuit Analysis Procedure using the Laplace Transform
- Inverse Laplace Transform using partial fraction expansion
- s-domain circuit element models and circuit analysis
- s-domain circuit analysis examples
- Filters, bandwidth, resonant peak/frequency

ENGR 100

1. Course number and name

Engineering Orientation – ENGR 100

2. Credits, contact hours, and categorization.

0 credit hours. 50 min/week contact hours, Engineering

3. Instructor's or course coordinator's name

Simona Narubin (Full session instructor); Department Break-out Sessions: Zhinan Wang (BioE), Alan Zdunek (ChE), Hossein Ataei (CME), Patrick Troy (CS), Vahe Caliskan (ECE), Saeed Manafzadeh (MIE)

4. Textbook, title, author, and year

Sample resumes, handouts and presentation materials available on Blackboard

5. Specific course information

a. Brief description of the content of the course (catalog description)

General Orientation course on skills pertinent to students for successful completion of their undergraduate degree. Required of all engineering students. Course Information: Satisfactory/Unsatisfactory grading only. No graduation credit. Should be taken in the first semester after acceptance into the College of Engineering.

b. Prerequisites or co-requisites

Admission to the College of Engineering.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Establish a strong foundation for academic and professional success in the College of Engineering.	(7)
CLO2: Introduce students to essential engineering mindsets and learning strategies needed for long-term growth.	(1), (7)
CLO3: Connect students with peers, mentors, organizations, and campus resources to build community and belonging	(3), (5)
CLO4: Provide early exposure to engineering majors, careers, and departmental expectations to support informed decision-making.	(4), (2), (5)
CLO5: Support a smooth transition into university life by reinforcing habits that promote confidence and persistence.	(5), (7)

7. Brief list of topics to be covered

- Engineering Career Center presentations
- Academic advising tutorial

- Introduction to engineering majors and departments
- Time management and study strategies for engineering students
- Professional communication and teamwork basics
- Engineering student organizations and involvement opportunities
- Campus resource overview (tutoring, labs, cultural centers, etc.)
- Introduction to engineering ethics and professional expectations
- Department-specific technical project modules
- Online learning modules on academic success and engineering mindset

A.3. Mathematics and Basic Science Courses.

CHEM 116

1. Course number and name

Honors and Majors General and Analytical Chemistry I

2. Credits, contact hours and categorization

5 credit hours, 120 min/week, lecture/discussion, 180 min/week lab period

3. Instructor's or course coordinator's name

Stephen Gramsch

4. Text book, title, author, and year

- a. Atkins, P.; Jones, L.; Laverman, L.; Patterson, J.; Young, K. *Chemical Principles, 8th Edition*. (McMillan Learning, 2023; ISBN 978-1-319-13798-4), with ACHIEVE for online problems and assessments. The link to obtain the textbook and ACHIEVE is provided on the Blackboard course site.
- b. UIC Chemistry Staff. *Chemistry 116 Laboratory Manual*. (McMillan Learning, 2025; ISBN 978-1-5339-7345-0). The laboratory manual is available electronically through the UIC bookstore. other supplemental materials

5. Specific course information

- a. Brief description of the content of the course (catalog description)
General and analytical chemistry with laboratory. Coverage of the fundamentals of chemistry including stoichiometry and equilibrium. Coverage of the principles of analytical chemistry, including the use of instrumentation.
- b. Prerequisites or co-requisites
Superior performance on the UIC chemistry placement examination
- c. Indicate whether a required, elective, or selected elective
Required for Honors students

6. Specific goals for the course

- a. Specific outcomes of instruction
The lecture portion of the course will address the fundamental principles of chemistry including atomic and molecular structure, chemical periodicity, stoichiometry, chemical reactivity, the behavior of gases, and the properties of liquids and solids. The laboratory portion of the course will address the principles of analytical chemistry, including the use of instrumentation and treatment of experimental data.
- b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. Brief list of topics to be covered

Discussion/Lecture Topics

- 1: The Nature of Chemistry Measurement Fundamentals
- 2: Atomic Structure
- 3: Elements and Compounds, The Mole Concept
- 4: Chemical Equations, Reaction Stoichiometry
- 5: Solution Chemistry
- 6: Patterns in Chemical Reactivity
- 7: Energy and Chemical Reactions, The Enthalpy Function
- 8: Electromagnetic Radiation
- 9: Development of Quantum Theory, Electronic Structure of Atoms
- 10: Chemical Bonding Fundamentals, Lewis Pictures and Electron Counting
- 11: Molecular Shapes, Electronic Structure of Molecules
- 12: Ideal Gas Equation of State, Behavior of Gases
- 13: Intermolecular Forces, The Liquid State
- 14: Phase Transitions and Phase Diagrams
- 15: Fundamentals of the Solid State, X-Ray Diffraction

Laboratory Topics

- 1 Data and Regression Analysis
- 2 Gravimetric Determination of Calcium as $\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$
- 3 Synthesis of Ammonium Decavanadate
- 4 Determination of the Charge on the Electron
- 5 Spectrophotometric Determination of a Hydrate
- 6 Spectrophotometric Determination of Iron in Vitamin Tablets
- 7 Acid-Base Titration
- 8 Determination of Ca^{2+} and Mg^{2+} in Natural Waters by EDTA Titration
- 9 Properties of CO_2 and Determination of NaHCO_3
- 10 Calorimetry and Hess's Law
- 11 Enthalpy and Entropy Changes for Borax Dissolution
- 12 Analysis of Ammonium Decavanadate

CHEM 118

1. Course number and name

Honors and Majors General and Analytical Chemistry II

2. Credits, contact hours and categorization

5 credit hours, 120 min/week, lecture/discussion, 180 min/week lab period

3. Instructor's or course coordinator's name

Stephen Gramsch

4. Textbook, title, author, and year

- a. Atkins, P.; Jones, L.; Laverman, L.; Patterson, J.; Young, K. *Chemical Principles, 8th Edition*. (McMillan Learning, 2023; ISBN 978-1-319-13798-4), with ACHIEVE for online problems and assessments. The link to obtain the textbook and ACHIEVE is provided on the Blackboard course site.
- b. UIC Chemistry Staff. *Chemistry 118 Laboratory Manual*. (McMillan Learning, 2026; ISBN 978-1-319-13798-4). The laboratory manual is available electronically through the UIC bookstore.

5. Specific course information

- a. Brief description of the content of the course (catalog description)
- b. General and analytical chemistry with laboratory. Coverage of the fundamentals of chemistry including atomic and molecular structure, thermodynamics, and kinetics. Coverage of principles of analytical chemistry, including the use of instrumentation.
- c. Prerequisites or co-requisites
- d. Grade of C or better in CHEM 116.
- e. Indicate whether a required, elective, or selected elective
Required for Honors students

6. Specific goals for the course

- a. Specific outcomes of instruction
The lecture portion of the course will address the fundamental principles of chemistry including intermolecular forces and the properties of liquids and solids, the properties of solutions, chemical thermodynamics, chemical kinetics, acid-base and solubility equilibria, electrochemistry, and nuclear chemistry. The laboratory portion of the course will address the principles of analytical chemistry, including the use of instrumentation and treatment of experimental data.

- b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. Brief list of topics to be covered

Discussion/Lecture Topics

- 1 Intermolecular Forces, Liquids and Solids
- 2 The Entropy Function
- 3 The Free Energy Function
- 4-6 Chemical Equilibrium
- 7-8 Chemical Kinetics
- 9-10 Acid-Base Chemistry
- 11 Solubility Equilibria
- 12-13 Electrochemistry
- 13 Coordination Chemistry
- 14 Nuclear Chemistry

Laboratory Topics

1. Safety Refresher
2. EDTA Titration of Ca^{2+} and Mg^{2+}
3. Enthalpy and Entropy Changes for Borax Dissolution
4. Ion Exchange Separation of Ni and Co
5. Determination of an Equilibrium Constant
6. Chemical Kinetics: Decomposition of H_2O_2 by Iodide
7. Mn^{2+} Standardization by EDTA Titration
8. Manganese in Steel by Spectrophotometry
9. Manganese in Steel by Atomic Absorption
10. Acid Dissociation Constant of an Indicator
11. Analysis of an Acid-Base Titration: The Gran Plot
12. Iodometric Titration of Vitamin C
13. pH Dependence of the Course of a Redox Reaction

CHEM 122

1. Course number and name

Matter and Energy – CHEM 122

2. Credits and contact hours

3 hours with a total of 4 contact hours per week

3. Instructor's or course coordinator's name

Department of Chemistry Faculty

4. Textbook, title, author, and year

Chemistry: An Atom's First Approach, 4th edition.

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

Brief description of the content of the course (catalog description)

Chem 122 is an introductory, one-semester college-level course that explores the fundamental principles of chemistry. Topics include the quantum model of the atom, periodic trends, chemical bonding, types of chemical reactions, solutions, stoichiometry, thermochemistry, and intermolecular forces, providing a strong foundation for further study in chemistry and related fields.

Prerequisites or co-requisites

Grade of C or better in CHEM 101 or appropriate score on the department placement test.

Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Understand and critically evaluate information and concepts in the natural and mathematical sciences	(1)
CLO2: Appreciate the value of and difference between scientific laws, theories, hypotheses, and speculation	(1)
CLO3: Use scientific and mathematical reasoning to make relevant distinctions among ideas	(1)
CLO4: Analyze quantitative information and draw conclusions from these analyses	(6)

7. Brief list of topics to be covered

- Introduction to Chemistry and Matter
- Measurements, Units, and Scientific Notation
- States and Properties of Matter
- Atomic Structure and Subatomic Particles
- Isotopes, Atomic Mass, and the Periodic Table
- Electromagnetic Radiation and Quantum Theory
- Electron Configurations and Periodic Trends
- Chemical Bonding: Ionic, Covalent, Metallic
- Lewis Structures, VSEPR, and Molecular Geometry
- Valence Bond Theory and Molecular Orbital Theory
- Intermolecular Forces and Physical Properties
- Chemical Reactions and Stoichiometry
- Solutions, Concentrations, and Dilutions
- Acid-Base Chemistry and Neutralization Reactions
- Precipitation and Redox Reactions
- Gas Laws and Kinetic Molecular Theory
- Thermochemistry: Energy, Heat, Work, and Enthalpy
- Calorimetry, Hess's Law, and Standard Enthalpies of Reaction

CHEM 123

1. Course number and name

Foundations of Chemical Inquiry I – CHEM 123

2. Credits and contact hours

2 hours, with online lectures and 3 hours of lab per week.

3. Instructor's or course coordinator's name

Department of Chemistry Faculty

4. Textbook, title, author, and year

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

Brief description of the content of the course (catalog description)

Chem 123 is a first-semester general chemistry laboratory course focused on experimental techniques and quantitative analysis. Students will perform experiments involving measurements of mass, volume, temperature, and spectrophotometric methods to investigate the properties of chemical substances and reactions, including the determination of composition and characteristic properties of various compounds.

Prerequisites or co-requisites

Grade of C or better in CHEM 101; and concurrent registration or Grade of C or better in CHEM 122.

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Analyze experimental data to draw conclusions about chemical systems	(6)
CLO2: Measure chemical quantities and compute chemical amounts based on measurements	(6)
CLO3: Classify chemical systems according to type and using symbolic representations	(1)
CLO4: Demonstrate an understanding of chemical composition and change using atomic level and symbolic concepts	(1)
CLO5: Perform chemical experiments with technical proficiency	(2)

7. Brief list of topics to be covered

- Introduction to chemical inquiry and laboratory safety
- Measurement of mass, volume, and temperature
- Use of spectrophotometric methods
- Quantum model of the atom
- Stoichiometry and mass calculations
- Periodicity and elemental properties
- Types of chemical reactions
- Acid-base chemistry and pH determination
- Intermolecular forces and physical properties
- Data analysis: linear regression, averages, standard deviation
- Proper laboratory techniques and instrumentation
- Experimental design, observation, and reporting

CHEM 124

1. Course number and name

Chemical Dynamics

2. Credits, contact hours and categorization

3 credit hours, 150 min/week lecture, 60 minutes/week discussion: Basic science

3. Instructor's or course coordinator's name

Department of Chemistry Faculty

4. Textbook, title, author, and year

- a. Electronic draft version of *Chemistry, Life, the Universe and Everything* (CLUE) by Melanie M. Cooper and Michael W. Klymkowsky provided at no cost.

5. Specific course information

- a. Brief description of the content of the course (catalog description)
A second semester course in chemical principles including chemical thermodynamics, phase transitions, spontaneity/equilibrium, electrochemistry, kinetics, bonding, molecular spectroscopy, coordination compounds, and buffer solutions.
- b. Prerequisites or co-requisites
Grade of C or better in CHEM 116; or Grade of C or better in CHEM 122 and Grade of C or better or concurrent registration in CHEM 123.
- c. Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course

- a. Specific outcomes of instruction
This course is a second semester course in chemical principles including thermodynamics, behaviors of solutions, acid/base reactions (Arrhenius, Bronsted, and Lewis), reaction mechanisms, oxidation reduction reactions, kinetics, equilibrium, and buffers/networked reactions. This course continues CHEM 122 Matter and Energy (General Chemistry I) with fundamental topics in introductory Physical Chemistry and Analytical Chemistry. These topics are necessary and required for studying Organic Chemistry, Biochemistry, Chemical Engineering, Medicinal Chemistry, and many other fields.

- b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. Brief list of topics to be covered

Chapter 5 Systems thinking

Chapter 6 Solutions

Chapter 7 A field guide to chemical reactions

Chapter 8 How far? How fast?

Chapter 9 Reaction systems

- Explain and model how the existence of atoms leads to the conservation of matter.
- Use appropriate models and theories to describe chemical and physical phenomena.
- Construct representations of chemical species and use them to predict chemical and physical properties.
- Explain how and why the atomic-molecular structure affects the properties of a substance, and vice versa.
- Predict and explain the energy changes associated with interactions of atoms, molecules, and ions.
- How to apply systems thinking to both molecular level and macroscopic systems.
- Identify common types of reactions and predict the products
- Understand the factors that affect the rate and extent of reactions.
- Predict and explain the outcome of coupled reactions.

CHEM 125

1. Course number and name

Foundations of Chemical Inquiry II

2. Credits, contact hours and categorization

2 credit hours, 170 minutes/week lab session, 60 minute/week discussion: Basic science

3. Instructor's or course coordinator's name

Department of Chemistry Faculty

4. Textbook, title, author, and year

- Lab Manual: CHEM 125 manual (ISBN: 9781533973474, Follette Bookstore) is available at the UIC bookstore and includes pages for your laboratory notebook.
- Labflow Software

5. Specific course information

a. Brief description of the content of the course (catalog description)

Laboratory in general chemistry including chemical thermodynamics, spontaneity, chemical equilibrium, acid-base equilibrium, electrochemistry, kinetics, bonding, order/symmetry in condensed phases, coordination compounds, and spectroscopy.

b. Prerequisites or co-requisites

Grade of C or better in CHEM 116; or Grade of C or better in CHEM 122 and Grade of C or better in CHEM 123; or Grade of B or better in CHEM 115 and department consent; and Grade of C or better or concurrent registration in CHEM 124: Chemical Dynamics

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

Analyze experimental data to draw conclusions about chemical systems, construct and interpret pH titration curves, making analyses related to acid-base reactions and their equilibrium, design, prepare, and evaluate buffer solutions, measure thermodynamic quantities and relate these to one another and to chemical equilibria, demonstrate the ability to interpret spectroscopic data in terms of atomic and molecular structure, demonstrate how to design and interpret experiments based on electrochemical phenomena, perform experiments to determine the kinetics of chemical reactions. As the course proceeds, students will also learn important practical techniques, such as safe handling and disposal of chemicals in the laboratory, proper use of bench-top instrumentation and lab techniques for chemical analysis, and the analysis of scientific data from chemical systems to draw conclusions.

- b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. Brief list of topics to be covered

13 in-lab experiments:

- Analysis of Glass Cleaners Parts I-III
- K_a of an Indicator
- Stabilization of pH with Buffers
- Enthalpy and Entropy Changes for Borax Dissolution
- Determination of the Charge on the Electron
- Electrochemical Measurement and Batteries Parts I-III
- Electrochemical Measurement and Batteries Parts IV-VI
- Iodide Ion Catalyzed Decomposition of H_2O_2 : An Initial Rates Kinetics Approach
- Kinetics of Dye Bleaching
- Kinetics Experiment of the Reaction of Coumarin-102 with Base; Parts 1 & II
- Kinetics Experiment of the Reaction of Coumarin-102 with Base: Part III
- Fluorescence, Quenching, and DNA Part I
- Fluorescence, Quenching, and DNA Parts II & III

CHEM 222

1. Course number and name

Analytical Chemistry

2. Credits, contact hours and categorization

4 credit hours, Lecture 100 min/week; two, 3 hr. lab sessions per week; Basic science

3. Instructor's or course coordinator's name

Emil Tripa

4. Textbook, title, author, and year

- Online Macmillan Learning ACHIEVE® access (includes e-Textbook and e-homework) The e-Textbook is Quantitative Chemical Analysis (10th edition) by Harris and Lucy.
- other supplemental materials: e-Homework: Use the "Achieve" button on Blackboard to access the required materials homework and e- textbook. Lab Manuals and materials.

5. Specific course information

- Brief description of the content of the course (catalog description)
Theory and application of chemical equilibria and instrumentation in quantitative analysis. Includes two weekly three-hour laboratories.
- Prerequisites or co-requisites
Course Information: Prerequisite(s): Grade of C or better in CHEM 114; or Grade of C or better in CHEM 124 and Grade of C or better in CHEM 125; or Grade of C or better in CHEM 118 or the equivalent.
- Indicate whether a required, elective, or selected elective
Required

6. Specific goals for the course

a. Specific outcomes of instruction:

This course will support your development in practical ways: learning new Chemistry lab skills, ability to solve problems, and thinking about how the broad practical questions generate the Analytical Chemistry method. In the lab, you will learn to solve analytical problems using correct technique, analysis methods, and drawing appropriate conclusions. In the lecture, we will examine the underlying chemical principles and relate what we are learning to real-world problems.

Laboratory Objectives:

- Conduct experiments in quantitative and qualitative analysis of atoms, ions, molecules and their mixtures.
- Use the scientific method to: Collect and record quantitative and qualitative data and observations during experiments. Process data according to the best modern practice in analytical chemistry, Apply theories or models to interpret

the results and observations of experiments. Demonstrate validity of known scientific laws and theories of analytical chemistry.

- Create and maintain an effective laboratory notebook / record. Gain laboratory experience in preparation of samples for analysis, qualitative analysis, and quantitative analysis.
- Gain hands-on experience in techniques and skills of a modern chemical analyst: gravimetric analysis, volumetric analysis, titrimetry, electrochemical analysis (pH, ion sensors), photochemical analysis (spectrophotometry and atomic absorption).
- Learn and use data-processing and graph-making software (spreadsheets) to process the raw data and compute uncertainties in the collected data.
- Assess the observed uncertainties against accepted uncertainties documented in the literature for an analytical method.
- Use mathematics to model trends in data for various analytical tasks such as calibration, interpolation, extrapolation.
- Communicate the results in a written lab report which documents all the data and lab notes as evidence that supports the results and the interpretation provided.

Lecture Objectives:

- Learn and understand terms and concepts specific to the field of Analytical Chemistry related to the analytical process, method, instrumentation, and modern theories.
- Learn and understand operation, specifications, best practice, and limitations of the chemistry lab tools associated with the classical methods of analysis (gravimetric, volumetric, titrimetric).
- Learn and understand the operation, theory and applications of modern electrochemical and spectrophotometric instruments in analytical chemistry.
- Describe the mathematical formalism and solve conceptual and numerical problems in the following fields commonly used in the analytical methods and models:
 - >Statistics (uncertainties, confidence interval, quality assurance)
 - >Chemical equilibria (acid-base, solubility, complexation, redox, activities)
 - >Titration curves (inflexion points, stoichiometry, logarithmic scale)
 - >Electrochemical analysis by potentiometry
 - >Signal vs concentration calibration curves, trends, fitting equation, interpolation, extrapolation, integration, derivation

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO #3: an ability to communicate effectively with a range of audiences.

SLO #5: An ability to function effectively in a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

7. Brief list of topics to be covered:

Lecture:

1. The Analytical Process
2. Chemical Measurements
3. Basic Analytical Tools
4. Experimental Error
5. Statistics
6. Quality Assurance
7. Chemical Equilibrium
8. Titrations
9. Thermodynamic activity of electrolytes
10. Monoprotic and Polyprotic acid-base equilibria
11. Acid/Base Titration Curves
12. EDTA titrations
13. Electrochemical analysis
14. Electrodes and Potentiometry
15. Redox titrations
16. Spectrophotometry and applications
17. Atomic Spectroscopy

Lab:

- Exp. 1 Calibration of Volumetric Glassware Experimental Error
- Exp. 2 - Gravimetric Determination of Ca as $\text{CaC}_2\text{O}_4 \times \text{H}_2\text{O}$
- Exp. 3 - Statistical Evaluation of Acid-Base Indicators
- Exp. 4 - Spectrophotometric Determination of Iron in Vitamin Tablets
- Exp. 5 - Preparing Standard Acid and Base
- Exp. 6 - Using a pH Electrode for an Acid-Base Titration
- Exp. 7 - EDTA Titration of Ca^{2+} and Mg^{2+} in Natural Waters
- Exp. 8 - Microscale Spectrophotometric Determination of Iron in Foods
- Exp. 9 - Ion-Exchange Separation and Spectrophotometric Determination of Nickel and Cobalt
- Exp. 10 - Determination of Mn in Steel (Standard Addition)
- Exp. 11 - Iodimetric Titration of Vitamin C
- Exp. 12 - Determination of I^- in Salt (Ion-Selective Electrode, Standard Addition)

CHEM 232

1. Course number and name

Organic Chemistry I, Structure and Function

2. Credits, contact hours and categorization

3 credit hours, 150 min/week lecture; 50 min/week discussion; Basic science

3. Instructor's or course coordinator's name

Maria Yermolina, Justin Mohr

4. Text book, title, author, and year

Organic Chemistry, Klein, 4th or 5th Edition, or digital e-text (Wiley Plus Access)

a. other supplemental materials

WileyPLUS online HW access card

iClicker2 (www.iclicker.com)

Molecular model set (Prentice Hall, HGS, or anything similar)

5. Specific course information

a. Brief description of the content of the course (catalog description)

First semester of a one-year sequence. Structure, reactivity, and synthesis of organic molecules.

b. Prerequisites or co-requisites

Grade of C or better in CHEM 116; or Grade of C or better in CHEM 122 and Grade of C or better in CHEM 123.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

The emphasis of the course is in understanding the structure, reactivity, and synthesis of organic molecules.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

1. Molecular Representation
2. Acids & Bases
3. Alkanes & Cycloalkanes
4. Stereoisomerism
5. Infrared Spectroscopy
6. NMR Spectroscopy
7. Chemical Reactivity & Mechanism

8. Alkyl Halides (start)
9. Addition to Alkenes
10. Alkynes
11. Radical Reactions
12. Synthesis
13. Alcohols & Phenols
14. Ethers & Epoxides

CHEM 233

1. Course number and name

Synthesis Techniques Laboratory

2. Credits, contact hours and categorization

2 credit hours; one, 3 hour and 50-minute lab session per week; Basic science

3. Instructor's or course coordinator's name

Maria Yermolina

4. Text book, title, author, and year

Experimental Organic Chemistry: A Miniscale and Macroscale Approach, Gilbert & Martin, 6th Edition, 2016 Brooks/Cole.

a. other supplemental materials

Organic Chemistry Laboratory Manual, Landrie, McQuade, Yermolina 9th Edition
Chemistry Lab Notebook – 100 pages (<http://www.hmpublishing.com/products/organic-chemistry-spiral-bound.html>), ISBN 978-1-930882-46-1).

5. Specific course information

a. Brief description of the content of the course (catalog description)

Introductory organic chemistry laboratory. Basic techniques (distillation, crystallization, chromatography, MP and BP), reactions (substitution, elimination, Diels-alder, oxidation-reduction), instrumentation (gas and liquid chromatography, IR.).

b. Prerequisites or co-requisites

Grade of C or better in CHEM 230 or Grade of C or better or concurrent registration in CHEM 232; and Grade of C or better in CHEM 122 and Grade of C or better in CHEM 123; or Grade of C or better in CHEM 116 and Grade of C or better in CHEM 118; or consent of the instructor.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction:

- Employ appropriate and safe laboratory techniques to perform experiments, collaborating with partners.
- Interpret the results of spectroscopic and chromatographic instruments to analyze reactions and products.
- Employ laboratory skills and content knowledge to determine the identity of unknown compounds.
- Minimize risk to self and others by adhering to documented and verbalized laboratory safety policies.
- Operate instrumentation, such as an gas-liquid chromatographer, infrared spectrometer, melting point device, bench NMR and polarimeter, independently to acquire data relevant to an experiment.

- Assemble glassware apparatuses to perform techniques such as distillation, extraction and chromatography.
- Document laboratory procedures, report the results and observations, analyses and conclusions in a laboratory notebook according to scientific standards.
- Performing quantitative calculations and interpreting the results.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO #3: An ability to communicate effectively with a range of audiences.

SLO #5: An ability to function effectively in a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

Brief list of topics to be covered:

Lab 1: Safety, IR Primer

Lab 2: Chromatographic Methods and Thin-layer chromatography

Lab 3: Simple & Fractional Distillation GC Analysis

Lab 4: Steam Distillation: Chirality and Stereoisomerism

Lab 5: Acid-Base Extraction, Recrystallization, Solubility

Lab 6: Identification of Unknown Organic Compounds by IR and NMR

Lab 7: Preparation of Alkyl Halides by SN1 & SN2 reactions

Lab 8: Dehydrohalogenation reactions: E1 and E2 reactions

Lab 9: Rearrangement of carbocations (pinacol). Synthesis of semicarbazone derivative

Lab 10: Reduction of ketone with NaBH₄

Lab 11: Williamson Ether Synthesis: Phenacetin and acetaminophen

Lab 12: Lab Practical: Synthesis Chemiluminol and Chemiluminescence test

CHEM 234

1. Course number and name

Organic Chemistry II: Chemical Synthesis

2. Credits, contact hours and categorization

3 credit hours, 150 min/week lecture; 50 min/week discussion; Basic science

3. Instructor's or course coordinator's name

Maria Yermolina and Laura Anderson

4. Textbook, title, author, and year

Organic Chemistry, Klein, 4th Edition (2013) Wiley, ISBN-13: 978-1119659594

b. other supplemental materials

iClicker2 (www.iclicker.com)

Molecular model set (Prentice Hall, HGS, or anything similar)

5. Specific course information

a. Brief description of the content of the course (catalog description)

Second semester of a one-year sequence. Structure, reactivity, and synthesis of organic molecules.

b. Prerequisites or co-requisites

Grade of C or better in CHEM 232; and Grade of C or better in CHEM 124 and Grade of C or better in CHEM 125; or Grade of C or better in CHEM 118.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

Chemical synthesis is the art of applying chemical reactions to form new bonds in an optimal order to achieve the construction of target molecules. In this course, you will learn basic transformations for forming C–C, C–O, and C–N bonds to make organic molecules. You will also learn electron-pushing mechanisms that describe the flow of electrons in these reactions that make them thermodynamically favorable and allow you to distinguish similarities in reactivity patterns. This foundational knowledge will then be combined in multi-step questions for you to learn about functional group compatibility and how to think backwards to design a synthesis when provided with the desired molecular target. Finally, you will also learn common analytical techniques such as IR and NMR spectroscopy, which are used in chemical synthesis to identify reaction products. While some nomenclature and structural properties of molecules will be reviewed, these concepts were covered in Chem 232 and are considered a prerequisite for this course.

- Develop a functional understanding of basic organic reactions and how they can be used in sequences to synthesize organic molecules.

- Be able to critically reason how to convert functional groups, how to install a new functional group on a given compound, and to build more complex molecules from simple reagents.
- Analyze a given reaction and be able to sketch an arrow pushing mechanism to explain how electrons flow between electrophiles and nucleophiles to create new bonds to give the product.
- Become familiar with common transition metal catalysts used routinely for organic reactions.
- Apply 3-dimensional models to the interaction of molecules in solution to predict the kinetically favored stereoisomer.
- Analyze experimental spectroscopy data to determine the outcome of a reaction and predict the structure of an unknown compound.
- Understand the fundamental addition and elimination reactivity that governs carbonyl chemistry and many biochemical reactions.
- Identify the common organic functional groups in biomolecules such carbohydrates, peptides, and nucleic acids that direct their reactivity following the same principles as small molecules.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Brief list of topics to be covered

1. Alcohols, Ethers and Epoxides
2. IR spectroscopy and Mass Spectrometry
3. ^1H NMR and ^{13}C Spectroscopy
4. Conjugated π systems and pericyclic reactions
5. Aromatic compounds
6. EAS and SnAr
7. Organometallics
8. Aldehydes and Ketones
9. Carboxylic Acids & Derivatives
10. Enolate Chemistry
11. Amines
12. Carbohydrates
13. Amino acids, Peptides and Proteins
14. Lipids
15. Polymers

CHEM 342

1. **Course number and name**
Physical Chemistry I
2. **Credits, contact hours and categorization**
3 credit hours, 150 min/week lecture; 50 min/week discussion; Basic science
3. **Instructor's or course coordinator's name**
George Papadantonakis
4. **Text book, title, author, and year**
Physical Chemistry, 5th edition by Silbey, Alberty, Papadantonakis, Bawendi Wiley
 - a. other supplemental materials
General Physics and Calculus textbook, Handouts
5. **Specific course information**
 - a. **Brief description of the content of the course (catalog description)**
Thermodynamics of gases, solutions, reaction equilibria, and phase transitions.
 - b. **Prerequisites or co-requisites**
A grade of C or better in MATH 181; and a grade of C or better in CHEM 118 or concurrent registration, or a grade of C or better in CHEM 222; and a grade of C or better in PHYS 142; and a grade of C or better or concurrent registration in MATH 210.
 - c. **Indicate whether a required, elective, or selected elective**
Required
6. **Specific goals for the course**
 - a. **Specific outcomes of instruction**
The emphasis of the course is in understanding thermodynamics of gases, solutions, reaction equilibria, and phase transitions. We will also cover the thermodynamics of macromolecules, such as proteins. We will study terms and ideas you have learned and used before, such as q , w , ΔG , ΔH , ΔS , K_{eq} .
 - b. **Explicitly indicate which of the student outcomes are addressed by the course.**
SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
7. **Brief list of topics to be covered**

From Physical Chemistry, 5th edition by Silbey, Alberty, Papadantonakis, and Bawendi

Chapter 1: in its entirety

Chapter 17: from 17.1 through 17.6 and the handout

Chapter 2: in its entirety

Chapter 3: in its entirety

Chapter 4: in its entirety

Chapter 5: 5.1 through 5.9

Chapter 6: 6.1 – 6.8 and 6.9

Chapter 7: in its entirety

Topics:

1. Properties of gases
2. The first law of thermodynamics
3. Thermochemistry
4. The second law of thermodynamics
5. Chemical equilibrium
6. Phase equilibrium
7. The thermodynamics of solids
8. Thermodynamics of nonelectrolytic solutions
9. Thermodynamics of electrolytic solutions

CHEM 346

1. Course number and name

Physical Chemistry II

2. Credits, contact hours and categorization

3 credit hours, 150 min/week lecture; 50 min/week discussion; Basic science

3. Instructor's or course coordinator's name

Preston T. Snee

4. Textbook, title, author, and year

"Free Energy" an open educational resource textbook for Physical Chemistry. ISBN 979-8-218-34940-0

a. other supplemental materials

- Other useful books: Atkins, Physical Chemistry, Zemansky, Heat and Thermodynamics

5. Specific course information

a. Brief description of the content of the course (catalog description)

Kinetic and molecular theory of gases; introduction to the principles of quantum mechanics with application to model systems, multi-electron atoms, diatomic molecules, and bonding.

b. Prerequisites or co-requisites

Grade of C or better in CHEM 342 and Grade of C or better in MATH 210.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

c. Specific outcomes of instruction

The emphasis of the course is in understanding kinetic and molecular theory of gases and the principles of quantum mechanics.

d. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

- The Kinetic Theory of Gas
 - Probability vs. Probability Distribution
 - The Boltzmann Distribution
 - Average and RMS Velocities
 - Average Relative Velocity and Collision Frequency
- Boltzmann Statistics
 - The Blackbody Radiator

- Heat Capacity of Solids
- Introduction to the Schrödinger Equation
 - Einstein's Theory of Relativity
 - The Schrödinger Equation
 - Born Interpretation
 - The Eigenvalue Equation and Operators
 - The Freewave Potential
- Potential Surfaces and the Heisenberg Uncertainty Principle
 - Potential Energy Surfaces
 - Complex Potential Energy Surfaces: Vibration
 - Uncertainty and Superposition: Wavefunctions as Waves
 - Perturbation Theory
- Multidimensional Quantum Mechanics and Rotation
 - Multidimensional Free waves and the Particle in a Cube
 - Rotational Quantum Mechanics: Introduction
 - Wavefunctions
 - Spin Angular Momentum
 - Angular Momentum Operators
 - Addition of Angular Momentum and Term Symbols
- The Hydrogen Atom
 - The Bohr Model
 - The Hydrogen Schrödinger Equation
 - Hydrogen Radial Wavefunctions
 - Spin-Orbit Coupling
 - Spectroscopy
 - Multielectron Atoms and Exchange
- Kinetics
 - Rate Laws and 1st Order Kinetics
 - Rate Laws and 2nd Order Kinetics
 - Consecutive Reactions
 - Reaction Mechanisms
 - Biological Kinetics

MATH 180

1. Course number and name

Calculus I – MATH 180

2. Credits and contact hours

4 hours, with a total of 5 contact hours per week (3 hours of lecture and 2 hours of Discussion/Recitation)

3. Instructor's or course coordinator's name

Duha Hamed, Will Ash, McKinley Meyer

4. Instructional Materials

Briggs, Calculus Early Transcendentals, 3rd Edition Digital Update, William L. Briggs, Lyle Cochran, Bernard Gillett, Eric Schulz.

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

Brief description of the content of the course (catalog description)

Math 180 is an introductory calculus course that provides students with a foundational understanding of the mathematical study of change. The course covers key concepts including limits, rates of change, differentiation, and basic integration techniques, along with applications of these topics in various contexts. Students will engage in active problem solving, collaborative learning, and critical thinking to develop their analytical skills. The course is designed to support students from diverse mathematical backgrounds, ensuring all students gain the knowledge and skills necessary to succeed in subsequent calculus courses and STEM-related fields.

Prerequisites or co-requisites

Grade of C or better in MATH 121 or appropriate performance on the department placement test.

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Compute limits, derivatives, and integrals.	(1)
CLO2: Analyze functions using limits, derivatives, and integrals.	(1)
CLO3: Recognize the appropriate tools of calculus to solve applied problems	(1), (7)
CLO4: Understand and critically evaluate information and concepts in the natural and mathematical sciences.	(1)
CLO5: Analyze quantitative information and draw conclusions from these analyses.	(1)

7. **Brief list of topics to be covered**

- The Idea of Limits
- Definitions of Limits
- Techniques for Computing Limits
- The Squeeze Theorem
- Infinite Limits
- Limits at Infinity
- Continuity
- Introducing the Derivative
- Working with Derivatives
- Rules of Differentiation
- Product and Quotient Rules
- Derivatives of Trigonometric Functions
- Derivatives as Rates of Change
- The Chain Rule
- Implicit Differentiation
- Related Rates
- Derivatives of Logarithmic and Exponential Functions
- Derivatives of Inverse Trigonometric Functions
- Maxima and Minima
- The Mean Value Theorem
- What Derivatives Tell Us
- Graphing Functions
- Optimization Problems
- Linear Approximation and Differentials
- L'Hôpital's Rule
- Anti-derivatives
- Approximating Areas Under Curves
- Definite Integrals
- The Fundamental Theorem of Calculus
- Working with Integrals

MATH 181

1. Course number and name

Calculus II – MATH 181

2. Credits and contact hours

4 hours, with a total of 5 contact hours per week (3 hours of lecture and 2 hours of Discussion/Recitation).

3. Instructor's or course coordinator's name

Martina Bode, John Steenbergen, Victor Protsak.

4. Instructional Materials

Early Transcendentals (3rd Edition, Digital Update)

by William L. Briggs, Lyle Cochran, Bernard Gillett, and Eric Schulz

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

Brief description of the content of the course (catalog description)

Math 181, the second course in the three-semester calculus sequence, builds on concepts from Calculus I and deepens the study of single-variable calculus. The course covers advanced integration techniques and applications, introduces sequences and series, and explores curve analysis using parametric equations and polar coordinates.

Prerequisites or co-requisites

Grade of C or better in MATH 180.

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Students will be able to apply various techniques of integration to evaluate definite and indefinite integrals.	(1)
CLO2: Students will be able to compute arc length and volumes of solids of revolution using integral methods.	(1)
CLO3: Students will be able to use integration in solving applied problems from geometry and physical sciences	(1)
CLO4: Students will be able to represent and analyze curves using parametric equations and polar coordinates.	(1)
CLO5: Students will be able to determine the convergence or divergence of infinite sequences and series.	(1)
CLO6: Students will be able to represent functions such as power series and find intervals of convergence.	(1)

7. Brief list of topics to be covered

- Review of Definite Integrals

- Review of the Fundamental Theorem of Calculus
- Working with Integrals
- Substitution
- Velocity and Net Change
- Regions between Curves
- Volumes by Slicing
- Length of Curves
- Physical Applications
- Integration by Parts
- Trigonometric Integrals
- Trigonometric Substitutions
- Partial Fractions
- Numerical Integration
- Improper Integrals
- Overview of Sequences and Series
- Sequences
- Infinite Series
- Divergence and p-test
- Comparison Tests
- Alternating Series and Absolute Convergence
- Ratio and Root Tests
- Approximating Functions with Polynomials
- Properties of Power Series
- Taylor Series
- Working with Taylor Series
- Parametric Equations
- Polar Coordinates

MATH 210

1. Course number and name

Calculus III – MATH 210

2. Credits and contact hours

3 hours, with a total of 4 contact hours per week(3 hours for lecture and 1 hour for Discussion).

3. Instructor's or course coordinator's name

Whyte, Van Limbeek, Dumas

4. Instructional Materials

Calculus, Early Transcendentals, by W. Briggs and L. Cochran, 3rd edition.

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

Brief description of the content of the course (catalog description)

Math 210, focuses on multi-dimensional analysis, building on the one-dimensional concepts learned in Calculus I and II. Students explore vectors, vector fields, functions of several variables, partial derivatives, line integrals, and multi-variable integrals, gaining a foundation in vector calculus applicable to engineering, physics, mathematics, biology, and beyond.

Prerequisites or co-requisites

Grade of C or better in MATH 181.

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Understand and critically evaluate information and concepts in the natural and mathematical sciences.	(1)
CLO2: Analyze quantitative information and draw conclusions from these analyses.	(6)

7. Brief list of topics to be covered

- Vectors in the plane and in space
- Distance and spheres
- Dot product and cross product
- Work of a force, torque
- Vector and parametric equations of a line

- Equations of planes
- Distance from a point to a line
- Cylinders and quadratic surfaces
- Vector-valued functions
- Calculus of vector-valued functions
- Motion concepts: velocity, acceleration, speed, motion in a gravitational field
- Arc length (Cartesian coordinates)
- Functions of 2 variables: graphs, level curves
- Functions of 3 variables: level surfaces
- Multivariable calculus: limits, two-path test
- Partial derivatives (first and higher order)
- Clairaut's Theorem
- Differentiability
- Chain Rule and implicit differentiation
- Gradient and directional derivative
- Tangent planes
- Linear approximation and differential
- Local extrema, critical points, second derivative test
- Absolute optimization
- Lagrange Multipliers and optimization problems
- Double integrals: over rectangles, general regions, changing order, volumes, area
- Double integrals in polar coordinates
- Triple integrals: volumes, masses, cylindrical coordinates, spherical coordinates
- Center of mass formulas
- Plane transformations, Jacobian, change of variables
- Vector fields: radial, gradient, potential
- Line integrals of scalar functions
- Integrals of vector fields: circulation, flux, work of force
- Conservative fields, potentials, path independence, FTC for vector fields
- Green's Theorem: circulation, flux, area
- Divergence and curl in 3D
- Surface integrals of scalar functions
- Surface area elements (spherical, cylindrical, graph cases)
- Flux of a vector field through a surface
- Stokes' Theorem
- Divergence Theorem

MATH 220

1. Course number and name

Introduction to Differential Equations – MATH 220

2. Credits and contact hours

3 hours, with a total of 4 contact hours per week (3 hours of lecture and 1 hour of discussion).

3. Instructor's or course coordinator's name

Irina Nenciu, Victor Protsak, Andrew Shulman

4. Instructional Materials

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

Brief description of the content of the course (catalog description)

MATH 220 covers techniques and applications of differential equations, first and second order equations, Laplace transforms, series of solutions, graphical and numerical methods, and partial differential equations.

Prerequisites or co-requisites

Grade of C or better in MATH 210.

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Understand and critically evaluate information and concepts in the natural and mathematical sciences involving differential equations.	(1)
CLO2: Analyze quantitative information based on calculus and differential equation techniques and draw conclusions from these analyses.	(6)
CLO3: Recognize basic types of differential equations and choose an appropriate method of solving them or analyzing them qualitatively or numerically.	(1)
CLO4: Follow calculus-based algorithms for solving differential equations of special types explicitly or numerically and perform all necessary calculations by hand.	(1)
CLO5: Identify mathematical models used in several applications to physics, engineering, and biological sciences and apply them to analyze word-based scientific scenarios quantitatively, using differential equations.	(1)

7. Brief list of topics to be covered

- First-Order Differential Equations
- Second-Order Differential Equations
- Systems of Differential Equations

- Laplace Transforms
- Series Solutions and Eigenvalue Problems
- Partial Differential Equations
- Applications and Mathematical Modeling
- Graphical and Computational Methods

PHYS 141

1. Course number and name

General Physics I (Mechanics) – PHYS 141

2. Credits and contact hours

4 hours, with a total 4 contact hours per week (2 hours of lecture and 2 hours of lab).

3. Instructor's or course coordinator's name

Ryan Hazelton

4. Instructional Materials

University Physics, Volume 1 from OpenStax

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

Brief description of the content of the course (catalog description)

Physics 141 is a calculus-based course focused on classical mechanics. It is the first in a 2-semester introductory sequence; the second course covers electricity and magnetism and some optics. This course uses calculus at an intermediate level. The second semester uses calculus at a substantially greater depth. For mechanics, basic knowledge of polynomial and trigonometric differentiation and integration will be used.

Prerequisites or co-requisites

Grade of C or better or concurrent registration in MATH 180; or approval of the department; and Grade of C or better in PHYS 100 or adequate performance on the departmental placement test.

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Understand and critically evaluate information and concepts in the natural and mathematical sciences.	(1)
CLO2: Use and understand scientific method to analyze ideas and obtain knowledge	(1), (6)
CLO3: Appreciate the value of and difference between scientific laws, theories, hypotheses, and speculation	(1)
CLO4: Use scientific and mathematical reasoning to make relevant distinctions among ideas	(1)
CLO5: Think critically about contemporary issues in science and technology	(4)
CLO6: Analyze quantitative information and draw conclusions from these analyses	(6)

7. Brief list of topics to be covered

- Kinematics in one and two dimensions
- Vectors and vector operations
- Newton's laws of motion
- Linear momentum and collisions
- Work and kinetic energy
- Potential energy and conservation of energy
- Rotational kinematics and rotational energy
- Rotational dynamics and torque
- Static equilibrium
- Simple harmonic motion

PHYS 142

1. Course number and name

General Physics II (Electricity and Magnetism) – PHYS 142

2. Credits and contact hours

4 hours, with a total 4 contact hours per week (2 hour of lecture and 2 hours of lab).

3. Instructor's or course coordinator's name

Ryan Hazelton

4. Instructional Materials

University Physics, Volumes 2+3 from OpenStax

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

Brief description of the content of the course (catalog description)

Physics 142 is a calculus-based course focused on electricity and magnetism, as well as some waves and optics. It is the second in a 2-semester introductory sequence; the first course covers mechanics. This course uses calculus at a fairly deep level. Knowledge of differentiation and integration of 1-variable functions is assumed, as well as simple 1-D differential equations.

Prerequisites or co-requisites

Grade of C or better in PHYS 141; and Grade of C or better or concurrent registration in MATH 181.

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Understand and critically evaluate information and concepts in the natural and mathematical sciences.	(1)
CLO2: Use and understand scientific method to analyze ideas and obtain knowledge	(1), (6)
CLO3: Appreciate the value of and difference between scientific laws, theories, hypotheses, and speculation	(1)
CLO4: Use scientific and mathematical reasoning to make relevant distinctions among ideas	(1)
CLO5: Logically and clearly communicate experimental results and observations to others.	(3)
CLO6: Analyze quantitative information and draw conclusions from these analyses	(6)

7. Brief list of topics to be covered

- Electrostatics (Coulomb's law, electric fields, electric potential)
- Electric currents and DC circuits (Ohm's law, Kirchhoff's rules, energy in circuits)

- Magnetic fields and magnetic materials
- Electromagnetic induction (Faraday's law, Lenz's law)
- AC circuits (RLC circuits, resonance, phase relationships)
- Maxwell's equations and electromagnetic waves
- Wave phenomena (interference, diffraction, polarization)
- Geometrical optics (lenses, mirrors, reflection, refraction)

A.4. English Courses

ENGL 160

1. Course number and name

Academic Writing I – ENGL 160

2. Credits and contact hours

3 credit hours, with a total of 3 contact hours per week.

3. Instructional Materials

Dr. Angela Dancey

4. Instructional Materials

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

Brief description of the content of the course (catalog description)

This course develops students' rhetorical awareness by having them write in multiple genres and tailor communication to specific audiences. Students learn effective writing strategies through genre analysis, drafting, and substantial revision. Emphasis is placed on critical thinking, clear and concise writing, and constructing coherent paragraphs. The course also aims to replace unproductive writing habits with effective, transferable skills while fostering a more confident and enjoyable writing experience.

Prerequisites or co-requisites

Eligibility as determined by performance on the Department placement test.

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Demonstrate rhetorical awareness of audience through different genre-based assignments.	(3)
CLO2: Read texts in a variety of disciplines and genres, using critical reading strategies.	(1), (3)
CLO3: Draw connections at the level of ideas across multiple texts.	(1), (3)
CLO4: Defend a position in relation to the range of ideas surrounding a topic.	(1), (3)
CLO5: Demonstrate knowledge of writing as a process, including peer/instructor feedback, from draft to final revision.	(3), (5), (7)
CLO6: Demonstrate control over organization and sentence structure.	(3)

7. Brief list of topics to be covered

Students complete multiple writing projects and reflective exercises that develop skills in audience awareness, genre conventions, and revision. Each section covers different topics but adheres to the following framework:

- Personal Narrative (WP1)
 - Writing a first-person story about a meaningful life experience.
 - Focus: Show, not tell; narrative arc; dialogue; reflection.
- Listicle/Curated List (WP2)
 - Writing a list-based article of at least five items around a theme.
 - Focus: Organization, audience engagement, persuasive language, ranking (optional).
- Open Letter (WP3)
 - Writing a public letter addressing a person/group to argue a position.
 - Focus: Argumentation, rhetorical appeals (ethos, pathos, logos), source integration.
- Final Reflection (WP4)
 - Reflecting on personal growth and learning during the semester.
 - Focus: Connecting course learning outcomes to personal writing experiences.
- Supporting Activities
 - Weekly Blackboard journals to develop reflection and critical thinking.
 - Peer review and instructor feedback to strengthen drafting and revision skills.
 - Writing Center tutoring sessions to improve clarity and organization.

ENGL 161

1. Course number and name

Academic Writing II – ENGL 161

2. Credits and contact hours

3 credit hours, with a total of 2.5 contact hours per week.

3. Instructor's or course coordinator's name

Eman Elturki

4. Instructional Materials

Kessler, J., Bennett, M., & Primeau, S. (Eds.). (2023). Writing for inquiry and research. Windsor & Downs Press

Greene, S., & Lidinsky, A. (2012). From inquiry to academic writing: A text and reader (2nd ed.). Macmillan.

Course materials, discussion worksheets, and additional resources will be provided through Blackboard.

5. Specific course information

Brief description of the content of the course (catalog description)

This course examines sustainability on university campuses, exploring the environmental impact of campus facilities, operations, and community actions. Students engage in a semester-long research project related to UIC's five sustainability commitments: carbon neutrality, zero waste, net zero water, biodiversity, and transformative scholarship.

Course assignments guide students through the research process, including an annotated bibliography, research proposal, literature review, and a final research paper. Students will also deliver an oral presentation of their findings. The course emphasizes critical reading, academic writing, research skills, synthesis of information, and effective communication of results.

Prerequisites or co-requisites

ENGL 160 or the equivalent

6. Specific goals for the course

Course Learning Objective	Mapped ABET Outcomes
CLO1: Conduct academic research drawing from multiple sources in multiple media.	(7) , (3)
CLO2: Read texts in a variety of disciplines and genres, using critical reading strategies.	(1), (3)
CLO3: Defend a position in relation to the range of ideas surrounding a topic.	(1), (3)
CLO4: Construct a logically supported argument.	(1), (3)
CLO5: Demonstrate knowledge of writing as a process, including consideration of peer and instructor feedback, from initial draft to final revision.	(3) , (7 d)
CLO6: Demonstrate control over organization and sentence construction.	(3)

7. Brief list of topics to be covered

Students complete multiple writing projects and reflective exercises that develop skills in research, argumentation, synthesis, and revision. Each section covers different topics but adheres to the following framework:

- Annotated Bibliography (WP1)
 - Summarize, evaluate, and reflect on 6 sources related to sustainability.
 - Focus: Source credibility, summary clarity, critical reflection, proper citation.
- Research Proposal (WP2)
 - Present a research question, methodology, and significance of a study related to campus sustainability.
 - Focus: Clear research question, structured plan, persuasive rationale.
- Literature Review (WP3)
 - Synthesize a minimum of 5 scholarly sources, identifying patterns, gaps, and relevance.
 - Focus: Organization, synthesis, comparative analysis, thematic connections.
- Research Paper (WP4)
 - Write an 8–10 page essay presenting a clear thesis supported by at least 6 sources.
 - Focus: Argument development, evidence integration, formal academic style.
- Oral Research Presentation (WP5)
 - Share findings from the research paper using slides and structured 5–7 minute presentation.
 - Focus: Clear communication, visual support, engaging delivery, audience awareness.

Appendix B – Faculty Vitae

Faculty Vitae: Ezinne C. Achinivu-Ibagere

1. Education

Ph.D., Chemical and Biomolecular Engineering, North Carolina State University, 2014

M.S., Chemical and Biomolecular Engineering, North Carolina State University, 2012

B.S., Chemical Engineering, University of Maryland, College Park, 2010

2. Academic Experience

Assistant Professor, University of Illinois Chicago, 2022–Present

3. Non-Academic Experience

AAAS Science & Technology Policy Fellow, U.S. Department of Energy, 2020–2021

Postdoctoral Fellow, Sandia National Laboratories, 2018–2020

Guest Postdoctoral Fellow, Lawrence Berkeley National Laboratory, 2018–2020

Fulbright Postdoctoral Fellow, Chaire ABI, 2017–2018

Research Scientist/Chemist, 525 Solutions, 2015–2016

4. Certifications, or professional registrations

None

5. Current membership in professional organizations

6. Honors and Awards

AAAS Science & Technology Policy Fellowship

Fulbright Postdoctoral Fellowship

7. Selected service activities

Advisory Board Member, U.S. Department of Energy Agile BioFoundry, 2025–Present

UIC Honors College Fellow, 2023–Present

BETO BRIDGES National Review Board, 2022–2024

LSAMP / DuSable / SROP Faculty Affiliate, UIC, 2022–Present

Peer Reviewer: DOE, NSF, NIST, ACS, RSC Journals

NSBE ELEVATE National STEM Day Panelist, 2024

8. Selected Publications and presentations

1. Taiwo Z. Adesanya, Jessica Fajardo, Matthew Gaughan, Ogechikanma Ihenacho, Hussein Rezk, **Ezinne Ibagere**, High-Performance Biobased Packaging: Ionic Liquid-Assisted Cellulose-Lignin Bioplastics as Sustainable Alternatives to LDPE, Chemical Engineering Journal, *Chemical Engineering Journal*, Volume 529, 1 February 2026, 173080. <https://doi.org/10.1016/j.cej.2026.173080>.
2. Saliu, O. D.; Adesanya, T. Z.; Acquah, A.; Akobe, B.; Aganyi, D. D.; Abubakar, O. D.; **Achinivu-Ibagere, E.** Transformation of Sugar Cane Lignin into Renewable Fuel-Range Cyclo-Alkanes: In-Situ Hydrogen Release Using Earth-Abundant AlCl₃ and Biobased

Mannose Triflate Support. *Ind. Eng. Chem. Res.* **2025**, *64* (1), 179–189. <https://doi.org/10.1021/acs.iecr.4c03341>.

3. **Ezinne C Achinivu-Ibagere**, Taiwo Zacchaeus Adesanya, Valentin Larzilliere, Erika Clavijo Rivera, Florent Allais. Charge mediated nano-filtration for recovering sinapic acid from a mustard bran hydrolysate, *Journal of Membrane Science*, 2025, 713, 123333; <https://doi.org/10.1016/j.memsci.2024.123333>.
4. **Ezinne C Achinivu**, Brian W Blankenship, Nawa Raj Baral, Hemant Choudhary, Ramu Kakumanu, Mood Mohan, Edward EK Baidoo, Corinne D Scown, Anthe George, Blake A Simmons, John Gladden. Biomass pretreatment with distillable ionic liquids for an effective recycling and recovery approach, *Chemical Engineering Journal*, 2023, 479, 147824; <https://doi.org/10.1016/j.cej.2023.147824>.
5. Paul Wolski, Brian W Blankenship, Athiyya Umar, Mica Cabrera, Blake A Simmons, Kenneth L Sale, **Ezinne C Achinivu**. Factors that influence the activity of biomass-degrading enzymes in the presence of ionic liquids—a review, *Front. Energy Res.*, 2023, 11, 1212719; <https://doi.org/10.3389/fenrg.2023.1212719>
6. **Ezinne C Achinivu**, Mica Cabrera, Athiyya Umar, Minliang Yang, Nawa Raj Baral, Corinne D Scown, Blake A Simmons, John M Gladden. In Situ Synthesis of Protic Ionic Liquids for Biomass Pretreatment, *ACS Sustainable Chem. Eng.*, 2022, 10, 37, 12090–12098; <https://doi.org/10.1021/acssuschemeng.2c01211>
7. **Ezinne C. Achinivu**, Julia L. Shamshina, Robin D. Rogers. Chitin extracted from various biomass sources: It's not the same, *Fluid Phase Equilibria*, 2022, 552; <https://doi.org/10.1016/j.fluid.2021.113286>
8. **Ezinne C. Achinivu**, Skye Frank, Nawa Raj Baral, Lalitendu Das, Mood Mohan, Peter Otoupal, Emara Shabir, Sean Utan, Corinne D. Scown, Blake A. Simmons and John Gladden. Alkanolamines as Dual Functional Solvents for Biomass Deconstruction and Bioenergy Production. *GreenChem.*, 2021, 23, 8611-8631; <https://doi.org/10.1039/D1GC02667D>
9. **Ezinne C. Achinivu**, Mood Mohan, Hemant Choudhary, Lalitendu Das, Kaixuan Huang, Harsha D. Magurudeniya, Venkataramana R. Pidatala, Anthe George, Blake A. Simmons, and John M. Gladden, A predictive toolset for the identification of effective lignocellulosic pretreatment solvents: a case study of solvents tailored for lignin extraction. *Green Chemistry*, 2021, 23, 7269-7289; <https://doi.org/10.1039/D1GC01186C>
10. Lalitendu Das, **Ezinne C. Achinivu**, Carolina Araujo Barcelos, Eric Sundstrom, Bashar Amer, Edward E. K. Baidoo, Blake A. Simmons, Ning Sun and John M. Gladden. Deconstruction of Woody Biomass via Protic and Aprotic Ionic Liquid Pretreatment for Ethanol Production. *ACS Sustainable Chem. Eng.* 2021, 9, 4422–4432. <https://doi.org/10.1021/acssuschemeng.0c07925>
11. Carolina A. Barcelos, Asun M. Oka, Jipeng Yan, Lalitendu Das, **Ezinne C. Achinivu**, Harsha Magurudeniya, Jie Dong, Simay Akdemir, Nawa Raj Baral, Chunsheng Yan, Corinne D. Scown, Deepti Tanjore, Ning Sun, Blake A. Simmons, John Gladden, and Eric Sundstrom, High-Efficiency Conversion of Ionic Liquid-Pretreated Woody Biomass to Ethanol at the Pilot Scale, *ACS Sustainable Chem. Eng.* 2021, 9, 4042–4053; <https://doi.org/10.1021/acssuschemeng.0c07920>

12. **Ezinne C. Achinivu**, Amandine L. Flourat, Fanny Brunissen, Florent Allais. Valorization of waste biomass from oleaginous “oil-bearing” seeds through the biocatalytic production of sinapic acid from mustard bran. *Biomass and Bioenergy*, 2021, 145; <https://doi.org/10.1016/j.biombioe.2020.105940>.
13. **Ezinne C. Achinivu**, Protic Ionic Liquids for Lignin Extraction—A Lignin Characterization Study. *International Journal of Molecular Sciences*, 2018, 19, 428; <https://doi.org/10.3390/ijms19020428>.
14. **Ezinne C. Achinivu**, Reagan M. Howard, Guoqing Li, Hanna Gracz and Wesley A. Henderson, Lignin extraction from biomass with protic ionic liquids. *Green Chemistry*, 2014, 16, 1114-1119; <https://doi.org/10.1039/C3GC42306A>

Relevant Intellectual Property

- US Appl 17/547,134; Dec 9, 2021; Use of polyamines in the pretreatment of biomass.
- Provisional; US 63/035,508; Jun 5, 2020; Use of "in-situ" ionic liquid (IL) and deep eutectic solvent (DES) synthesis using chemically synthesized or biomass-derived ions in the pretreatment of biomass.
- Provisional; US 63/023,770; May 12, 2020; Use of alkanolamines for lignin extraction in the pretreatment of biomass.
- Provisional; US 62/978,771; Feb 19, 2020; Use of distillable volatile salt for the pretreatment of biomass.

Presentations

- **Achinivu, E. C.**, et al *ACS National Meeting*, Fall 2020, Fall 2021, Fall 2022, Fall 2023, Spring 2024
- **Achinivu, E. C.**, et al *AIChE Annual Meeting*, Fall 2020, Fall 2021, Fall 2022, Fall 2023, Fall 2024, Fall 2025
- **Achinivu, E. C.**, et al *Symposium on Biomaterials, Fuels and Chemicals* 2024, 2025 (Session Chair)
- **Achinivu, E. C.**, et al *Congress of Ionic Liquids (COIL)*, March 2019

9. **Professional Development Activities**

Not applicable

Faculty Vita: Said Al-Hallaj

1. Education

B.Sc. Chemical Engineering, Jordan University of Science and Technology, Jordan, 1990
M.Sc. Chemical Engineering, Jordan University of Science and Technology, Jordan, 1994
PhD Chemical Engineering, Illinois Institute of Technology, Chicago, IL, USA, 1999

2. Academic Experience

9/09-now Visiting Research Professor, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL
5/00-5/09 Research Professor, Department of Chemical and Biological Engineering, Coordinator of IIT Renewable Energy Programs, IIT, Chicago, IL
10/99-8/00 Senior Research Associate: Li-Battery/PEM fuel cell project at IIT, sponsor: U.S. Army Research Office
3/99-9/99 Associate Visiting Researcher, Department of Applied Chemistry, College of Engineering, Tohoku University, Sendai, Japan: thermal characterization and calorimetric studies for Li-battery research

3. Non-Academic Experience

6/14-now Founder/CEO NETenergy, Chicago, IL
3/22-now Chief Battery Scientist, Beam Global (NYSE: BEEM), (acquired AllCell)
9/08-7/20 Chairman/CEO, All Cell Technologies LLC, Chicago, IL

4. Certification or professional registrations

None

5. Current membership in professional organizations

Electrochemical Society

6. Honors and Awards

- Stanford Top 2% researchers in Chemical Engineering
- Top 10 papers at the MIT A+B 2024 Energy Symposium
- Illinois Tech Alumni Professional Achievement Award, 2022
- Advisory Board member, Armour College of Engineering, Illinois Tech
- Industry Advisory Board, Department of Mechanical and Industrial Engineering, UIC
- Shell GameChanger Accelerator Powered by NREL (GCxN) 2019
- Wells Fargo Innovation Incubator (IN2), NREL, 2016
- New Venture Challenge, 2nd place, Booth School of Business, University of Chicago, 2015
- Clean Energy Challenge, 1st place award, 2015
- Chicago Magazine 2012 Green Award
- Embassy of France, Young Entrepreneur Initiative Award (YEI), 2007

7. Selected service activities

- Course development: developed and teaching a graduate level course entitled “ChE 451 Renewable Energy Technologies” and co-authored course textbook “Hybrid Hydrogen Systems”
- Engineering and Entrepreneurship: developed and teaching a course entitled “ChE 427 Entrepreneurship in Chemical Engineering”
- Technology Commercialization: currently serving as Commercialization Fellow for UIC Office of Technology Management and successfully spun off two university-based tech start-up companies

8. Selected Publications and presentations

- 1- M Qasem, M Haddadin, Y Yassin, S Ratrou, C Chen, S Stoyanov, S Al-Hallaj, and M Krishnamurthy, “Real-Time Electrochemical Model-Based BMS Control for Mitigating Li-Plating and Extending Battery Life”, IEEE Transactions on Transportations Electrifications, Vol. 11, No. 3, June 2025
- 2- S Stoyanov, T G. Ritter, S Pappu, S Al-Hallaj, and R Shahbazian-Yassar, “Effect of Separator Type, Formation Cycle, and Fluoroethylene Carbonate on the long-Term Cycling Performance of Lithium” Metal Half-Cells, Chemistry Select 2025, 10, e202501481
- 3- Yi Zenga, R J Clark, Y Galazutdinova, A Odukamaiya, S Al-Hallaj, M M Farid, S Kaur, J Woods, “Open-cycle thermochemical energy storage for building space heating: Practical system configurations and effective energy density”, Applied Energy 376 (2024) 124218
- 4- Yana Galazutdinova 1, Ruby-Jean Clark 1, Said Al-Hallaj 1, Sumanjeet Kaur 2 and Mohammed Farid. “New Thermochemical Salt Hydrate System for Energy Storage in Buildings”, Energies 2024, 17, 5228. <https://doi.org/10.3390/en17205228>
- 5- M Qasem, S Stoyanov, S Ratrou, M Haddadin, Y Yassin, C Chen, S Al-Hallaj, and M Krishnamurthy, “Synthetic Data-Integrated Li-ion Battery Modeling for eVTOL Energy Systems”, IEEE Access, Vol. 12, 2024
- 6- R.J. Clark, G. Gholamibozanjani, J. Woods, S. Kaur, A. Odukamaiya, S. Al-Hallaj, M. Farid, “Experimental screening of salt hydrates for thermochemical energy storage for building heating application”, Journal of Energy Storage, 51, PP 104415, 2022
- 7- A Goyal, E. Kozubal, J. Woods, M. Nofal, S. Al-Hallaj, “Design and performance evaluation of a dual-circuit thermal energy storage module for air conditioners”, Applied Energy 292, 116843, 2021
- 8- Y. Galazutdinova, S. Ushak, M. Farid, S. Al-Hallaj, M. Grágeda, “Development of the inorganic composite phase change materials for passive thermal management of Li-ion batteries Application”, Journal of Power Sources 491, (2021) 229624
- 9- Y Galazutdinova, S Al-Hallaj, M Grágeda, S Ushak, “Development of the inorganic composite phase change materials for passive thermal management of Li-ion batteries: material characterization”, International Journal of Energy Research 44 (3), 2011-2022, 2020
- 10-
- 11- “Energy Storage for Smart Grid Applications,” Said Al-Hallaj, Keynote Speaker: 2nd International Conference on Smart Grid and Renewable Energy, November 19-21, 2019, Doha, Qatar
- 12- “Thermal Management and Safety of Li-ion Batteries,” Said Al-Hallaj, Invited Speaker: AIChE Midwest Regional Conference, 2019, UIC, Chicago, IL USA

- 13- “Thermal Management and Safety Concerns for Li-ion Batteries in Electric Mobility and Autonomous Vehicles Applications,’ Said Al-Hallaj, Keynote Speaker: IEEE Vehicle and Propulsion Conference, August 27-30, 2018, Chicago, IL, USA
- 14- Numerical modeling of transient heat transfer in a phase change composite thermal energy storage (PCC-TES) system for air conditioning applications, Aljehani, LC Nitsche, S Al-Hallaj, Applied Thermal Engineering 164, 114522, 2020
- 15- Development of the inorganic composite phase change materials for passive thermal management of Li-ion batteries: material characterization, Y Galazutdinova, S Al-Hallaj, M Grágeda, S Ushak, International Journal of Energy Research, 2019
- 16- Exploring the charge reactions in a Li–O₂ system with lithium oxide cathodes and nonaqueous electrolytes T Zhang, R Amine, X Bi, Y Qin, M Li, S Al-Hallaj, F Huo, J Lu, K Amine Journal of materials chemistry A 7 (26), 15615-15620, 2019
- 17- Liu, Y Yuan, J Wen, K He, X Bi, Y Guo, Nano Letters 17 (5), 2959-2966, 2017
- 18- Preventing thermal runaway propagation in lithium ion battery packs using a phase change composite material: An experimental study, S Wilke, B Schweitzer, S Khateeb, S Al-Hallaj, Journal of Power Sources 340, 51-59, 2017
- 19- Semi-Empirical Modeling of Capacity Fade: A Practical Approach for Battery Pack Manufacturers, S Wilke, B Schweitzer, S Khateeb, S Al-Hallaj, ECS Transactions 73 (1), 109-119, 2016
- 20- Modeling of compression curves of phase change graphite composites using Maxwell and Kelvin models, MF Alzoubi, S Khateeb, S Al-Hallaj, Journal of Composite Materials 50 (8), 1123-1135, 2016

9. **Professional development activities**

- Leadership: Board Member, Armour College of Engineering at Illinois Tech, and Industry Advisory Board for Mechanical Engineering at UIC
- Technology Commercialization: serving as Commercialization Fellow for UIC Office of Technology Management and successfully spun off two university-based tech start-ups.
- Engineering and Entrepreneurship: mentoring startups at University of Chicago, Resurgence/Polsky Center, UIC, Argonne National Lab, University of Auckland
- Jordan Solar EV projects: worked closely with the Jordan parliament and USAID to draft and pass legislation to exempt EVs from import tariffs to boost EV adoption and catalyze sales growth. Forbes magazine recently published an article stating that in 2024, 65% of new cars in Jordan were electric, which is the second highest share of EV sales in the world, after Norway, despite many obstacles.
- Community Service: member of Chicago-Amman Sister Cities Committee and founding and board member of Connect.Jo, a USAID funded initiative that focuses on connecting Jordanian diaspora professionals with researchers and startups from Jordan.

Faculty Vitae: Anh Ngo

1. Education

B.S. Theoretical Physics, Hanoi University of Science, Vietnam National University, 2000

M.S. The ICTP Advanced Diploma Theoretical Condensed Matter Physics, ICTP, Trieste, Italy. 2002

PhD Department of Physics and Astronomy, Ohio University, USA, 2010

2. Academic Experience

Professor : Department of Chemical Engineering, University of Illinois- Chicago and Joint Appointment with Argonne National Laboratory : August 2025 – present.

Associate Professor : Department of Chemical Engineering, University of Illinois- Chicago and Joint Appointment with Argonne National Laboratory : August 2020 – 2025.

Staff scientist: Materials Science Division, Argonne National Laboratory, March 2017 — 2020.

Postdoctoral Fellow: Materials Science Division and Advanced Photon Source, Center for Nanoscale Materials, Argonne National Laboratory, March 2014 — February 2017.

Postdoctoral Fellow: Department of material sciences, University of Wisconsin-Madison, USA, May 2012– March 2014.

Postdoctoral Fellow: Department of Physics and Astronomy, Louisiana State University, USA, 2010— May 2012

3. Non-Academic Experience

Staff scientist: Materials Science Division, Argonne National Laboratory, March 2017 — 2020.

Postdoctoral Fellow: Materials Science Division and Advanced Photon Source, Center for Nanoscale Materials, Argonne National Laboratory, March 2014 — February 2017.

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers

6. Honors and awards

Not applicable

7. Service activities

Not applicable

8. Selected Publications

1. Selva Chandrasekaran Selvaraj, Volodymyr Koverga, **Anh T. Ngo**. Exploring Li-Ion transport properties of Li_3TiCl_6 : a machine learning molecular dynamics study. *J. Electrochem. Soc.* 171 050544. (2024).
2. Guo-Xing Li, et, al. , **Anh T. Ngo**, Donghai Wang. Enhancing lithium-metal battery longevity through minimized coordinating diluent. **Nature Energy** 9, 817–827 (2024).
3. Weiran Zhang, et, al. **Anh T. Ngo**, Chunsheng Wang. Single-phase local-high-concentration solid polymer electrolytes for lithium-metal batteries. **Nature Energy** 9, 386–400 (2024).
4. Guo-Xing Li, et, al. , **Anh T Ngo**, Gunther Brunklaus, Donghai Wang. Interfacial solvation-structure regulation for stable Li metal anode by a desolvation coating technique. **Proceedings of the National Academy of Sciences** 121 (4), e2311732121, (2024).
5. Tolulope M Ajayi, et, al. , **Anh T. Ngo**, Saw-Wai Hla. Characterization of just one atom using synchrotron X-rays. **Nature** 618, 69–73 (2023).
6. Jijian Xu, Volodymyr Koverga, et, al., **Anh T. Ngo**, Chunsheng Wang. Revealing the anion–solvent interaction for ultralow temperature lithium metal batteries. *Advanced Materials* 36 (7), 2306462, (2024).
7. Xiaozhao Liu, Volodymyr Koverga, Hoai T. Nguyen, **Anh T. Ngo**, Tao Li. Exploring solvation structure and transport behavior for rational design of advanced electrolytes for next generation of lithium batteries. *Appl. Phys. Rev.* 11, 021307 (2024).
8. Huu Do, Alex Lee Hyowon Park and **Anh T. Ngo**, Delocalized polaron and Burstein-Moss shift induced by Li in $\alpha\text{-V}_2\text{O}_5$: A DFT+DMFT study. *Physical Review B* 108 (20), 205122, (2023).
9. Alex Lee, Hyowon Park and **Anh T. Ngo**. Effect of off-diagonal elements in the Wannier Hamiltonian on DFT+DMFT for low-symmetry materials: Study of Li_2MnO_3 . *Phys. Rev. B* 108, 205146, (2023).
10. Yue Gao, Tomas Rojas, Ke Wang, Shuai Liu, Daiwei Wang, Tianhang Chen, Haiying Wang, **Anh T. Ngo**, Donghai Wang. ” Low-temperature and high-rate-charging lithium metal batteries enabled by an electrochemically active monolayer-regulated interface”. **Nature Energy** 5, 534–542 (2020).
11. Pallab Barai, **Anh Ngo**, Badri Narayanan, Kenneth Higa, Larry A Curtiss, Venkat Srinivasan. “ The role of local inhomogeneities on dendrite growth in LLZO-based solid electrolytes” *Journal of The Electrochemical Society*, 167 100537 (2020).
12. Hyowon Park, Ravindra Nanguneri, and **Anh T. Ngo**. “DFT DMFT study of spin-charge-lattice coupling in covalent LaCoO_3 ” *Phys. Rev. B* 101, 195125 (2020).
13. Zhe Zhang, Yiming Li, Bo Song, Yuan Zhang, Xin Jiang, Ming Wang, Ryan Tumbleson, Changlin Liu, Pingshan Wang, Xin-Qi Hao, Tomas Rojas, **Anh T. Ngo**, Jonathan L Sessler, George R Newkome, Saw Wai Hla, Xiaopeng Li . “ Intra-and intermolecular self-assembly of a 20-nm-wide supramolecular hexagonal grid” **Nature Chemistry** 12, 468–474 (2020).
14. Yuan Zhang, Jan Patrick Calupitan, Tomas Rojas, Ryan Tumbleson, Guillaume Erbland, Claire Kammerer, Tolulope Michael Ajayi, Shaoze Wang, Larry A Curtiss, **Anh T. Ngo**, Sergio E Ulloa, Gwénaél Rapenne, Saw Wai Hla. “ A chiral molecular propeller designed for unidirectional rotations on a surface” **Nature Communications** 10, 3742 (2019).

15. Yipeng Cao, Milind J Gadre, **Anh T. Ngo**, Stuart B Adler, Dane D Morgan. “Factors controlling surface oxygen exchange in oxides” **Nature Communications** 10, 1346 (2019).
16. M. Asadi, B. Sayahpour, P. Abbasi, **Anh T. Ngo**, Klas Karis, Jacob R Jokisaari, C. Liu, B. Narayanan, Marc Gerard, Poya Yasaei, X. Hu, A. Mukherjee, Kah Chun Lau, R. S Assary, Fatemeh Khalili-Araghi, Robert F Klie, Larry A Curtiss and Amin Salehi-Khojin. “A lithium–oxygen battery with a long cycle life in an air-like atmosphere”. **Nature** 555, 502–506 (2018).
17. Yang Li, **Anh T. Ngo**, Andrew DiLullo, Kyaw Zin Latt, Heath Kersell, Brandon Fisher, Peter Zapol, Sergio E Ulloa, Saw-Wai Hla. “ Anomalous Kondo resonance mediated by semiconducting graphene nanoribbons in a molecular heterostructure ” **Nature Communications** 8, 946 (2017).
18. **Anh T. Ngo**, Eugene H. Kim, and Sergio E. Ulloa “Single-atom gating and magnetic interactions in quantum corrals” *Phys. Rev. B* 95, 161407(R) (2017).
19. Philippe Debray, SMS Rahman, J Wan, RS Newrock, M Cahay, **Anh T. Ngo**, SE Ulloa, ST Herbert, Mustafa Muhammad, M Johnson “ All-electric quantum point contact spin-polarizer” **Nature Nanotechnology** 4, 759–764 (2009).

9. Professional Development

Not applicable

Faculty Vitae: Vikas Berry

1. Education

BS Chemical Engineering, Indian Institute of Technology-Delhi, 1999
MS Chemical Engineering, University of Kansas, 2003
PhD Chemical Engineering, Virginia Tech, 2006

2. Academic Experience

University of Illinois at Chicago, Chicago, IL

- Aug 2022 – Present Dr. Satish C. & Asha Saxena Professor
- Nov 2016 – Present Department Head
- Aug 2019 – Present Professor
- Aug 2015 – Nov 2016 Interim Department Head
- Aug 2014 – Aug 2019 Associate Professor

Kansas State University, Manhattan, KS

- Nov 2011 – Aug 2014 William H. Honstead Professor
- Jul 2012 – Aug 2014 Associate Professor
- Jan 2007 – Jul 2012 Assistant Professor

Virginia Tech, Blacksburg, VA

- Jan 2013 – Dec 2016 Graduate Research Assistant (PhD)

University of Kansas, Lawrence, KS

- Aug 2000 – Dec 2002 Graduate Research Assistant (MS)

3. Non-Academic Experience

Grapherry Inc. Mar 2024 – Present. Co-Founder and Act. Chief Executive Officer
Viroscope Inc. Oct 2023 – Present Co-Founder
Cadila Pharmaceuticals, India Aug 1999 – Jul 2000 Pharmaceutical Process Scientist

4. Certification or professional registrations

None

5. Current membership in professional organizations

- AIChE, Electrochemical Society, Omega Chi Epsilon Honor Society of Chemical Engineering, Member, Sigma Xi Scientific Research Society, Member, 2007

6. Honors and Awards

- Listed in the Top 2% cited researchers since 1960 (Stanford Rankings), 2021

- Louis Stokes Alliances for Minority Participation-NSF International Center of Excellence (LSAMP-NICE) Faculty Advisor Award, 2019
- Rudolph A. Marcus Award, 2017
- NSF-CAREER Award, 2011-17
- William H. Honstead Professorship, 2011 – 2014
- Sigma Xi Outstanding Junior Scientist Award, 2010
- Big-XII Fellow - 2009, Kansas State University
- Stroebeel Scholarship Recipient, University of Kansas 2000-02

7. Selected service activities

- 8 research group alumni currently hold faculty positions
- Editorial Service
 - Scientific Reports, Editorial Board Member, 2012 – Present
 - Science Advances Letters, Associate Editor, 2014 – Present
 - Advanced Carbon, Scientific Advisory Board
 - Journal of Nanomaterials and Molecular Nanotechnology, Scientific Advisory Board

8. Selected Publications and Presentations

SELECT PUBLICATIONS (CITATIONS = 9332) *Indicates corresponding authorship

1. Mehran Arzani, Hamidreza Mahdavi and Vikas Berry, “Engineering porous liquids for enhanced mobility and stable battery electrolytes”, **ACS Energy Letters**, 10, 3259, 2025
2. Hisham Maddah, Vikas Berry and Sanjay Behura, “Optimal design of spin-coated layers of TiO₂ for maximum open-circuit voltage in anthocyanin-based dye-sensitized solar cells”, **Energy Conversion and Management**, 346, 120478, 2025
3. Kartikey Sharma, Songwei Che, Sanjay Behura, Naveen Dandu, Anika Shinde, Anh Ngo and Vikas Berry, “h₆ Organometallic Functionalization of Hexagonal Boron Nitride: Implications for Multifunctional Molecular and Nanoscale Interfaces for Electronic Applications”, **ACS Applied Nano Materials**, 1, 233, 2024
4. Ahmad Jaradat, Musawenkosi K Ncube, Ilias Papailias, Nikhil Rai, Khagesh Kumar, Volodymyr Koverga, Roshan Y Nemade, Chengji Zhang, Nannan Shan, Hessam Shahbazi, Arash Namaeighasemi, Pardis Seraji, Shahriar Namvar, Vikas Berry, Jordi Cabana, Arunkumar Subramanian, Anh T Ngo, Larry A Curtiss, Amin Salehi-Khojin, “Fast Charge-Transfer Rates in Li-CO₂ Batteries with a Coupled Cation-Electron Transfer Process”, **Advanced Energy Materials**, 14 (15). 2303467, 2024
5. Roshan Nemade, Sheldon Cotts, and Vikas Berry, “Graphene Fermi Level-Guided Attachment of Single Exoelectrogens and Induced interfacial Doping”, **ACS Applied Materials Interfaces**, 16 (5). 5548 2024

6. Sungjoon Kim, Vikas Berry, Jessica Metcalfe, and Anirudha Sumant, “Thin Film Charged Particle Detectors”, *Journal of Instrumentation*, 18, 07, P07047, 2023
7. Nishithan C Kani, Ngoc Hoang Lan Nguyen, Kyle Markel, Rajan R Bhawnani, Benjamin Shindel, Kartikey Sharma, Sungjoon Kim, Vinayak P Dravid, Vikas Berry, Joseph A Gauthier, Meenesh R Singh, “Electrochemical Reduction of Nitrates on CoO Nanoclusters-Functionalized Graphene with Highest Mass Activity and Nearly 100% Selectivity to Ammonia”, **Advanced Energy Materials**, 13, 17, 2204236
8. Hisham A Maddah, Vikas Berry, Sanjay K Behura, “Simple harmonic oscillation model explaining MA torsional locking in surface passivated MAPbI₃ crystal”, **Chemical Physics Letters**, 806. 139967, 2022
9. Bijentimala Keisham, Sungjoon Kim, Sheldon Cotts, Vikas Berry, “Graphene Chemo-Phonics for Biosensor Applications: An Interfacial Raman Transducer”, **Advanced Materials Interfaces**, 9, 30, 2200773, 2022
10. Roshan Nemade, Sheldon Cotts, Vikas Berry, “Plasma treatment for enhanced microbe-electrode interfaces: A bio-electronic sink”, **Journal of Power Sources**, 544, 231834, 2022
11. Alireza Ahmadianyazdi, Ngoc Hoang Lan Nguyen, Jie Xu, Vikas Berry, “Glucose measurement via Raman spectroscopy of graphene: Principles and operation”, **Nano Research**, 15, 10, 8697 2022
12. Ngoc Hoang Lan Nguyen, Sungjoon Kim, Garrett Kindemann and Vikas Berry, “COVID-19 Spike Protein Induced Phononic Modification in Antibody-Coupled Graphene for Viral Detection Application”, **ACS Nano**, 15, 11743, 2021
13. Sanjay Behura*, Chen Wang, Yu Wen, and Vikas Berry*, "*Graphene/Semiconductor Heterojunction Sheds Light on Emerging Photovoltaics*", **Nature Photonics**, 13, 312–318, 2019
14. N. Mohanty, D. Moore, Z. Xu, T. S. Sreeprasad, A. Nagaraja, A. A. Rodriguez and V. Berry*, "*Nanotomy Based Production of Transferrable and Dispersible Graphene-Nanostructures of Controlled Shape and Size*", **Nature Communications**, 3, Article number: 844, 2012

PATENTS:

GRANTS

15. “*Direct and sequential formation of monolayers of boron nitride and graphene on substrates*”, Michael R. Seacrist, Vikas Berry and Phong T. Nguyen, **US Patent US9029228 (2023)**
16. “*Direct formation of graphene on semiconductor substrates and structures prepared thereby*”, M.R. Seacrist and Vikas Berry, **US Patent US20160233305 (2023)**
17. “*Optical COVID-19 Detection System*”, Vikas Berry, Lan Nguyen and Sungjoon Kim, **US Patent 20220299506 (2022)**
18. “*Production of Graphene Nanoribbons and Quantum Dots with Controlled Dimensions and Crystallographic Orientation*”, Vikas Berry, Nihar Mohanty and David Moore, **US Patent**

US9272911 (2016)

19. “*Direct formation of graphene on semiconductor substrates*”, Michael R. Seacrist and Vikas Berry, **Patent Number US8884310 (2014)**
20. “*Fabrication of Ultra Long Necklace of Nanoparticles*”, Ravi Saraf, S. Niu, V. Berry, V. Maheshwari, **Patent Number: US7749561 (2010)**

9. Professional Development Activities

- Plenary and Keynote Lectures
 - Plenary: Graphene Conference (Rome, 2019), *Tsinghua University* (China 2011)
 - Keynote: TechConn (2020), RRR (2019), *ISFM* – (India 2018); *ICREA* – *Graphene Biosensors* (Spain - 2015); *IMST* (India, 2012)
- Founded two startups (Grapherry Inc., Viroscope Inc.) advancing graphene materials and airborne virus detection technologies.
- Mentored 3 postdocs, 18 PhD and 6 MS students; 8 former trainees now faculty members worldwide.
- Organizer of departmental workshops, and industrial advisory engagements.
- Implemented major curriculum innovations, including development of new BS concentrations, professional development modules, and practice-oriented graduate training pathways.
- Initiated and led research collaboration programs with Argonne National Laboratory, the University of Chicago, and multiple international academic partners.

Faculty Vitae: Betul Bilgin

1. Education

- Ph.D. Chemical Engineering, Michigan State University, East Lansing, MI, 2014
M.S. Biomedical Engineering, University of Minnesota, Minneapolis, MN, 2008
B.S. Chemistry, Fatih University, Istanbul, Turkey, 2005

2. Academic Experience

- Department of Chemical Engineering, University of Illinois at Chicago, Clinical Professor, 2025-Present, Full-Time
- Department of Chemical Engineering, University of Illinois at Chicago, Clinical Associate Professor, 2021-2025, Full-Time
- Department of Chemical Engineering, University of Illinois at Chicago, Clinical Assistant Professor, 2015-2021, Full-Time
- Department of Chemical Engineering, University of Illinois at Chicago, Visiting Lecturer, 2014-2015, Part-Time

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

- UIC Inclusive Education Scholars' Certificate Program (IESCP) Spring 2023
- UIC Equity Advocates Certificate Spring 2022
- Entrepreneurship Minded Learning (EML) June 16th 2019
Tampa, FL
- Faculty Summer Institute May 20th -22nd 2019
Urbana-Champaign, IL
- BASF Faculty Workshop on Process Safety July 23rd-26th 2018
Wyandotte, MI
- ChemE Summer School July 29th-August 3rd 2017
NC State University, Raleigh, NC
- National Effective Teaching Institute – 1 May 21st-23rd 2017
Philadelphia, PA

5. Current membership in professional organizations

American Institute of Chemical Engineers (AIChE)
American Society for Engineering Education (ASEE)

6. Honors and awards

- *UIC Teaching Scholar (Spring 2017)*
- *American Institute of Chemical Engineers (AIChE) 35 under 35 award in education category (August 2017)*
- *American Society for Engineering Education (ASEE) 20 under 40 award (June 2018)*
- *UIC Teaching Sustainability Initiative Spring (2020 and 2024)*
- *UIC Teaching Recognition Program (TRP) Award (Spring 2021)*
- *UIC COE Harold Simon Award (Spring 2022)*

- *UIC COE Advising Award (Spring 2023)*
- *UIC COE Teaching Award (Spring 2022-24)*
- *ASEE Ray Fahien Award (June 2024)*

7. Service activities

- *AIChE Education Division 2nd Vice Chair (November 2025-present)*
- *ASEE Chemical Engineering Division Director (July 2024-present)*
- *AIChE Education Division Teaching Focus Faculty Leadership Board (Summer 2023-present)*
- *COE Summer Camp Chemical Engineering Advisor (Summer 2022-2023-2024)*
- *CHE Senior Design Showcase (April 2022-2024)*
- *AIChE Student Chapter Advisor (January 2024-present)*
- *Department Advisory Committee member (September 2023-present)*
- *Departmental Action Team (DAT) and CHEME Connec (January 2023-present)*
- *Summer CHEM-E Club and ACCESS Club- Founder/Chair (May 2020-2022)*
- *NSF S-STEM Project- An Integrated Program for Recruitment, Retention, and Graduation of Academically Talented Low-Income Engineering Students – co-PI (September 2019-present)*
- *COE Non-Tenure Track Faculty Committee-Member (Fall 2018-present)*
- *Instructional Delivery Planning (IDP) Committee – Member (May-June 2020)*
- *Diversity, Recruitment and Retention Committee-Chair (Fall 2017-present)*
- *Industry and Public Relation Committee- Chair (Fall 2015-present)*
- *Student and Alumni Development (Spring 2015-present)*
- *Search Committees*
 - *Clinical Faculty Hire Search Committee (Summer 2020)*
 - *Associate Dean of Undergraduate Affairs (Summer 2020)*

8. Selected Publications

- **Bilgin, Betul**, Christopher VH-H. Chen, and Stephanie Butler Velegol. "Generative AI in Chemical Engineering Education: Rebuilding Thermodynamics, Material and Energy Balances and Kinetics Courses with AI and Chemical Engineering Students' Perception of AI." In *2025 ASEE Annual Conference & Exposition*. 2025.
- Revelo, Renata A., Melissa Espindola, **Betul Bilgin**, and Houshang Darabi. "WIP Exploring the Interplay Between Social Identities and Engineering Identity Formation." In *2025 ASEE Annual Conference & Exposition*. 2025.
- **Bilgin, Betul**. "CHE CALCULATOR®: Empowering Students with Computational Tools for Real-World Engineering Challenges." In *2025 ASEE Annual Conference & Exposition*. 2025.
- **Bilgin, Betul**, Dania Nazimuddin, Louie Edano, and Nadia Nikolova. "Fostering Early Engineering Identity Through K-12 Outreach: Insights from a Three-Week Summer Program." In *2025 IEEE Global Engineering Education Conference (EDUCON)*, pp. 1-7. IEEE, 2025.
- **Betul Bilgin**. "Fostering Diversity, Equity, and Inclusion in Engineering Education- A Case Study of UIC Chemical Engineering Department" In *2024 ASEE Annual Conference & Exposition*, 2024.

- **Betul Bilgin**, Daisaq Cyrilla, Naomi Elisabeth Groza, Ruben Lopez. “Assessing Chemical Engineering Students' Perspectives on Data Science and Its Integration in the Academic Curriculum: Implications for Industry Readiness” In *2024 ASEE Annual Conference & Exposition*, 2024.
- **Betul Bilgin**, Hasiya Najmin Isa, Emily Seriruk, Cody Mischel, James Pellegrino. “Bridging the Gap between Industry and Academia and Developing Students’ Engineering Identity.” In *2023 ASEE Annual Conference & Exposition*, 2023.
- Houshang Darabi, Rezvan Nazempour, Peter C Nelson, Renata A Revelo, Shanon Marie Reckinger, Didem Ozevin, **Betul Bilgin**, Anthony E Felder. “Board 375: Reporting the Progress and Performance Evaluation of an Ongoing Integrated Program for Recruitment, Retention, and Graduation of High-Achieving, Low-income Engineering Students.” In *2023 ASEE Annual Conference & Exposition*, 2023.
- **Betul Bilgin**, Cody Mischel, Lewis Wedgewood, James Pellegrino, and Vikas Berry. "Impacts of implementing up-to-date industry problems on engineering identity development." In *2022 ASEE Annual Conference & Exposition*. 2022.
- Nazempour, Rezvan, Houshang Darabi, **Betul Bilgin**, Anthony Felder, Shanon Reckinger, Renata Revelo, Peter Nelson, Jeremiah Abiade, and Didem Ozevin. "An Integrated Program for Recruitment, Retention, and Graduation of Academically Talented Low-Income Engineering Students: Lessons Learned and Progress Report." In *2022 ASEE Annual Conference & Exposition*. 2022.
- **Betul Bilgin**, Cody Mischel, Lewis Wedgewood, James Pellegrino, and Vikas Berry. "Developing Professional Identity: Integrating Academic and Workplace Competencies within Engineering Programs." In *2022 ASEE Annual Conference & Exposition*. 2022.
- **Bilgin, Betul**, Anthony E. Felder, Houshang Darabi, Rezvan Nazempour, Shanon Reckinger, Renata A. Revelo, and Didem Ozevin. "Looking Ahead: Structure of an Industry Mentorship Program for Undergraduate Engineering Students." *Advances in Engineering Education* 10, no. 3 (2022).

9. Professional Development

I regularly attend and present at major engineering and education conferences, including the *ASEE Annual Conference*, *AIChE Annual Meeting*, *IEEE EDUCON*, and *AIChE Chicago Section* events, focusing on engineering education, sustainability, electrochemistry, and generative AI in teaching and learning.

I applied for *ABET Program Evaluator Training (PEV)* and regularly participate in UIC faculty development seminars and national workshops on course redesign, inclusive teaching, assessment, safety, instrumentation, and proposal development (NSF IUSE, RFE, REU). My DEI-focused professional development includes participation in the certificate programs and ongoing leadership training as Chair of the departmental DEI Committee.

My professional growth is supported by active research collaborations across UIC and nationally, including work on engineering identity, data-driven learning, sustainability, and the *NSF ERC AMaZED* initiative. I also develop innovative pedagogical tools such as *AI-integrated curriculum*, CHE CALCULATOR®-based assignments, and K–12 outreach programs.

I regularly engage with industry partners through Senior Design mentorship coordination, strengthening my understanding of current engineering applications and industry needs.

Faculty Resume: Michael Caracotsios

1. Education

Ph.D. in Chemical Engineering - January 1986 at the **University of Wisconsin-Madison**

Thesis Advisor: Pr. Warren E. Stewart

Thesis Title: *Model Parametric Sensitivity Analysis and Nonlinear Parameter Estimation. Theory and Applications.*

Minor: Computer Science and Mathematics - GPA: 4.0/4.0

M.S. in Chemical Engineering - May 1981 at the **University of Kentucky-Lexington**

GPA: 4.0/4.0 Rank # 1 in Graduating Class

Thesis Title: *Carbonylation of Methanol Catalyzed by Rhodium-Exchanged Zeolites.*

B.S. in Chemical Engineering - May 1979 at the **University of Thessaloniki-Greece**

GPA: 9.3/10.0 Rank # 1 in Graduating Class

Thesis Title: *Estimation of the Alloying and Deoxidization Efficiency for 50tn Electric Furnace of a Steelmaking Plant in Northern Greece.*

2. Academic Experience

09/2015 to Present: Clinical Associate Professor- UIC IL.

Currently teaching undergraduate level courses in Chemical Engineering Thermodynamics, Chemical Process Control and Process Simulation.

01/2007 to 08/2015: Adjunct Professor- Northwestern University, Evanston IL.

Currently teaching undergraduate and graduate level courses in Computer Aided Process Design and Optimization, Applied Numerical Methods and Statistical Data Analysis in Chemical and Biological Engineering.

05/1986 to 12/2006: Adjunct Professor- Illinois Institute of Technology, Chicago IL.

Taught undergraduate and graduate level courses in Computer Aided Process Design and Optimization, Applied Numerical Methods and Statistical Data Analysis in Chemical Engineering, Fluid Mechanics and Heat Transfer. Received the Excellence in Teaching Award in the 1991-92 academic year.

3. Non-Academic Experience

08/1996 to 09/2015: Research Fellow - UOP, Optimization and Advanced Control Applications, Process Information and Control Division, Solutions and Services Division, Modeling and Optimization Division, Gas Processing Division.

03/1986 to 08/1996: Senior Research Engineer - Amoco Chemical Company, Research and Development Department, Aromatics Oxidation Division, Polymers Catalysis and Process Division, Computer Aided Process Engineering Division.

05/1984 to 09/1984: Research Engineer, Chevron Oil Field Research Corporation,
Reservoir Engineering Division. Developed software on Cray supercomputers for steam-flood simulations in oil exploration.

4. Certifications or professional registrations

Green belt six-sigma certification

5. Current membership in professional organizations

American Institute of Chemical Engineers

6. Honors and awards

Honoring your Professor UIC Award (2017)
Honeywell Team Performance Award, for designing state-of-the-art catalyst regeneration systems (2010)
BP Helios Award for developing state-of-the-art optimization technology (2000)
William M. Langdon Award For Excellence in Teaching, IIT, Chicago (1991-1992)
First prize award, National Competition in Mathematics, Greece (1974)
Five awards for outstanding performance (ranked # 1) in Chemical Engineering from National Fellowship Foundation, Greece (1974-79).
Three awards for outstanding performance (ranked # 1) in the entire College of Engineering, from the Technical Chamber, Greece (1974-79).

7. Service activities

Freshmen Engineering Success Program Committee, participant, 2017-2019
Chemical Engineering Advisory Committee, August 2016-present
Presenter, High School District 203 Career Night, Engineering, Industrial Technology, College recruitment activities; (2014-2019)

8. Selected Publications

Kinetic Modelling of Water Inhibited Methane Oxidation Reactions over a Pd/Al₂O₃ Wash-coated Monolith using Spatially and Temporally Resolved Experimental Data, Ciaran Coney, Cristina Stere, Paul Millington, Agnes Raj, Sam Wilkinson, Geoffrey McCullough, Michael Caracotsios, Chris Hardacre, Kevin Morgan, David Thompsett, Alexandre Goguet, ACS Catalysis (2020)

Nonlinear mixed-effects models for kinetic parameter estimation with batch reactor data, Daniel A. Hickman, Michael J. Ignatowich, Michael Caracotsios, James D. Sheehan and Fabio D'Ottaviano, *Chemical Engineering Journal* (2019)

Robustness of coke burning from catalyst for light paraffin dehydrogenation process, US Patent, US 8,895,464 B. Egolf, L. Leonard, M. Caracotsios (2014)

Computer Aided Modeling of Chemically Reactive Systems, co-authored with W. E. Stewart, *John Wiley and Sons, Inc.* , 2008

Innovations for refinery-wide modeling, simulation and optimization, *Hydrocarbon Asia*, 2001

Non-traditional optimization for isomerization unit improves profits, with T.C. Crowe *et al.*, *Oil and Gas Journal*, May 11, 1998

Polymerization of alpha-Olefins, (An advanced control method is disclosed for the polymerization of an alpha-olefin in a substantially horizontal, quench-cooled, stirred bed reactor) with A. Buchelli, *United States Patent*, Patent Number: 5,504,166, April 2, 1996.

Polymerization of alpha-Olefins, with A. Buchelli, *World Intellectual Property Organization*, International Bureau, International Publication Number WO 93/24533.

Sensitivity Analysis of Initial-Boundary-Value Problems with Mixed PDEs and Algebraic Equations. Applications to Chemical and Biochemical Systems, with W. E. Stewart, *Computers and Chemical Engineering*, Vol. 19, No. 9, pp. 1019-1030 (1995).

Sorption Kinetics and Axial Diffusion in Binary Solute Affinity Membrane Bio separations, with M. R. Etzel, *Chemical Engineering Science*, Vol. 48, No. 10, pp. 1801-1812 (1993)

Theoretical Modeling of Amoco's Gas Phase Horizontal Stirred Bed Reactor for the Manufacturing of Polypropylene Resins, *Chemical Engineering Science*, Vol. 47, No. 9-11, pp. 2591-2596, 1992.

9. Professional Development

Industrial Multivariable Process Control

Computational Fluid Dynamics and Combinatorial Chemistry.

Real Time On-Line Optimization with ASPEN RT-OPT

Listening, Interpersonal and Communication Skills, Influencing without Direct Authority

Concepts in Quality Manufacturing and Technology Transfer

Management Principles and Techniques and Managing Personal Growth

Artificial Intelligence and Expert Systems in Engineering

Polymer Reaction Engineering, Poly-olefins Manufacture Featuring Metallocene Catalysts

Advance Modeling and Control of Polymerization Reactors

Applications of Pinch Technology to Chemical Industries

Principles of Heterogeneous Catalysis and Organo-metallic Chemistry

Advanced Programming with Microsoft Visual Basic 3.0 for Windows, C and PASCAL

10. Open-Source Textbook

Michael Caracotsios, *An Introduction to Chemical Engineering Thermodynamics. Theory and Applications*, Spring 2026

Faculty Vita: Brian P. Chaplin

1. Education

- B.S. Civil Engineering, University of Minnesota, 1999
- M.S. Civil Engineering, University of Minnesota, 2003
- Ph.D. Environmental Engineering, University of Illinois at Urbana-Champaign, 2007

2. Academic Experience

- Department of Chemical Engineering, University of Illinois at Chicago
Director of Graduate Studies, 2024–present. Full-Time
- Department of Chemical Engineering, University of Illinois at Chicago
Professor, 2022–present. Full-Time
- Department of Chemical Engineering, University of Illinois at Chicago
Associate Professor, 2018–2022. Full-Time
- Department of Chemical Engineering, University of Illinois at Chicago
Assistant Professor, 2013–2018. Full-Time
- Department of Civil and Environmental Engineering, Villanova University
Assistant Professor, 2010–2013. Full-Time
- Department of Chemical and Environmental Engineering, University of Arizona,
Postdoctoral Research Associate, 2008–2010. Full-Time
- Department of Civil and Environmental Engineering, University of Illinois at Urbana-
Champaign, Postdoctoral Research Associate, 2007–2008. Full-Time

3. Non-Academic Experience

- U.S. Geological Survey, Mounds View, MN, Hydrologist, 1999–2003. Full-Time

4. Certifications or professional registrations

- E.I.T. certified (Minnesota since 1999)

5. Current membership in professional organizations

- American Institute of Chemical Engineers
- Association of Environmental Engineering and Science Professors

6. Selected Honors and awards

- 2019 *Environmental Science & Technology and Environmental Science & Technology Letters Virtual Issue: Early Career Scientists*
- 2018 *Environmental Science & Technology Best Paper Award*
- *NSF Early CAREER Development Award* (2015–2020)
- *Environmental Science & Technology Excellence in Review Award* (2017)
- *Environmental Science & Technology Letters Excellence in Review Award* (2016)
- UIC College of Engineering Research Award (2015)
- Summer Research Fellowship/Research Support Grant, Villanova University (2011)

- Environmental Water Resources Institute's 2011 Samuel Arnold Greeley Award (Best Paper). **Chaplin** et al., (2009). *Journal of Environmental Engineering-ASCE*. **135**(8), 666-676.

7. Selected service activities

- Associate Editor, *ACS ES&T Engineering*, 2024-present
- Director of Graduate Studies, Department of Chemical Engineering, 2024-present
- Editorial Advisory Board, *Environmental Science and Technology*, 2020-2024
- Member, *Chemical Engineering Departmental Advisory Board*, Department of Chemical Engineering 2018-present
- Chair, *Safety Committee*, Department of Chemical Engineering, 2018-present
- Chemical Engineering Department Seminar Series Coordinator, 2018-present.
- NSF CAREER panel reviewer, CBET-Env. Eng. Program, Nov 2015, Sept 2016
- NSF panel reviewer, CBET-Env. Eng. Program, March 2016, February 2017

8. Selected Publications from the last 5 years

1. Raghavan, S.; **Chaplin, B. P.**; Mehraeen, S. Elucidating the Impact of the Electrocatalyst Surface and Applied Potential on the Decarboxylation of the PFOA Radical on the Ti₄O₇ Facet Using DFT Simulations. *J. Phys. Chem. C* **2025**, *129* (23), 10234–10248. DOI: 10.1021/acs.jpcc.5c01234.
2. Zheng, X.; Gupta, N.; He, H.; Bindra, J. K.; Hossain, S.; Vizuet, J. P.; Nadeali, A.; Zangeneh, D.; Singh, R. P.; Klie, R. F.; **Chaplin, B. P.**; Brezinsky, K.; Poluektov, O. G.; Niklas, J.; Zapol, P.; Glusac, K. D. A Light-Responsive Metal–Organic Framework with Perchlorinated Nanographene Ligands. *J. Am. Chem. Soc.* **2025**, *147* (19), 16420–16428. DOI: 10.1021/jacs.5c02844.
3. Saffar-Avval, S.; Modiri Gharehveran, M.; Alvarez Ruiz, R.; Lee, L. S.; **Chaplin, B. P.** Matrix Effects on Electrochemical Oxidation of Per- and Polyfluoroalkyl Substances in Sludge Centrate. *Environ. Sci. Technol.* **2025**, *59* (16), 8263–8273. DOI: 10.1021/acs.est.4c13720.
4. Mohamed, M. S.; **Chaplin, B. P.**; Abokifa, A. A. Screening of Transition Metals for PFAS Adsorption: A Comparative DFT Investigation. *Chem. Eng. Sci.* **2025**, *307*, 121363. DOI: 10.1016/j.ces.2025.121363.
5. Khalid, Y. S.; Islam, S. M. M.; Mehraeen, S.; **Chaplin, B. P.** Reactive-Transport Modeling of Oxidation Pathways of Insensitive High Munitions in Porous Flow-Through Electrodes. *ACS ES&T Eng.* **2025**, DOI: 10.1021/acsestengg.4c00895.
6. Mohamed, M. S.; **Chaplin, B. P.**; Abokifa, A. A. Ab Initio Investigation of Per- and Polyfluoroalkyl Substances (PFAS) Adsorption on Zerovalent Iron (Fe⁰). *ACS Omega* **2024**, *9* (44), 44532–44541.
7. Hafeez, S.; Khanam, A.; Cao, H.; **Chaplin, B. P.**; Xu, W. Novel Conductive and Redox-Active Molecularly Imprinted Polymer for Direct Quantification of Perfluorooctanoic Acid. *Environ. Sci. Technol. Lett.* **2024**, *11* (8), 871–877.
8. Misal, S. N.; Li, D.; Kim, S.; **Chaplin, B. P.** Effect of Solution Conditions and Applied Potential on Ion Transport in TiO₂ Nanopores. *ACS ES&T Eng.* **2024**, *4* (10), 2495–2505.

9. King, J. F.; **Chaplin, B. P.** Electrochemical Reduction of Per- and Polyfluorinated Alkyl Substances (PFAS): Is it Possible? Applying Experimental and Quantum Mechanical Insights from the Reductive Defluorination Literature. *Curr. Opin. Chem. Eng.* **2024**, *44*, 101014.
10. Mohamed, M. S.; **Chaplin, B. P.**; Abokifa, A. A. Adsorption of Per- and Poly-fluoroalkyl Substances (PFAS) on Ni: A DFT Investigation. *Chemosphere* **2024**, *357*, 141849.
11. Islam, S. M. M.; **Chaplin, B. P.** Electrochemical Destruction of Insensitive High Explosives using Magneli Phase Titanium Oxide Reactive Electrochemical Membranes. *ACS ES&T Eng.* **2024**, *4* (5), 1240–1252.
12. Almassi, S.; Ren, C.; Dandu, N. K.; Ngo, A.; Liu, J.; **Chaplin, B. P.** Mechanistic Study of Electrocatalytic Perchlorate Reduction using an Oxorhenium Complex Supported on a Ti₄O₇ Support. *ACS Catal.* **2024**, *14*, 2597–2608.
13. Raghavan, S.; **Chaplin, B. P.**; Mehraeen, S. Small-Molecule Adsorption Energy Predictions for High-Throughput Screening of Electrocatalysts. *J. Chem. Inf. Model.* **2023**, *63* (17), 5529–5538.
14. Plunkett, S. T.; Kondori, A.; Chung, D. Y.; Wen, J.; Wolfman, M.; Lapidus, S. H.; Ren, Y.; Amine, R.; Amine, K.; Mane, A. U.; Asadi, M.; Al-Hallaj, S.; **Chaplin, B. P.**; Lau, K. C.; Wang, H.; Curtis, L. A. A New Cathode Material for a Li-O₂ Battery Based on Lithium Superoxide. *ACS Energy Lett.* **2022**, *7* (8), 2619–2626.
15. Almassi, S.; Ren, C.; Liu, J.; **Chaplin, B. P.** Electrocatalytic Perchlorate Reduction Using an Oxorhenium Complex Supported on a Ti₄O₇ Reactive Electrochemical Membrane. *Environ. Sci. Technol.* **2022**, *56* (5), 3267–3276.
16. Khalid, Y. S.; Misal, S. N.; Mehraeen, S.; **Chaplin, B. P.** Reactive-Transport Modeling of Electrochemical Oxidation of Perfluoroalkyl Substances in Porous Flow-through Electrodes. *ACS ES&T Eng.* **2022**, *2* (4), 713–725.
17. Samonte, P. R. V.; Li, Z.; Mao, J.; **Chaplin, B. P.**; Xu, W. Pyrogenic Carbon-Promoted Haloacetic Acid Decarboxylation to Trihalomethanes in Drinking Water. *Water Res.* **2022**, *210*, 117988.
18. Plunkett, S. T.; Zhang, C.; Lau, K. C.; Kephart, M. R.; Wen, J. G.; Chung, D. Y.; Phelan, D.; Ren, Y.; Amine, K.; Al-Hallaj, S.; **Chaplin, B. P.**; Wang, H.; Curtis, L. A. Electronic Properties of Ir₃Li and Ultra-nanocrystalline Lithium Superoxide. *Nano Energy* **2021**, *90*, 106549.

9. Professional development activities

Science Leadership Council, *Current*, Chicago, IL, 2017-present

Faculty vitae: Naveen Dandu

1. Education

Dec 2016 Ph.D. in Chemistry, North Dakota State University, North Dakota, USA
May 2013 M.S. in Computer Science, North Dakota State University, North Dakota, USA
Jun 2007 M.S. in Chemistry, University of Hyderabad, Hyderabad, TG, INDIA
May 2003 B.S. in Chemical Technology, Loyola Academy, Hyderabad, TG, INDIA

2. Academic Experience

2 - present Research Assistant Professor, Chemical Engineering, University of Illinois
 at Chicago, Illinois, USA
9 - 02/2022 Postdoctoral Appointee, Materials Science Division, Argonne National
 Lab, Lemont, Illinois, USA
7 - 02/2019 Postdoctoral Researcher, Department of Chemistry, University of Nevada,
 Reno, Nevada, USA
4 - 08/2014 Summer Intern, Los Alamos National Lab, Los Alamos, New Mexico,
 USA
3 - 08/2013 Summer Intern, Los Alamos National Lab, Los Alamos, New Mexico,
 USA
9 - 01/2012 Research Assistant, Center for Biopharma. R&D, Fargo, North Dakota,
 USA
3 - 05/2005 Research Assistant, Indian Institute of Chemical Technology, Hyderabad,
 TG, INDIA

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers

Association of Environmental Engineering and Science Professors

6. Selected Honors and awards

Not applicable

7. Selective Service Activities

- Serving as peer reviewer for Computational Materials Science Journal at Elsevier Publishers, Polymers for Advanced Technologies, ACS Catalysis, The Journal of Physical Chemistry,

iScience, Scientific Reports – open access, and Journal of Chemical Information and Modeling. (so far reviewed 40 articles)

- Teaching graduate and undergraduate level course titled "molecular and macromolecular engineering" (01/2023 to 05/2023).
- Served as a judge at 2020 National Science Bowl (01/2020)
- Served as a judge at Graduate Student Association Spring Awards – Research Grant Competition, University of Nevada, Reno, NV (03/2017)
- Volunteered in North Dakota Science Olympiad Program as an assistant organizer in forensic science division, North Dakota State University, Fargo, ND (04/2016)

8. Selected Publications Last 5 Years

Naveen K. Dandu, Alex Taekyung Lee, Sergio Ulloa, Larry Curtiss, Saw Wai Hla and Anh T. Ngo. Exploring the Coexistence of Spin States in $[\text{Fe}(\text{tpy-Ph})_2]^{2+}$ Complexes on Au(111) Using DFT Calculations. *J. Phys. Chem. A*, 129, 40, 9185–9194 (2025).

Mohammed Lemaalem, Selva Chandrasekaran Selvaraj, Ilias Papailias, Naveen K. Dandu, Arash Namaeighasemi, Larry A. Curtiss, Amin Salehi-Khojin and Anh T. Ngo. Interactive multiscale modeling to bridge atomic properties and electrochemical performance in Li-CO₂ battery design. *Applied Energy*, Volume 401, Part B (2025).

Ajayi, T. M.; Shirato, N.; Rojas, T.; Wieghold, S.; Cheng, X.; Latt, K. Z.; Trainer, D. J.; Dandu, N. K.; Li, Y.; Premarathna, S.; Sarkar, S.; Rosenmann, D.; Liu, Y.; Kyritsakas, N.; Wang, S.; Masson, E.; Rose, V.; Li, X.; Ngo, A. T.; Hla, S.-W. Characterization of Just One Atom Using Synchrotron X-Rays. *Nature* **2023**, 618 (7963), 69–73. <https://doi.org/10.1038/s41586-023-06011-w>.

Dandu, N. K.; Ward, L.; Assary, R. S.; Redfern, P. C.; Curtiss, L. A. Accurate Prediction of Adiabatic Ionization Potentials of Organic Molecules Using Quantum Chemistry Assisted Machine Learning. *The Journal of Physical Chemistry A* **2023**, 127 (28), 5914–5920. <https://doi.org/10.1021/acs.jpca.3c00823>.

Dandu, N. K.; Ward, L.; Assary, R. S.; Redfern, P. C.; Curtiss, L. A. Accurate Prediction of Adiabatic Ionization Potentials of Organic Molecules Using Quantum Chemistry Assisted Machine Learning. *The Journal of Physical Chemistry A* **2020**, 124 (28), 5804–5811. <https://doi.org/10.1021/acs.jpca.3c00823>.

Dandu, N. K.; Assary, R. S.; Redfern, R. C.; Ward, L.; Foster, I. and Curtiss, L. A. Improving the Accuracy of Composite Methods: A G4MP2 Method with G4-like Accuracy and Implications for Machine Learning. *J. Phys. Chem. A* **2022**, 126, 27, 4528–4536. <https://pubs.acs.org/doi/abs/10.1021/acs.jpca.0c01777>

Hemmat, Z.; Ahmadiparidari, A.; Wang, S.; Kumar, K.; Zepeda, M.; Zhang, C.; Dandu, N.; Rastegar, S.; Majidi, L.; Jaradat, A.; Ngo, A.; Thornton, K.; Curtiss, L. A.; Cabana, J.; Huang, Z.; Salehi-Khojin, A. Unprecedented Multifunctionality in 1D Nb_{1-x}Ta_xS₃ Transition Metal

Trichalcogenide Alloy. *Advanced Functional Materials* **2022**, 32 (34).
<https://doi.org/10.1002/adfm.202205214>.

Ward, L.; Dandu, N. K.; Blaiszik, B.; Narayanan, B.; Assary, R.; Redfern, P.C.; Foster, I. and Curtiss, L. A. Graph-Based Approaches for Predicting Solvation Energy in Multiple Solvents: Open Datasets and Machine Learning Models. *J. Phys. Chem. A* **2021**, 125, 27, 5990–5998.
<https://doi.org/10.1021/acs.jpca.1c01960>

Ward, L.; Sivaraman, G.; Pauloski, J. G.; Babuji, Y.; Chard, R.; Dandu, N. K.; Redfern, P.C.; Assary, R.; Chard, K.; Curtiss, L. A.; Thakur, R. and Foster, I. Colmena: Scalable Machine-Learning-Based Steering of Ensemble Simulations for High Performance Computing. 2021 IEEE/ACM Workshop on Machine Learning in High Performance Computing Environments (MLHPC), St. Louis, MO, USA, **2021**, pp. 9-20. doi: 10.1109/MLHPC54614.2021.00007.
<https://ieeexplore.ieee.org/abstract/document/9653177>

9. Selected Presentations

Year	Title of the talk	Conference Details
	State Calculations of Lanthanum and Europium based Critical Materials (Invited talk)	<i>Critical Materials workshop, Athens, Ohio</i>
	Time-Dependent DFT: Concepts and Applications (Invited talk)	<i>INDO-POLISH international workshop</i>
	Accurately Predicting Energies of Organic Molecules using Quantum Chemistry Informed Machine Learning	<i>Telluride Scientific Research Center Telluride, CO</i>
	State computation of the spin state energetics of transition metal complexes. Role of exchange-correlation functional, scalar-relativistic and solvent effects	<i>ACS conference, Boston, MA</i>
2018	Development of descriptors for methane-to-methanol conversion and side-reactions on zeolite-supported tri-metal tri-oxo catalysts	<i>ACS conference, New Orleans, LA</i>
	Performance of density functional theory for predicting hydrogenation and dimerization of ethylene using transition metal hydride catalysts supported on zeolites	<i>ACS conference, New Orleans, LA</i>

10. Professional Development Activities

Not applicable

Faculty Vitae: Sangil Kim

1. Education

B.S. Polymer Science and Engineering, Chungnam National University South Korea, 1997
M.S. Polymer Science and Engineering, Chungnam National University South Korea, 1999
M.S. Chemical Engineering, University of Cincinnati, Cincinnati, OH, 2003
PhD Chemical Engineering, Virginia Tech, 2007

2. Academic Experience

Department of Chemical Engineering, University of Illinois, Chicago, Associate Professor,
2022 – present, Full-Time
Department of Chemical Engineering, University of Illinois, Chicago, Assistant Professor,
2015 – 2022, Full-Time.

3. Non-Academic Experience

Lawrence Livermore National Lab, Livermore, CA, Guest Researcher, 2015 – present, Part-Time
Lawrence Livermore National Lab., Livermore, CA, Staff Scientist, 2011– 2015, Full-Time
Porifera Inc, Hayward, CA, Research Scientist, 2009– 2011, Full-Time
University of California Davis/Lawrence Livermore National Lab., Livermore, CA,
Post-doctoral Associate, 2007 – 2009, Full-Time
Korea Institute of Science and Technology (KIST), Seoul, South Korea, Researcher, Complex
Fluid and Membrane Laboratory, 2001, Full-Time

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

Electrochemical Society, Materials Research Society, American Institute of Chemical Engineers

6. Honors and awards

Not applicable

7. Service activities

- 2021-present: Editorial Board Member, *Membranes*
- Reviewer for journals (e.g., Langmuir, Journal of Membrane Science, Nanotechnology, Journal of Colloid and Interface Science, Journal of Microfluidics and Nanofluidics, Journal of Physics:Condensed Matter)
- Proposal reviewer for DOE-EERE and NSF

8. Selected Publications

Patents

- 1) J Ye, AC Baumgaertel, J Biener, MM Biener, S Kim, “Three-dimensional nanoporous membrane having multiple independent, continuous pore systems”, U.S. Patent 11,103,832, 2021
- 2) M. Worley, P. Campbell, S. Kim, and M. Merrill, "Molten Hydroxide Membrane for Separation of Acid Gases from Emissions", U.S. Patent 10,464,015, 2019.

- 3) PD. Hoeprich and S. Kim, “Electrochemical flow-cell for hydrogen production and nicotinamide dependent target reduction, and related methods and systems”, U.S. Patent 10,151,037, 2018.
- 4) S. Kim, P. Naraghi-Arani, L. Megan, “Filtration Device for Rapid Separation of Biological Particles from Complex Matrices”, U.S. Patent 9,861,939, 2018.
- 5) M. Worsley, J. Satcher, S. Kucheyev, S. Charnvanichborikarn, J. Colvin, T. Felter, S. Kim, M. Merrill, C. Orme, “Nanoporous Metal-Carbon Composites”, U.S. Patent 9,844,762, 2018.
- 6) R. Revanur, V. Lulevich, I. Roh, J. Klare, S. Kim, A. Noy, and O. Bakajin, “Membranes having aligned 1-D nanoparticles in a matrix layer for improved fluid separation”, U.S. Patent, 9,216,391, 2015.
- 7) O. Bakajin, A. Noy, F. Fornasiero, H. Park, J. Holt and S. Kim, “Membranes with Functionalized Carbon Nanotube Pores for Selective Transport”, U.S. Patent 8,940,173, 2015.
- 8) E. Marand and S. Kim, “Method for Making Oriented Carbon Nanotube/Polymer Nano-Composite Membranes”, U.S. Patent 7,931,838, 2011.

Book chapters

- 1) O. Bakajin, A. Noy, F. Fornasiero, C. P. Grigoropoulos, J. K. Holt, J. In, S. Kim, H. Park, Nanofluidic Carbon Nanotube Membranes: Applications for Water Purification and Desalination, Nanotechnology for Water Quality, Eds: N. Savage, et al., William Andrew Publishers, 2008.

Peer-reviewed publications in last 5 years (*co-corresponding author)

- 1) K. Wang, N. Hwee, W. Degraff, S. M. Biener, S. Huang, L. Feng, S. Watts, W. L. Smith, B. Wang, S. Chatterjee, T. W. Heo, J. Ye, G. Cheng, J. Biener, S. Kim*, “Zwitterionic Photocurable Resin for High-Resolution 3D Printing of Ultralow-Fouling Microstructures”, *Small Methods*, e01222, 2025
- 2) KM. Lwin, T. Kim, J. Lee, S. Han, J. Bae, S. Park, S. Kim*, K. Kim*, “Cost-effective thin metal–polyphenol coating for enhancing fouling resistance of ultrafiltration membranes”, *Sustainable Materials and Technologies*, e01509, 2025
- 3) K. Wang, V. Koverga, N. Maslekar, F. Wu, R. Kuphl, X. Lyu, P. Deshpande, H. Guo, H. Seol, W. Degraff, J. L. Schaefer, C. Fang, T. Li, G. Cheng, A. T. Ngo, S. Kim, “Novel Zwitterionic Polyurethane-in-Salt Electrolytes with High Ion Conductivity, Elasticity, and Adhesion for High-Performance Solid-State Lithium Metal Batteries,” *Advanced Energy Materials*, 2405676, 2025
- 4) L. Feng, S. Huang, D. Li, K. Wang, S. Kim, TW. Heo, J. Biener, “Holistic Microstructure Control Strategies in Photopolymerization-Induced Phase Separation of Acrylate Systems”, *Small Structures*, 2500081, 2025
- 5) S. Misal, D. Li, S. Kim*, B. Chaplin*, “Effect of Solution Conditions and Applied Potential on Ion Transport in TiO₂ Nanopores”, *ACS ES&T Engineering*, 2495-2505, 2024
- 6) S. Cetindag, A. Pendse, R. J. Castellano, P. Rehak, M. Howard, K. Wang, J. Yi, R. Praino, P. Kral, L. Liu, S. Kim*, J. Shan*, “Anomalous Diffusive and Electric-field-driven Ion Transport in Boron-Nitride Nanotubes”, *ChemRxiv*, 2023 (currently 2nd revision at *Nature Nanotechnology*)
- 7) K. Wang, H. Seol, A. Cheng, N. McKeague, M. Carlson, W. Degraff, S. Huang, and S. Kim, “Simple Bioparticle Filtration Device Based on an Ultralow-Fouling Zwitterionic

- Polyurethane Membrane for Rapid Large-Volume Separation of Plasma and Viruses from Whole Blood”, *Membranes*, 13, 524, 2023
- 8) A. Pendse, S. Cetindag, K. Wang, D. Li, R. J. Castellano, T. Wang, J. W. Shan, and S. Kim, “Intrinsic Ion Transports of Highly Charged Sub-3-nm Boron Nitride Nanotube”, *Materials Today*, 60, 79-90, 2022
 - 9) K. Wang, T. Arado, A. Huner, H. Seol, X. Liu, H. Wang, L. Hassan, K. Suresh, S. Kim*, G. Cheng*, “Thermoplastic zwitterionic elastomer with critical antifouling properties”, *Biomaterials Science*, 10, 2892-2906, 2022
 - 10) T. Wang, J. Lee, X. Wang, K. Wang, C. Bae, and S. Kim, “Surface-Engineered Nafion/CNT Nanocomposite Membrane with Improved Voltage Efficiency for Vanadium Redox Flow Batteries”, *Journal of Applied Polymer Science*, 51628, 2021
 - 11) K. Wang, H. Seol, X. Liu, H. Wang, G. Cheng*, and S. Kim*, “Ultralow-Fouling Zwitterionic Polyurethane-Modified Membranes for Rapid Large-Volume Separation of Plasma from Whole Blood”, *Langmuir*, 37, 10115, 2021
 - 12) T. Wang, X. Wang, A. Pendse, Y. Gao, K. Wang, C. Bae*, and S. Kim*, “High-efficient multifunctional electrochemical membrane for lithium polysulfide redox flow batteries”, *Journal of Membrane Science*, 636, 119539, 2021
 - 13) A. Pendse, S. Cetindag, P. Rehak, S. Behura, H. Gao, N.H.L. Nguyen, T. Wang, V. Berry, P. Kral, J. Shan*, and S. Kim*, “Highly Efficient Osmotic Energy Harvesting in Charged Boron-Nitride-Nanopore Membranes”, *Advanced Functional Materials*, 31, 2009586, 2021 (*co-corresponding author)
 - 14) T. Wang, J. Han, K. Kim, J. Lee, A. Muenchinger, Y. K. Choi, KD Kreuer, C. Bae*, and S. Kim*, "Novel Sulfonated Aromatic Polymer Ion Exchange Membranes for Breaking the Conventional Proton Selectivity/Conductivity Trade-off Limitation", *Materials Advances*, 1, 2206, 2020 (*co-corresponding author)
 - 15) T. Wang, S. Liang, Z. Qi, M. Biener, M. A. Worsley, T. Braun, Y. M. Wang, J. Biener, S. Kim*, and J. Ye*, "Nanoengineered 3D nm-thick Biomimetic Membrane for Ultrafast Selective Mass Transportation", *Materials Horizons*, 7, 2422 - 2430, 2020 (†co-corresponding author)
 - 16) T. Wang, J. Y. Jeon, J. Han, J. H. Kim, C. Bae*, and S. Kim*, "Poly(terphenylene) Anion Exchange Membranes with High Conductivity and Low Vanadium Permeability for Vanadium Redox Flow Batteries (VRFBs)", *Journal of Membrane Science*, 598, 117665, 2020 (*co-corresponding author)

9. Professional Development

Not applicable

Faculty Vitae: Ying Liu

i. Education

B.S. Engineering Mechanics, Tsinghua University, 2001

M.A. Mechanical and Aerospace Engineering, Princeton University, 2004

Ph.D. Mechanical and Aerospace Engineering, Princeton University, 2007

ii. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago

Assistant Professor, 2008-2015

Associate Professor, 2015-2021

Professor, 2021-2022

Dr. Satish C. & Asha Saxena Professor, 2022-current

Department of Biopharmaceutical Sciences, University of Illinois-Chicago

Adjunct professor, 2008-present

Department of Bioengineering, University of Illinois-Chicago

Adjunct professor, 2018-present

Department of Chemistry, University of Chicago

Post Doctoral Researcher, July 2007- July 2008

iii. Non-Academic Experience

Not applicable.

iv. Certifications or professional registrations

Not applicable.

v. Current membership in professional organizations

American Association for the Advancement of Science (AAAS)

American Institute of Chemical Engineers (AIChE)

American Physical Society (APS)

American Association of Pharmaceutical Science (AAPS)

Sino-American Pharmaceutical Association

vi. Honors and awards

Sino-American Pharmaceutical Professionals Association Leadership Award, 2023

Chicago Biomedical Consortium, Catalyst Award, 2020

UIC Graduate Mentor Award, UIC, IL, 2017

UIC Researcher and Scholar of the Year, Rising Star Award, UIC, IL, 2016

CAREER Award, National Science Foundation, Jan. 2014 – Jan. 2019

College of Engineering Research Award, UIC, 2013

OTM POC Award, UIC, 2013

Chancellor Discovery Award, UIC, 2011

Technology for Developing Regions Fellowship, Princeton University, 2006

Howard Crathorne Phillips Graduate Fellowship, Princeton University, 2003

Engineering Fellowship, Princeton University, 2002
Rong Hong National Fellowship, Tsinghua University, 2000

vii. Service activities

Interim Associate Dean of Research, 2024 – present
Engineering Dean Search Committee, 2023 – 2024
Director of Graduate Study, Department of Chemical Engineering, 2023 – 2024
UIC faculty senate, 2021 – 2024
College of Engineering, Executive Committee, UIC, 2016-2020, 2022 – present
Fellow of Honor's college, 2015 – present
Campus Research Board (CRB), 2016 – present
Member, Graduate Committee, Department of Chemical Engineering, University of Illinois at Chicago, 2008-2018
Student Sponsor for Illinois Math and Science Academy, 2010-2016
Mentor, Girls' Electronic Mentoring in Science, Engineering and Technology, 2009-present
Member, Editorial Board, Advances in Materials Science and Engineering, 2013-2019
Member, Editorial Board, Journal of Oncology and Biomarker Research, 2013-2018

viii. Selected Publications Last 5 Years (Selected from 33 publications)

1. Han Xiao, Bailin Feng, Dongdong Gao, Shuangni Shi, Yiqing Yang, Yiwei Zhang, Fengyang Wang, Qi Yao, Haiqing Song, Ying Liu*, Gang Cheng*, A naturally derived lipopeptide lipid nanoparticle platform enabling multiple nucleic acids delivery, *Bioactive Materials*, 54, 829 – 849, 2025.
2. Bailin Feng, Shiyu Du, Yiqing Yang, Yuli Zhu, Nash McKeague, Binhua Lin, Wei Bu, Gang Cheng, Ying Liu*, Molecular Effects of Zwitterionic Peptide on Monolayer Lipid Membranes upon Enzyme-catalyzed Degradation, *Langmuir*, 41 (5), 3402-3412, 2025. PubMed PMID: 39895134.
3. Yuli Zhu; Lionel Porcar; Thirupathi Ravula; Krishna C. Batchu; Tera Lavoie; Ying Liu, Ursula Perez-Salas, Unexpected asymmetric distribution of cholesterol and phospholipids in equilibrium model membranes, *Biophysical Journal*, 123 (22), 3923-3934, 2024.
4. Kaige Ma+, Tiep Pham+, Jun Wang, InSug O-Sullivan, Amy DiCamillo, Shiyu Du, Fackson Mwale, Zeba Farooqui, Gina Votta-Velis, Benjamin Bruce, Andre J. van Wijnen, B. Duncan X. Lascelles, Ying Liu*, Hee-Jeong Im*, Nanoparticle-based Inhibition of Vascular Endothelial Growth Factor Receptors Resolves Osteoarthritis-Associated Pain and Cartilage Damage, *Sciences Advances*, 10 (7), eadi5501, 2024.
5. Alex Cheng, Ying Liu, and Haiqing Song, Elevating nucleic acid delivery via a stable anionic peptide dextran ternary system, *Biointerphases*, 18, 051001, 2023.
6. Hui Tang, Maryam Zaroudi, Yuli Zhu, Alex Cheng, Lei Qin, Bin Zhang, and Ying Liu*, Toroidal-Spiral Particles as A CAR-T Cell Delivery Device for Solid Tumor Immunotherapy, *Journal of Controlled Release*, 362, 620 – 630, 2023.
7. Chang Liu, Sherry Gong, and Ying Liu*, Micelle Morphology Phase Diagram in a Phospholipid, PEGylated Lipid, and Peptide Amphiphiles Ternary System, *Chemical Engineering Research and Design*, 181, 354-362, 2022.
8. Tiep Pham, Paola Leon Plata, Pin Zhang, Anand Vellara, Wei Bu, Binhua Lin, Gang Cheng, and Ying Liu*, Knowledge-based Design of 5-Fluorouracil Prodrug Liposomal Formulation:

Molecular Packing and Interaction Revealed by Interfacial Isotherms and X-ray Scattering Techniques, *Molecular Pharmaceutics*, 18, 12, 4331-4340, 2021.

9. Paola Leon Plata, Ludwig C. Nitsche, and Ying Liu*, Manipulation of Toroidal-Spiral Particles Internal Structure by Fluid Flow, *Physics of Fluids*, 33, 054102, 2021.

10. Paola Leon-Plata, Maryam Zaroudi, Chun-Yin Lee, Colin Foster, Ludwig C. Nitsche, Peter D. Rios, Yong Wang, Jose Oberholzer, and Ying Liu*, Heterogeneous Toroidal Spiral Particles for Islet Encapsulation, *Biomaterials Science*, 9, 3954, 2021.
511 (1), 415-423, 2016.

ix. Professional Development

Not applicable

Faculty Vitae: Shafigh Mehraeen

1. Education

B.Sc. Mechanical Engineering, University of Tehran, Iran 2001
M.Sc. Mechanical Engineering, Stanford University, CA, 2006
PhD Mechanical Engineering, Stanford University, CA, 2011

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago,
Assistant Professor, 2016 – 2024, Full-Time
Associate Professor, 2024 – Present, Full-Time
Massachusetts Institute of Technology, MA, Chemistry Research Fellow, 2013-2016, Full-Time
Georgia Institute of Technology, GA, Chemistry Research Associate, 2011-2013, Full-Time

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

2015, Certificate of Teaching, MIT Teaching Program
2014, Certificate of Teaching, Vanderbilt Teaching Program

5. Current membership in professional organizations

American Institute of Chemical Engineering (AIChE)
American Chemical Society (ACS)

6. Honors and awards

2018, 2019, 2023, College of Engineering Teaching Award, UIC,
2014, Ascher Shapiro Award, MIT

7. Service activities

- Session Organizer, AIChE Midwest Regional Conference, Division of Self-Assembly of Nanostructures, Chicago, (2020 – Present)
- Session Organizer, APS, Division of Fluid Dynamics, Chicago (2020)
- Academic Editor, IntechOpen, London, UK (2019 – Present)
- Undergraduate Research Mentor at UIC (2016- on going)
- Graduate Admission Committee Member: Responsible for reviewing applications for M.S. and Ph.D. programs, reviewing instructional units and graduate courses concerning general policies for graduate education at UIC (2016- on going)
- Reviewer: NSF panel reviewer in CBET program

- Journal reviewer: Physical Review Letters, Physical Review E, Physical Review B, Organic Electronics, Physical Chemistry Letters, Physical Chemistry C, Langmuir (2016 - on going)

8. Selected Publications

1. S. Raghavan, B.P. Chaplin, and S. Mehraeen, Elucidating the impact of electrocatalyst surface and applied potential on the decarboxylation of PFOA radical on Ti₄O₇ [112] facet using DFT Simulations, Journal of Physical Chemistry, in press, (2025).
2. Y.S. Khalid, S.M.M. Islam, S. Mehraeen, and B.P. Chaplin, Reactive-transport modeling of oxidation pathways of insensitive high munitions in porous flow-through electrodes, ACS ES&T Engineering, in press, (2025).
3. S. Raghavan, B.P. Chaplin, and S. Mehraeen, Small-molecule adsorption energy predictions for high- throughput screening of electrocatalysts, Journal of Chemical Information and Modeling, 63(17), 5529- 5538 (2023).
4. D. Christiansen, and S. Mehraeen, Prediction of zwitterion hydration and ion association using machine learning, Soft Matter, 19, 3179-3189 (2023).
5. Z. Luo, and S. Mehraeen, Unraveling the impact of template geometry and confinement on template- assisted self-assembly of nanoparticles, Journal of Molecular Systems Design and Engineering, 8(8), 1075-1086 (2023).
6. B. K. Choi, J. Kim, Z. Luo, T. Hyeon, S. Mehraeen, S. Park, and J. Park, Shape Transformation Mechanism of Gold Nanoplates, ACS Nano, 17(3), 2007- 2018 (2022).
7. Z. Luo, and S. Mehraeen, Free energy contributions to the template-assisted self-assembly of nanopar- ticles using steered molecular dynamics simulations: Implications for molecular- specific imaging and sensing, Journal of Applied Nano Materials, 6(1), 283-295 (2022).
8. D. Christiansen, and S. Mehraeen, Hydration and ion interactions of zwitterionic homopolymers with varying carbon spacer lengths, Journal of Chemical Engineering Research and Design, 186, 174-183 (2022).

9. Professional Development

Mentor training workshop, UIC (2025)

Faculty Vitae: Roshan Yashwant Nemade

1. Education

B. E. Chemical Engineering, University of Mumbai, India, 2018

Ph.D. Chemical Engineering, University of Illinois, Chicago, IL, 2023

Post-doc Chemical Engineering, University of Illinois, Chicago, IL, 2024

2. Academic Experience

Department of Chemical Engineering, University of Illinois, Chicago, IL,
Research Assistant Professor, 2024 – Present

3. Non-Academic Experience

Grapherry, Inc., Chicago, Illinois, Co-founder,
2024-Present

Gharda Chemicals, India, In-plant trainee
06/2016 – 07/2016

4. Certifications or professional registrations

None

5. Current membership in professional organizations

None

6. Honors and awards

Postdoc Small Scholarship Award at University of Illinois Chicago, 2024.

Award for Graduate Research (AGR) at University of Illinois Chicago, 2022.

7. Service activities

1. Reviewer: Chemical Engineering Research and Design
2. Undergraduate Research Mentoring, University of Illinois Chicago
Provided supervision and skill-building guidance for undergraduate researchers each semester and summer, including experimental safety training, data interpretation, and support for research presentations and reports.

8. Publications

1. Bhawnani, Rajan R., Michael L. Barsoum, Ankit Dhakal, Vamsi V. Gande, Roshan Y. Nemade, Prem KR Podupu, Roberto dos Reis et al. "Dynamic structural and morphological transformations in high-entropy metal-organic frameworks." *Materials Today Chemistry* 49 (2025): 103009.
2. Jaradat, Ahmad, Musawenkosi K. Ncube, Ilias Papailias, Nikhil Rai, Khagesh Kumar, Volodymyr Koverga, Roshan Y. Nemade et al. "Fast Charge-Transfer Rates in Li-CO₂ Batteries with a Coupled Cation-Electron Transfer Process." *Advanced Energy Materials* (2024): 230346
3. Mukhopadhyay, Arani, Sungjoon Kim, Anish Pal, Roshan Nemade, Sreya Sarkar, Vikas Berry, and Constantine M. Megaridis. "Graphene-Enhanced Metal Condensers in Wick-Free Vapor Chambers for Thermal Management in Electronics." *In 2024 23rd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm)*, pp. 1-6. IEEE, 2024.
4. Nemade, R., Cotts, S., & Berry, V. "Graphene Fermi-Level Guided Attachment of Single Exoelectrogens and Induced Interfacial Doping", *ACS Appl. Mater. Interfaces* 2024, 16, 5, 5548–5553

5. Nemade, R., Cotts, S., & Berry, V. (2022). "Plasma treatment for enhanced microbe-electrode interfaces: A bio-electronic sink" *Journal of Power Sources*, 544, 231834. Perspective article
6. Dighe, A. V., Nemade, R., & Singh, M. R. (2019). "Modeling and simulation of crystallization of metal-organic frameworks." *Processes*, 7(8), 527. Selected for Journal Cover
7. Dr Ilias Papailias, Arash Namaeighasemi, Musawenkosi Ncube, Roshan Nemade, Dr Naveen Dandu, Nikhil Rai, Syed Ibrahim Gnani Peer Mohamed, Hessam Shahbazi, Suchit Sarin, Ahmad Jaradat, Shahriar Namvar, Pardis Seraji, Vikas Berry, Arunkumar Subramanian, Jeffrey Shield, Siamak Nejati, Dr Anh Ngo, Dr Larry Curtiss. "Pathways for Sustainable Reaction Kinetics in Li-CO₂ Batteries" *Nature Sustainability*, submitted: under consideration.
8. Mr Arani Mukhopadhyay , Dr Sungjoon Kim , Mr Anish Pal , Dr Roshan Nemade , Dr Sreya Sarkar , Vikas Berry. "Synergistic integration of graphene on laser-textured copper for thermal management of high-heat-flux electronics" *Nature Electronics*, submitted: under consideration.

Patents

1. Berry, Vikas, Roshan Nemade, Sungjoon Kim, and Sheldon Cotts. "ELECTRON-TUNNELING OR HOPPING BASED AERIAL VIRAL DETECTOR."

9. Professional Development Activities

1. NSF I-Corps Technology Commercialization Training (Summer 2022)
Participated as Co-Entrepreneurial Lead in a competitive national innovation program focused on customer discovery, value-proposition development, and commercialization pathways for engineering technologies.
2. Industrial Research Collaboration with Anheuser-Busch InBev (2020–2023)
Designed and assembled a scale-up CVD coating system in partnership with industrial R&D staff, reinforcing skills in engineering translation and manufacturing environments.
3. Grant Development & Sponsored Research Activities
Developed federal and state proposals as PI/Co-I/Key Personnel (DOE-LBNL, ISTC, NSF), strengthening capabilities in funding acquisition and lab sustainability.
U.S. Patent Application# 18/743,238
4. Berry, Vikas, Roshan Nemade, and Namrita Berry. " Method and Procedure to decoupled heating and electron injection to convert carbon content into graphene and other carbon allotropes"

Patent Application# 63/690,881
Berry, Vikas, Roshan Nemade, Kartikey Sharma. "Nano diagnosis of Skin Cancer with graphene based device", UIC Disclosure Ref. No. 2022-023
5. Berry, Vikas, Roshan Nemade, Kartikey Sharma. "RAPID-GR: Rapid, Large-Area Graphene Growth via Resistive Heating of Catalysts", UIC Disclosure Ref. No. 2024-082
6. Berry, Vikas, Roshan Nemade, Kartikey Sharma. "Remote Graphene Detector", UIC Disclosure Ref. No. 2022-009

Faculty Vitae: Vivek Sharma

1. Education

- B. Tech. Textile Technology, Indian Institute of Technology Delhi, New Delhi, India, 2001
M.S. Polymer Science, University of Akron, Akron, OH, 2003
M.S. Chemical Engineering, Minor: Nonlinear Sciences & Chaos, Georgia Institute of Technology, Atlanta, GA 2006
Ph.D. Polymer Science & Engineering, Minor: Physics, Georgia Institute of Technology Atlanta, GA, 2008
Post-Doc Mechanical Engineering & Hatzopoulos Microfluids Lab, Massachusetts Institute of Technology Cambridge, MA 2012

2. Academic Experience

Department of Chemical Engineering, University of Illinois Chicago, Professor, 2025-present, Full-Time

Department of Chemical Engineering, University of Illinois Chicago, Associate Professor, 2019-2025, Full-Time

Department of Chemical Engineering, University of Illinois Chicago, Assistant Professor, 2012-2019, Full-Time

Visiting Professor Appointments

SIMM: Soft Matter Sciences and Engineering Laboratory, ESPCI Paris – PSL, France. Visiting Professor (Chaire Paris Sciences) Nov 2025-Dec 2025

Pritzker School for Molecular Engineering (PME). University of Chicago, Hyde Park, Chicago, IL. Visiting Associate Professor, Jan 2023-June 2023

Chemical Engineering, Stanford University, Palo Alto, CA, Visiting Associate Professor, Sept 2022-Dec 2022.

Institute for Molecular Engineering, University of Chicago, Chicago IL, Visiting Assistant Professor, Sept 2017- Dec 2017, Part-Time

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers, Society of Rheology, American Physical Society (APS), Adhesion Society, American Chemical Society

6. Honors and awards

John H. Dillon Medal from the American Physical Society, 2023.

3M Non-tenured Faculty Award, 2019.

Best Presentation Award for talk on “Computational analysis of pinch-off dynamics and printability of simple and complex fluids”, *Flow-3D Users Meeting*, Santa Fe, NM 2017 (co-author Jelena Dinic).

Distinguished Young Rheologist Award, TA Instruments, 2015.

UIC College of Engineering (COE) Faculty Award For Teaching. 2025

UIC College of Engineering (COE) Faculty Award For Teaching. 2023

UIC College of Engineering (COE) Faculty Award For Advising. 2022

UIC College of Engineering (COE) Faculty Award For Teaching. 2017

UIC HOPE (Honoring Our Professor's Excellence) Award. 2017.

7. Service activities

Member of the graduate committee, chemical engineering (2014-2016 & 2017-2019, 2021-2022)

Seminar organizer for the department of chemical engineering, (2015-2019)

Website committee for the department of chemical engineering (2015-current)

Chair, Molecular Engineering Strategic Committee, for chemical engineering (2016-current)

Faculty search committee (2015-2016 & 2019-2020)

Outside UIC

Member-at-Large, APS-DPOLY, 2023-2026

Chair, SOR Centennial Committee, Society of Rheology (2022-2024)

Member, SOR Centennial Committee, Society of Rheology (2024-)

Member, Education Committee, APS DPOLY (2022-2023)

Member, ACS Colloids Summer Symposium Organizing Committee (2022-2028)

Publicity committee, APS DPOLY (2022-)

Chair, Membership Committee, GSOF (2016-2020); Helped in GSOF-DSOF transition!

Membership Committee, Society of Rheology (2017-2022)

Social Media Chair, Society of Rheology (2017-2022)

Programming Committee, Area 1J: Fluids, AIChE (2017-2025)

Reviewer: Soft Matter, Journal of Fluid Mechanics, Physical Review Letters, Advanced Materials, Journal of Rheology, Rheologica Acta, Physical Review E, International Journal of Multiphase Flow, Langmuir, Applied Rheology, International Journal of Polymer Science, Journal of Material Chemistry, Macromolecules, Lab Chip, Polymer Chemistry, RSC Advances, and Experiments in Fluids

Short Course Organizer or Instructor/ Workshop Organizer

Summer School on Soft Matter Systems: from Fundamentals to Sensory Perception, Instructor and co-organizer of demos, Cargese, Corsica, France, July 2026

Workshop on Molecular Engineering of Soft Matter, co-organizer with Shikha Nangia, and Harm-Anton Klok. Telluride, May 2025.

Summer School on Soft Matter Systems: from Fundamentals to Food, Instructor and co-organizer of demos, Cargese, Corsica, France, July 2024

Workshop on Molecular Engineering of Soft Matter, co-organizer with Shikha Nangia, and Harm-Anton Klok. Telluride, June 2023.

Short Course on Macromolecular Engineering of Formulations, co-organized with Thomas O'Connor, under the aegis of DPOLY at the *APS March Meeting*, 2021.

Short course on Structure and Rheology of Emulsions and Foams co-organized and co-taught with Sibani Lisa Biswal, at the *Society of Rheology Meeting*, Houston, TX.

8. Selected Publications

M-A. Fardin, M. Hautefeuille, and V. Sharma, “Radical scaling: Beyond feet and fingers”, *Soft Matter*, 2025. <https://doi.org/10.48550/arXiv.2507.02631>

L. Hassan, K. Al Zahabi[#], N.N. Nikolova, M. W. Boehm, S. K. Baier, and V. Sharma, “Rheology of animal and plant milks featuring dripping-onto-substrate rheometry”, *Phys. Fluids* (2025).

R. B. Reboucas, N. N. Nikolova, V. Sharma, “Modeling drop deformations and rheology of dilute to dense emulsions”, *Current Opinions in Colloid and Interface Science*, invited review, (2025).

C. Xu, Y. Zhang, and V. Sharma*, “Spatiotemporal mapping of nanotopography and thickness transitions of ultrathin foam films”, *Soft Matter*, 20, 3719-3727 (2024)

N. Nikolova[#], C. D. V. Martínez Narváez, L. Hassan, R. A. Nicholson, M. W. Boehm, S. K. Baier, and V. Sharma, “Rheology and dispensing of real and vegan mayo: the chickpea or egg problem”, *Soft Matter*, 19, 9413-9427 (2023). (Collaboration with Motif)

L. N. Jimenez, C. D. V. Martínez Narváez, and V. Sharma*, “Solvent Properties Influence Rheology and Pinching Dynamics of Polyelectrolyte Solutions: Thickening the Pot with Glycerol and Cellulose Gum”, *Macromolecules*, 55, 8117-8132 (2022).

J. Merchiers[†], C. D. V. Martínez Narváez[†], C. Slykas, N.K. Reddy and V. Sharma*, “Evaporation and rheology chart a processability map for centrifugal force spinning”, *Macromolecules*, 54 (23), 11061-11073 (2021). (w/ Hasselt)

C. Martinez Narvaez, X. Yu, C. Wang, R. Rock, H. Sun, and V. Sharma, “Rheology and pinching dynamics of associative polysaccharide solutions”, *Macromolecules*, 54, 6372-6288 (2021). (w/ PPG)

C. Ochoa, S. Gao, S. Srivastava, and V. Sharma*, “Foam film stratification studies probe intermicellar interactions”, C. Ochoa, S. Guo, S. Srivastava, and V. Sharma*, *Proceedings of the National Academy of Sciences of the USA*, 118 (25), e2024805118 (2021). (w/ UCLA)

J. Dinic and V. Sharma*, “Macromolecular Relaxation, Strain and Finite Extensibility Determine Visco-elastocapillary Thinning and Extensional Viscosity of Polymer Solutions”, *Proceedings of National Academy of Sciences*, 116(18), 8766-8774 (2019).

9. Professional Development

Chaired /organized sessions (>30) at ACS Colloids, APS-DFD, AIChE, SOR, APS March, ACS, and Adhesion Society Meetings.

Faculty Vitae: Meenesh Singh

1. Education

B.S.: B.E., Chemical Eng., Sardar Patel University, Gujarat, India, August 2001-05, GPA: 8.41/10

M.S.: MTech., Chemical Eng., Indian Institute of Technology Bombay, India, August 2006-08
GPA: 9.18/10

Ph.D.: Chemical Engineering, Purdue University, USA, Jan 2009-March 2013 GPA: 4.0/4.0

2. Academic Experience

Full Professor Department of Chemical Engineering, University of Illinois Chicago	Aug 2025 – Present
Joint Appointment Argonne National Laboratory	April 2024 – present
Visiting Faculty Department of Chemical Engineering, Indian Institute of Technology Roorkee	Jan 2023 – present
Associate Professor Department of Chemical Engineering, University of Illinois Chicago	Aug 2023 – July 2025
Associate Professor Department of Chemical Engineering, University of Illinois Chicago	Aug 2016 – July 2023

3. Professional Experience

Chief Technology Officer eN-RAMPS LLC.	July 2021 – present
Affiliate Scientist Joint Center for Artificial Photosynthesis, Lawrence Berkeley National Lab	Aug 2016 – Nov 2017
Postdoctoral Scholar Joint Center for Artificial Photosynthesis, U C Berkeley, Lawrence Berkeley National Lab Title: Artificial Photosynthesis for CO ₂ Reduction Supervisor: Alexis T. Bell, Adam Z. Weber	Jan 2014 – Aug 2016
Postdoctoral Scholar Joint Center for Artificial Photosynthesis, U C Berkeley, Lawrence Berkeley National Lab Title: Analysis of Solar-Fuel Generators Supervisor: Rachel A. Segalman	March 2013 – Jan 2014
Research Intern, Abbott Labs: Global Pharma R&D Title: Experimental Investigation of Crystal Morphology during Dissolution and Growth Supervisors: Nandkishor Nere, Hsien-Hsin Tung and Shailendra Bordawekar	May 2011 – July 2011
Graduate Research Assistant, Purdue University Title: Towards Control of Crystal Shape and Morphology Distributions in Crystallizers	Jan 2009 – March 2013

Advisor: Doraiswami Ramkrishna

Research Assistant, IIT-B

July 2008 – Nov 2008

Title: Monte Carlo Simulations for Growth of Nanoparticles in Microemulsion

Supervisors: Anurag Mehra, Rochish Thaokar and Jayesh Bellare

Graduate Research Assistant, IIT-B

Jan 2007 – May 2008

Title: Cryogenic Grinding for Synthesis of Nanoparticles and Extraction of Protein and DNA from Cells

Advisors: Jayesh Bellare and Sandip Roy

4. Certifications or professional registrations

Editor	Chemical Engineering Research & Design (Elsevier)	2020 – Present
Associate Editor	Energy Lab (Lab Academic Press)	2022 - Present
Advisory Board	Challenges (MDPI)	2020 - Present
Guest Editor	Special issue on “Modeling and Control of Crystallization” in Processes	2019
Guest Editor	Special issue on “Electroreduction of Activated Nitrogen Compounds” in Communications Chemistry (Nature Portfolio Journal)	2024

5. Current membership in professional organizations

Member	American Institute of Chemical Engineers	2009 – Present
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6. Honors and awards

1.	Faculty Entrepreneur Leadership Program (FELP) Cohort Member	Aug 2025
2.	UIC’s Outstanding Global Scholar	Sep 2024
4.	UIC Proof-of-Concept Phase-II Award	Jan 2023
5.	UIC CTRI Award	May 2023
6.	UIC CoE Advising Award	Sep 2022
8.	UIC Proof-of-Concept Phase-I Award	Jan 2021
9.	UIC CTRI Award	May 2021
10.	UIC CoE Research Award	May 2021
11.	George Klinzing Best PhD Award, AIChE	Nov 2015
13.	Faculty Lectureship Award, School of Chemical Engineering, Purdue University	March 2014
14.	AIChE Process Development Division Student Paper Award	Nov 2013
15.	Outstanding Graduate Student Research Award, College of Engineering, Purdue University	March 2013
16.	AIChE Separation Division Graduate Student Research Award	Oct 2012
17.	McDonnell Douglas Fellowship	Fall 2012

7. Service activities

- Member of 4 college-level committees
- AIChE Chem-E-Car and ChemE Cube advisor at UIC
- Editorial board member for 5 scientific journals
- Organized 34 technical sessions at international conferences

8. Selected Publications

1. Emani, Pavan S., Hisham Maddah, Arjun Rangoonwala, Songwei Che, Aditya Prajapati, Meenesh R. Singh, Dieter M. Gruen, Vikas Berry, and Sanjay K. Behura. "Organophilicity of Graphene Oxide for Enhanced Wettability of ZnO Nanorods," *ACS Appl. Mater. Interfaces*, 12 (35), 39772 – 39780, 2020.
2. Nishithan C. Kani, Aditya Prajapati, Brianna A. Collins, Jason D. Goodpaster, and Meenesh R. Singh, "Competing Effects of pH, Cation-Identity, H₂O Saturation, and N₂ Concentration on the Activity and Selectivity of Electrochemical Reduction of N₂ to NH₃ on Electrodeposited Cu at Ambient Conditions," *ACS Catalysis*, 10 (24), 14592 – 14603, 2020.
3. Ayankola O. Ayansiji, Anish V. Dighe, Andreas A. Linninger, and Meenesh R. Singh, "Constitutive Relationship and Governing Physical Properties for Magnetophoresis," *PNAS*, 117 (48), 30208-30214, 2020.
4. Aditya Prajapati, Brianna A. Collins, Jason D. Goodpaster, and Meenesh R. Singh, "Fundamental Insights into Electrochemical Activation of Methane towards Methanol over Transition Metal Oxides," *PNAS*, 118 (8), e2023233118, 2021.
5. Paria Coliaie, Manish S. Kelkar, Marianne Langston, Chengxiang Liu, Neda Nazemifard, Daniel Patience, Dimitri Skliar, Nandkishor K. Nere, and Meenesh R. Singh, "Advanced Continuous-Flow Microfluidic Device for Parallel Screening of Crystal Polymorphs, Morphology and Kinetics at Controlled Supersaturation," *Lab on a Chip*, 21 (12), 2333-2342, 2021.
6. John Cavin, Alireza Ahmadiparidari, Leily Majidi, Arashdeep Singh Thind, Saurabh N. Misal, Aditya Prajapati, Zahra Hemmat, Sina Rastegar, Andrew Beukelman, Meenesh R Singh, Kinga A. Unocic, Amin Salehi-Khojin, Rohan Mishra, "Two-Dimensional High-Entropy Transition Metal Dichalcogenides for Carbon Dioxide Electrocatalysis," *Adv. Mater.*, 33, 2100347, 2021.
7. Nishithan C. Kani, Joseph Gauthier, Aditya Prajapati, Isha Bordawekar, Windom Shields, Mitchell Shields, Aayush R. Singh, Meenesh R. Singh, "Solar-Driven Electrochemical Synthesis of Ammonia with >10% Solar-to-Fuel Efficiency at Ambient Conditions," *Energy Environ. Sci.*, 14, 6349-6359, 2021.
8. Aditya Prajapati, Rohan Sartape, Tomas Rojas, Naveen K. Dandu, Pratik Dhakal, Amey S. Thorat, Jiahua Xie, Ivan Bessa, Miguel T. Galante, Marcio H. S. Andrade, Robert T. Somich, Marcio V. Rebouças, Gus T. Hutras, Nathalia Diniz, Anh T. Ngo, Jindal Shah and Meenesh R. Singh, "Migration-assisted, moisture gradient process for ultrafast, continuous CO₂ capture from dilute sources at ambient conditions," *Energy Environ. Sci.*, 15, 680-692, 2022.
9. Anish V. Dighe, Prem K. R. Podupu, Paria Coliaie, and Meenesh R. Singh, "Three-Step Mechanism of Antisolvent Crystallization," *Cryst. Growth Des.*, 22 (5), 3119-3127, 2022.
10. Aditya Prajapati, Rohan Sartape, Nishithan Kani, Joseph A. Gauthier, and Meenesh R. Singh, "Chloride-Promoted, High-Rate Ambient Electrooxidation of Methane to Methanol on a Patterned Cu-Ti Bimetallic Oxides," *ACS Catalysis*, 12 (22), 14321- 14329, 2022.
11. Aditya Prajapati, Rohan Sartape, Miguel T. Galante, Jiahua Xie, Samuel Leung, Ivan Bessa, Marcio H. S. Andrade, Robert T. Somich, Marcio V. Rebouças, Gus T. Hutras, Nathalia Diniz,

- and Meenesh R. Singh, “Fully Integrated Electrochemical System that Captures CO₂ from Flue Gas to Produce Value-Added Chemicals at Ambient Conditions,” *Energy Environ. Sci.*, 15, 5105 - 5117, 2022.
12. Anish V. Dighe, Prem K. R. Podupu, and Meenesh R. Singh, “Emulsification of Supersaturated Solutions Amplifies Induction Time Variation in Crystallization,” *Cryst. Growth Des.*, 23 (9), 6290 – 6297, 2023
 13. Rohit Chauhan, Rohan Sartape, Rashmi Mishra, Jindal K. Shah, and Meenesh R. Singh, High-throughput Measurements of CO₂ Permeance and Solubility in Ionic Liquid Reveal a Synergistic Role of Ionic Interactions and Void Fractions, *Chem. Eng. J.*, 496, 153697, 2024.
 14. Ishita Goyal, Nishithan C. Kani, Samuel A. Olusegun, Sreenivasulu Chinnabattigalla, Rajan R. Bhawnani, Ksenija D. Glusac, Aayush R. Singh, Joseph A. Gauthier, and Meenesh R. Singh, “Metal Nitride as a Mediator for the Electrochemical Synthesis of NH₃,” *ACS Energy Lett.*, 9, 4188-4195, 2024.
 15. Samuel A. Olusegun, Yancun Qi, Nishithan C. Kani, Meenesh R. Singh, and Joseph A. Gauthier, “Understanding activity trends in electrochemical dinitrogen oxidation over transition metal oxides,” *ACS Catalysis*, 14, 16885-16896, 2024.
 16. Rajan R. Bhawnani, Rohan Sartape, Vamsi V Gande, Michael L. Barsoum, Elias M. Kallon, Roberto dos Reis, Vinayak P. Dravid, and Meenesh R. Singh, “Non-aqueous Electrochemical CO₂ Reduction to Multivariate C2-products over Single Atom Catalyst at Current Density up to 100 mA/cm²,” *Small*, 21 (18), 2408010, 2024.
 17. Prem K.R. Podupu, Vamsi V. Gande, Ragavendra Hari, Akshay Korde, Manish S. Kelkar, Nandkishor K. Nere, Meenesh R. Singh, “Analytical solution to the discretized population balance equation for pure breakage with application to kernel identification,” *Chem. Eng. Res. Des.*, 217, 295-303, 2025.
 18. Rohan Sartape, Rohit Chauhan, Venkata S.R.P. Yadavalli, Ishita Goyal, Ishaku Amos, Yancun Qi, Vamsi V. Gande, Abdul M. Sarkar, Ksenija D. Glusac, and Meenesh R. Singh, “Tetraethylammonium Cation Activates Fe for Selective Electroreduction of CO₂ to Oxalate,” *ChemElectroChem*, e202500093, 2025.
 19. Rohit Chauhan, Huda Usman, Nitin Minocha, Mehdi Molaei, Tagbo H.R. Niepa, and Meenesh R. Singh, “Predictive Modeling and Experimental Validation of Magnetophoretic Delivery of Magnetic Nanocultures,” *ACS Mater. Lett.*, 7, 2679 – 2685, 2025.
 20. Prem K.R. Podupu, Prince Verma, Ankit Dhakal, Rajan R. Bhawnani, Meagan Phister, Anish V. Dighe, Kevin H. Stone, Gaurav Giri, and Meenesh R. Singh, “Mechanistic Insights into Metal-Organic Framework Thin Film Growth from Microkinetic Analysis of In-Situ X-Ray Scattering Data,” *Matter*, 102430, 2025.
 21. Ishita Goyal, Najmin Isa, Vamsi V. Gande, Rohit Chauhan, and Meenesh R. Singh, “Stabilizing Calcium Nitride for Efficient, Long-Term Electrochemical Ammonia Synthesis,” *PNAS*, 2025.

9. Professional Development

- Awarded total project funding ~ \$9.2 MM and about \$3.7 MM to his lab.
- Research lab expenditures >\$500 K per year.
- Strong collaboration with 7+ multinational companies with continued funding support.
- Chief Technology Officer at eN-RAMPS LLC. (UIC Start-up)
- Actively involved in prototyping and scale-up of 4 technologies developed in PI’s lab.
- Raised ~\$1M in industrial funding to scale up technologies.

Faculty vitae: Christos G. Takoudis

1. Education

Diploma in Chemical Engineering, National Technical University of Athens, Greece, 1977.
Ph.D., Department of Chemical Engineering, University of Minnesota, 1982.

2. Academic Experience

- University of Illinois Chicago, Department of Chemical Engineering, Professor 8/1996 -
- University of Illinois Chicago, Department of Bioengineering, Professor 2/2005 -
- University of Illinois Chicago, Department of Restorative Dentistry, Adjunct Professor 2/2014 -
- University of Illinois Chicago, Department of Chemistry, Adjunct Professor 1/2025 -
- Purdue University, School of Chemical Engineering, Professor (8/1995 - 7/1998), Associate Professor (8/1987 - 7/1995), Assistant Professor (12/1981 - 7/1987).
- University of Minnesota, Department of Chemical Engineering, Instructor (1/1981 - 12/1981).
- UNESP – Universidade Estadual Paulista, Bauru, Brazil (6/2015),
- Biomedical Institute of Research and Technology, Greece, Visiting Professor (4/2008-8/2008)
- University of Cambridge, Department of Chemical Engineering, Visiting Professor (2/1990 - 6/1990).
- University of Leeds, School of Chemistry, Visiting Professor (1/1990 - 2/1990).

3. Non-Academic Experience

- Applied Materials, Epi Applications Lab, Santa Clara, Visiting Professor, (1/1988 - 2/1988).
- British Petroleum Research Centre, Sunbury-on-Thames, England, Research Engineer, (7/1975 - 9/1975).

Certifications or professional registrations

None

5. Current membership in professional organizations

American Institute of Chemical Engineers (AIChE)
The Electrochemical Society (ECS)
American Chemical Society (ACS)
American Physical Society (APS)
American Vacuum Society (AVS)
Technical Chamber of Greece (TEE)

6. Honors and awards

- 2002-2003 Advisor of the Year Nominee, University of Illinois at Chicago.
- 2000-2001 Teaching Recognition Program Award, University of Illinois at Chicago.

- Member of the Editorial Board of the Journal: "*Chaos, Solitons and Fractals. Applications in Science and Engineering*," Pergamon Press, (1990 - 2006).
- ARCO Faculty Fellow Award (1982), Purdue University.
- Institute of Technology Fellow (1979), University of Minnesota
- Bush Foundation Fellow (1978 - 1981), University of Minnesota
- Ph.D. Dissertation Fellow (1977), University of Minnesota
- Kondilion Award (1977), Greece
- Thomaidion Award (1977), Greece
- Chryssovergon Award (1977), Greece
- First Award of the Technical Chamber of Greece (1977)

7. Selected service activities

- 236 refereed journal publications
- Graduated 40 Ph.D. students
- Graduated 31 M.S. students
- 8 ex-students are professors

University and Departmental Service

Ch.E. Graduate Committee (July 1983 - June 1985).

Purdue ASEE Chapter (July 1983 - June 1991).

Ch.E. Seminar Committee (July 1984 - June 1985 ; July 1987 - June 1988).

Ch.E. Undergraduate Committee (July 1985 - June 1987 and July 1989 - Present; Chairman of the Committee: August 1993 - August 1996).

Engineering Library Coordination Committee (August 1990 - 1996).

Andrews Fellowships Committee (1992 and 1993).

New Directions Curriculum Committee (1992 - 1996).

Chemical Engineering Safety Committee (January 1994 - 1996).

Purdue-ChE Strategic Planning Committee (August 1995 - May 1996)

UIC-ChE, Advisory Committee (August 1996 - August 1999, August 2003 – October 2004)

UIC-ChE, Graduate Committee (August 1996 – October 2002; August 2003 – August 2004)

Bio-Natural Sciences and Engineering Advisory Committee (January 1997 – December 2007)

UIC-ChE, Director of Graduate Studies (September 1997 - August 1999; September 2001 – October 2002)

UIC-ChE, Undergraduate Committee (August 2004 – March 2005)

UIC-BioE Curriculum Committee (March 2005 – August 2006)

UIC-BioE Awards and Scholarship Committee (Chair, August 2006 – 2012)

UIC-BioE Advisory Committee (August 2006 – 2009)

UIC-BioE Faculty Search Committee (November 2009 – May 2010)

UIC-COE Space Allocation Committee (April 2006 – March 2007)

UIC-COE Scholarship and Awards Committee (August 2006 – 2012)

UIC-BioE Faculty Search Committee (November 2011 – 2015)

UIC-BioE Graduate Committee (August 2011 - present)

UIC-BioE Seminar Committee (August 2014 – present)

UIC-COE College Executive Committee (August 2011 – August 2013)

UIC Internal Grants Review Panel Committee (August 2011 – August 2012)
UIC Chancellor's Committee on the Status of Persons with Disabilities (August 2012 – August 2017; August 2018 - present; Co-Chair: August 2015 – August 2017)
UIC Diversity Advisory Committee (August 2015 – August 2017)

International Service

Evaluator of the Chemical Engineering Department, University of Patras, Patras, Greece;
November 1118, 2015

Evaluator of the Physics/SFI Program, Dublin City University, Dublin, Ireland; October 12-15, 2016

Evaluator of the Chemistry Department, Kapodistrian University of Athens, Athens, Greece;
February 1824, 2019

Evaluator of at least one Department of either Chemical Engineering or Chemistry or Physical Sciences per year, in Greece, after 2019.

8. Selected Publications

- Radhe Agarwal, Yogesh Sharma, Siliang Chang, Krishna C. Pitike, Changhee Sohn, Serge M. Nakhmanson, Christos G. Takoudis, Ho Nyung Lee, Rachel Tonelli, Jonathan Gardner, James F. Scott, Ram S. Katiyar, and Seungbum Hong, "Room-temperature relaxor ferroelectricity and photovoltaic effects in SnTiO_x/Si thin film heterostructures," Physical Review B 97, 054109/1-7 (2018).
- Siliang Chang, Christos G. Takoudis, "Growth behavior and structural analysis of atomic layer deposited Sn_xTi_{1-x}O_y films," J. Vac. Sci. Technol. A 36, 01A121-1/-8 (2018).
- Sathees Kannan Selvaraj, Jaya Parulekar and Christos G Takoudis, "Selective atomic layer deposition of zirconia on copper patterned silicon substrates using ethanol as oxygen source as well as copper reductant", Journal of Vacuum Science and Technology A, 32, 010601 (2014)
- Multi-metal films, alternating film multilayers, formation methods and deposition system, Takoudis, Christos G.; Singh, Manish; Selvaraj, Sathees Kannan, Patent US-1,021,4817 issued in February 2019
- Sarah Hashemi Astaneh, Gregory Jursich, Cortino Sukotjo, Christos G. Takoudis, "Surface and Subsurface film growth of titanium dioxide on polydimethylsiloxane by atomic layer deposition," Applied Surface Science 493, 779-786 (2019)

9. Professional Development Activities

Served as a member of the National Research Council committee evaluating graduate programs
390 research presentations at conferences/meetings

Over 200 invited seminars

Over 180 REU-NSF undergraduate students; Over 25 RET-NSF teachers

Consulting:

Applied Materials, Santa Clara, CA, Air Liquide, Countryside, IL & Newark, DE

BP, Naperville, IL, Mobil, Paulsboro, NJ, Motorola, Austin, TX, Delco Electronics, Kokomo,

IN, Amoco Oil Company, Naperville, IL, National Semiconductor, Santa Clara, CA

Faculty vitae: Korosh Torabi

1. Education

Ph.D. Chemical Engineering Purdue University, West Lafayette, IN	Dec. 2011
M.S. Chemical Engineering Illinois Institute of Technology, Chicago, IL	July 2007
M.S. Energy Systems Engineering Sharif University of Technology, Tehran, Iran	June 2005
B.S. Chemical Engineering University of Isfahan, Isfahan, Iran	July 2003

2. Academic Experience

Clinical Assistant Professor University of Illinois Chicago, Chicago, IL Department of Chemical Engineering	Jan. 2023- Present
Assistant Professor Wayne State University, Detroit, MI Department of Chemical Engineering and Materials Science	Aug. 2014- Aug. 2022
Postdoctoral Fellow Northwestern University, Evanston, IL Department of Chemistry and Nonequilibrium Energy Research Center	Jan 2012-Aug 2014

3. Non-Academic Experience

Not applicable

4. Certification or professional registrations

None

5. Current membership in professional organizations

- American Institute of Chemical Engineers
- The Biophysical Society
- American Chemical Society

6. Honors and Awards

- Not applicable

7. Selected service activities

Course development:

- Advanced Thermodynamics (UIC, CHE 501)
- Fluid Phase Equilibria (UIC, CHE 502)
- Transport Phenomena I (UIC, CHE 311)
- Transport Phenomena II (UIC, CHE 312)
- Chemical Reaction Engineering (UIC, CHE 321)
- Advanced Thermodynamics (WSU, CHE-MSE 7300)
- Thermodynamics (WSU, CHE 3300)
- Advanced Engineering Mathematics (WSU, CHE-MSE 7100)
- Materials Chemistry and Engineering (WSU, CHM 6060-7060)

Ph.D. Dissertations

Mojtaba Jokar, A statistical mechanical model for predicting stability and pairing-selectivity of coiled-coil protein structures, Aug. 2019. (PI, main advisor)

Zhixia Liu, Computational studies of allostery and dynamics of Myosin motor proteins, Expected April 2022. (PI, main advisor)

Master Theses

Deepak Joshi, A computational study of heterogeneous crystallization of nanoparticles via phase field crystal modeling, May 2017. (PI, main advisor)

Academic Administration and Service:

- ChemE Connect committee, UIC
- Department Action Team committee, UIC
- College of Engineering Academic Operations Committee, WSU
- College of Engineering Budget Committee, WSU
- College of Engineering Academic Standards Committee, WSU
- Undergraduate Curriculum and ABET Committee, WSU
- Graduate Program Committee, WSU

8. Selected Publications and presentations

- Kilani, M.; Torabi, K.; Mao, G. Application of Virtual Laboratories and Molecular Simulations in Teaching Nanoengineering to Undergraduate Students. *Computer Applications in Engineering Education* **2018**, 26 (5), 1527–1538.
- Soroush Barhaghi, M.; Torabi, K.; Nejahi, Y.; Schwiebert, L.; Potoff, J. J. Molecular Exchange Monte Carlo: A Generalized Method for Identity Exchanges in Grand Canonical Monte Carlo Simulations. *The Journal of Chemical Physics* **2018**, 149 (7), 72318.
- Zhou, Y.; Zhou, X.; Park, D. J.; Torabi, K.; Brown, K. A.; Jones, M. R.; Zhang, C.; Schatz, G. C.; Mirkin, C. A. Shape-Selective Deposition and Assembly of Anisotropic Nanoparticles. *Nano Letters* **2014**, 14 (4).
- Torabi, K.; Schatz, G. C. Tensile Mechanics of α -Helical Polypeptides. *Macromolecules* **2013**, 46 (19), 7947–7956.
- Torabi, K.; Corti, D. S. Toward a Molecular Theory of Homogeneous Bubble Nucleation: I. Equilibrium Embryo Definition. *The Journal of Physical Chemistry B* **2013**, 117 (41), 12479–12490.

- Torabi, K.; Corti, D. S. Toward a Molecular Theory of Homogeneous Bubble Nucleation: II. Calculation of the Number Density of Critical Nuclei and the Rate of Nucleation. *The Journal of Physical Chemistry B* **2013**, *117* (41), 12491–12504.

Selected Conference Presentations, Seminars, and Invited Talks

- Department of Chemical Engineering Seminar, April 2019. University of Toledo, Toledo, OH. Computational Studies of protein structure and dynamics.
- Biophysical Society Annual Meeting, Winter 2018, San Francisco, CA. Zhixia Liu and Korosh Torabi, Allosteric Network in Myosin Motor Protein.
- AIChE Annual Meeting, Fall 2017, Minneapolis, MN. Mojtaba Jokar and Korosh Torabi, Electrostatic Interactions and Coiled-coil Coupling Specificity.
- 19th International Union of Pure and Applied Biophysics, Summer 2017, Edinburgh, UK. Korosh Torabi, Electrostatic Interactions and Coiled-coil Coupling Specificity
- Biophysical Society Annual Meeting, Winter 2017, New Orleans, LA. Mojtaba Jokar and Korosh Torabi, Tensile mechanics of Coiled-coil peptides.
- Biophysical Society Annual Meeting, Winter 2016, Los Angeles, CA. Mojtaba Jokar and Korosh Torabi, Tensile mechanics of Coiled-coil peptides.

9. Professional development activities

- The Biophysical Society Education Committee
- I have served as a member of the Education Committee of the Biophysical Society for a three-year term beginning July 1, 2019, and ending June 30, 2022.
- Engineering Technology Advisory Board at Henry Ford Community College
- I served as a member of the advisory board of the Engineering Technology at Henry Ford Community College from 2015 to 2018.
- AIChE Annual Meeting Session chair
- Nucleation and Growth I and III: AIChE Annual Meeting, Fall 2012, Pittsburgh, PA

Faculty Vitae: Lewis E. Wedgewood

1. Education

Ph.D. Chemical Engineering, University of Wisconsin-Madison, 1988,
Major Professor: R. Byron Bird,
Research Area: Macromolecular Fluid Mechanics
Thesis Title: *A Theory of Chainlike Polymers: This thesis investigated the effects of hydrodynamic interaction on flow material properties*
B.S. Chemical Engineering, Cum Laude with highest distinction, University of Illinois-Urbana, 1982

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago
Associate Professor, 1998-Present
Department of Chemical Engineering, University of Wisconsin-Madison,
Visiting Faculty, 1997-1998
Department of Chemical Engineering, University of Washington,
Assistant Professor, 1990-1998
Department of Chemical Engineering, Wisconsin-Madison,
Research Associate, 1989-1990
NATO/NSF Postdoctoral Fellow, Albert-Ludwigs University, Freiburg, West Germany,
1988-1989

3. Non-Academic Experience

None

4. Certifications or professional registrations

None

5. Current membership in professional organizations

American Institute of Chemical Engineers 1982-present
The Society of Rheology 1982-present

6. Honors and awards

Award for Excellence in Teaching / University of Illinois at Chicago, 2010-2011
2009 Faculty Teaching Award, UIC College of Engineering
Teaching Recognition Program Award 2008-2007
2008 Faculty Advising Award, UIC College of Engineering
2007 Faculty Teaching Award, UIC College of Engineering
2007 Faculty Advising Award, UIC College of Engineering
Harold Simon Teaching Award, 2004
Teaching Recognition Program Award 2000-2001
Outstanding Instructor Award / University of Wisconsin, 1998
Excellence Award - Exceptional Teacher/ University of Washington, 1990-1998
3M Company Non-Tenured Faculty Award, 1997
3M Company Non-Tenured Faculty Award, 1996
NATO/NSF Postdoctoral Fellowship, 1987

7. Service activities

Member of Advisory Committee, UIC Department of Chemical Engineering, 2003-present
Member of Undergraduate Committee, UIC Department of Chemical Engineering, 2018-present

8. Selected Publications Last 10 Years

- (1) Mischel, Cody Wade; Wedgewood, Lewis Edward, Pellegrino, James W.; Bilig, Betul "Developing Professional Identity: Integrating Academic and Workplace Competencies within Engineering Programs" 2022 ASEE Annual, Paper ID #38296: JUN 2022
- (2) Mischel, Cody Wade; Wedgewood, Lewis Edward, Pellegrino, James W.; Bilig, Betul "Impacts of implementing up-to-date industry problems on engineering identity development" 2022 ASEE Annual, Paper ID #37955: JUN 2022
- (3) Dubina, Sean Hyun; Wedgewood, Lewis Edward "Application of nonuniform magnetic fields in a Brownian dynamics model of ferrofluids with an iterative constraint scheme to fulfill Maxwell's equations" Physics Of Fluids Volume: 29 Issue: 9 Article Number: 092001 Published: SEP 2017
- (4) Dubina, Sean Hyun; Wedgewood, Lewis Edward "A Brownian dynamics study on ferrofluid colloidal dispersions using an iterative constraint method to satisfy Maxwell's equations" Physics Of Fluids Volume: 28 Issue: 7 Article Number: 072001 Published: JUL 2016

9. Professional Development

Not applicable

Faculty Vitae: Alan D. Zdunek

1. Education

B.A. Chemistry, Knox College, 1982
M.S. Chemical Engineering, Illinois Institute of Technology, 1986
Ph.D. Chemical Engineering, Illinois Institute of Technology, 1990

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago
Clinical Associate Professor, Director of Undergraduate Studies,
Director of Engineering Success Program, 2020-present.

Department of Chemical Engineering, University of Illinois-Chicago
Clinical Assistant Professor, Director of Undergraduate Studies,
Director, Freshmen Engineering Success Program, 2014-present

Department of Chemical Engineering, University of Illinois-Chicago,
Adjunct Professor, 2010-2014

Department of Chemical Engineering, Illinois Institute of Technology
Research Associate Professor/Adjunct Faculty, 2007-2009

3. Professional Experience

S&C Electric, Chicago, IL, Consultant. 2014, part-time
Advanced Diamond Technologies, Romeoville IL, Consultant, full-time 2012
HOH Water Technology, Palatine IL, Consultant. 2008-2009, part-time
Argonne National Laboratory, CSE Division, Lemont IL, Consultant, 2008-2009, part-time
American Air Liquide, Countryside, IL, Group Expert. 2003-2006, full-time
American Air Liquide, Countryside IL, Senior Scientist. 1999-2003, full-time
American Air Liquide, Countryside IL, Scientist, 1995-1999, full-time
BIRL Industrial Research Laboratory, Evanston IL, Scientist, 1990-1994, full-time

4. Certifications or professional registrations

Project Management Certificate, Loyola University, 2009.
Hazards Material Technician, 1997

5. Current membership in professional organizations

American Institute of Chemical Engineers, American Chemical Society

6. Honors and awards

College of Engineering 2025 Teaching Award
College of Engineering 2024 Advising Award
College of Engineering 2021 Teaching Award
College of Engineering 2018 Teaching Award
College of Engineering 2017 Teaching Award
College of Engineering 2016 Teaching Award
Keynote Speaker, Conference on International Exchange of Talents & Projects
Discussion of Yellow River Delta High-Efficiency Economic Zone, Dongying City,
China, November 2011.

7. Service activities

College of Engineering Education Policy Committee: 2014-present
Student recruitment and enrollment activities: Open house events, department information sessions, Intent-to-enroll events, HS presentations, Women in Engineering Scholarship

interviews: 2015- present
 Engineering Success Program Committee, Director, 2016-present
 ENGR100 Faculty instructor: CHE breakout sessions: 2018-present
 Faculty Advisor, AIChE Student Chapter, UIC. 2011-2012, 2014-2024.
 Faculty Advisor: AIChE North-Central Regional Conference, 2023
 Faculty/Staff Search Committees: 2020-2024
 IES Abroad Dean of Engineering search committee: 2023
 Chair, Electrochemical Society Chicago Section, 2015-2017
 Von Steuben HS student research project for CPS City Science Fair, 2016, 2018
 Editor, Journal of Bio- and Tribo-Corrosion, 2015-2018
 AIChE Midwest Regional Conference planning committee, 2016-2017
 UIC Electrochemistry Collaborative Group (2015-2017)
 Career Services Advisory Council, 2017-2019
 Reviewer, DPI Seed Funding Program, Summer 2019
 Chemical Engineering Advisory Committee, August 2016-present
 Chemical Engineering Safety Committee, 2015-2017
 Chemical Engineering Undergraduate Committee, 2014-present
 Chemical Engineering Public Relations Committee, 2015-present
 Chemical Engineering Classroom Upgrade Ad-Hoc Committee, 2015-2016
 Chemical Engineering Club Development, 2018-2018
 Initiated and Organized Chemical Engineering Industry Day, 2015-present
 ChemE Forum and Chemical Engineering Alumni Mixer, 2016-present
 Booth organizer and attendee at ProFood Expo, Chicago, IL March 2019
 Chemical Engineering/Engineering Career Center Ad-Hoc Committee, 2014-2015
 HS Senior Project Mentor, North Lawndale College Prep High School, 2015-2016

8. Selected Publications

M. H. Mustafa, A. Zdunek, “Supercapacitor Nanofiber Electrodes Graphene-Based”, Int. J. Electrochem. Sci, 14 (2017).
 X. Cui, A. Zdunek, Gregory Jursich, and C. Takoudis, “Nanostructured Ni-YSZ Anode by Atomic Layer Deposition”, ECS Journal of Solid State Science and Technology, 4(12), P1-P7, 2015.
 R. Sparks, G. Jursich, A. Zdunek, Jorge Ivan Rossero, C. Takoudis, “Tunability and Electrical Properties of Yttria-doped-ceria Films Fabricated via Atomic Layer Deposition”, The Journal of Undergraduate Research at the University of Illinois at Chicago, Vol.7 No. 1, (2014).

9. Professional Development

One UIC Alignment for Enrollment Success Conference, 2025
 IES Abroad Global Institute, Barcelona, Spain, 2024
 IES Abroad Engineering Consortium, Dublin, Ireland, 2022.
 2017 ABET Symposium, Baltimore, MD April 2017
 Faculty Summer Institute, University of Illinois Urbana Champaign, Urbana, IL, May 2015

Appendix C – Equipment

Equipment List for Chemical Engineering Laboratory (ChE 381/382) Experimental Demonstration Apparatus:

1. Distillation
 - a. Continuous Distillation column: Armfield UOP3CC (10L, 8 tray)
 - b. Control Module with computer interface/data acquisition
 - c. Anton-Parr Snap-41 Hand-held Digital Alcohol Density Meter, ATAGO Digital Hand-held pocket Refractometer: PAL-RI, Analog Refractometer; Bausch & Lomb Model ABBE-3L
2. Liquid-Liquid Extraction
 - a. Liquid-liquid extractor; Denver Corp. Model SEM-SU
3. Fluid Flow in Pipe Systems
 - a. Hampden Model H-6920 Pipe Friction Demonstrator
4. Ultrafiltration
 - a. Ultrafiltration column and apparatus; UIC Designed/Built w/ Koch Membrane Systems PM50 filter column
5. Membrane Separation
 - a. Membrane Air Separation Apparatus; Perfected Experiments, Inc. Model ChE06
6. Heat Transfer
 - a. Constant Stirred Tank Reactor (CSTR); UIC Designed/Built
7. Viscosity of Liquids
 - a. Capillary Tube Viscometer; UIC Designed/Built
8. Fluidized Bed
 - a. Two-Column fluidized bed apparatus; UIC Designed/Built
9. Humidification
 - a. Falling Liquid Film humidification apparatus; UIC Designed/Built
10. Evaporation
 - a. Single-stage glass evaporator, vacuum pressure; UIC Designed/Built
11. Gas Absorption
 - a. Armfield Model UOP7 Gas Absorption Column
12. Process Control
 - a. Armfield PCT50-Level Control
 - b. Armfield PCT53-Pressure Control
 - c. Armfield PCT54- Industrial PID Controller
 - d. Armfield PCT55- Programmable Logic Controller

General Lab & Support Equipment

1. Laboratory space
 - a. Dedicated workspaces for chemical process demonstrators with utility hook-up
 - b. 9 moveable lab benches with lab cabinet storage underneath, 5 with wood countertops, 4 with chemical-resistant countertops, overhead storage areas and electric power outlets
 - c. 2 wall-service chemical resistant countertop lab benches with overhead storage and cabinets
2. Classroom space
 - a. 4 conference tables/chairs for lab team collaboration and lab analysis
 - b. WiFi connectivity, ethernet connectivity
3. Chemical/gas storage
 - a. Two (2) flammable liquid cabinets (under fume hoods)
 - b. Acids/base storage (under fume hoods)
 - c. Compressed gas cylinder storage
 - d. 55-gallon drum storage for white mineral oil
4. Laboratory Utilities:
 - a. Two horizontal sash fume hoods with automatic face velocity control
 - b. Four exhaust snorkels
 - c. Two laboratory sinks (hot/cold water, purified water)
 - d. Steam boiler; Lattner Model 48 BF electric boiler, 100 psi max
 - e. Compressed air; house, several taps throughout lab, max pressure 120 psi
 - f. Industrial chilled water; various taps throughout lab
 - g. Industrial cooling water inlet/return (two locations)
 - h. Purified (deionized) water; three taps in lab, two in sink, one overhead
 - i. Hot purified water (Tube-in-tube steam heat exchanger)
 - j. 5 floor drains
 - k. Electrical; 120V and 220V, individual circuit breakers on specific equipment
 - l. Vacuum line; in fume hoods
5. Miscellaneous:
 - a. Three Digital balances, Ohaus Scout (300g), Valor 3000 Xtreme (100 g), Mettler PJ (12kg)
 - b. One mechanical scale; Fisher Scientific, 5 kg
 - c. pH meters (2)
 - d. Magnetic stirrers/burettes and burette holders
 - e. Flasks, beakers, buckets, pipettes, rulers, stopwatches
 - f. Digital tachometer, digital refractometer, digital hygrometer, digital barometer, analog refractometer, analog hygrometers, electric mixer (for viscosity experiment), Heating/cooling temperature bath, gas-diffusion apparatus

- g. Two (2) Fisherbrand Isotemp Hotplate Stirrers (Cat. No. S88857200, Serial No. M2720015122121431)
 - h. Extech Instruments 382275 Adjustable Switching Mode DC Power Supply (1–30 V, 20 A)
 - i. Two (2) The Giant Pump Co. laboratory diaphragm vacuum pump (Serial No. EE1185353)
6. Safety equipment:
- a. Two (2) Eyewash stations
 - b. One (1) Safety Shower
 - c. Safety gloves; heat-resistant, chemical-resistant, sharp-resistant, general nitrile gloves.
 - d. Safety glasses; 30+ count
 - e. Ear protection
 - f. 10 industrial hard hats
 - g. Two (2) First aid kits
 - h. Two (2) Chemical spill kits, oil-spill pads, drip-trays
 - i. MSDS Binder, Laboratory Safety Plan

Appendix D – Institutional Summary

The Institution

a. University of Illinois Chicago

2800 University Hall (m/c 102), 601 S. Morgan Street, Chicago, IL 60607-7128

b. *Chief Executive Officer:*

Marie Lynn Miranda, Chancellor

University of Illinois Chicago

2800 University Hall (m/c 102), Chicago, IL 60607-7128

c. Miiiri Kotche, Associate Dean for Undergraduate Affairs

University of Illinois Chicago

851 S. Morgan St. (m/c 154), Chicago, IL 60607-7128

d. The University of Illinois Chicago is accredited by the Higher Learning Commission (HLC), an institutional accreditor recognized by the U.S. Department of Education. HLC is located at 230 South LaSalle Street, Suite 7-500, Chicago, Illinois 60604-1411. Students and the general public can contact the HLC by phone at (312) 263-0456 or by sending an inquiry to info@hlcommission.org. The University of Illinois Chicago (UIC) has been continuously accredited by HLC since 1970 and received its [most recent reaffirmation of accreditation](#) from the Commission in July 2024. UIC will undergo its next 10-year comprehensive review in 2033-2034, with the 4-year mid-cycle virtual Assurance Review scheduled for 2027-2028. Verification of UIC's institutional accreditation is communicated through the [Mark of Accreditation Status](#) on the institution's website. UIC's undergraduate, graduate, and professional programs, including all programs in the College of Engineering, are approved by the Illinois Board of Higher Education, 1N Old State Capitol Plaza, Suite 333, Springfield, Illinois 62701, (217) 782-2551.

History of Institution

Following World War II, the University of Illinois expanded its presence in Chicago by establishing a temporary two-year branch campus at Navy Pier. Known as the Chicago Undergraduate Division, it primarily served student veterans on the G.I. Bill, allowing students to complete their first two years locally before transferring to Urbana-Champaign. This facility, renamed the University of Illinois at Chicago Circle, moved to its present location in 1965, when it opened its doors as a four-year university. UIC was born in 1982 when the Circle and Medical Center campuses consolidated to form a comprehensive university campus with seven health science colleges and an academic medical center. Consolidation helped UIC reach elite Carnegie Research I status.

In 1993, then-UIC Chancellor (and later UI President) James Stukel launched the Great Cities Initiative to join UIC teaching and research with community, corporate and government partners in tackling urban challenges. Renamed the Great Cities Commitment at its 10-year mark, the signature program integrates research with genuine community engagement.

In the 2000s, UIC's South Campus development brought student housing, retail stores, restaurants, and private residences to the historic Maxwell Street neighborhood. The increase in faculty and staff families, along with the expansion of student housing, helped UIC transition from a daytime commuter campus into a vibrant, 24-hour academic community. Today, one-third of UIC first-year students and about one-fifth of undergraduates live on campus.

In July 2019, UIC opened its newest campus living facility. The 10-story Academic and Residential Complex, which is part of a public-private partnership, offers 550 beds in a mix of traditional dorm rooms and suite-style units. The facility's academic areas feature three large lecture halls, four classrooms, several small group study rooms, a tutoring center, computer stations, and collaboration spaces.

The Engineering Innovation Building, which also opened in July 2019, is the first new academic building on the east side of campus since 1991. The 50,000-square-foot facility houses instructional space, research labs and faculty and staff offices. It also features the university's first high-bay structural research laboratory, where researchers can conduct a wide range of tests on large-scale structural components. In August 2019, UIC acquired the John Marshall Law School and formed Chicago's only public law school.

UIC's mission centers on providing broad access to intellectual excellence and having an impact on students, the community, the city, and the world. With 94 undergraduate majors, 84 minors, and 165 graduate and professional programs currently open to enrollment across 16 colleges and schools, students can explore academic interests and engage in service learning, study abroad programs, internships, and other cocurricular activities.

UIC is 1 of only 3 universities in the Chicago area that meet the criteria for Carnegie's highest research classification, and it is the only public institution in the city to do so.

The College of Engineering was one of four colleges at Navy Pier, along with the Colleges of Commerce and Business Administration, the College of Liberal Arts and Sciences, and the College of Physical Education. Back then, the College of Engineering at Navy Pier was quite small, with approximately 400 students attending on average each year.

In 1965, when the University of Illinois at Chicago Circle was established, the College of Engineering reorganized into "functional" departments of Materials Engineering, Energy

Engineering, Information Engineering, and Systems Engineering. Degree programs in the College were identified by functions that engineers perform, such as Mechanical Analysis and Design, Systems Analysis, or Communications Engineering. The College grew rapidly at Chicago Circle, reaching nearly 2,000 undergraduate students by 1968. In 1973, a number of the College's programs received their first accreditation from the Engineers Council for Professional Development (now ABET).

Although the non-traditional engineering degree programs administered by the College in the 1970's were accredited, they were not well understood by the public, and it was clear that our industrial constituents preferred to recruit students from traditional degree programs, so that the backgrounds and qualifications of their employees could be more easily predicted. In 1982, the University made the decision to reorganize the College into traditional departments and traditional degree programs, essentially as they are offered now.

The College of Engineering at the University of Illinois Chicago (UIC) is one of the largest and fastest-growing engineering colleges in the Midwest, serving a diverse student population in a major urban research institution. The college combines academic excellence with a strong commitment to access and social mobility, reflecting UIC's broader mission of providing high-quality education to students from varied backgrounds. With an emphasis on experiential learning, research engagement, and applied problem-solving, the college prepares graduates to meet the technological and societal challenges of an increasingly complex global environment.

The college is organized into six academic departments: Biomedical Engineering; Chemical Engineering; Civil, Materials, and Environmental Engineering; Computer Science; Electrical and Computer Engineering; and Mechanical and Industrial Engineering. These departments offer a comprehensive portfolio of undergraduate and graduate degree programs, ranging from traditional engineering fields to emerging areas such as computer science, data science, energy engineering, materials engineering, and biomedical innovation. Programs are designed to provide a strong theoretical foundation while integrating laboratory work, design experiences, and industry-aligned skills.

College of Engineering

The College of Engineering is deeply committed to research and innovation. UIC is classified as an R1 institution with very high research activity, and engineering faculty contribute significantly to the university's research output through funded projects, publications, patents, and interdisciplinary collaborations. Many faculty integrate their research with student learning, providing opportunities for undergraduate and graduate students to participate in cutting-edge research in areas such as data science, health technologies, energy systems, robotics, cybersecurity, and advanced materials.

Enrollment in the college has grown steadily over the past decade, with more than 5,000 undergraduate students and a rapidly expanding graduate population. This growth reflects the college’s rising reputation, strong demand for engineering and computing degrees, and commitment to creating pathways for transfer students and underrepresented groups. The college also hosts numerous student organizations, design teams, mentorship programs, and partnerships with industry, which together enhance student engagement, professional development, and workforce readiness.

Overall, the College of Engineering at UIC provides a comprehensive and rigorous engineering education supported by a strong research environment, modern facilities, and a commitment to student success. Its programs are continuously reviewed and improved to meet accreditation standards and evolving workforce needs, ensuring that graduates are well prepared for careers in engineering, technology, and innovation-driven industries.

Student Body

In Fall 2025, total enrollment at UIC increased from 33,906 (2024) to 35,869 (2025), an increase of 5.8 %. While 81.5% of the total student body hailed from Illinois, 64% of the entering first-time, full-time undergraduate student body were graduates of Chicago Public Schools and Cook County in Illinois. Overall, UIC students represented all 50 states, four U.S. territories, and 109 countries worldwide.

Many full-time students also hold part-time jobs, both on and off campus. Additionally, a large number of students find time to participate in one or more of the approximately 233 campus student organizations. With a growing residential population, UIC houses students on the east, west, and south sides of campus. Approximately 3,800 students live on campus, including more than half of first-year undergraduate students.

Table D.1 Total Enrollment Fall 2018-2025

	2018	2019	2020	2021	2022	2023	2024	2025
Undergraduate	20,783	21,641	21,921	22,279	21,807	22,107	22,495	24,260
Graduate	7,673	7,572	7,442	7,697	7,798	7,315	7,252	7,341
Professional	3,227	4,177	4,155	4,223	4,142	4,100	4,159	4,268
Total	31,683	33,390	33,518	34,199	33,747	33,522	33,906	35,869

UIC enrollment data for the past 8 years are shown above. The overall campus student population has increased from 29,120 (2016) to 35,869 (2025) students over the past decade.

In Fall 2025, UIC enrolled 24,260 undergraduate students, of whom 53% are female, and 47% are male. Notably, 84% of these students are under the age of 23 years. The undergraduate student body reflects a rich diversity of cultural backgrounds: approximately 20% identified as Asian, 9%

as Black, 36% as Latinx, and 22% as White. UIC’s commitment to serve “a wide range of students” has been recognized by its federal designations as a Minority-Serving Institution, an Asian American and Native American Pacific Islander-Serving Institution, and a Hispanic-Serving Institution.

Over the past decade, UIC has maintained one of the most ethnically diverse undergraduate populations in the country. The data (Table D.2) show a steady increase in students identifying as Hispanic, with a slight but consistent decline in students identifying as White.

Table D.2 Ethnic Diversity of Undergraduate Students Fall 2018-2025

Race/Ethnicity	2018	2019	2020	2021	2022	2023	2024	2025
Am. Indian and Alaska Native	8	9	8	7	9	10	11	13
	0.04%	0.04%	0.04%	0.03%	0.04%	0.05%	0.05%	0.05%
Black/African American	1,630	1,637	1,684	1,654	1,643	1,744	1,829	2,141
	7.84%	7.56%	7.68%	7.42%	7.53%	7.89%	8.13%	8.83%
Asian	4,339	4,444	4,521	4,650	4,545	4,522	4,680	4,885
	20.88%	20.54%	20.62%	20.87%	20.84%	20.46%	20.80%	20.14%
Hispanic	6,977	7,297	7,531	7,837	7,853	7,999	8,023	8,844
	33.57%	33.72%	34.36%	35.18%	36.01%	36.18%	35.67%	36.46%
White	5,822	5,671	5,612	5,435	5,047	4,988	4,987	5,300
	28.01%	26.20%	25.60%	24.40%	23.14%	22.56%	22.17%	21.84%
International/U. S. Non-Resident	1,270	1,576	1,534	1,598	1,642	1,712	1,809	1,825
	6.11%	7.28%	7.00%	7.17%	7.53%	7.74%	8.04%	7.52%
Native Hawaiian and Pacific Islander	13	8	7	4	5	7	8	6
	0.06%	0.04%	0.03%	0.02%	0.02%	0.03%	0.04%	0.02%
Multi-Race	627	637	645	657	639	653	672	735
	3.02%	2.94%	2.94%	2.95%	2.93%	2.95%	2.99%	3.03%
Unknown	97	362	379	437	424	472	476	511
	0.47%	1.67%	1.73%	1.96%	1.94%	2.14%	2.12%	2.11%
UIC Total	20,783	21,641	21,921	22,279	21,807	22,107	22,495	24,260

Consistent with UIC’s role as a public institution serving Chicago and Illinois, 89% of UIC’s undergraduate students in Fall 2025 were from Illinois, with approximately 35% of first-time, full-

time undergraduates being graduates of Chicago Public Schools. The table below shows the geographic origins of incoming freshmen. Chicago and Cook County make up the majority, 64%, of UIC's incoming freshmen.

Table D.3 The Geographical Location of entering Freshmen

	2018	2019	2020	2021	2022	2023	2024	2025
Chicago	1,781	1,820	1,427	1,798	1,888	1,962	1,773	2,262
	42.82%	41.30%	40.30%	43.05%	44.49%	42.22%	40.12%	41.85%
Cook County	2,771	2,953	2,367	2,896	2,850	3,110	2,921	3,680
	66.63%	67.01%	66.85%	69.33%	67.15%	66.92%	66.10%	68.09%
Illinois	3,748	4,016	3,267	3,783	3,802	4,227	4,011	5,015
	90.12%	91.13%	92.26%	90.57%	89.59%	90.96%	90.77%	92.78%
Out-of-State	180	171	168	222	190	187	171	198
	4.33%	3.88%	4.74%	5.31%	4.48%	4.02%	3.87%	3.66%
Foreign	229	218	106	182	264	251	233	187
	5.51%	4.95%	2.99%	4.36%	6.22%	5.40%	5.27%	3.46%

Admission to the University of Illinois at Chicago is competitive, with the average enrolled freshman student being in the top 25% of his or her high school class, as shown in the following table for freshmen enrolled over the past 8 years. While the high school rank has been steady over this period, the average ACT score of entering freshmen has gradually increased by 1 point over this period, rising from 23.1 to 24.1.

Table D.4 Median ACT Scores and Average High School GPA of First-time, First-year students: Fall 2018-2025

	2018	2019	2020	2021	2022	2023	2024	2025
ACT Composite	27.2	27	27	27	27	27	27	28
Average GPA*	3.31	3.35	3.38	3.40	3.41	3.47	3.50	3.55

*Source: Common Data Set Tables

Table D.5 Undergraduate Enrollment at College of Engineering 2018-2025

Engineering	2018	2019	2020	2021	2022	2023	2024	2025
Undergraduate	3,861	4,127	4,227	4,421	4,448	4,934	5,050	5,424

Undergraduate engineering enrollment at UIC increased steadily from 3861 students in 2018 to 5,424 students in 2025, indicating a 40.5 percent growth over the past 8 years. Consistent with overall UIC patterns, 86 percent of the engineering undergraduates were Illinois residents, while 12 percent were international students.

Figure D-1 shows the College of Engineering enrollment since 2021 leading to undergraduate enrollment of approximately 5425 for Fall 2025.

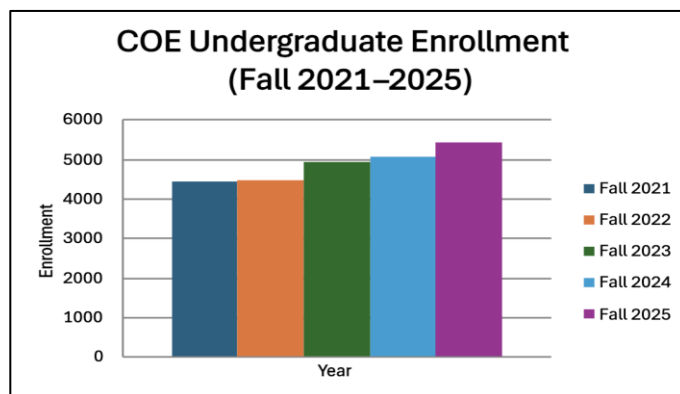


Figure D-1 College of Engineering Undergraduate Enrollment in Fall semester from 2021 to 2025

Table D.2 College of Engineering Undergraduate Enrollment shown by program from 2021 to 2025. Figure D-2 shows the distribution of enrollments among the College’s programs during that period. The Bioengineering, Computer Science, and Mechanical Engineering programs have grown significantly during this time, while other programs have had stable or decreasing enrollment.

Table D.2 College of Engineering Undergraduate Enrollment shown by program from 2021 to 2025

Department	Fall 2021	Fall 2022	Fall 2023	Fall 2024	Fall 2025
Biomedical Engineering	332	323	335	319	358
Chemical Engineering	245	200	177	165	188
Civil Engineering	406	381	383	380	432
Environmental Engineering	0	2	22	45	56
Computer Science	1713	1801	1888	1832	1597
Computer Science & Design	0	36	101	124	136
Data Science	27	74	146	184	243
Computer Engineering	355	343	526	544	584
Electrical Engineering	353	314	313	372	468
Engineering Physics	16	17	16	29	23
Engineering Management	29	26	20	29	29
Industrial Engineering	90	88	91	71	82
Mechanical Engineering	727	676	706	800	1000
Engr-Undecided	128	167	210	156	229

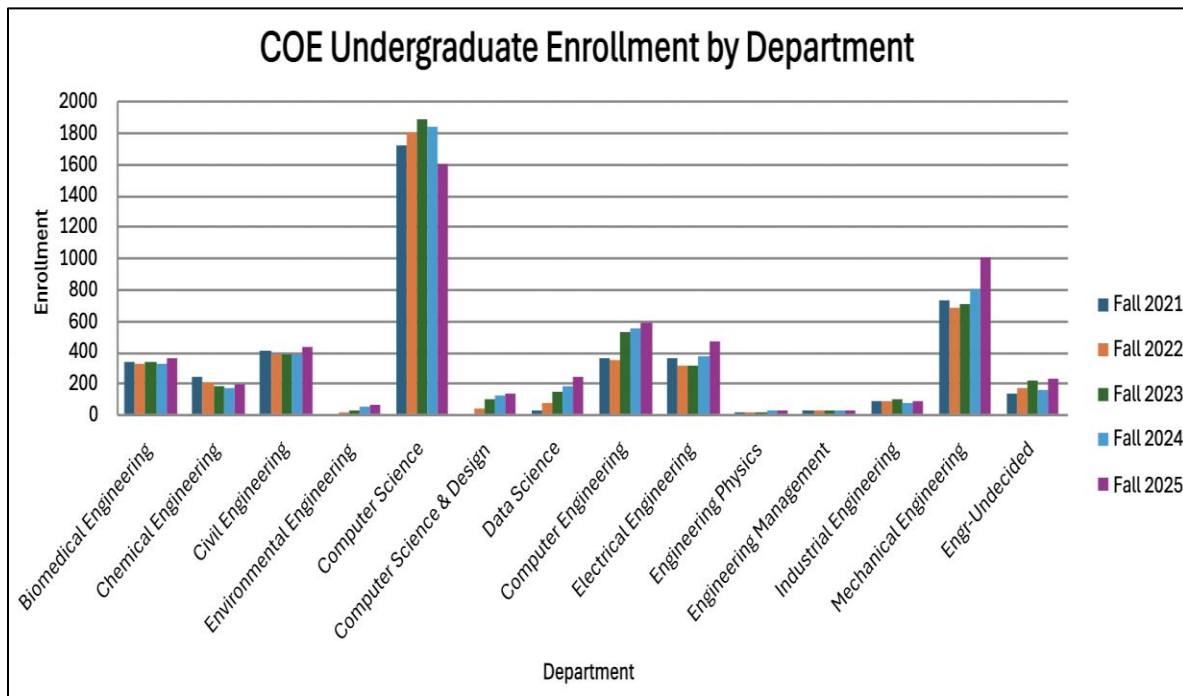


Figure D.2 College of Engineering Undergraduate Enrollment shown by program from 2021 to 2025

COE Freshmen Demographics

Both male and female freshman enrollments have steadily increased over the last decade. However, the growth in enrollment for males exceeds that of female students. The incoming freshmen class increased from 108 female students in 2016 to 245 female students in Fall 2025. Despite this rise, females still account for only 23 percent of the total entering freshman class in 2025. Hispanic students showed the most significant growth, increasing from 106 in 2016 to 367 in 2025, and became the largest ethnic group among engineering freshmen. Asian students also experienced a steady increase in numbers, although the increase among students identifying as Black was relatively small. Overall, the number of White students grew, but their proportion in the incoming freshman class decreased as diversity increased from 2016 to 2025.

Table D.5 Gender and Race of Freshmen Entering Students 2016-2025

Gender										
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
F	108	148	134	177	145	194	218	335	227	245
M	284	455	361	496	411	648	662	875	694	815
Grand Total	392	603	495	673	556	842	880	1,210	921	1,060

Race										
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
AIAN				2						1
Asian	93	175	139	185	163	239	270	344	263	280
Black/African American	20	26	21	25	24	58	54	90	76	78
Hispanic	106	158	101	169	173	283	245	386	299	367
International	19	98	109	108	50	79	109	128	94	87
Multi-Race	12	14	16	15	10	20	32	30	19	30
NHPI	1						1			
Unknown			2	8	10	12	18	20	17	30
White	141	132	107	161	126	151	151	212	153	187
Grand Total	392	603	495	673	556	842	880	1,210	921	1,060

Table D.5 Ethnicity of Freshmen Engineering Students

Race/Ethnicity	2018	2019	2020	2021	2022	2023	2024	2025
AIAN	0	2	0	0	0	0	0	1
	0.0%	0.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.09%
Asian	139	185	163	239	270	344	263	280
	28.08%	27.49%	29.32%	28.38%	30.68%	28.43%	28.56%	26.42%
Black/African American	21	25	24	58	54	90	76	78
	4.24%	3.71%	4.32%	6.89%	6.14%	7.44%	8.25%	7.36%
Hispanic	101	169	173	283	245	386	299	367
	20.4%	25.11%	31.12%	33.61%	27.84%	31.9%	32.46%	34.62%
International	109	108	50	79	109	128	94	87
	22.02%	16.05%	8.99%	9.38%	12.39%	10.58%	10.21%	8.21%
Multi-Race	16	15	10	20	32	30	19	30
	3.23%	2.23%	1.8%	2.38%	3.64%	2.48%	2.06%	2.83%
NHPI	0	0	0	0	1	0	0	0
	0.0%	0.0%	0.0%	0.0%	0.11%	0.0%	0.0%	0.0%
Unknown	2	8	10	12	18	20	17	30
	0.4%	1.19%	1.8%	1.43%	2.05%	1.65%	1.85%	2.83%
White	107	161	126	151	151	212	153	187
	21.62%	23.92%	22.66%	17.93%	17.16%	17.52%	16.61%	17.64%
Total	495	673	556	842	880	1210	921	1060

Table D.6 Gender Distribution

	2018	2019	2020	2021	2022	2023	2024	2025
Male	4098	4159	4152	4300	4363	4622	4724	4967
Female	1231	1263	1289	1349	1380	1590	1662	1702
Undefined	7	3	3	2	3	8	12	17
COE Total	5336	5425	5444	5651	5746	6220	6398	6686

The gender distribution at UIC shows a steady increase in total enrollment from 4,705 students in 2016 to 6,686 in 2025 - a growth of about 42% over the decade. Male enrollment has consistently remained higher, increasing from 3,626 to 4,967 students. Female enrollment has also shown notable growth, rising from 1,021 in 2016 to 1,702 in 2025, reflecting the college's ongoing efforts to encourage gender diversity in engineering fields. The "undefined" category remains very small throughout the period, indicating clear gender identification reporting. Overall, the data suggest sustained expansion and gradual improvement in female representation within the College of Engineering.

2. Type of Control

A. The University of Illinois Chicago (UIC), one of three distinct universities within the University of Illinois System, is a mission-driven, research-intensive public university. Under Illinois law, the University of Illinois System is subject to the authority of the state legislature. Under the University of Illinois Statutes, UIC is governed by the Board of Trustees and is a non-profit institution without obligation to investors, parent organizations, or external interests. The board is the governing body for the 3 universities within the System.

B. Per the University of Illinois *Statutes*, the Board of Trustees has delegated the execution of policies to its administrative agents acting under its supervision, with the chancellor designated as UIC's chief executive officer. The chancellor serves as a vice president of the University of Illinois System and reports directly to the System president. The chancellor oversees vice chancellors and administrative heads, who are responsible for academic affairs; health affairs; administrative services; finance; student affairs; research; strategic marketing and communications; innovation and strategic partnerships; diversity, equity and engagement; and intercollegiate athletics.

3. Educational Unit

The organizational structure of the College of Engineering at the University of Illinois Chicago is shown schematically in Figure D.3. The Directors of Undergraduate Study responsible for the B.S. programs presented in this document, report to the Department Heads of their respective departments, who report, in turn, to the Dean of the College. The Dean is supported in his activity by Associate Deans with responsibilities for Research and Graduate Studies, Corporate Relations (also supervises the Engineering Career Center and Co-op Program), On-line Master of

Engineering Program and International Programs (all graduate level programs), Administration (primarily budget and financial issues), and Advancement.

The authority lines for the Associate Dean for Undergraduate Affairs are shown in more detail in Figure 1-3, because these personnel address the programs reviewed in this report. The Associate Dean for Undergraduate Affairs supervises the Director of the Minority Recruiting and Retention Program (MERRP). The Assistant Dean is supported by Assistant Directors for both the recruiting program and the MERRP program. The recruiting activities are carried out throughout the Chicago region to continuously enhance the number and quality of students entering the College's programs. The MERRP program (described in more detail elsewhere in this report) is in place to recruit and support minority enrollments in the College of Engineering.

The Director of Engineering Admission and Records also reports to the Associate Dean for Undergraduate Affairs. The Director of Admission and Records reviews the records of student applicants to the College of Engineering and makes admission decisions. The office also maintains the records of continuing students and reviews them for compliance with College policies and graduation standards. It is supported by three academic counselors who meet with individual students to advise them on policies and requirements and also conduct orientation and advising sessions with new freshmen and transfer students. The Director is also supported by an Office Manager, who is usually the first responder for students with problems. Together with the Associate Dean for Undergraduate Affairs, he sets policies for special petitions and enforces probation and academic drop rules.

Organizational Structure

The authority lines for the Associate Dean for Undergraduate Affairs are shown in more detail below, because these personnel address the programs reviewed in this report. The Associate Dean for Undergraduate Affairs supervises the Assistant Dean for Undergraduate Recruiting, who is also the Director of the Minority Recruiting and Retention Program (MERRP). The Assistant Dean is supported by Assistant Directors for both the recruiting program and the MERRP program. The recruiting activities are carried out throughout the Chicago region to continuously enhance the number and quality of students entering the College's programs. The MERRP program (described in more detail elsewhere in this report) is in place to recruit and support minority enrollments in the College of Engineering. The following flowchart (Figure D3) shows the organizational relationships among the offices of the College of Engineering.

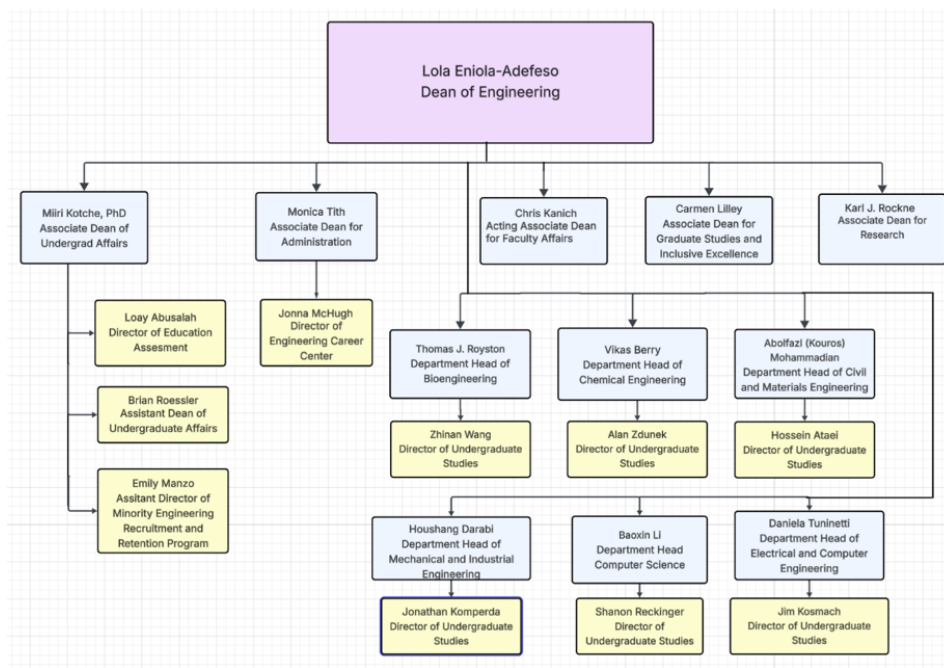


Figure D.3 The flowchart shows the organizational relationships among the offices of the College of Engineering.

The Director of Engineering Admission and Records also reports to the Associate Dean for Undergraduate Affairs. The Director of Admission and Records reviews the records of student applicants to the College of Engineering and makes admission decisions. His office also maintains the records of continuing students and reviews them for compliance with College policies and graduation standards. In this, he is supported by three academic counselors who meet with individual students to advise them on policies and requirements and also conduct orientation and advising sessions with new freshmen and transfer students. He is also supported by an Office Manager, who is usually the first responder for students with problems. Together with the Associate Dean for Undergraduate Affairs, he sets policies for special petitions and enforces probation and academic drop rules. The list of college of engineering associate deans and directors could also be found on our website <http://engineering.uic.edu/bin/view/COE/Staff>. The organizational structure for the University of Illinois at Chicago is shown in Figure D4. Note that Provost Colley will retire on June 30. Chancellor Miranda has decided to suspend the national search for a permanent provost at this time. Kathryn Chval will serve as the Interim Provost and Vice Chancellor for Academic Affairs starting July 1st.

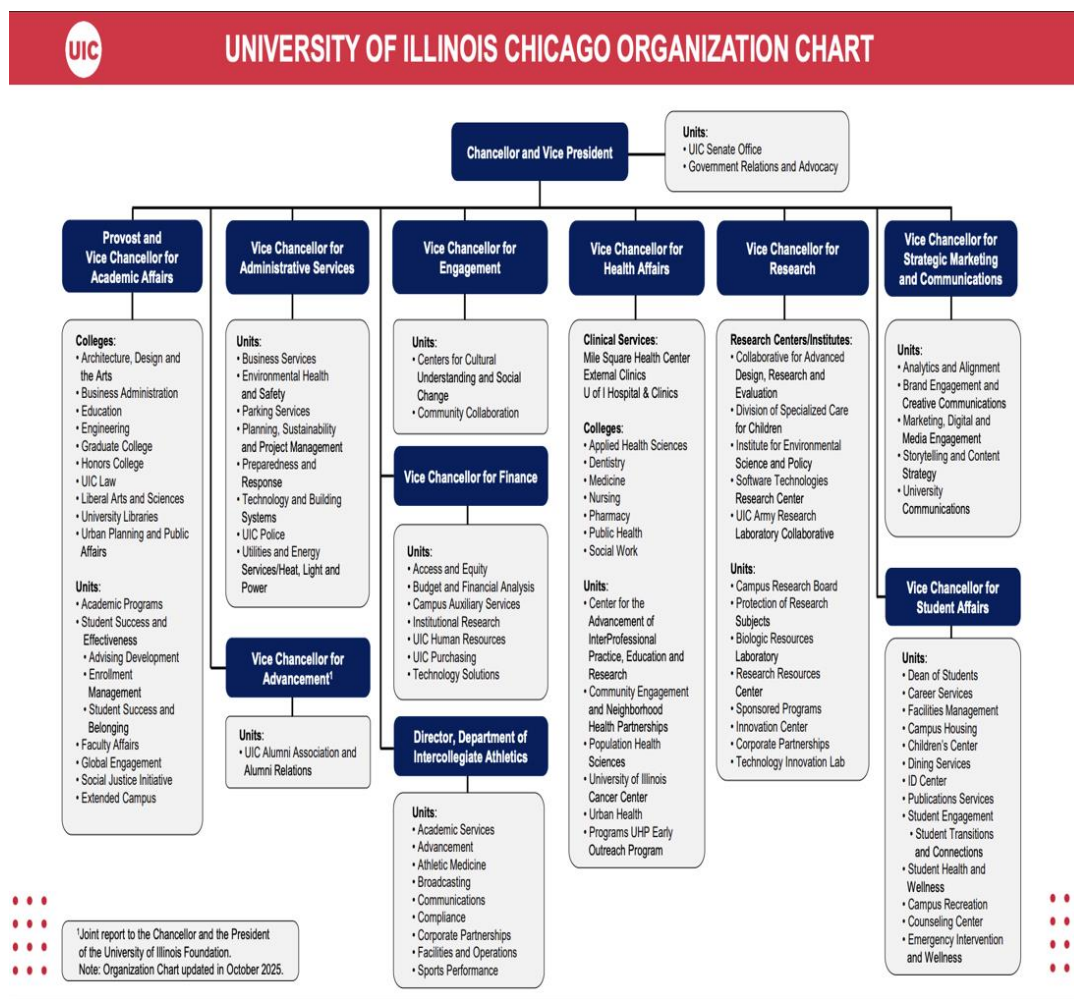


Figure D.4 The organizational structure for the University of Illinois at Chicago.

4. Academic Support Units

Required courses for College of Engineering programs are taught by faculty from several departments in the College of Liberal Arts and Sciences. These include:

Department of Mathematics, Statistics, and Computer Science

Department of Mathematics, Statistics, and Computer Science, which teaches required mathematics courses. The department partners with the College of Engineering Data Science Major. The MSCS Department is housed at:

Department of Mathematics, Statistics, and Computer Science
University of Illinois Chicago
322 Science and Engineering Offices (M/C 249)
851 S. Morgan Street
Chicago, IL 60607-7045

Department Head: Julius Ross, 312 SEO; (312) 996-2171; juliusro@uic.edu

The MSCS Department comprises 84 faculty members and 115 teaching assistants, who are responsible for offering mathematics courses to students in the College of Engineering and other programs that rely on mathematics as a component of their knowledge base.

Department of Chemistry

The Department of Chemistry teaches required chemistry courses. It is located at:

Department of Chemistry MC 111
4500 SES (Science and Engineering South)
University of Illinois Chicago
845 W. Taylor Street
Chicago, IL 60607

Department Head: Wonhwa Cho, 4508 SES; (312) 996-4883; wcho@uic.edu

The Department of Chemistry houses 39 research and teaching faculty and several graduate teaching assistants who are responsible for offering chemistry courses taken by students in the College of Engineering and in other programs that rely on chemistry as a component of their knowledge base.

Department of Physics

The Department of Physics, which teaches the required physics courses. The Department of Physics is located at:

Department of Physics
2244 SES (Science and Engineering South)
University of Illinois Chicago
845 W. Taylor St. M/C 273
Chicago IL 60607-7059

Department Head: Robert Klie, 2374 SES; (312)- 996-6064 ; rfklic@uic.edu

The Department of Physics comprises 38 faculty members and several teaching assistants, who are responsible for offering physics courses to students in the College of Engineering and other programs that rely on physics as a component of their knowledge base.

Department of English

The Department of English teaches the required English composition courses. The Department of English is located at:

Department of English College of Liberal Arts and Sciences
2027 University Hall
601 S. Morgan Street
Chicago, Illinois 60607

Department Head: Peter Coviello, 2033 University Hall; (312) 413-2204;
coviello@uic.edu

The [Department of English](#) has 49 faculty members and 52 lecturers responsible for teaching English composition courses to students in the College of Engineering and other undergraduate students at the University of Illinois Chicago.

Additionally, several other academic departments, including the [Department of Biological Sciences](#) and the [Department of Earth and Environmental Sciences](#), offer courses that are utilized in specific College of Engineering programs. Courses that satisfy the Humanities and Social Sciences requirements for College of Engineering programs are selected from a list of courses that fulfill the [General Education requirements](#) of the University of Illinois Chicago and are offered by various departments within the University.

5. Non-Academic Supporting Units

[Library](#)

The University of Illinois Chicago is supported by an extensive library system with sites at several locations around the campus. The library is administered by:

University Librarian and Dean of University Libraries, Rhea Ballard-Thrower;
rheabt@uic.edu

Richard J. Daley Library
801 S. Morgan, M/C 234
Chicago, IL 60607

Reference materials relevant to the College of Engineering courses are most commonly found at either:

The [Richard J. Daley Library](#), 801 S. Morgan Street, Chicago

The Science Library, SES; 845 W. Taylor Street

Or

The [Library of Health Sciences](#) – Chicago, 1750 W. Polk Street, Chicago, IL 60612

In addition, the library maintains extensive collections of electronic resources and access to web-based journals and abstracting services that are frequently used by College of Engineering students and faculty. More information is available at: <http://library.uic.edu>

[Technology Solutions](#)

Previously known as the Academic Computing & Communications Center, Technology Solutions (TS) oversees the information technology functions for UIC. TS oversees various technologies and services, including email and calendaring systems, computer labs, print services, login credentials, UIC Wi-Fi, equipment lending, helpdesk support, and software. The [Learning Technology Solutions](#) (LTS) department of TS provides support to students, faculty, and staff for integrating new technologies to enhance

teaching, learning, and research. Student services supported by TS include [Educational Technology](#) (e.g., Blackboard Introduction, Ally File Transformer, iClicker, and Respondus Monitor), [Email](#), [Online Learning Tools](#) (e.g., Virtual Computer Lab), [Equipment Lending](#), [Printing](#), [Software and Licensing](#) (e.g., productivity tools, Github, Qualtrics, web conferencing, Box, Google Drive), and [On-Campus](#) (e.g., computer labs, walk-in Tech support, wireless internet) technology services. Additionally, [Artificial Intelligence \(AI\) Resources](#) (e.g., Microsoft 365 Copilot, Grammarly, Adobe Creative Cloud, Box AI, Zoom AI Companion) are made available to the campus community. Each tool provided has its own unique data security measures to ensure a safe and productive environment for all users. The TS [leadership team](#) is led by:

Matt Riley, Associate Vice Chancellor for Innovation and Chief Information Officer. Additional information about services is available through the [Technology Solutions webpage](#).

Tutoring and Academic Resources: The College of Engineering offers tutoring and other academic support to its students through various campus organizations. These include:

[Academic Center for Excellence](#)

The Academic Center for Excellence (ACE), one of UIC's largest success units, offers comprehensive first-year programs along with learning and support services to help all UIC students achieve their academic goals. By building supportive relationships, coaching, and developing academic skills, transition coaches and academic specialists help students navigate UIC, establish campus connections, and acquire the skills necessary for college success and beyond. Students can access intensive support through the [Academic Skills Program](#) and work with academic coaches to develop effective self-management and learning strategies. [The First-year Initiatives](#) provides personalized support to students as they transition from high school to college, focusing on key skills such as time management, study techniques, navigating campus resources, exploring career options, and engaging with the broader UIC community.

[The Honors College](#)

The [Burnham Hall Tutoring Center](#) offers all UIC students free one-on-one peer tutoring services from Honors College students, including assistance with class assignments for over 100 courses each semester. Covered disciplines include Math, Chemistry, English, Biology, and Physics courses relevant to College of Engineering students.

[Math & Science Learning Center](#) (MSLC)

The MSLC offers free academic support to students enrolled in introductory courses in Biological Sciences, Chemistry, Mathematics, and Physics at UIC. Students are encouraged to bring their questions, homework, lab reports, and projects to peer learning assistants, faculty, teaching assistants, and staff for assistance through both in-person and virtual sessions. Dedicated academic support is available for [Biological Sciences](#), [Chemistry](#), [Math](#), and [Physics](#) courses through

scheduled in-person or online sessions. The MSLC maintains a repository of additional academic support opportunities in STEM on its [Campus Academic Support](#) webpage.

The Writing Center

The Writing Center provides individualized tutoring services to [students](#). The center invests in tutoring conducted by peers, focusing on tutors taking an interest in the content of writing and creating opportunities for fellow students to make choices about their own writing. The Writing Center complements classroom instruction by providing tutorials, resources, workshops, and publication opportunities for writers, educators, and writing groups from all disciplines and experience levels.

Tutoring

UIC offers a wide variety of tutoring services to help students with their academic work. The Office of Student Success and Belonging provides a list of these resources organized by [units offering peer tutoring](#), [subject area tutoring support](#), and [tutoring services by course number](#).

Equity and Inclusion in Engineering Program (EIEP)

The EIEP program within the College of Engineering aims to empower historically underrepresented groups, first-generation students, and other minoritized populations, as well as all students in the College of Engineering, to succeed in engineering and technology by supporting their persistence in the field. [Services](#) provided include tailored guidance on academic pathways, personal growth, and professional development. Additionally, support is provided through personalized advising to help students navigate their academic journeys, offering one-on-one guidance to address specific issues related to academic progress, persistence, and accessing additional resources within the College of Engineering and the broader campus. [Graduate assistants and undergraduate student workers](#) provide mentorship, valuable peer-to-peer support, and maintain a sense of community through the EIEP meeting space activities. Resources include information about [EIEP student organizations](#). The EIEP contact is:

Elsa Soto, Director, EIEP
851 S. Morgan Street,
Chicago, IL 60607-7045
esoto3@uic.edu

Preparation for Majoring in Engineering (PREP-ME)

This program is for UIC freshmen and community college students in engineering. Prep-ME Plus is a six week program designed to introduce new UIC engineering students to college-level math and engineering as a discipline and profession. The primary focus is math instruction. Students also attend seminars on goal setting, study skills, and time management. Students gain an understanding of college requirements and expectations. Qualified students enroll in a freshman English course and a math workshop.

Women in Engineering Program (WIE)

845 West Taylor Street (MC 180)
Office 205D Science Learning Center, SES
Chicago, IL 60607

The goal of the UIC Women in Science and Engineering Program, WIE, is to increase the number of women students pursuing and graduating in science, technology, engineering and math, STEM, disciplines, and to promote the recruitment, retention and advancement of women who have chosen academic careers. Nationally, retention rates for women STEM students fall far behind those of their male counterparts, and women consistently make up less than 20% of tenured STEM faculty. WIE supports women undergrad/graduate students and faculty in STEM by sponsoring activities that foster a positive educational and professional environment, and enable excellence in scholarship, teaching and service. More information about the WIE program is available at: <https://wiep.uic.edu/>

African American Academic Network (AAAN)

AAAN promotes success and high-impact engagement for students of African descent through comprehensive advising, developmental programming, student advocacy, and structured learning experiences within an inclusive community. Services provided by AAAN include educational advising and [peer tutoring](#) support through the [Learning Resource Center](#). More information is available at:

African American Academic Network (AAAN)
1200 West Harrison Street
SSB Suite 2800
Chicago, IL 60607
aaan@uic.edu

Latin American Recruitment and Educational Services Program (LARES)

The LARES program has been in operation since 1975. It was created to help UIC recruit Latino students and provide academic support to boost their chances of success. [Services](#) available to students include academic advising, financial aid advising, tutoring, and academic skills programs for English and Math courses. The [programs](#) offered include the LARES Summer Institute, Leaders Academy, SEEDS Career Initiative, Latina Leadership Institute, Project HOMBRES, and Peer Mentorship programs. More information is available through:

Luis Duarte, Executive Director, LARES, luisd@uic.edu
1200 West Harrison Street
SSB Suite 2640
Chicago, Illinois 60607

[Native American Support Program \(NASP\)](#)

For over 50 years, UIC's Native American Support Program (NASP) has provided resources to support the growth, development, leadership skills, and academic achievement of Native American, Alaska Native, Native Hawaiian, Pacific Islander, and other Indigenous students. Services offered by NASP include academic coaching, financial assistance, leadership and professional development, hosting cultural events, and conducting community outreach. Additional information is available at:

NASP
1200 West Harrison Street
SSB Suite 2700
Chicago, Illinois 60607
nasp@uic.edu

[Disability Resource Center](#)

UIC's Disability Resource Center is a vital and valuable resource that provides academic accommodations for UIC students. These include materials in alternate formats, utilizing assistive technology, note-taking assistance, testing accommodations, facilitating ASL interpretation for classes and events, CART live captioning services, support with class relocation, and paratransit services. The DRC supports over 3000 undergraduate, graduate, professional, and health-science students across UIC's 16 colleges and provides resources for prospective and current students, faculty, and staff. The DRC team is led by:

Sophia Hamilton, Director of DRC, shami3@uic.edu
1200 West Harrison Street
SSB Suite 1070
Chicago, Illinois 60607
drc@uic.edu

[Counseling Center](#)

The UIC Counseling Center provides a [comprehensive range of mental health services](#) to UIC students, including assessment and diagnosis of mental health concerns, group and individual therapy, [outreach and educational programming](#), consultation services, virtual drop-in spaces, workshops, peer support groups, and 24/7 [crisis intervention](#) services. [Online resources](#) covering mental health, suicide prevention and education, and coping with recent events are accessible to all UIC students through the Counseling Center webpages. Additionally, the Counseling Center provides professional training and development through its advanced psychotherapy externship, an APA-accredited doctoral internship, and a postdoctoral fellowship. The Counseling Center is located at:

UIC Counseling Center
1200 West Harrison Street, 2010 SSB (MC333), Chicago, IL 60607
(312) 996-3490

Early Access to Paid Internships Program (EAPIP)

College of Engineering
851 S. Morgan St. MC 159
Chicago, IL 60607
uic-engr@uic.edu

The goal of the program, Early Access to Paid Internships Program, is to continue helping students gain internship experience early in their college career. According to Handshake and the Business-Higher Education Forum, junior year is the most common year to complete an internship; therefore, an internship completed in freshman or sophomore year is considered an early internship.

Students who meet the requirements will be granted access to an exclusive internship database, where faculty and other supervisors will recruit for on-campus internships. Students will need to apply with a resume and interview for these internships.

Career Services

Career Services assists UIC students and alumni in identifying their career goals, preparing for future careers, and [connecting with Chicago's diverse range of employers](#). The office offers [career advising and assessments](#), [job interview preparation](#), [resume and cover letter](#) reviews, [career events](#), job search resources, [internship opportunities](#), and support in selecting graduate schools. The UIC Career Services website connects potential [employers](#) with opportunities to post jobs and internships, schedule on-campus interviews, and organize information sessions and tables to promote their organizations. Career Services is directed by:

Jean Riordan, Executive Director, Career Services
1200 W. Harrison
Room 3050 SSB (MC 099)
Chicago, IL 60607
Phone: 312-996-2300; jriordan@uic.edu

Engineering Career Center (ECC)

The Engineering Career Center (ECC) offers valuable [resources](#) specifically designed for UIC College of Engineering students and alumni to help them successfully secure internships, co-ops, and jobs. More information is available through ECC:

Jonna (Cottrell) McHugh | Interim Director
800F SEO | 312-413-3873 | jonnac@uic.edu
851 S. Morgan Street
Chicago, IL 60607

6. Credit Unit

As outlined in the Contact/Credit Hour policy, one credit hour of instruction is defined as a 50-minute period of classroom or direct faculty instruction each week plus a minimum of two 50-minute periods of out-of-class student work weekly (for a total of 150 minutes per week) for 15 weeks, or the equivalent amount of work over a different amount of time. For example, a typical

3-credit-hour undergraduate lecture/discussion course offered in the fall or spring semester at UIC—if scheduled in a 3-day block—would meet for three 50-minute periods (150 minutes) each week for 15 weeks and would require a total of six 50-minute periods (300 minutes) of out-of-class student work per week over the course of the semester, for a course total of 6,750 minutes. The 16th week of the semester is reserved for final examinations. The required instructional time varies depending on the type of instruction. For example, two 50-minute periods of laboratory or practicum per week, combined with at least two 50-minute periods of out-of-class student work each week (for a total of 200 minutes per week) for 15 weeks equate to one semester hour of credit. Meanwhile, three 50-minute periods of studio work for 15 weeks and a minimum of two 50-minute periods out-of-class student work each week for 15 weeks (for a total of 250 minutes per week) is required to award one credit hour per UIC's credit hour policy. It is important to note that, across all these instructional formats, the expectation for 100 minutes of out-of-class student work per credit hour remains consistent; only the time associated with instruction and/or supervised experience varies. This approach ensures compliance with the university's definition of a credit hour and aligns with the U.S. Department of Education's guidelines.

Instructional Modes

All programs in the College of Engineering are provided through traditional face-to-face on-campus instruction. Students in the Co-op program may spend one or more semesters off campus pursuing engineering internships, but they still complete their coursework on campus through traditional classes.

Grade-Point Average

In order to calculate the grade point average: Take the grades for each course taken and determine the grade points per hour: A=4, B=3, C=2, D=1, F=0. Multiply the grade points per hour for each course to get the grade points for each course. Add the points for each course to get the total number of grade points for the semester. Add the semester hours taken for each course to get the total number of semester hours. Divide the total number of grade points for the semester by the total number of semester hours taken to get the grade point average.

In order to receive a degree from the College of Engineering at the University of Illinois Chicago, a student must present a minimum grade point average of 2.00/4.00 in all work taken in the major. In addition, the student must satisfy the University requirement of a 2.00/4.00 grade point average in two categories: (1) all work taken at UIC; (2) all work taken at UIC and all other two- and four-year institutions combined.

Faculty Workload

The teaching load in most departments in the College of Engineering is four classroom courses per year or faculty who have some research activity (some graduate students and some research publications). Faculty who are active in research have reduced loads of three courses per year.

Faculty who have no M.S. or PhD. Students should teach one additional course per semester in exchange for their lack of individual graduate instruction. Hence, faculty who have no research activities, (no graduate students, no publications, no funding) will be expected to teach six classroom courses per year. Departments have buy-out policies for reducing the teaching loads. A common model is for faculty to pay 1/9 of their academic year salary for every course reduction. In addition, faculty members receive course buy-outs for significant administrative loads, such as Department Heads, Director of Graduate Studies, Director of Undergraduate Studies, Associate Deans, etc.

7. Tables

Below are Tables D1 and D2

Table D-1. Program Enrollment and Degree Data
Chemical Engineering

	Academic Year	Enrollment Status	Enrollment Year					Total Undergrad	Total Grad	Degrees Awarded			
			1st	2nd	3rd	4th	5th			Associate s	Bachelors	Masters	Doctorates
Current Year	25-26	FT	30	38	36	63		167	49	-	27	7	5
Current Year	25-26	PT	2	2	2	15		21	7	-	-	-	-
1	24-25	FT	19	22	48	50		139	50	-	44	7	5
1	24-25	PT	0	4	4	18		26	6	-	-	-	-
2	23-24	FT	23	32	29	68		152	51	-	46	8	5
2	23-24	PT	0	1	4	20		25	4	-	-	-	-
3	22-23	FT	27	31	39	83		180	44	-	51	9	5
3	22-23	PT	0	1	6	13		20	8	-	-	-	-
4	21-22	FT	35	32	50	104		221	51	-	65	16	12
4	21-22	PT	2	4	5	13		24	9	-	-	-	-

Give official fall term enrollment figures (head count) for the current and preceding four academic years and undergraduate and graduate degrees conferred during each of those years. The "current" year means the academic year preceding the on-site visit.

FT—full-time, PT—part-time

Table D-2. Personnel**Chemical Engineering**Year¹: 2025

Employment Category	HEAD COUNT		FTE ¹
	FT	PT	
Administrative ¹	1		1
Faculty (tenure-track) ³	9	2	10.03
Other Faculty (excluding student assistants)	5		5
Student Teaching Assistants ⁴		9	4.50
Technicians/Specialists	1		1
Office/Clerical Employees	2	.50	2.50
Others ⁵			

¹ Data on this table should be for the fall term immediately preceding the visit. Updated tables for the fall term when the ABET team is visiting are to be prepared and presented to the team when they arrive.

² Persons holding joint administrative/faculty positions or other combined assignments should be allocated to each category according to the fraction of the appointment assigned to that category.

³ For faculty members, 1 FTE equals what your institution defines as a full-time load.

⁴ For student teaching assistants, 1 FTE equals 20 hours per week of work (or service). For undergraduate and graduate students, 1 FTE equals 15 semester credit-hours (or 24 quarter credit-hours) per term of institutional course work, meaning all courses — science, humanities and social sciences, etc.

⁵ Specify any other category considered appropriate or leave blank.

Appendix E- Lab Manuals and Textbooks

Textbooks

The textbook titles or reading materials used in lieu of a textbook for all the chemical engineering courses are listed below. The Program Evaluator will have access to hard copies or hyperlinks of the textbooks. When applicable, this material will be organized in a web format presentation system that facilitates efficient examination and cross-checking of the material.

CHE201

Fundamentals of Engineering Thermodynamics, Moran/Shapiro/Boettner/Bailey, 9th Edition (Zybooks version is used)

CHE205

Numerical Methods for Chemical Engineers Using Excel, VBA, and MATLAB, by V.J. Law. ISBN: 978-1-4665-7534-9, CRC Press, 2013.

CHE210

Introduction to Chemical Processes: Principles, Analysis, Synthesis by Regina M Murphy, McGraw-Hill, 1st edition, New York, 2007

CHE230

Materials Science and Engineering: An Introduction, Willam D. Callister Jr and David G. Rethwisch, Wiley

CHE301

Fundamentals of Chemical Engineering Thermodynamics, Themis Matsoukas, Prentice Hall, 2013.

- a. other supplemental materials
 - Chemical Engineering Thermodynamics Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020
 - J. M. Smith, H. C. Van Ness and M. M. Abbott, Introduction to Chemical Engineering Thermodynamics, 7th Edition, McGraw-Hill, 2005.
 - S. I. Sandler, Chemical and Engineering Thermodynamics, 3rd Edition, John Wiley & Sons, Inc., 1999.

CHE311

R.B.Bird, W.E. Stewart & E. N. Lightfoot, Transport Phenomena, 2nd Ed., Wiley (2006)

CHE312

Required (KT) –Transport Phenomena, Revised 2nd Edition by R. B. Bird, W. E. Stewart and E. N. Lightfoot. John Wiley & Sons, Inc (2006).

Required (CT) -- Theodore L. Bergman, Adrienne S. Lavine, *Fundamentals of HEAT and MASS Transfer*, Wiley, Eighth Edition, 2019

CHE313

Wankat, P. C., Separation Processes Engineering: Includes Mass Transfer Analysis, 5th edition, Prentice Hall, 2022. -OR-

Christie John Geankoplis, Allen Hersel, Daniel H. Lepek, Transport processes and Separation Process Principles, 5th edition, Prentice Hall, 2018

a. Other supplemental materials

Seader, J.D et al., Separation Processes Principles, 3rd edition, John Wiley & Sons, New Jersey, 2011.

CHE321

“Elements of Chemical Reaction Engineering”,
Fifth, Fourth or Third Edition, by Fogler, H. Scott

CHE330

Polymer Chemistry, Second Edition / Edition 2 by Paul C. Hiemenz, Timothy P. Lodge, Tim Lodge

CHE341

Process Dynamics and Control, D.E. Seborg, T.F. Edgar, D.A. Mellichamp, and F.J. Doyle, Wiley, 2011.

a. other supplemental materials

- Chemical Process Control Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020
- T.E. Marlin, Process Control: Designing Processes and Control Systems for Dynamic Performance, McGraw-Hill, 2000; B.A. Ogunnaike and W.H. Ray, Process Dynamics, Modeling and Control, Oxford, 1994; J.B. Riggs and M. N. Karim, Chemical and Bio-

Process Control, Ferret Publishing, Austin, Texas, 2016; G. Stephanopoulos, Chemical Process Control: An Introduction to Theory and Practice, Prentice Hall, 1984.

- SOFTWARE: Athena Visual Studio, EXCEL based Thermodynamic Calculator CHE301 Calculator.xlsm; EXCEL based Process Control tool CHE341 Loop Simulator.xlsm

CHE381

Unit Operations of Chemical Engineering, W.L. McCabe, J.C. Smith, P. Harriott, 7th Edition, McGraw-Hill, 2005.

Other supplemental materials: Provided online in the UIC Blackboard course folder to help aid in successfully preparing and carrying out the experiments. Material includes Lab report templates, lab presentation templates, laboratory safety information, technical writing guidelines, error analysis methodology, specific articles and files related to the various experiments including vendor information on the experimental apparatus and components, theory and empirical correlations, engineering charts, and material and physical properties.

CHE382

Unit Operations of Chemical Engineering, W.L. McCabe, J.C. Smith, P. Harriott, 7th Edition, McGraw-Hill, 2005.

Other supplemental materials: Provided online in the UIC Blackboard course folder to help aid in successfully preparing and carrying out the experiments. Material includes; Lab report templates, lab presentation templates, laboratory safety information, technical writing guidelines, error analysis methodology, specific articles and files related to the various experiments including vendor information on the experimental apparatus and components, theory and empirical correlations, engineering charts, and material and physical properties.

CHE396

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012.

a. Other supplemental materials

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009

Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker

CHE397

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012.

a. Other supplemental materials

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009
Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker

CHE422

Bioprocess Engineering: Basic Concepts by Michael L Shuler, Fikret Kargi, Matthew DeLisa, Prentice Hall, 3rd Edition, 2017.

a. Other supplemental materials

Molecular Biology of the Cell, 6th Edition, Bruce Alberts, Alexander D. Johnson, Julian Lewis, David Morgan, Martin Raff, Keith Roberts, Peter Walter, Garland Science, 2014.

CHE425

Drug Delivery: Engineering Principles for Drug Therapy Topics in Chemical Engineering, W. Mark Saltzman, Oxford, 2001.

a. Other supplemental materials

Selected reading materials from high-impact journals and a few textbooks, including but not limited to, Science, Nature, JACS, PRL, Biomacromolecules, ACS Nano, Nano Letters, Journal of Colloids and Interfacial Science.

CHE426

- e) D. J. am Ende and M. T. am Ende (Editors), “Chemical Engineering in the Pharmaceutical Industry: Active Pharmaceutical Ingredients”, 2nd Edition, Hoboken, NJ, Wiley, 2019.
- f) D. J. am Ende and M. T. am Ende (Editors), “Chemical Engineering in the Pharmaceutical Industry: Drug Product Design, Development, and Modeling”, 2nd Edition, Hoboken, NJ, Wiley, 2019.
- g) F. Jameel, S. Hershenson, M. A. Khan, and S. Martin-Moe (Editors), “AAPS Advances in the Pharmaceutical Sciences Series: Quality by Design for Biopharmaceutical Drug Product Development (Vol 18)”, Springer, 2015.
- h) F. Jameel, and S. Hershenson (Editors), “Formulation and Process Development Strategies for Manufacturing Biopharmaceuticals”, Hoboken, NJ, Wiley, 2010.

Other supplemental materials

- c) N. K. Pandit and R. P. Soltis, “Introduction to the Pharmaceutical Sciences: An Integrated Approach”, 2nd Edition, Lippincott Williams & Wilkins, 2012.
- d) N. W. Warne and H-C. Mahler (Editors), “AAPS Advances in the Pharmaceutical Sciences Series: Challenges in Protein Product (Vol 38)”, Springer, 2015.

CHE427

No textbook required; Selected publications, reports and handouts are used.

CHE433

Process Simulation with Aspen Plus Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020.

a. other supplemental materials

- Ralph Schefflan, Teach yourself the Basics of Aspen Plus, 2nd Edition, Wiley. (2016).
- 2. Gavin Towler and Ray Sinnott, Chemical Engineering Design, 2nd Edition, Elsevier Ltd. (2013).
- Lorenz T. Biegler, Ignacio E. Grossmann and Arthur W. Westerberg, Systematic Methods of Chemical Process Design, Prentice Hall International Series, December (1999).
- T. F. Edgar and D. M. Himmelblau, Optimization of Chemical Processes, McGraw-Hill, Inc.(1988).
- A. W. Westerberg, H. P. Hutchison, R. L. Motard and P. Winter, Process Flowsheeting, 1st Edition, Cambridge University Press (1979).
- SOFTWARE: Athena Visual Studio from www.AthenaVisual.com; EXCEL, ASPEN Plus and optionally MATLAB and ASPEN Hysys.

CHE451

“Hybrid Hydrogen Systems for Stationary and Transportation Applications”, Said Al-Hallaj and Kristofer Kiszynski, 1st Edition, 2011, Springer Publishing, ISBN 978-1-84628-466-3.

CHE453

(Required) Bard, A. J.; Faulkner, L. R., *Electrochemical Methods: Fundamentals and Applications*. Second ed.; John Wiley and Sons: 2001; p 833.

c. Other supplemental materials

Bockris, J. O. M.; Reddy, A. K.; Gambr-Aldeco, M., *Modern Electrochemistry*. 2nd ed.; Plenum Press: New York, 2000.

Orazem, M. E.; Tribollet, B., *Electrochemical Impedance Spectroscopy*. John Wiley and Sons: 2008; p 525.

A. Lasia, Electrochemical Impedance Spectroscopy and Its Applications, In: *Modern Aspects of Electrochemistry*, B. E. Conway, J. Bockris, and R.E. White, Eds., Kluwer Academic/Plenum Publishers, New York, 1999, Vol. 32, p. 143-248.

CHE 454

No required textbook; Selected publications, reports and handouts used.

CHE455

“Introduction to Nanoscience”, S.M. LINDSAY. ISBN-10: 0199544212

d. Other supplemental materials

d) Introduction to Nanomaterials and Devices by Omar manasreh.

e) Introduction to Nanoscience and Nanotechnology, Chris Binns

f) Nanomaterials and Nanochemistry, by Bréchignac, C., Houdy, P., Lahmani, M

CHE 457

No required textbook; Selected publications, reports and handouts used.

CHE499

Professional development seminar speaker course. No textbook required. Selected speaker presentations and handouts are used.

Lab Manuals

The titles of the laboratory manuals for the twelve experiments performed in the two chemical engineering laboratory courses (CHE 381 and CHE 382) are listed below. The Program Evaluator will have access to the detailed laboratory manuals (pdf files) in a web format presentation system that will be available during the program review.

1. Batch Distillation Experiment
2. Continuous Distillation Experiment
3. Liquid-Liquid Extraction Experiment
4. Gas-Liquid Absorption Experiment
5. Membrane Air Separation Experiment
6. Heat Transfer to a Constant Stirred Tank Experiment
7. Fluidized Bed Experiment
8. Fluid Flow in Pipes Experiment
9. Humidification Experiment
10. Process Control: Tank Level Experiment
11. Non-Newtonian Viscosity using a Capillary Tube Viscometer
12. Evaporation Experiment